A Guide for New Reactome Curators

Overview of Curation

This set of slides (Appendix C1) provides a high-level overview of the curation process described below.

Quick Guide

This is a list of the most commonly used steps in the annotation process with links to the relevant guides, and is provided as PDF in Appendix C2.

Introduction

The goal of Reactome is to describe the known biochemical details of human biological pathways. A Reactome curator's job is to work with experts in different fields of biology to identify and curate suitable human pathways (see below) by breaking them down into subpathways and reactions and describing them in a format that is compatible with the Reactome data model. The purpose of this guide is to describe each step of that curation process to help the curator fully understand the steps involved. The guide is a useful reference for the experienced curator as it is nearly impossible to remember all of the steps and details of the curation process. Curators are actively encouraged to add to sections of the guide providing useful hints that may be missing for current and future curators. Regular use and addition of material to the guide will facilitate the process of curation and increase annotation consistency. If in doubt about any of these steps...do not hesitate to e-mail the internal list for clarification (and then contribute any helpful info that you get to save someone else the trouble of asking later!)

Here we provide basic guidelines as to how data for relevant pathways should be collected, organized and entered into Reactome via the Curator Tool. The process of converting a biological topic should maintain the referential integrity of the reactions, both newly added as well as those already present in Reactome. It is assumed that you (the reader of the guide) have some familiarity with Reactome, and have read the Reactome papers, including the very important paper describing naming conventions for both events and entities (xc).

Our publications are listed on the Publications page of the Reactome website (https://reactome.org/community/publications).

The entire dataset, website, curator, and author tools can be downloaded from the Reactome download page (https://reactome.org/download-data). Installation and configuration instructions are also provided here (https://reactome.org/download-data/reactome-curator-tool).

If you have any problems with the install or tools please contact us at help@reactome.org.

This rest of this guide is divided into the following sections:

1. Important general guidelines for new curators
2. Choosing a topic/pathway for curation

Important general guidelines for new curators

- Evidence and references:
  - When considering which type of evidence to use when creating events in Reactome, note that you should **NOT** include events based solely on microarray, proteomics or similar high throughput data, unless either 1) an expert author has indicated that they believe the results are correct and represent the current consensus, or 2) The results are a meta-data analysis that combines several independent studies to show the overlap between them. See guidelines describing External confidence criteria here (Appendix C3).
  - In general any time you make a statement in your summation you should include a reference. This is to point the user to the paper you found that proves the statement is true, otherwise the reviewer can't go and check. It also shows that it is an established fact, not your own finding or opinion.
  - Making inferences: If reactions (RLEs) have not been demonstrated experimentally, it is OK to include these as long as somewhere in subsequent events there is some kind of experimental data that suggests your inferred events must logically occur to get you to that point. It's essential that you explain the inference you have made (or others have suggested). You must include citations for the characterized mechanism that allows you to make the inference, i.e. why you believe the events you are describing are similar, despite lack of direct experimental proof in humans. It is best to have an expert's approval of this type of event before

including it. Note that you should NOT create a pathway in non-human species and use that to infer a human pathway. Instead, create individual non-human reactions and use those to infer the human RLEs where evidence for the RLE is missing. See the section below on Creating an inferred event.
    - Tools for literature reference searches: See Litball and slides describing how to use it.

- Updating existing content
    - Be extremely careful changing ANY released instance. This should almost never be done unless new information has become available. Be absolutely sure to check any/all referrers of the instance being changed (entity/event) and make sure that those changes are relevant in ALL referrers. Never change the compartment of an existing entity as that will change the compartment everywhere the entity is used.
    - Changing the species of an event or entity, once it has been committed to gk_central, will change its stableIdentifier. Please contact the managing editor if this has happened. Changing a species once an entity/event has been released will require the deletion of the instance and creation of a new one. The old deleted instance must point to the new one via the deletedInstance that is created. See section on deleting instances in this guide.
    - Adding an entity with a new species to an existing set (say a new pathogen entity is added to a set of entities from other pathogens to be used in an innate immune response event) can be a bit complicated. You have to then make sure that the species setting for the set includes the new pathogen species and that all referrers of that set are adjusted accordingly with the appropriate relatedSpecies). You can do this by looking for each referrer of the changed entity (all entities and ReactionLikeEvents) and then use the curatorTool species QA check to make sure there are no conflicts.

    - If you change an existing event, especially if it has been released, you need to change the summation to reflect this.

    - Substantive changes to released material (physical entities and events) are automatically tracked each release by the UpdateTracker script (see Appendix C4 for a description of changes that are caught by UpdateTracker)

- Naming events and entities:
    - Use the published guidelines (Jupe et al, 2014) on naming events and entities in Reactome. Note that, despite the statement in this paper,

current practice supports using coordinates for *all post-translational modifications, if known.

- Naming of growing complexes: Generally, for serial binding events the 'new' input is named after the existing complex, this gives a better sense that a complex is growing or changing.
- Generally, descriptive words like ‘complex’ or ‘receptor’ are not included in physical entity names except in cases where the common name uses those terms, ie RNA pol II complex, Mediator complex etc
- Use of the word 'activated': Activated has a special meaning in Reactome and should only be used to indicate a change of conformational state from inactive (this word is not used) to active or activated.
- Refer to the more detailed “Guidelines for naming Entities and Events” section in this guide.

References

1. Jupe S, Jassal B, Williams M, Wu G. A controlled vocabulary for pathway entities and events. Database (Oxford). 2014 Jun 20;2014:bau060. doi: 10.1093/database/bau060. PMID: 24951798; PMCID: PMC4064568.

- Naming people and consortia
  - *initial* is a mandatory attribute on a **Person** record, and refers to the first name initial. If no first name is available, enter a dash “-” as the *initial* AND *firstName* attribute
  - if a consortium is cited as an author in a **LiteratureReference**, the consortium name should be listed as the *surname* attribute, and a dash “-” should be listed as the *initial* attribute
- Use of compartments
  - Be careful to use the correct compartment. For example: Nucleus vs. nucleolus. Also, be as specific as possible. If you are talking about an event occurring inside the nucleus, use nucleoplasm not nucleus (Also use cytosol not cytoplasm).

Choosing a topic/pathway for curation

Focus first on:

1. Areas of biology that you know well (or one in which you have good contacts).
2. Pathways that are well characterized at the molecular level AND for which there is considerable HUMAN biochemical data and proteins that are new to Reactome.

Points to consider:

1. Give lower priority to topics that will require a large amount of inferences from other species.
2. Don't try to tackle huge pathways all at once.
3. Break large topics into *manageable* pieces. Experts are generally unwilling to commit to a project bigger than a dozen rxns and it is easy to get lost in the details and lose focus which is costly in terms of time.
4. Create a logical hierarchy for the events in the pathway that you are curating.
5. Work on several (preferably related) small projects simultaneously. Experts often fail to come through at critical times so it is good to have a few projects at different stages to keep work flowing through the pipeline.
6. Check the Editorial Calendar to ensure that your plans don't overlap with another curator. If in doubt, contact the curator. Discuss plans with Editor in Chief and managing editor.

Understanding and using the data model

Data Model Glossary

A detailed understanding of the Reactome data model is important for accurate and consistent curation. The best way to start learning about the different data classes and how to use them is to browse the Data Model Glossary (Appendix C5). Some classes are used infrequently but you still need to be aware of them. Some of the more important data classes are described in the annotation examples below.

More complex examples are cited in the advance curation section as well as the disease annotation guide.

Key Data Classes

PhysicalEntity

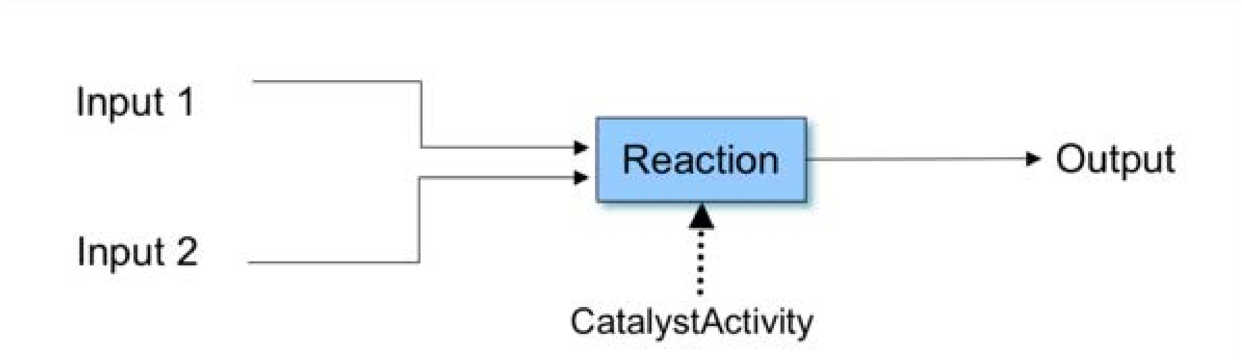
PhysicalEntities can be single entities, such as proteins, small molecules, RNA, DNA, carbohydrates, lipids, or sub-atomic particles. They can also be complexes consisting of a combination of any of the single entities, or polymers synthesized from the single entities. Related entities can be grouped into a set. The use of sets is described below. PhysicalEntities can be inputs, outputs, catalysts, regulators, or requirements in ReactionlikeEvents.

Events

The main concepts in the Reactome data model are Event and PhysicalEntity. At present there are two subclasses of Event: ReactionlikeEvent and Pathway.

ReactionLikeEvent

ReactionLikeEvent (RLE) in Reactome is a single step event in which input entities are converted to output entities by means of structural, chemical or compartmental changes.



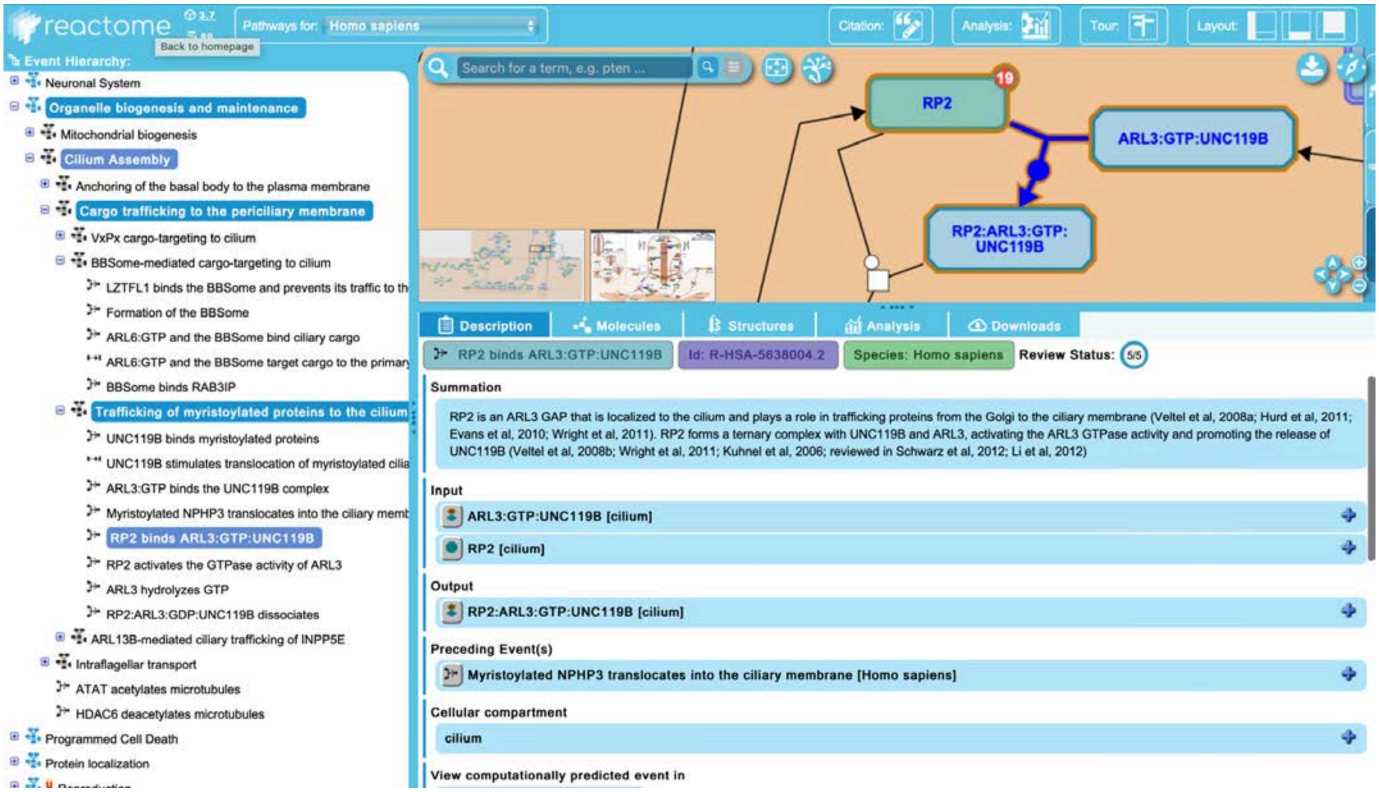
The Reactome data model robustly represents biology. The central concept in Reactome is the reaction, which is used together with pathways, macromolecules, small molecules, complexes, and catalyst activities to represent biological processes. The reaction itself is a single step biological event in which input entities are converted to output entities.
The ReactionLikeEvent subclass has 6 subclasses: Reaction, BlackBoxEvent, Depolymerisation, and Polymerisation, FailedReaction, and CellDevelopmentStep.
ReactionLikeEvents may be linked to other events that precede them, regulate or are regulated by them.
Participants of ReactionLikeEvents – inputs, outputs, catalysts, regulators and requirements – are PhysicalEntities.

Here are some common examples of different types of ReactionsLikeEvents. The compartment of an RLE that involves participants in more than one compartment (or movement of participants across multiple compartments), should reflect the compartments of all of the participants (inputs, outputs, catalyst).

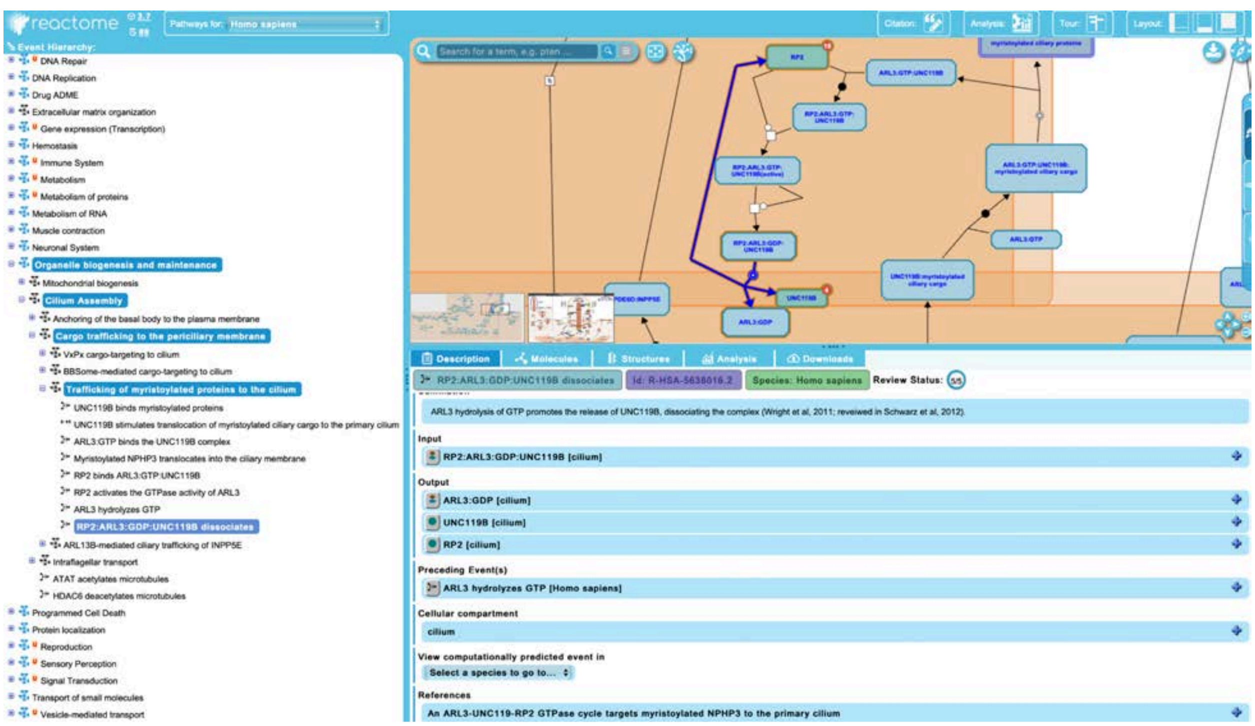
Reaction

A Reaction is an event that converts inputs to outputs in a single step. Any enzymatically-catalyzed event, or transport of physical entity to adjacent cell compartments, or protein-DNA binding or protein-protein complex formation or disassociation, for example, can be annotated as a reaction:

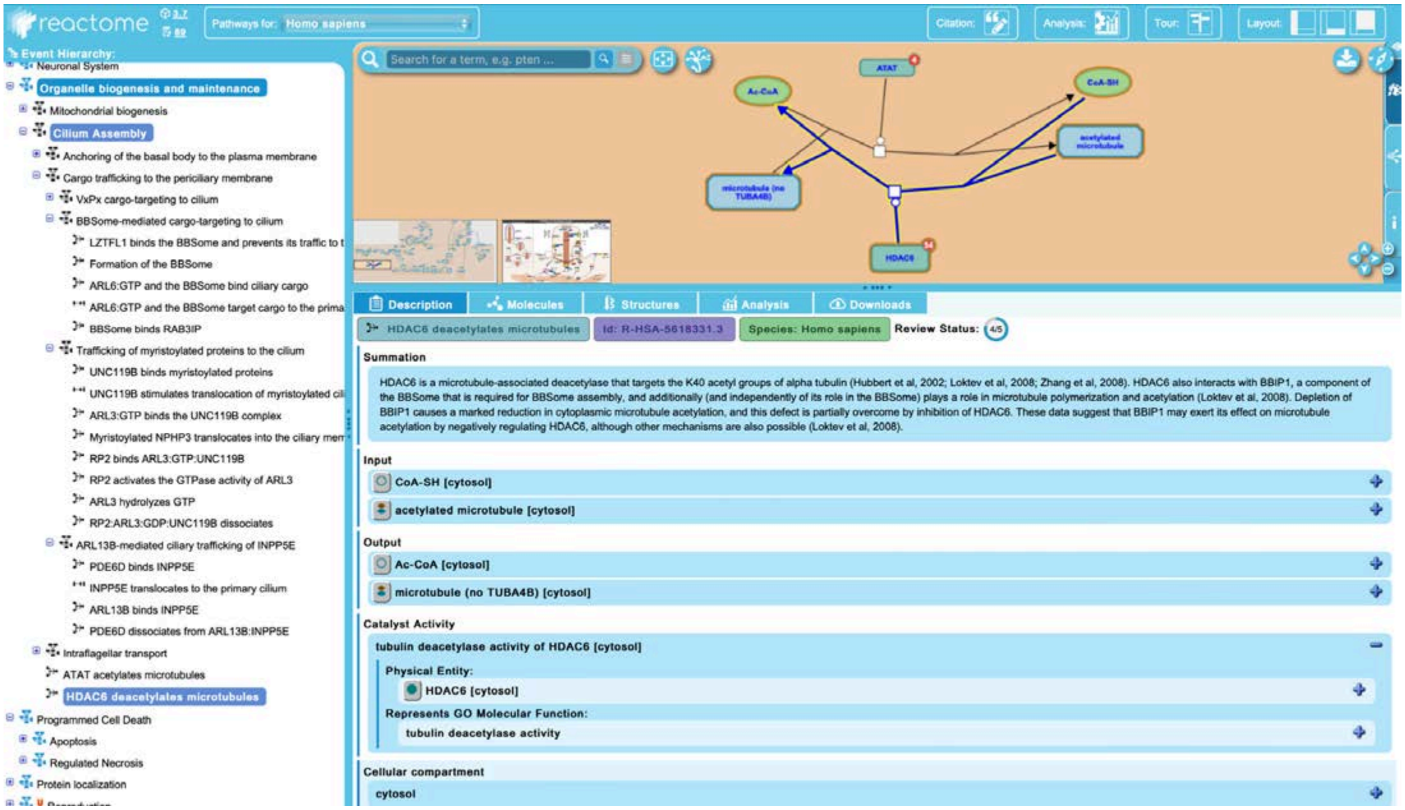
Complex Association:



Complex Dissociation:



Enzyme-catalyzed biochemical reactions:



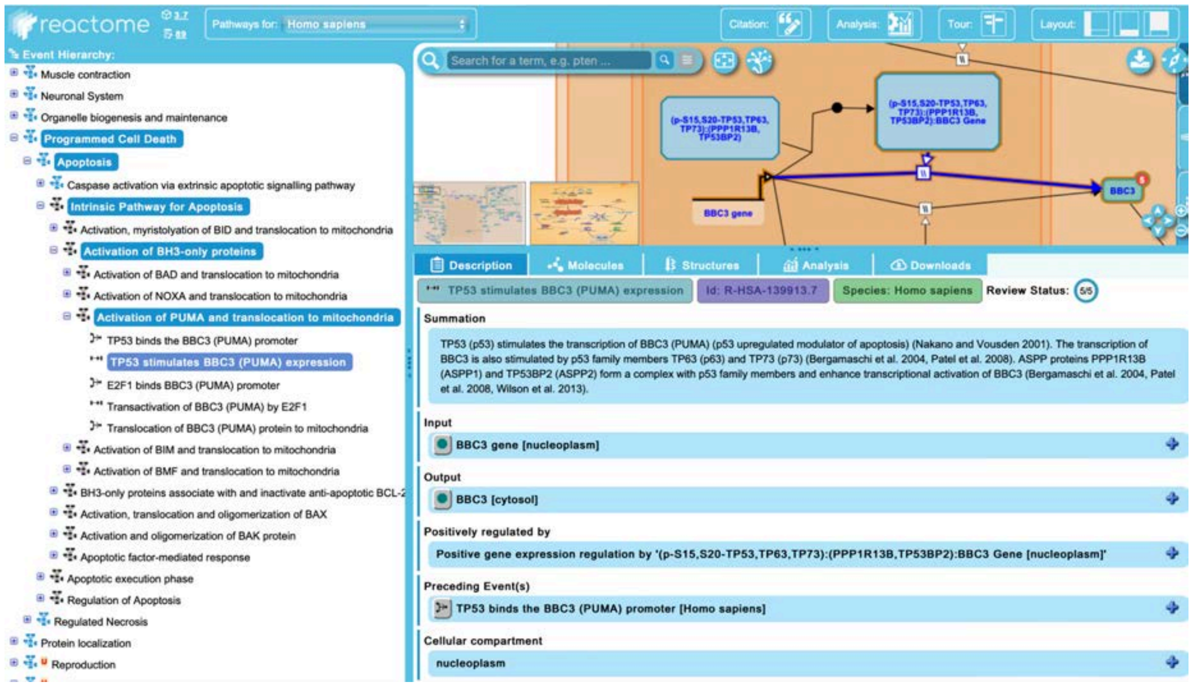
The compartment of a Reaction that involves participants in more than one compartment (or movement of participants across multiple compartments), should reflect the compartments of all of the participants (inputs, outputs, catalyst, regulator).

BlackBoxEvent

A BlackBox event converts inputs to outputs of a multiple step biological process in which intervening steps are not annotated because:

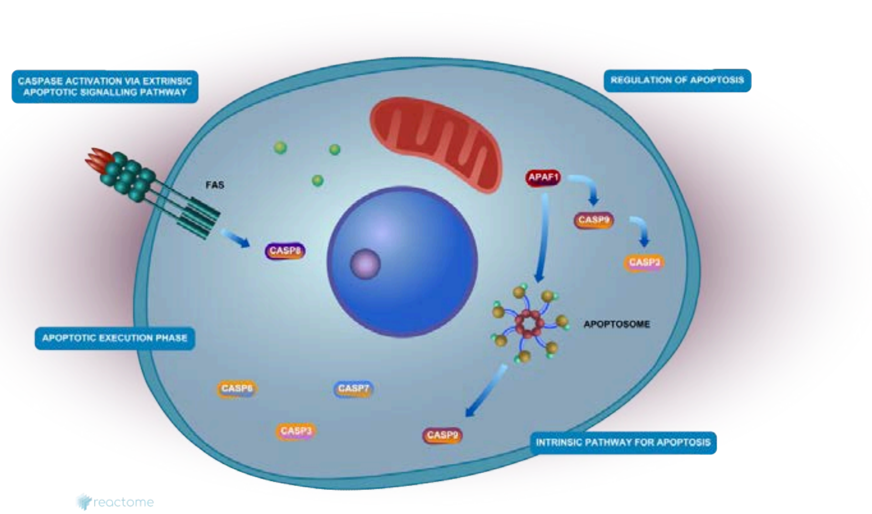
1. We don't know all of the intervening steps.
2. The intervening steps are known but we do not want to curate them for the purposes of the module. Examples are:
    1. The transactivation of a gene leading to production of a protein....we don't want to include all the steps of translation of that protein
    2. The degradation of a protein.
    3. Transport from one cellular organelle to another.

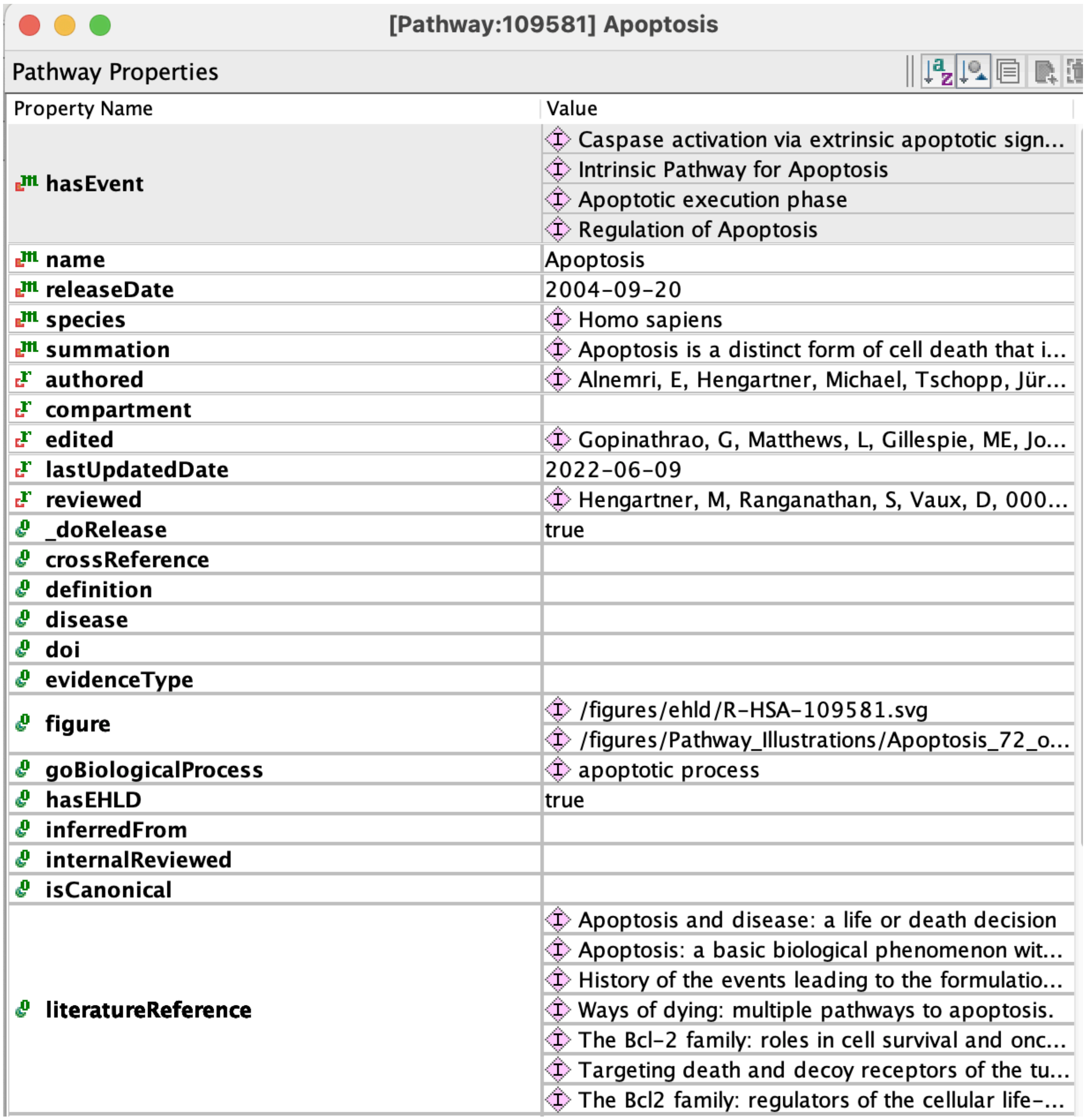
The compartment of a BlackBoxEvent that involves participants in more than one compartment (or movement of participants across multiple compartments), should reflect the compartments of all of the participants (inputs, outputs, catalyst).



FailedReaction
See Disease section of the curator guide here (Annotating loss-of-function reactions).

Pathway
Pathway in Reactome is a multiple step event which can be broken down to subpathways and/or ReactionLikeEvents. A concrete curated example is the Reactome Apoptosis pathway which consist of the subpathways "Caspase activation via extrinsic apoptotic signalling pathway”", "Intrinsic pathway for Apoptosis", "Apoptotic execution phase" and "Regulation of Apoptosis".





[Pathway:109581] Apoptosis

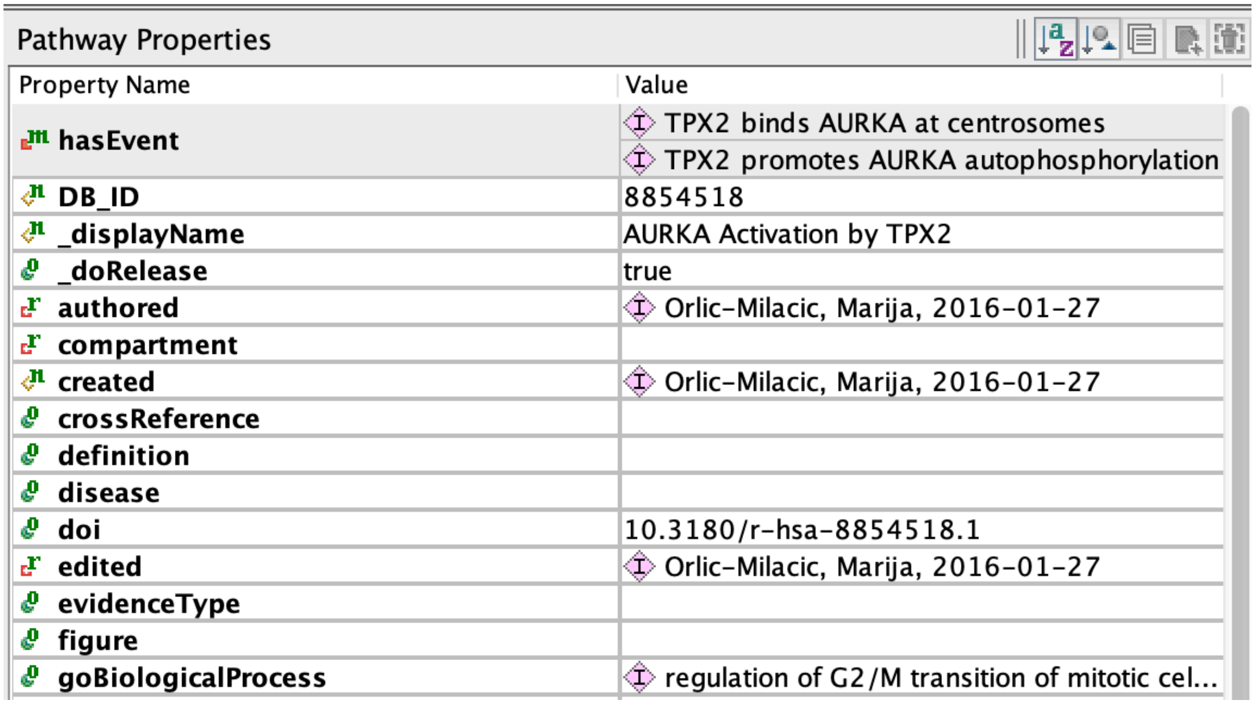
Pathway Properties

Property Name	Value
hasEvent	Caspase activation via extrinsic apoptotic sign...
	Intrinsic Pathway for Apoptosis
	Apoptotic execution phase
	Regulation of Apoptosis
name	Apoptosis
releaseDate	2004-09-20
species	Homo sapiens
summation	Apoptosis is a distinct form of cell death that i...
authored	Alnemri, E, Hengartner, Michael, Tschopp, Jür...
compartment	
edited	Gopinathrao, G, Matthews, L, Gillespie, ME, Jo...
lastUpdatedDate	2022-06-09
reviewed	Hengartner, M, Ranganathan, S, Vaux, D, 000...
_doRelease	true
crossReference	
definition	
disease	
doi	
evidenceType	
figure	/figures/ehld/R-HSA-109581.svg
	/figures/Pathway_Illustrations/Apoptosis_72_o...
goBiologicalProcess	apoptotic process
hasEHLD	true
inferredFrom	
internalReviewed	
isCanonical	
literatureReference	Apoptosis and disease: a life or death decision
	Apoptosis: a basic biological phenomenon wit...
	History of the events leading to the formulatio...
	Ways of dying: multiple pathways to apoptosis.
	The Bcl-2 family: roles in cell survival and onc...
	Targeting death and decoy receptors of the tu...
	The Bcl2 family: regulators of the cellular life-...

Mandatory, Required, and Optional properties (fields) for data classes

All the classes of data in Reactome have a number of properties (fields of information) that MUST be filled in before they can be 'released', i.e. made visible to the public. They also have required (where relevant) fields, optional fields, and non-editable fields.

When viewing an instance of a class in the curator tool, its properties all have a symbol that indicates the type, either a red box with a letter superimposed, or a yellow diamond with n superimposed. The yellow diamond with n indicates fields that are automatically completed. The red squares indicate, by letter,



- m - Mandatory - you must provide this information.
- r - Required - you must provide this information if it is possible - a good example of this is species, required for defined sets of proteins, but not required for defined sets of small molecules.
- o - Optional - not available for every instance, so cannot be “Mandatory” or 'Required' but in some circumstances may be a procedural requirement, e.g. inferredFrom must be completed for human reactions that are inferred from a model organism.

Guidelines for Naming Entities and Events

The guidelines were first set out in this paper (Jupe et al, 2014) describing “A controlled vocabulary for pathway entities and events”.
Please read the paper to learn how to name EWASs and events.

Here are some guidelines for the different classes of entities:

EWAS

Names are based on the UniProt gene symbol (the gene name), which is typically the same as the HGNC symbol. Post-translational modifications are represented by a prefix, cleavage events or differences from the canonical form are represented by a suffix.

Small Molecules

Please use the short name recommendations for small molecules found here (Appendix C6).

Complexes

These guidelines are an extension of the ideas for a pathway controlled vocabulary set out in this paper (Jupe et al, 2014). You should familiarize yourself with the rules for naming EWASs and simple entities.

In a typical scenario the names of sets and complexes are simply a joined list of the entities they contain, which in the Curator Tool can be copied to the name field from the hasComponent or hasMember fields.

Complexes should be named after the names of their components, separated by the colon symbol. Some examples: GRB2:SOS1 IL3:IL3RA:IL3RB:JAK2
IL4:p-Y-IL4R:JAK2:p-Y-IL2RG:JAK3:p-Y-JAK1:p-Y705-STAT3,p-Y641-STAT6
RXFP4:RXFP4 ligands

For cardinality (number of copies) use 2x, 3x etc. in front of the name. Is it OK to use dimer, tetramer etc. to indicate multimerization but NOT hetero- or homo- or the word complex.

When a complex is extended by the addition of further components, the name of the complex produced should show the 'added' component on the right end of the complex name.

For complexes larger than 4 components, it is acceptable, though less desirable, to use a descriptive name from the literature IF widely used, e.g. high-affinity IL3 receptor.

What about synonyms? You can add as many extra names as you like, listed after the 'correct' name in the name slot.

Authors/reviewers should NOT be allowed to dictate the name used. We have a controlled vocabulary designed to consistently name entities throughout the Reactome database. Authors may not like the name we use but that is usually because they have become used to a different name from the literature. Unfortunately there are often many alternative names in the literature and there is rarely 100% agreement on the best name to use. We can't make everyone happy, but we must pick one name from the alternatives. The only way we can justify not using names suggested by (highly biased) authors is to be 100% consistent and always use our controlled vocabulary!

Remember - if you diverge from the simple list of colon separated entity names, you **must** check your name is not the same as existing complexes!

Sets

Set names: Small sets (less than 5 members) should be named as a list of the members e.g. IL3,IL3RA,CSF2RB.

Candidate sets should include the candidates within parentheses, e.g. CCR2,(CCR4,CCR6) is a candidate set where CCR2 is a member and CCR4 and CCR6 are candidates.

For large sets (5 or more) it is acceptable, though less desirable, to use a descriptive name from the literature IF widely used, e.g. XXXXXX or a 'function-defining' name that indicates the usage of the set, e.g. 'RXFP4 ligands'. If the large set includes an entire family of proteins, it is OK to use the name of the family, e.g. CCRs (note the s suffix to indicate that there is more than one entity represented), there's no need to add the word family. If the set represents a range of members, use hyphen to indicate the range e.g. CCR2-5. For complicated sets that have diverse members, a descriptive name based on function may be the best option. Try to describe the attribute that is the basis for grouping the entities, their shared property or function, e.g. SLC receptors, VAV2-activated RhoGEFs.

What about sets that have a common name stem, but are not so simple? It's tempting to try and come up with complicated abbreviations around the gene name common prefixes, e.g. for the real example set of genes PLA2G1B PLA2G2A PLA2G2D PLA2G2E PLA2G2F PLA2G5 PLA2G10 PLA2G12A, the name might be PLA2G1B,G2A,G2D,G2E,G2F,G5,G10,G12A but don't - this is too hard to interpret.

Remember - if you diverge from the simple list of comma separated entity names, you **must** check your name is not the same as existing sets!

Sequence variants

Reactome's approach to annotating sequence variants of proteins is described in detail in the section on Disease Annotation, below (Annotating Disease Entities), and is informed by the Human Genome variation society description (http://www.hgvs.org/mutnomen/examplesAA.html) of sequence changes at protein level.

References

1. Jupe S, Jassal B, Williams M, Wu G. A controlled vocabulary for pathway entities and events. Database (Oxford). 2014 Jun 20;2014:bau060. doi: 10.1093/database/bau060. PMID: 24951798; PMCID: PMC4064568.

Using Data Inferred From Other Species

Sometimes there is not sufficient experimental support for a set of reactions in one species, simply because the experiments have not been done in the target species. The InferredFrom slot addresses this problem. If the experts (perhaps also the curator) knows that this reaction really occurs in the target species, the reaction or reactions can be constructed in the experimentally supported species and then the InferredFrom slot can be filled in for the reactions in the target species. In this case the experimental species *becomes* the support for the target species. See a more complete description here under "Creating an inferred event".

Providing evidence for assertions in Reactome

Principles for assigning literature references for reactions

The references associated with a reaction MUST provide direct experimental evidence for the occurrence of that reaction in the species you are annotating (i.e. human for human Reactome). If there is no direct experimental evidence in humans, but in other species, then you need to create an inferred human reaction as described in the section "Creating an inferred event", below. See Guidelines on evidence confidence here (Appendix C3).

Other evidence types

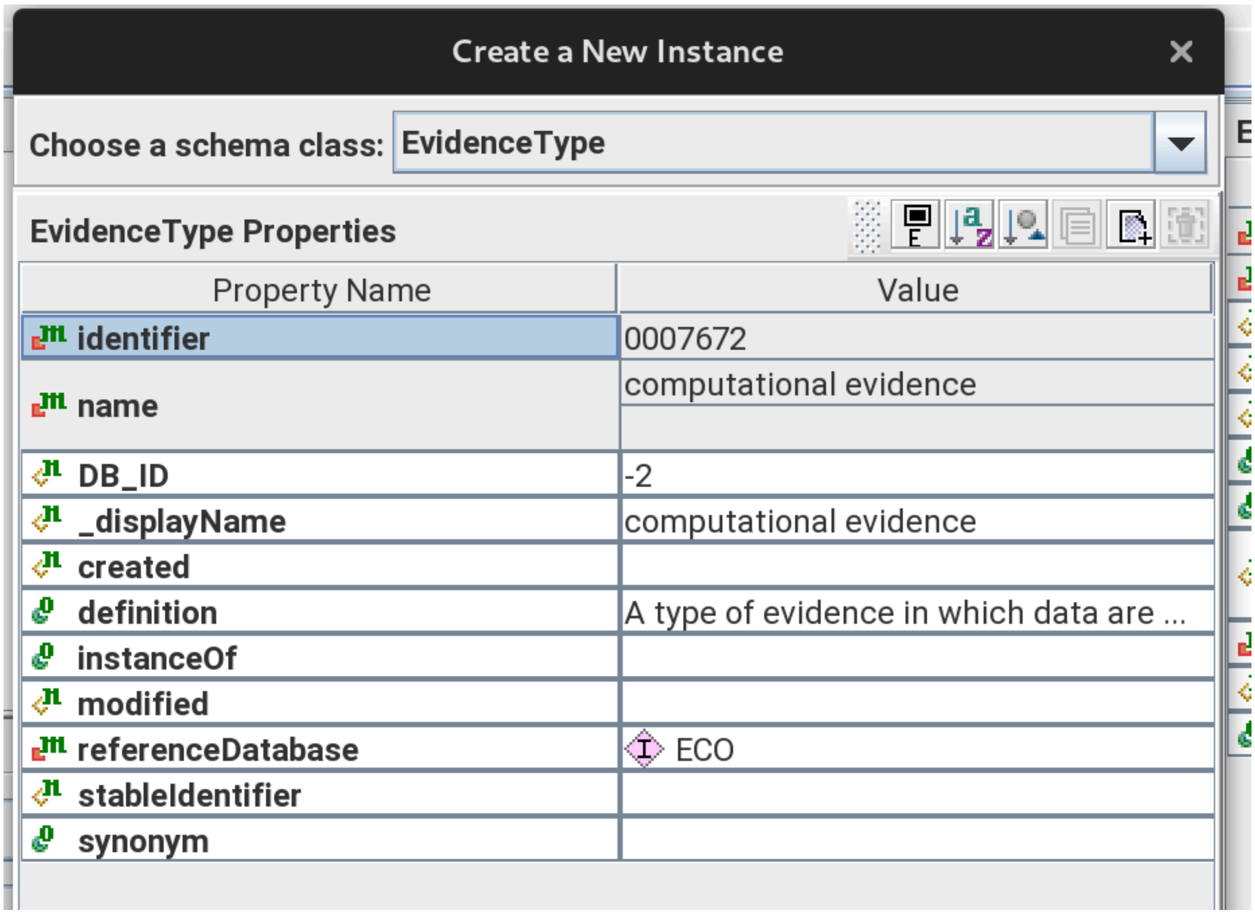
If applying the fundamental "direct evidence" rule leads to a case where a pathway just misses one or few reactions to be complete, it is possible to add missing reactions with less direct evidence. Such reactions MUST have the evidenceType slot filled and the summary MUST contain rationale on why the reaction was added, despite the weaker evidence.

It is also possible to use black box reactions when specific evidence is missing, in the following cases:

- localization reactions where the means of transport is unknown but the localization itself has direct evidence (e.g. microscopy)
- enzymatic reactions where the catalytic activity is unknown but the end product(s) themselves are supported by direct evidence (e.g. mass spectrometry, crystals)

How to assign an evidence type

1. In the curator tool, view your reaction and focus on the **evidenceType** slot
2. Set its value from a list of already existing values in your project, or values existing in the Reactome database, like with other slots
3. If you think a value is missing in Reactome database, create a new one by using a term from the Evidence & Conclusion Ontology (ECO) at https://www.ebi.ac.uk/ols/ontologies/eco
   In particular, the **name, definition, identifier, and referenceDatabase** slots need to be filled (see picture)



Citing biological databases

Generally, there shouldn't be a need to create ReferenceDatabase instances. Contact dev@reactome.org if you think you need a new ReferenceDatabase instance.

- Attributes: AccessURL- template used to form the URL that will be used to link to a particular record in an external database. Generally, this should not need to be touched. Contact the developers if you think any changes are necessary.
- URL- URL for the site which gives "summary information" about the database including a description of what information the db contains. This attribute should not need to be modified manually.

In certain cases, there are external databases that include information that is relevant to an annotated pathway.

One example is rare disease database such as RettBase, which describes Rett syndrome-related mutations in MECP2, FOXG1 and CDKL5: http://mecp2.chw.edu.au/. This database provides information about variants described in the disease pathway: Loss of function of MECP2 in Rett syndrome.
We are not able to cross-reference RettBase for individual MECP2 variants in the way that we cross-reference COSMIC for cancer variants because RettBase is not organized in a way that allows us to do this. However, we can mention RettBase in the pathway summation and cite their most recent publication (https://www.ncbi.nlm.nih.gov/pubmed/28544139) as a literatureReference for the Loss of function of MECP2 in Rett syndrome pathway. In addition, we can also create a second Publication instance of class "URL" and use this to cite the database itself.

NOTE that you should NOT cite disease databases at the reaction level **unless these disease databases capture biochemical/functional information.

Using the curator tool

Downloading, Installing, And Maintaining The Curator Tool

For instructions on downloading and installing the curator tool, see the public website here (https://reactome.org/download-data/reactome-curator-tool)

Windows:

Instructions

- After downloading, double-click install.zip to unzip it to an empty folder. This folder should be named "ReactomeCuratorTool" though any name can be used. Please use the latest version of WinZip if you use this application for unzipping.
- After unzipping, an application called ReactomeCuratorTool.exe should be found in the named folder. Double click this application to launch the Reactome Curator Tool.

Notes

- You need to install Java 11. You can download one from OpenJDK.

Linux:

Instructions

- After downloading open a shell, and cd to the directory where you downloaded the installer.
- At the prompt, type unzip install.zip to unzip the installer.

- Use ./ReactomeCuratorTool.sh to launch the Reactome Curator Tool.

Notes

- You need to install Java 11. You can download one from OpenJDK.

Mac:

Updated March 3, 2024 - Mac users now uses the Linux version of the curator tool. It is just too complicated to hack the native Mac installer. To run the curator tool under Mac, please following the following:

1) Download the installer for linux using a browser. After unzip it (it may be automatically unzipped after download), move it out from your Download folder to other place (e.g. Desktop).

2) Open a Mac OS terminal by double clicking “Terminal” in the Applications -> Utilities folder

3) Enter the following in the terminal to check if you have Java installed:

java --version

- If this fails ("The operation couldn’t be completed.", or something like that), install a Java JRE or JDK from https://adoptopenjdk.net (use Java 11, JDK. Make sure you choose the correct CPU architecture.)

4) At the same terminal after validating Java 11 is installed, issue command:

bash ReactomeCuratorTool.sh.

- The curator tool window should popup as usual.

5) Still another alternative for Mac users. After downloading and unzipping the Linux installer (as in step 1 above), assemble all the files into a single folder and name that folder CuratorTool (or any other unique and easy-to-remember name), and put that folder in a convenient directory on your Mac. (PD leaves it on the desktop). To launch the curator tool, open a terminal window on your Mac and cd to the CuratorTool folder:

cd desktop/CuratorTool

then, still in the terminal window:

bash ReactomeCuratorTool.sh

(The terminal window remains open while the curator tool is running.) The CuratorTool should open as usual. After you "quit" the CuratorTool, enter "exit" in the terminal window and quit the terminal window.

Adjusting the Display for the Curator Tool for Different Screen Sizes

While adjusting the curator tool for a new monitor, I had to adjust the font size and window size of the curator tool. I found some helpful internet info and used the "-Dsun.java2d.uiScale=" option in the command to start the tool. For example "-Dsun.java2d.uiScale=2" doubles the size of the curator tool window and font, while "-Dsun.java2d.uiScale=1.5" increases everything by 1.5X.

The command for linux in ReactomeCuratorTool.sh is:

java -Xmx8G -Dfile.encoding=UTF-8 -Dsun.java2d.uiScale=2 -cp ReactomeCuratorTool.jar launcher.Launcher

(Possibly also prepending GDK_SCALE=2 may make a difference)

If you are in Windows, you can put this command in a text file and save the file as a .bat file (executable file). Then you just need to double click the .bat file to start the tool.

If the .bat file is located outside the curator tool folder (e,g, on the Desktop), I found I needed to add line to tell the command line where to find the jar file and everything else, so the .bat file contains:

cd \Users\yourhomedirectory\path\to\curatortool

java -Xmx8G -Dfile.encoding=UTF-8 -Dsun.java2d.uiScale=2 -cp ReactomeCuratorTool.jar launcher.Launcher

Curator tool default settings

All routine curation uses a connection between your local curator tool and the gk_central database. These are the settings for that connection, which you make via the "database info" tab of "curator tool preferences". Other values of these settings can be used for various specialized purposes, but these are outside the scope of regular curation and can be dangerous.
database host: curator.reactome.org
database name: gk_central
database port: 3306
user name: [your user name]
user password: [your password]
Guanming Wu assigns usernames and passwords for new curators; yours will persist unchanged thereafter.
Some institutional firewalls see connecting to port 3306 of a remote server as a security risk and will block your access to it. If this happens, you, Guanming, and your institutional IT people will need to negotiate a workaround, e.g., whitelisting this port at curator.reactome.org for your individual computer.

Curator tool user interface

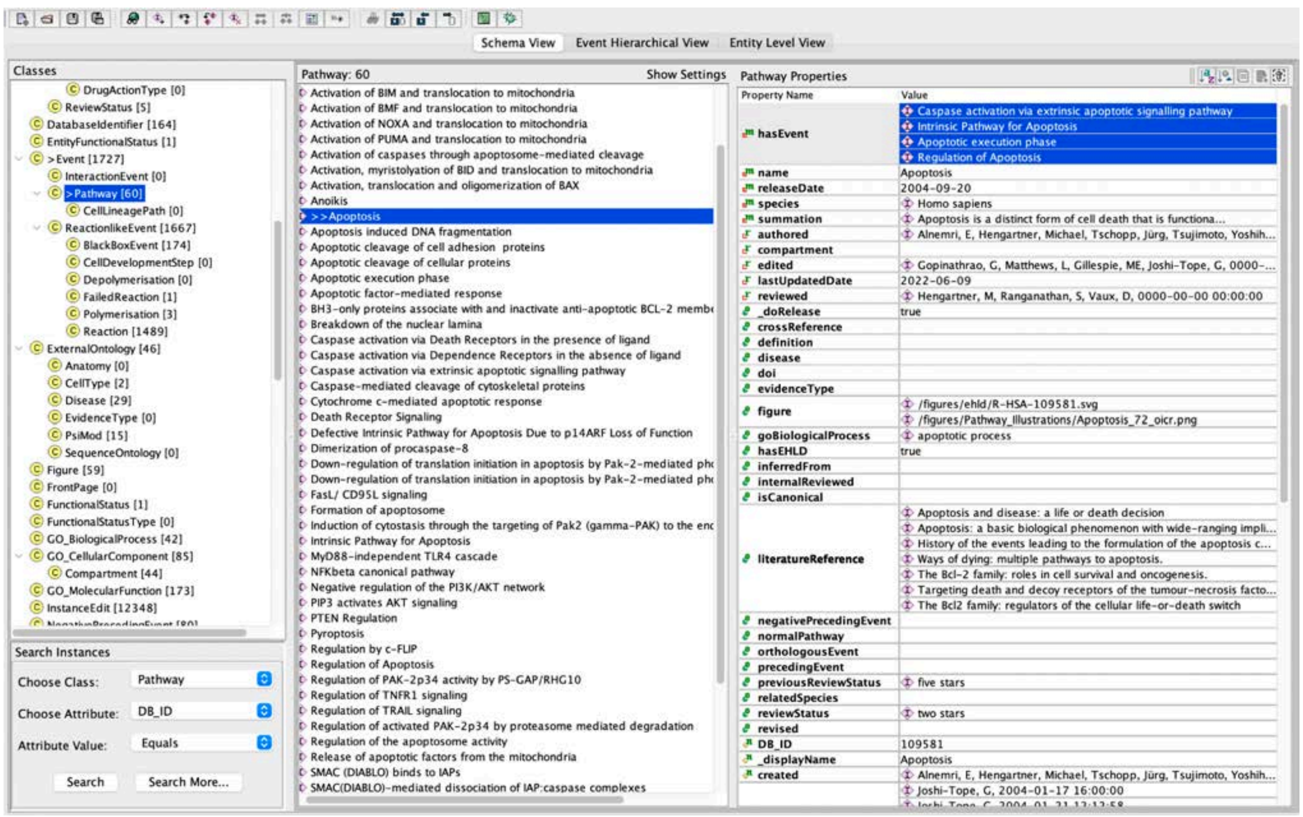
The curator tool has three panes to view the content. These different views can be switched using the tabs in the upper left hand corner of the curator tool window.

The Menu Bar

The menu bar contains a number of shortcut buttons, a scroll over message will tell you the function of each button.

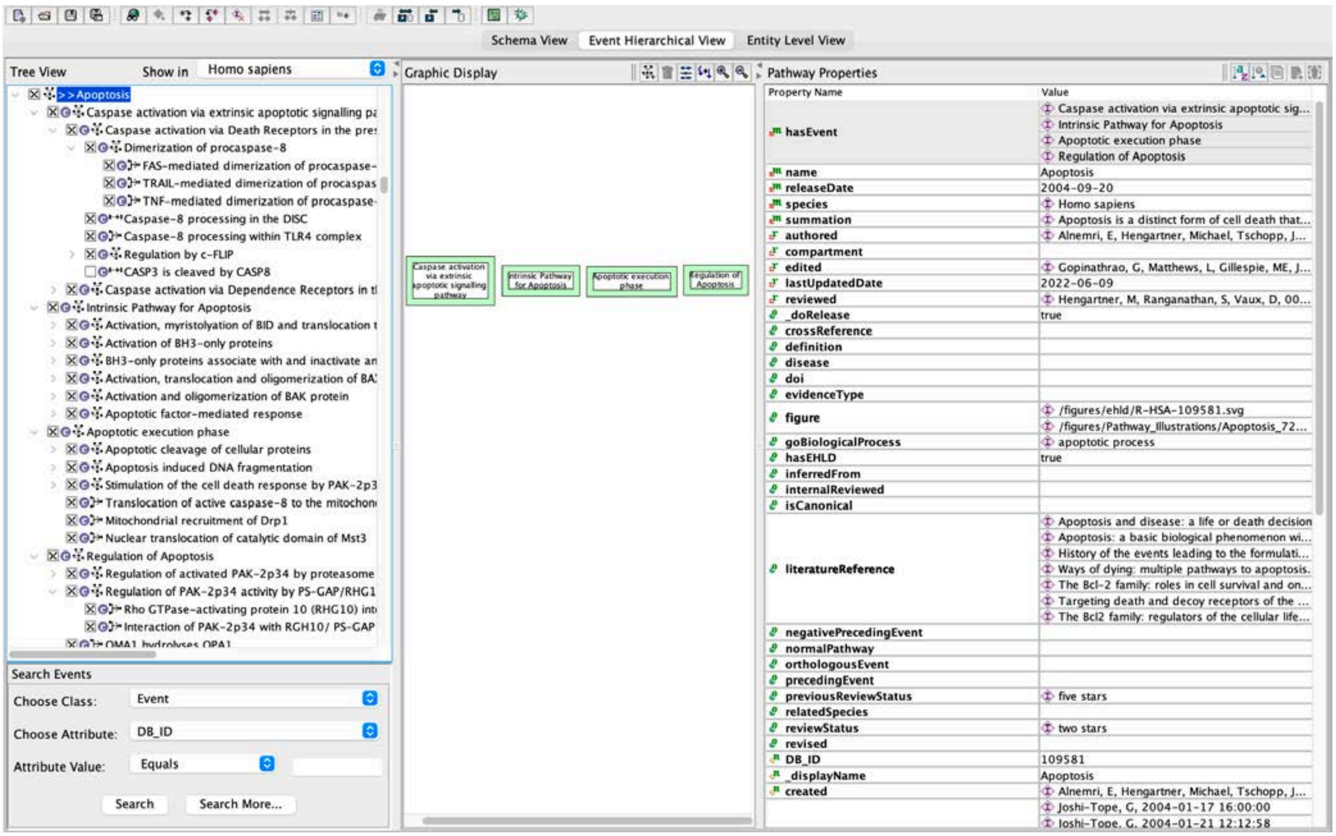
The Schema View

The Schema view provides a hierarchical list of the data classes. In this view you can search, open, edit and create instances of these classes. This panel is used most often for creating new items, such as an EWAS derived from a ReferenceGeneProduct.



The Event View

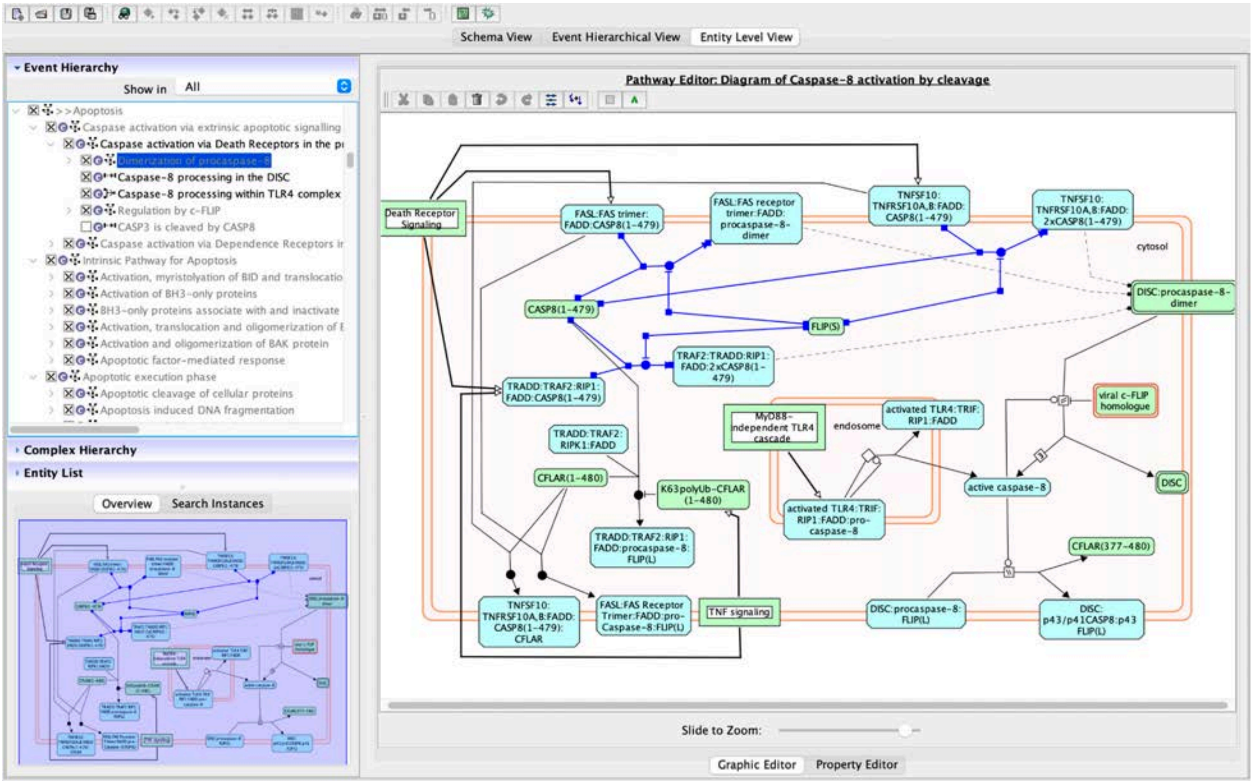
This view is where most curation is done. The event view displays a hierarchical, alphabetical list of the events in the local project on the left, a graphic in the middle that shows the relationship between objects for selected events, and on the right the details of selected events. Unfurling a pathway by clicking on the + symbol to the left of its name in this view reveals all of its component Reaction events. DON'T CONFUSE these symbols for the check box - clicking on an unchecked box marks the event as ready for release onto the public website; clicking on a checked box has the opposite effect. For pathways the check box selects the ENTIRE pathway, doing this by mistake can be time-consuming and tedious to reverse so be warned!



The Entity Level View (ELV)

The ELV is where pathway diagrams are created and subsequently represented. In this view the pathway is layed out graphically showing the inputs, outputs, catalyst where appropriate, and regulatory molecules, each in the correct cellular compartment.

For the usage please see the chapter on “Drawing a pathway diagram”, below.



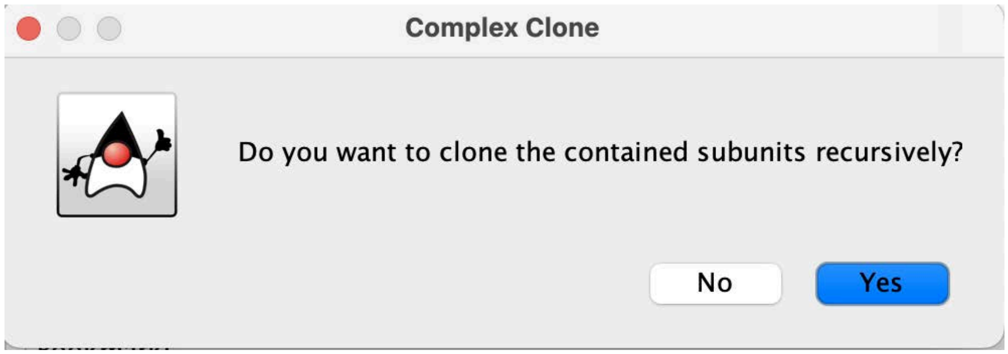
USING THE PRECEDINGEVENT SUGGESTION FEATURE IN THE CURATOR TOOL

See slides describing this process [here](#) (Appendix C7).

CLONING TOOL

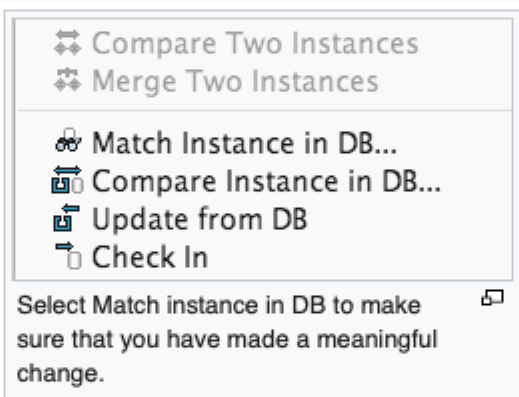
BE CAUTIOUS WHEN USING **THE CLONING TOOL.** Use it carefully and thoughtfully. The use of the cloning tool is restricted to making a copy of something that will be changed. When using the tool you must be absolutely sure that at least one of the defining attributes of that entity is different from the original.

Be particularly careful when cloning a complex or a set and are faced with the option to “clone contained subunits recursively” - the answer to this is almost always “no” since the danger of creating hard-to-track duplicate entities is high.



Once the cloning is done, immediately go back and change the defining attributes that need to be changed.

Once changed use the Select Match instance in DB to make sure that you have made a meaningful change.

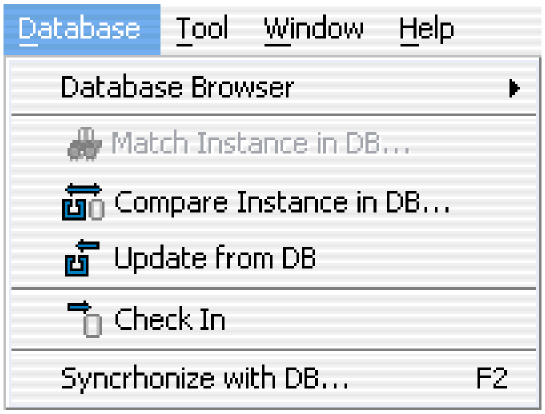


"Match instance in DB" feature to ensure that the instance is really different from the entity in gk_central. Do this by right clicking on the newly created instance.

Synchronizing local projects with the database

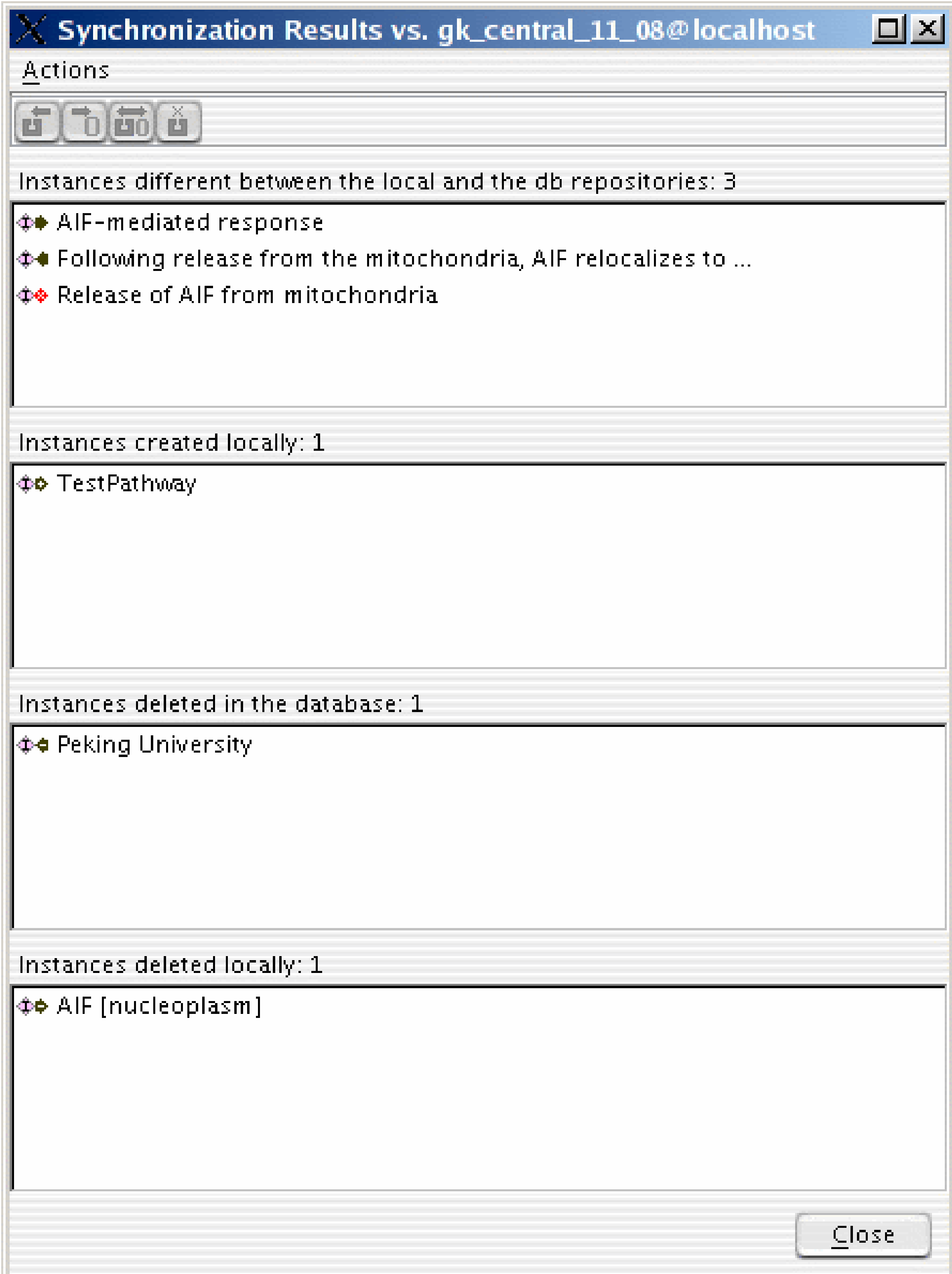
Synchronize at least once a day. It is good practice to synchronize your local project with the gk_central database to ensure that your work is not lost should the local copy become corrupted or lost due to hard drive failure. It is useful for other curators who may be able to reuse EWASs or other instances that you have created. It is perfectly acceptable to 'check-in' incomplete work unless it is marked as 'doRelease'.

To keep the data in the opened project and the database repository consistent, you need to synchronize the local project with the database periodically. Several actions are available for you to do synchronizing. The following is the "Database" menu for doing database-related actions:



"Match Instance in DB..." can find a matched instance in the database repository for the selected instance in the local repository. A matched instance has the same defining attributes as the local one. If a matched instance can be found, you can merge the local one to the database one. "Compare Instance in DB..." can compare a selected instance with the corresponding one in the database. Corresponding means the same DB_IDs. So you should not compare one instance checked from database A to an instance in database B because the same DB_ID might be assigned to different instances. "Update from DB" will update a checked out instance from the database. "Check In" is used to check in newly created or modified instances to the database. It is recommended that you should use "Compare Instance in DB..." first before "Update from DB" or "Check In". After a new instance or modified instance is checked into the database, the marker ">" will be removed to indicate that the local and the database copies are the same.

The "Synchronize with DB..." menu is used to check all instances in the selected schema class in the schema view or in the whole opened project if no class is selected. There are four possible inconsistency categories between the two repositories: instances different between the local and the db repositories resulting from instance modification either locally or in the database, instances created locally, instances deleted in the database by you or others, and instances deleted locally by you. You can choose appropriate action for the selected instances in different categories. Please be aware that if you do a multiple selection from different categories, the enabled actions are applied to all selected categories. For example, actions "Update from DB" and "Commit to DB" can be applied to instance "AIF-mediated response" in the first category, but only "Commit to DB" can be applied to instance "TestPathway" in the second category. If you select both "AIF-mediated response" and "TestPathway", only "Commit to DB" is enabled and "Update from DB" is disabled. Double clicking an instance in the first category will popup a comparison dialog of the local and database instances, while double clicking an instance in other categories will show the contents of the clicked instance. To deselect a single instance, hold the control key and click the selected instance.



There are three different cases in the first category, instances different between the local and the db repositories: instance is modified in the local project but not in the database, instance modified in the database but not in the local project, and instances are modified in both the local and database. The user can commit changes to the database or overwrite changes by updating from database for the first case. The user cannot commit changes for an instance in the second or third case, but can update from database for both cases.

The following is the list of icons used in the synchronization dialog:

Icon	Applicable actions	Note
	Update from DB, Commit to DB, Show Comparison	New changes in local instance only
	Update from DB, Show Comparison	New changes in database instance only
	Update from DB, Show Comparison	New changes in both local and database instances
	Commit to DB	New local instance
	Update from DB, Commit to DB	Local instance is deleted but still exists in the database
	Update from DB, Commit to DB, Clear Record	Database instance is deleted but still exists in the local

WHAT DO I DO IF THE CURATOR TOOL FLAGS INSTANCES AS BEING DUPLICATED IN GK_CENTRAL

You will need to evaluate each instance that is flagged as being a duplicate and compare it to the instance that is flagged in gk_central. This is easiest to do before synchronizing your project with the database as described below:
Search the local project for all database instances that contain ‘-‘, then highlight each one in turn, right-click to open a pop-up menu, and choose “match instance in DB”. Any results returned by that query are already-existing instances that, by the rules of gk_central, are duplicated by the new local instance. Mostly, that’s right and you should accept the option offered by the form, to replace the local instance in the local project with the already-existing one from gk_central. That cleanly gets rid of the duplication and preserves all references correctly in the local project and in gk_central. Sometimes (rarely) the form is wrong because, despite identical defining attributes, two instances are genuinely different, e.g., if a person instance has already been created for J(ohn) Doe and you have now created one for J(ane) Doe. In this case, you should refuse the offer made by the form and should make a note to force the new instance into gk_central at synchronize-and-commit time.

What to do if the curator tool reports an exception when committing a project

In the event that an exception is reported by the curator tool during the commit to gk_central, this may be the result of a dropped connection with the database. Try to rerun the synchronization. If any conflicts in instances are revealed (this should be rare), send Guanming the project and explanation of the problem. If, after resynchronizing and attempting to commit again, if you still get an exception, verify that you have a connection to the database. If no connection is available, try the commit again when the connection is reestablished. Contact Guanming if there are any problems.

Problem of commit size

Usually as a curator in a team, you want to make many small commits instead of one big chunk. The reason is to minimize the chance of other curators changing the database entities while you are working on them. Resolving these conflicts is much easier with small project files

Solution: selected rewrite

After checking a project file for correctness the Reactome editor synchronizes the local project with the database. Here conflicts with commits of other curators become visible, and a list of DB_IDs that are in conflict is presented. Instead of rewriting the whole project file, the external curator can follow these steps:

-
- Refresh your view of the gk_central (menu Database-->Database Browser-->...)
- Select the conflicting entities, one by one, in your project file (not in the database)
- Right-click on an instance with the conflict, select "Update from DB"—this resets the instance to the version in the current database
- Re-enter your work on that reset instance
- save the project file with a different name
- Repeat above three steps for each instance with a conflict.
- Commit project.

The Annotation Process for Traditional Reactome Pathways

Note that this section refers to “traditional” Reactome pathway. The annotation of a special class of pathways referred to as Cell Lineage Paths is described in detail in a separate section below.

Create the framework for the pathway(s) you intend to curate before filling in the details. Start by creating a pathway, add to this new or existing reactions in the correct order, complete the summations and literature citations, then identify the EWASs, Complexes, Sets etc. required and complete the details of the reactions consecutively. Cascading signaling processes can involve very complicated Complexes, in these circumstances the Graphic Display in Entity Hierarchical View is very useful as an overview of the order of events.

PRECURATION

1. Create a basic outline of pathway: Regulation of Apoptosis as an example. Further description of the annotation process will focus on the subpathway highlighted below in yellow.

I. Regulation of Apoptosis

II. Regulation of activated PAK-2p34 by proteasome mediated degradation

IIa Ubiquitination of PAK-2p34

IIb Proteasome mediated degradation of PAK-2p34

III. Regulation of PAK-2p34 activity by PS-GAP/RHG10

IIIa. Rho GTPase-activating protein 10 (RHG10) interacts with caspase-activated PAK-2p34

IIIb. Interaction of PAK-2p34 with RGH10/ PS-GAP results in accumulation of PAK-2p34 in the perinuclear region

IV. Role of DCC in regulating apoptosis

IVa. Caspase cleavage of DCC

IVb. DCC interacts with DIP13alpha

IVc. Caspase-9 binds DCC:DIP13alpha complex

IVd. Activation of caspase-3

IVe. Caspase cleavage of UNC5B

IVf. DAPK binds UNC5B

IVg Caspase cleavage of UNC5A

IVh UNC5A binds NRAGE

2. Flesh out outline with information including: molecules, compartment, species, text summary, references (PMIDs)

Text Summary — Species
Cell compartment — PMID
Inferenece — GO term

II. Regulation of activated PAK-2p34 by proteasome mediated degradation PMID:12853446

Summary
Stimulation of cell death by PAK-2 requires the generation and stabilization of the caspase-activated form, PAK-2p34 (Walter et al., 1998 PMID: 9786869 ;Jakobi et al., 2003 PMID: 12853446). Levels of proteolytically activated PAK-2p34 protein are controlled by ubiquitin-mediated proteolysis. PAK-2p34 but not full-length PAK-2 is degraded by the 26 S proteasome (Jakobi et al., 2003). It is not known whether ubiquitination and degradation of PAK-2p34 occurs in the cytoplasm or in the nucleus.

IIa Ubiquitination of PAK-2p34 [cytosol]

phospho-PAK-2p34 (Thr 402) [cytosol] + Ubiquitin [cytosol] => phospho-PAK-2p34:Ubiquitin [cytosol] complex
Catalyst: unknown ubiquitin ligase
GO:0004842 Ubiquitin protein ligase activity

Inferred: Oryctolagus cuniculus, Homo sapiens PMID: 12853446
phospho-PAK-2p34 (Thr 402)(Oryctolagus cuniculus) [cytosol] + Ubiquitin (human)[cytosol] => phospho-PAK-2p34 (Oryctolagus cuniculus):Ubiquitin (human) [cytosol] complex

Summary
PAK-2p34 is ubiquitinated prior to degradation (Jakobi et al., 2003PMID: 12853446). Here, ubiquitination of PAK-2p34 is described as occurring in the cytosol. However, to date it is not known whether this occurs in the nucleus or in the cytoplasm. Evidence for this reaction comes from experiments using both human and rabbit proteins.

IIb Proteasome mediated degradation of PAK-2p34 [cytosol]

PAK-2p34:ubiquitin [cytosol] =>ubiquitin [cytosol]
catalyst =ubiquitin ligase [cytosol]
GO:0004175 Endopeptidase activity

Inferred: Oryctolagus cuniculus, Homo sapiens PMID: 12853446
PAK-2p34:ubiquitin (human,Oryctolagus cuniculus) [cytosol] =>ubiquitin (human) [cytosol]
catalyst =ubiquitin ligase [cytosol]
GO:0004175 Endopeptidase activity

Summary
Proteolytically activated PAK-2p34, but not full-length PAK-2, is degraded rapidly by the proteasome (Jakobi et al., 2003PMID: 12853446). Here, degradation of PAK-2p34 is described as occurring in the cytosol. However, to date it is not known whether this occurs in the nucleus or in the cytoplasm.

3. Create table of of molecules participating in the pathway

-Each modified form of a protein as a separate entry
-Look up/enter corresponding UniProtID identifiers

	A	B
1	**Molecule**	**Uniprot ID**
2	Graf-2/ Arhgap10	A1A4S6
3	Death-associated protein kinase 3|DAPK3|DAPK3_HUMAN|DAPK3	O43293
4	DCC's fragment with the ADD domain (fragment)	P43146
5	DCC|DCC_HUMAN|Netrin receptor DCC	P43146
6	DCC's inhibitory C-terminal domain (fragment)	P43146
7	Death-associated protein kinase 1|DAPK1|DAPK1_HUMAN|DAPK1	P53355
8	phospho-PAK-2p34 (Thr 402)	Q13177
9	UNC5C	O95185-1
10	UNC5A fragment with death domain	Q6ZN44
11	UNC5A|Netrin receptor UNC5A|UN5A_HUMAN	Q6ZN44
12	UNC5B fragment with death domain	Q8IZJ1
13	UNC5B|Netrin receptor UNC5B|UN5B_HUMAN	Q8IZJ1
14	Death-associated protein kinase 2|DAPK2|DAPK2_HUMAN|DAPK2	Q9UIK4
15	DCC-interacting protein 13-alpha|DP13A_HUMAN|APPL1	Q9UKG1
16	NRAGE|Melanoma-associated antigen D1|MGD1_HUMAN|MAGED	Q9Y5V3
17	unidentified caspase	unknown
18	Ubiquitin ligase	unknown
19	DAPKs	Candidate set
20	ubiquitin	Defined set

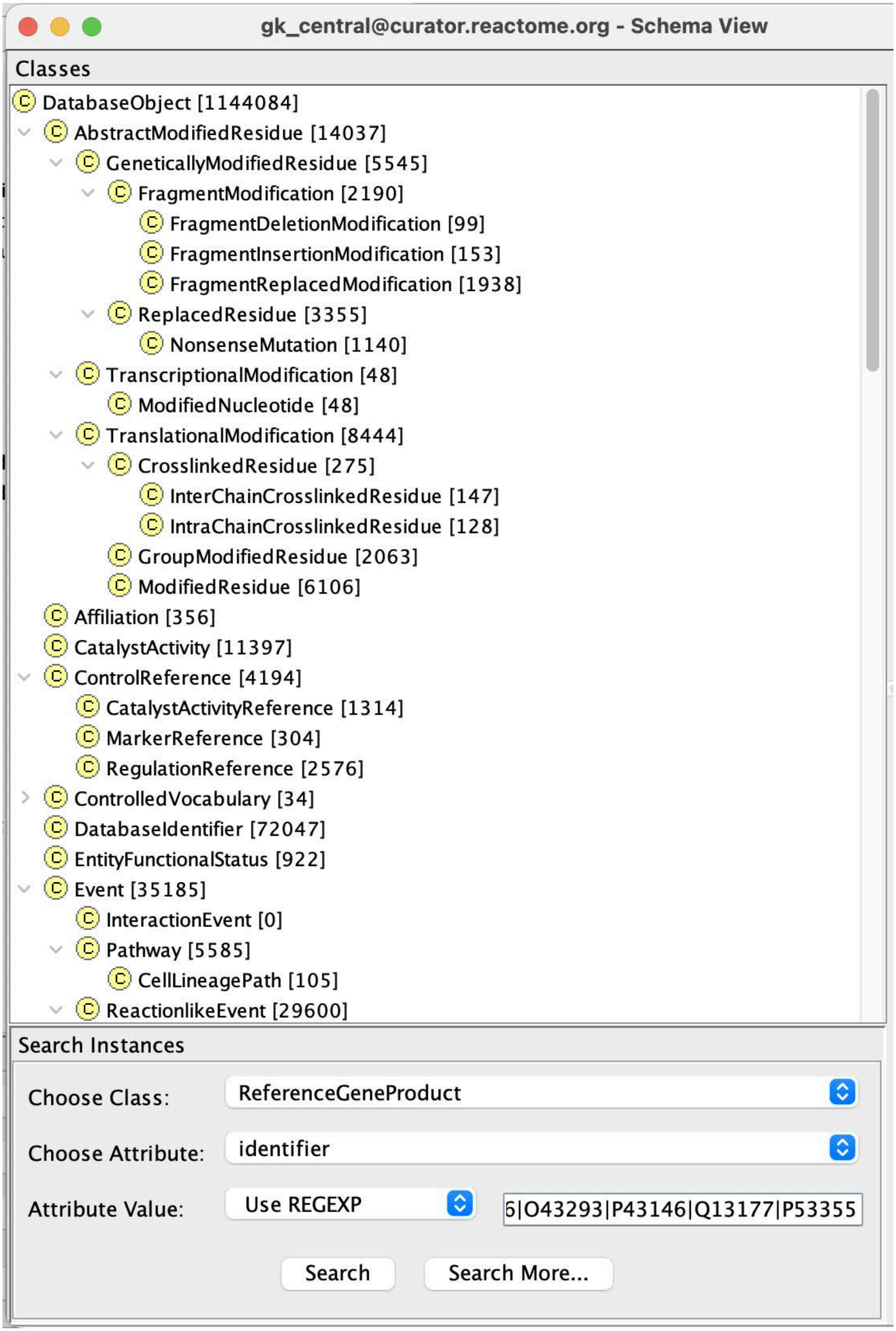
DATA ENTRY

IDENTIFY EXISTING PROTEINS/MOLECULES

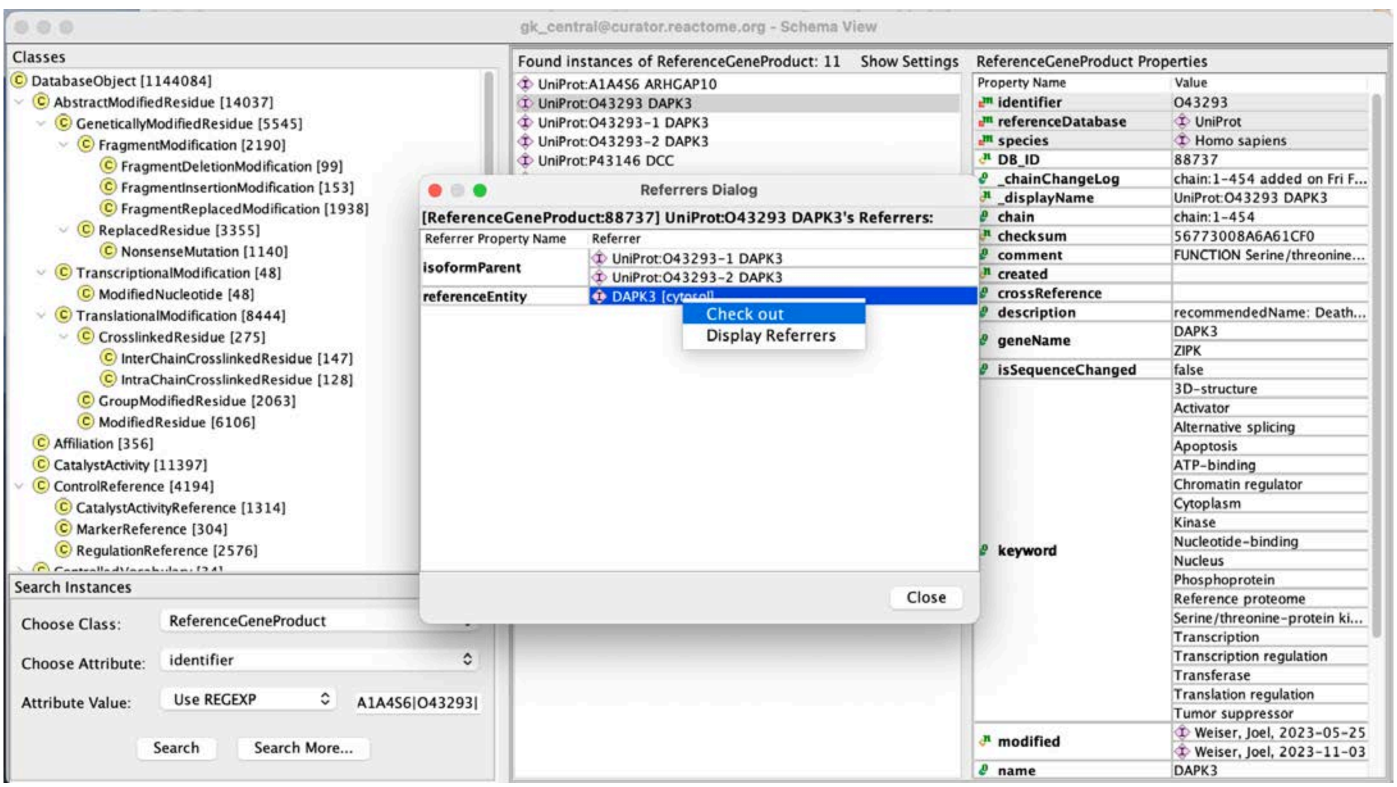
In many cases the proteins on your list will already exist in the database. You should make every effort to reuse existing instances wherever possible to avoid unnecessary and confusing duplications. Use the curator tool to search the ReferenceGeneProduct (RGP) class using Uniprot identifiers as follows:

Searching the database

Choose Class: ReferenceGeneProduct
Choose attribute: identifier
Attribute value: Use REGEXP
Enter your identifier in the search box. You can enter several as a pipe separated list (e.g A1A4S6|O43293|P43146|P43146....).

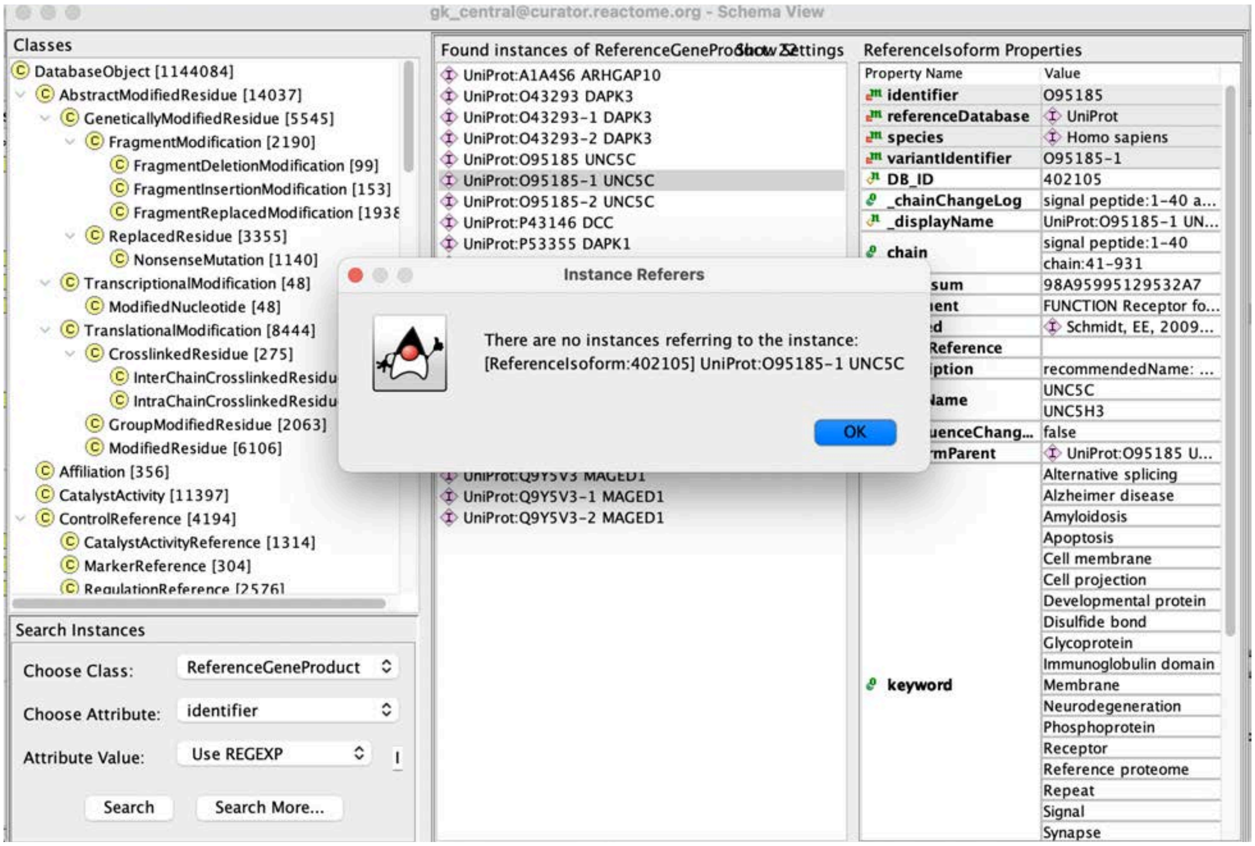

gk_central@curator.reactome.org - Schema View
Classes
DatabaseObject [1144084]
AbstractModifiedResidue [14037]
GeneticallyModifiedResidue [5545]
FragmentModification [2190]
FragmentDeletionModification [99]
FragmentInsertionModification [153]
FragmentReplacedModification [1938]
ReplacedResidue [3355]
NonsenseMutation [1140]
TranscriptionalModification [48]
ModifiedNucleotide [48]
TranslationalModification [8444]
CrosslinkedResidue [275]
InterChainCrosslinkedResidue [147]
IntraChainCrosslinkedResidue [128]
GroupModifiedResidue [2063]
ModifiedResidue [6106]
Affiliation [356]
CatalystActivity [11397]
ControlReference [4194]
CatalystActivityReference [1314]
MarkerReference [304]
RegulationReference [2576]
ControlledVocabulary [34]
DatabaseIdentifier [72047]
EntityFunctionalStatus [922]
Event [35185]
InteractionEvent [0]
Pathway [5585]
CellLineagePath [105]
ReactionlikeEvent [29600]
Search Instances
Choose Class:
ReferenceGeneProduct
Choose Attribute:
identifier
Attribute Value:
Use REGEXP
6|O43293|P43146|Q13177|P53355
Search
Search More...

Select the ReferenceGeneProduct (RGP) returned, if a list select them one at a time, right click and opt to "View referrers". The resulting Referrers Dialog box lists Referrers by property name. At the top of the list can be isoforms of the protein, if present in Uniprot. Do not use isoforms unless you are certain that only specific isoforms have the functionality you intend to represent. Isoforms may have their own referrers and you should check this - if someone took the trouble to create an isoform-specific EWAS they probably had good reason to do so. Items listed with the property name referenceEntity are EntitiesWithAccessionedSequence (EWASs), a Reactome identifier for specific forms/locations of a protein. Often there will be more than one EWAS for a single RGP, because post-translationally modified forms of proteins and proteins in different cellular compartments are each represented by a different EWAS. If any of the listed EWASs correspond to your needs, right click and opt to "Check Out" that referrer. If the correct molecular compartment or post-translationally modified form is not present, you can still check out an EWAS and later use the Curator tool to clone it and modify it to your needs.



Create new proteins/molecules

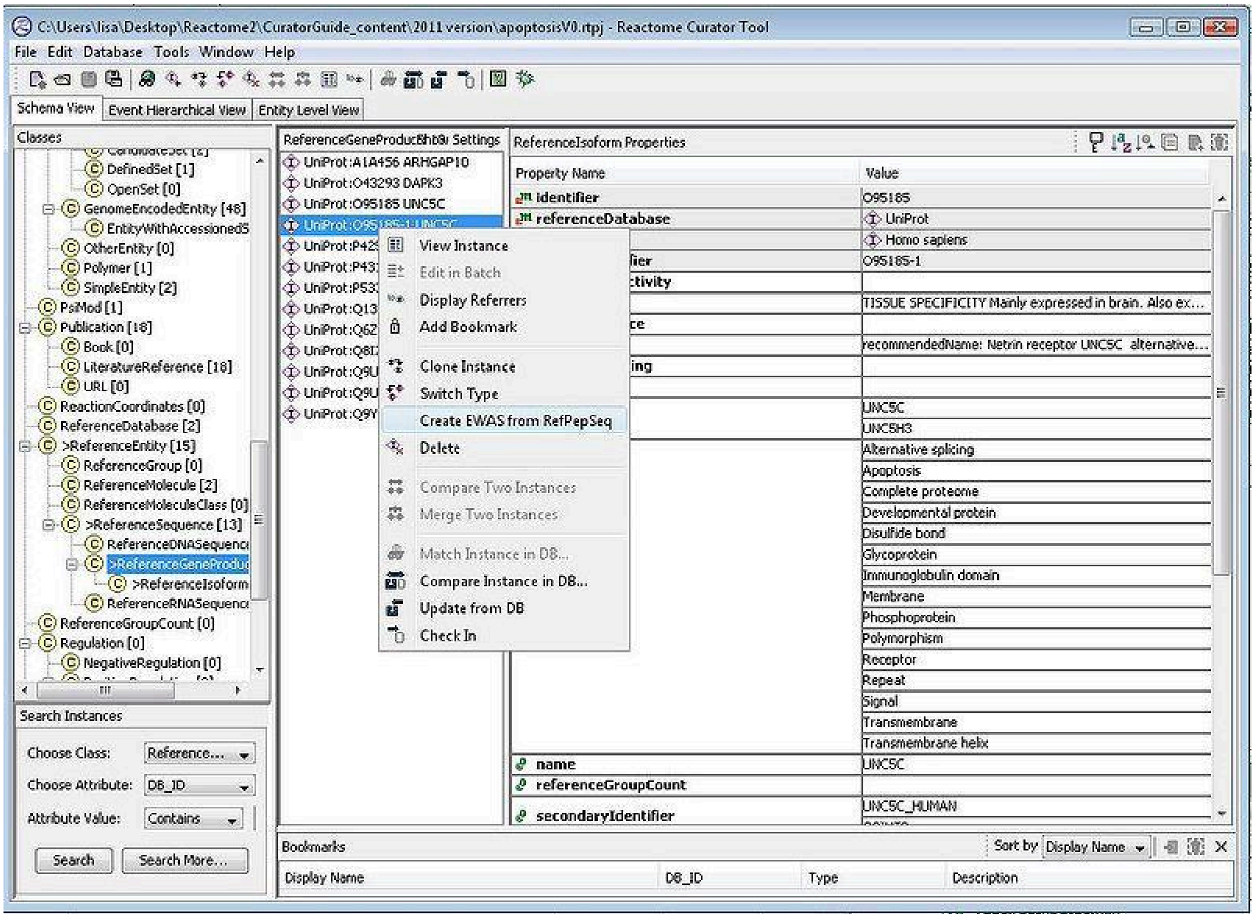
Please see the guidelines on naming entities in this paper (Jupe et al, 2014)! If you search for a RGP that does NOT have referrers in the database you will get this message:



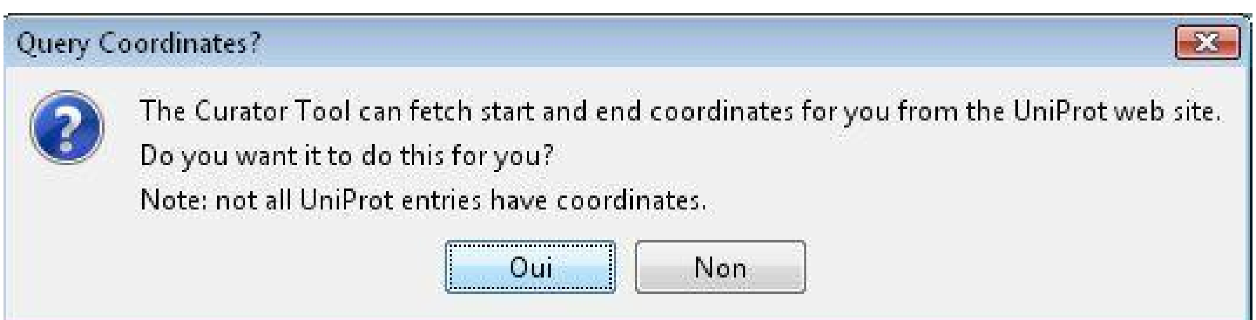
In this case, you will need to create the EWAS.

Creating a new EWAS

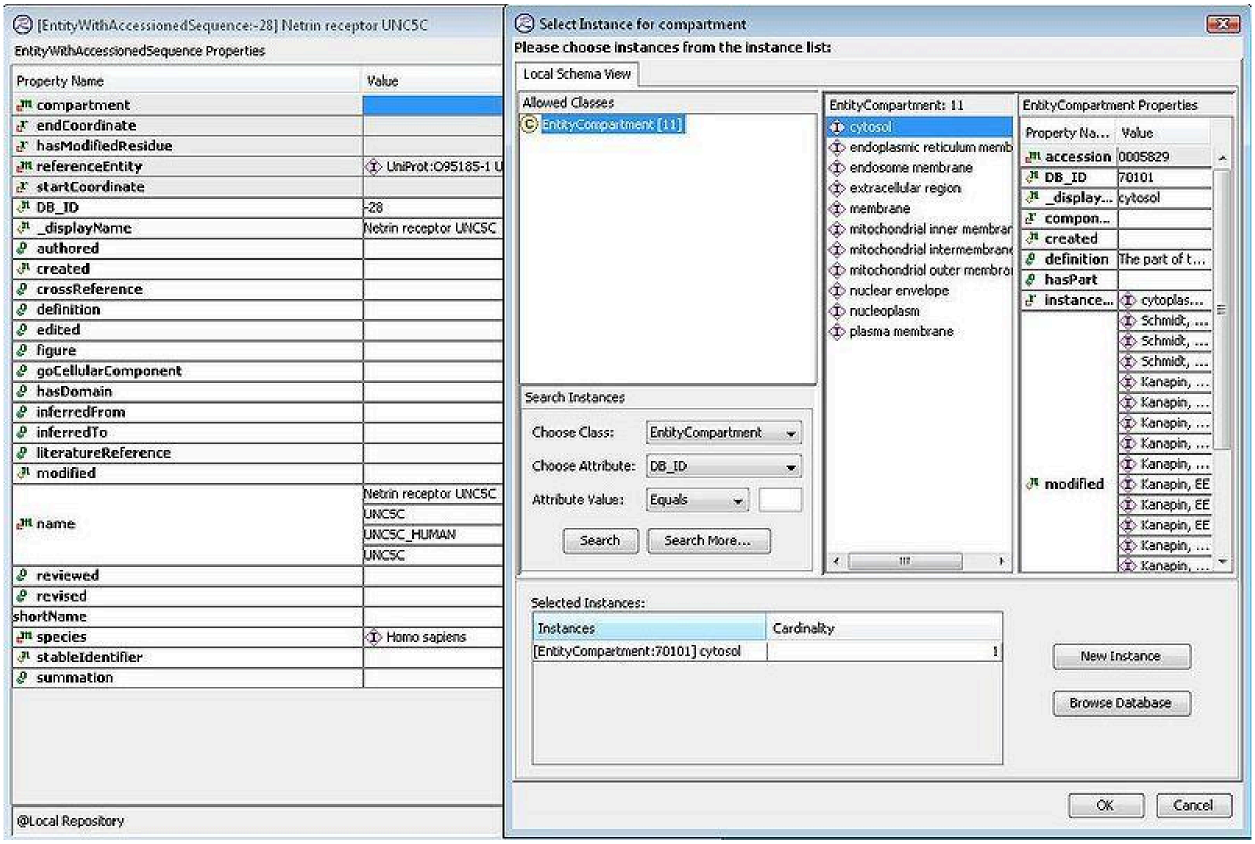
To do this, first check out the RGP of interest into your local project. Then, in the curator tool, select RGP in the class list, then scroll or search for the RGP of interest. Right click and opt to create EWAS from RGP.



It will ask if you want to accept the end coordinates described by Uniprot. Only say yes if you can confirm them to be accurate. If the end coordinates are not certain, the convention is to represent the start as 1, and end as -1.



In the newly created EWAS, the ReferenceEntity will have a name and species entered by default. You can add an alternative name if this was specified by the Author, do this by right-clicking on the existing name and select Add. You must define the compartment. To do this, right click in the compartment slot and select the correct compartment in your local repository. Select the compartment and hit OK. If you don't see the desired compartment in your local project , click the "Browse database" button to search in gk_central. ***If you still don't find the necessary compartment, DO NOT create a new one. Contact the managing editor. New compartment creation requires multiple steps as well as changes in the UI which need to be coordinated with developers.***



References

1. Jupe S, Jassal B, Williams M, Wu G. A controlled vocabulary for pathway entities and events. Database (Oxford). 2014 Jun 20;2014:bau060. doi: 10.1093/database/bau060. PMID: 24951798; PMCID: PMC4064568.

Translational modifications (see note below about how to make sure all PTMs register correctly in diagrams):

If the protein you want to represent with an EWAS is **post-translationally modified**, then the hasModifiedResidue slot of the EWAS is to be filled. Right clicking on the hasModifiedResidue slot of the EWAS will allow you to select a modification instance from one of the 3 child subclasses of the 'TranslationalModification' class, namely ModifiedResidue, GroupModifiedResidue and CrosslinkedResidue. You will be prompted to choose from your local project, browse gk_central or create a new modified residue instance. Almost always you will want to create a new instance, as modified residue instances are specific to the RGP and residue position. If EWAS instances are created from the same RGP but belong to two or more cellular compartments, then these EWASs, e.g., EWAS (cytosol) and EWAS (nucleoplasm), may share modified residue instances when modified at the same residue.

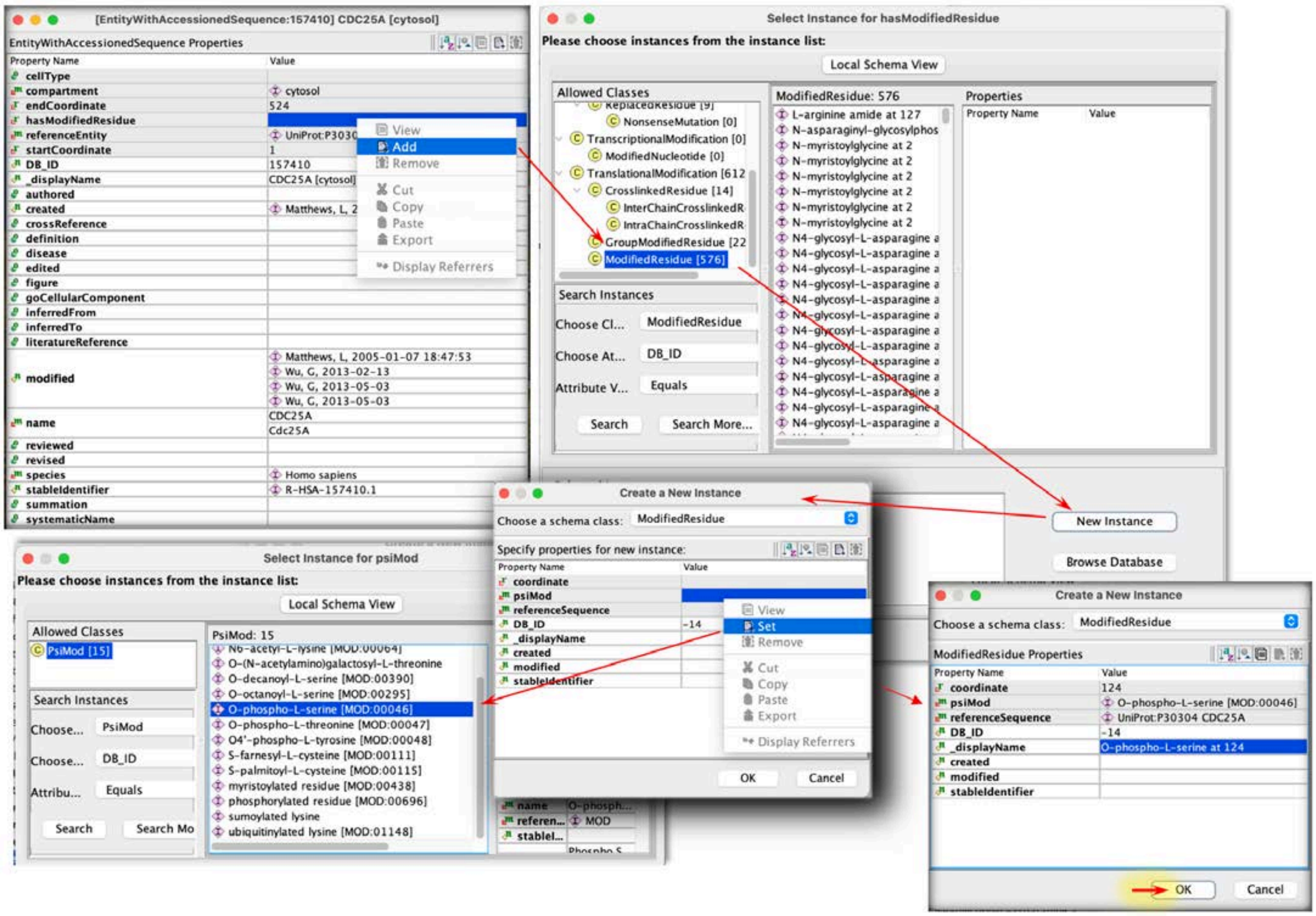
- *ModifiedResidue:*

This instance is added to EWAS to indicate that a chemical group such as phosphate or methyl group is added to an amino acid residue of a protein. Two mandatory slots need filling for a modifiedResidue instance - psiMod and referenceSequence. For example, to indicate that CDC25A protein has a phospho-serine modification at residue 124 you need to right click in the hasModifiedResidue slot of the EWAS, then select ‘Add’ from drop-down list and choose ModifiedResidue as a schema class in the pop-up window. Then, click on the ‘New Instance’ button to open a ‘Create a New Instance’ form. Right click in the PsiMod slot of the modification instance to select O-phospho-L-serine [MOD:00046] as a modification type and hit ok (a modification type can be found within the local repository or by searching the gk_central database).

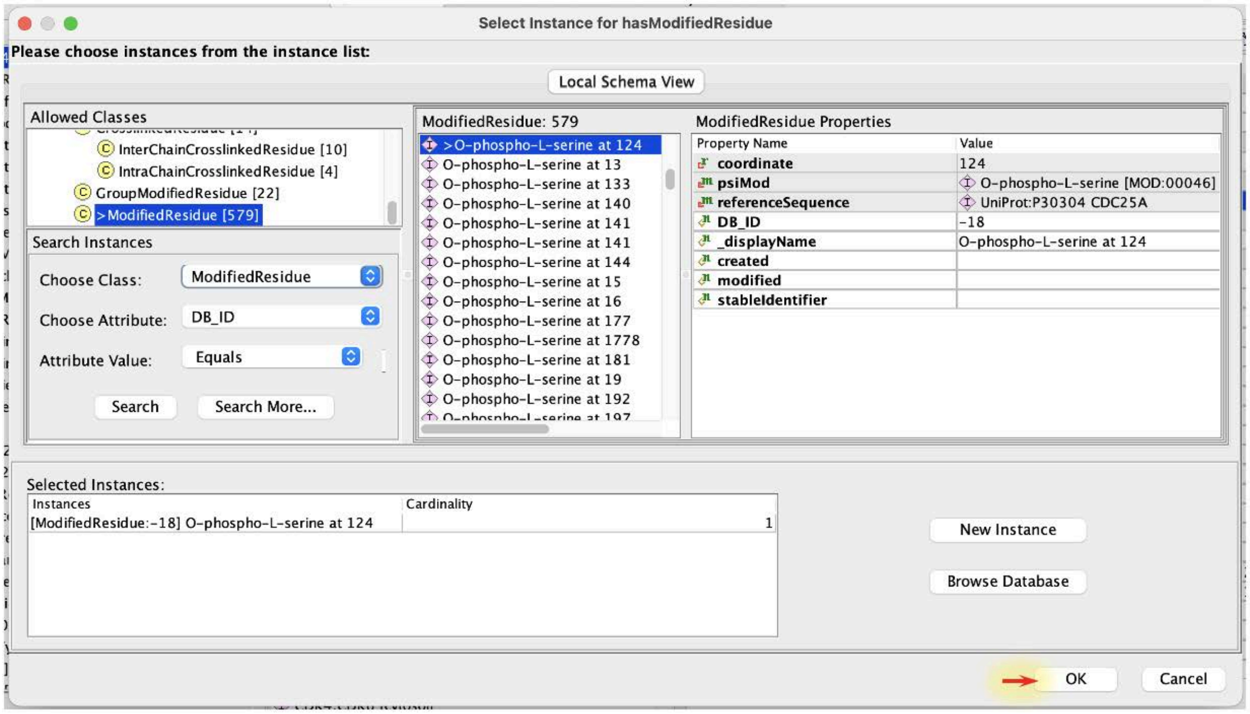
If a needed modification is not already present in gk_central, look up its identifier (a five-digit number) at the PSI-MOD website; copy-paste it into the "create new instance" form in the curator tool, and allow the wizard associated with the form to retrieve all other needed data from PSI-MOD. Once created, such an instance (like all other reference instances in gk_central) should not be modified. If an instance is needed that is not already in PSI-MOD, or if a PSI-MOD instance appears to be incorrect, contact PSI-MOD to resolve the problem.

Similarly, right clicking in the ReferenceSequence slot of the modification instance allows selecting RGP with Uniprot identifier for CDC25A protein. Finally, enter the residue number 124 in the coordinate slot and hit ok.

Note that the curator tool does not prevent you from adding a modifiedResidue instance with one ReferenceGeneProduct onto the EWAS record with a different ReferenceGeneProduct, and the confusion that results is substantial!



The modified residue instance can now be applied to the EWAS by clicking ok.



The modified EWAS is shown below. Make sure that the EWAS name is edited to reflect the modification, e.g., p-S124-CDC25A.

[EntityWithAccessionedSequence:69588] p-S124-CDC25A [...

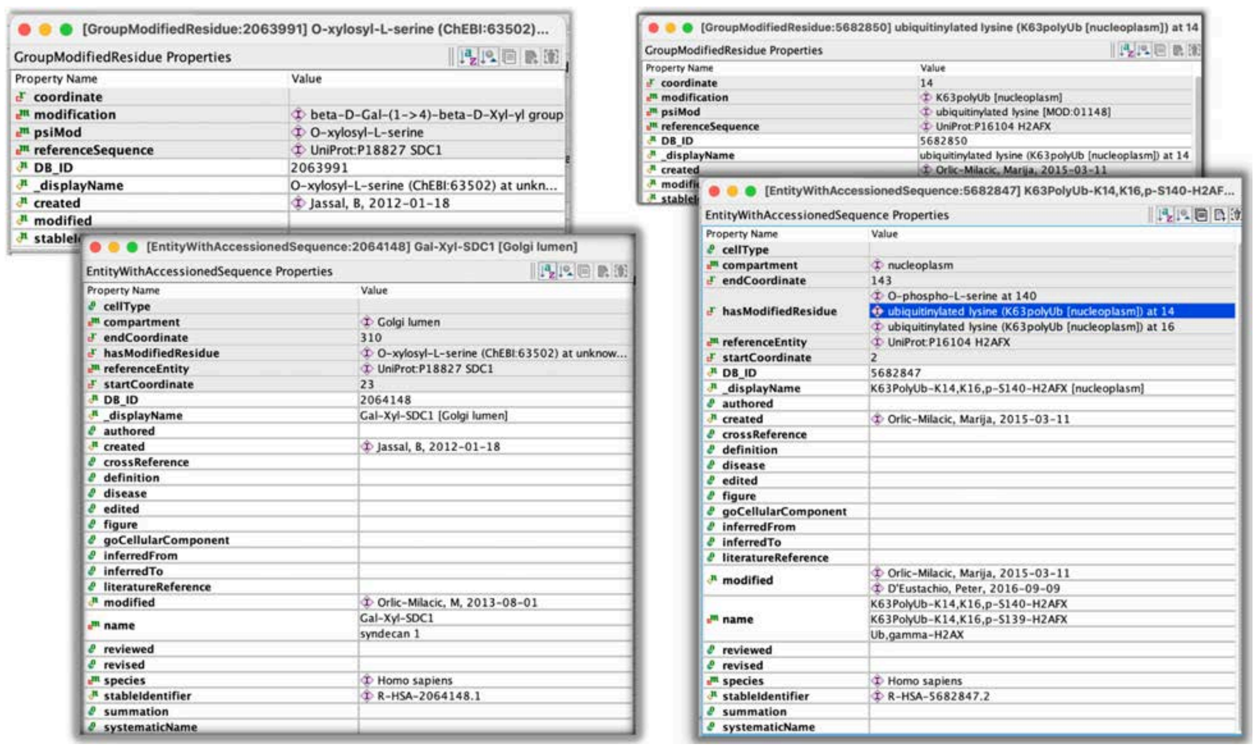
EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	524
hasModifiedResidue	O-phospho-L-serine at 124
referenceEntity	UniProt:P30304 CDC25A
startCoordinate	1
DB_ID	69588
_displayName	p-S124-CDC25A [cytosol]
authored	
created	
crossReference	
definition	
disease	
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	
modified	Schmidt, EE, 2004-06-09 05:34:08
	Matthews, L, 2004-12-30 22:28:30
	Wu, G, 2013-02-13
	Wu, G, 2013-05-03
	Wu, G, 2013-05-03
	D'Eustachio, Peter, 2022-12-28
name	p-S124-CDC25A
	phospho-Cdc25A
	M-phase inducer phosphatase 1
	Dual specificity phosphatase Cdc25A
reviewed	
revised	
species	Homo sapiens
stableIdentifier	R-HSA-69588.2
summation	
systematicName	

@Local Repository

- *GroupModifiedResidue (GMR)*

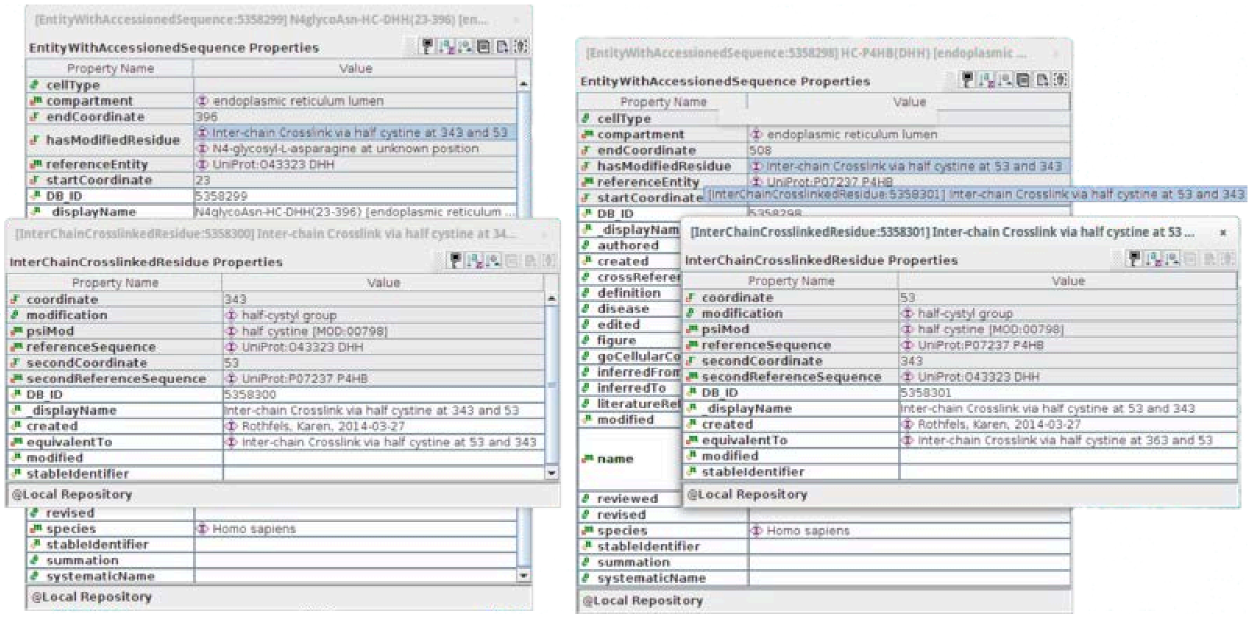
This instance is linked to EWAS instance to indicate that a protein has a modification formed by attachment of two (or more) either peptide or chemical moieties. 3 mandatory slots need filling in a GMR instance - modification, psiMod and referenceSequence. For example, GMR is extensively used to annotate EWASs in protein glycosylation process that involves stepwise addition of sugar moieties, attached to an amino acid residue of a protein. In the synthesis of several GAGs, the 4 sugar moieties, i.e., xylose (Xyl), 2 x galactose (Gal) and glucuronic acid (GlcA), are added in a stepwise fashion. The first reaction is the addition of Xyl to a Ser residue. The product, a protein with a Xyl-modified Ser residue, is represented by EWAS with a modifiedResidue instance where psiMod slot value is O-xylosyl-Ser [MOD:00814]. The second reaction requires Gal to be attached to O-xylosyl-Ser to form the protein product with Gal-Xyl-yl group attached to Ser. Right click in the hasModifiedResidue slot of the EWAS instance and select GroupModifiedResidue class. In the modification instance right click in the modification slot and add 'beta-D-Gal-(1->4)-beta-D-Xyl-yl group' [ChEBI:63502) as a referenceGroup. The psiMod slot still needs to be filled with O-xylosyl-Ser [MOD:00814]. Enter the Uniprot identifier for the ReferenceGeneproduct in the ReferenceSequence slot. GMR may also be linked to an EWAS instance to indicate that a protein is modified by attaching polypeptide chains such as polyUb or polySUMO. Polypeptide chain instances are selected from the Polymer class and added in the modification slot in GMR.



- *CrosslinkedResidue*

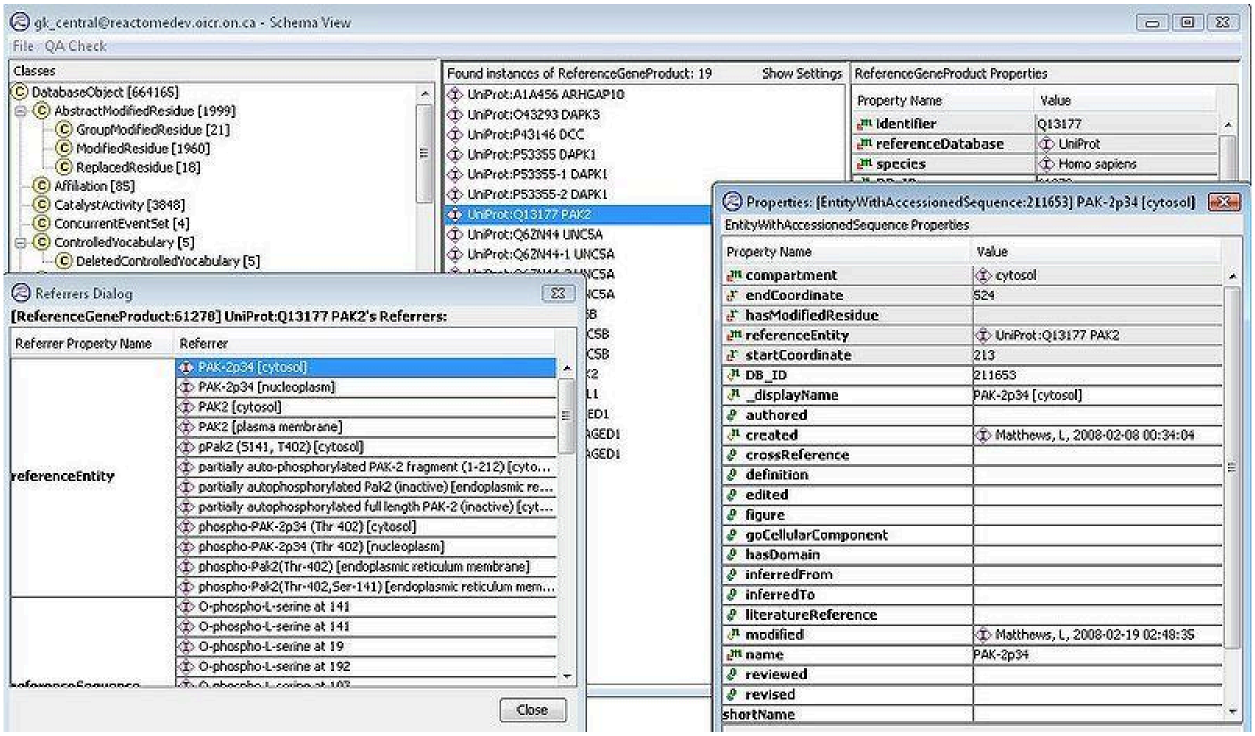
This instance is linked to a target EWAS instance to annotate disulfide, thioester, mono-sumoylated lysine, monoubiquitinated lysine or other protein crosslinks, where two coordinates describe a modification. The "crosslinkedResidues" subclass has two child classes - IntraChainCrosslinkedResidue and InterChainCrosslinkedResidue.

1. **IntraChainCrosslinkedResidue**
   is a value of hasModifiedResidues slots in EWAS records used to annotate crosslinking within the same amino acid sequence. The attributes and requirements are similar to those of ModifiedResidue, but it also includes a secondCoordinate slot.
2. **InterChainCrosslinkedResidue**
   To indicate that two different peptide chains are crosslinked you create an InterChainCrosslinkedResidue instance for each corresponding EWAS instance of these two proteins. The interchain modification instances are linked by having each other as values in their equivalentTo slots. There are 4 mandatory slots - psiMod, referenceSequence, secondReferenceSequence and equivalentTo. For example, to indicate that disulfide bond occurs between Cys343 of DHH and Cys53 of P4HB, two interChainCrosslinkedResidue instances are needed, one instance for modified DHH, the second instance for modified P4HB protein. Right click in the hasModifiedResidue slot of EWAS of modified DHH protein and select InterChainCrosslinkedResidue class. Then, right click in the referenceSequence slot of the modification instance and enter RGP with UniProt identifier 'UniProt:O43323 DHH'. A modified cysteine residue involved in the disulfide bond can be described as a half cystine (MOD:00798 , DB_ID 448182). Enter half cystine [MOD:00798] as a modification type in the psiMod slot. Enter half-cystyl group as a reference group value in the modification slot (optional). Enter 343 in the coordinate slot to state that DHH protein has a half cystyl group at position 343. Enter 53 in the secondCoordinate slot and RGP with identifier 'UniProt:P07237 P4HB' in the secondReferenceSequence slot to state the position on the second amino acid sequence. Repeat these steps for EWAS of modified P4HB but enter 53 as a coordinate value and select RGP with identifier 'UniProt:P07237 P4HB' in the referenceSequence slot, while secondCoordinate value is 343 and secondReferenceSequence value is RGP with the identifier 'UniProt:O43323 DHH'. Finally, right click in the equivalentTo slot of the first instance (linked to EWAS of modified DHH) and select the other one that is linked to EWAS of modified P4HB. Similarly, enter the first instance as a value in the equivalentTo slot of the second one.
3. In the event that you are creating a crosslink between two monomers of the same protein, create one InterChainCrosslinkedResidue instance and indicate that the referenceSequence and secondReferenceSequence are the same as are the coordinate and secondCooordinate.



Reactome does not aim to describe all disulfide bonds in all proteins. Curators may choose to annotate disulfide bonds at their discretion when, for example, the bonds or bonding process may be involved in a disease, may regulate the protein's activity, or may be required to hold a complex together.

If you are annotating a protein fragment, you can define the start and end coordinates of the fragment as shown below: Don't forget to change the default name to indicate that it is a fragment.



***To be sure that all PTMS get added correctly to the reactions in diagrams** you have to first make sure you have downloaded TranslationalModifications fully in your project. In the schema view, choose that class to get the whole list of these instances. Select all in the middle list and then select “Update Instances” from the popup menu. Then go to the diagram, right click, and select the option “update node features”. Say yes to the question “instances must be fully downloaded, do you want to continue, yes no”.

Creating a Complex

This is done in one of two ways. Either:

Go to the Schema view, select Complex, right-click and select Create Instance. The Create a New Instance dialog box appears. Enter a name in the field for Name. Typically you would also enter the *Compartment, Species and identify the entities (EWASs, sets or complexes) that make up this complex using the field hasComponent. All of these fields are completed by either double-clicking to type, or right clicking to select Add and identify the correct item from the local project.

- *We record two location attributes for every complex. The "core compartment" (the place most closely associated with the function of the complex) will be entered into the compartment slot. This slot will be made single-valued and will be filled manually. All of the "other" compartments of the participating components will be automatically added in the "includedLocation" attribute of the complex at the time of database slicing for release.

Example: Complexes that have component(s) localized to membrane as well as component(s) localized to other adjacent compartments should have the membrane listed in the "Compartment" slot . The other compartments would be assigned automatically.

Creating a Set

Reactome has several types of set - refer to the Glossary and User Guide for definitions.

The most commonly used sets are Defined Sets and Candidate Sets.

Defined Set members should be proven equivalents, i.e. all of them have been demonstrated to perform the function that is described by the event they participate in.

Candidate Sets have two categories of inclusion, members, equivalent to defined set members, and candidates, members that are not proven to be functionally equivalent, but are believed to be equivalent based on phylogeny, domain structure etc.

Creating sets is a similar process for all subtypes, select the appropriate type in the Schema view, right-click and select Create Instance, fill in the Name and Species, right click to add the set members.

- All members of a set must have the same compartment. The only time a set can have multiple compartment attributes is if its members themselves all have the same multiple compartment attributes, e.g., a set of membrane-spanning complexes with components explicitly located [on this side], in the membrane, and [on that side].
- The members or candidates of a set should (in most cases) belong to the same class...e.g all EWASs or all simpleEntities or all Complexes. However, there are exceptions under circumstances in which structurally different entities are really functionally indistinguishable (i.e either a monomer or dimer can function in a particular reaction). Since these exceptions are likely to be fairly rare, a QA check is in place to flag sets that have members that are from different classes. The ids of such exceptional acceptable sets may be forwarded to the editorial release manager to be put on a QA skip list.

Using sets

Representing outputs that include sets: When representing the output of a reaction:
Set [A, A’, A”] + EWAS B
Use the ‘set-to –complex’ approach: Set [A, A’, A”] + EWAS B → Complex (Set [A, A’, A”]:EWAS B), where no individual complex entities are created for the reaction. Entity of the individual complex (for example A:B) will be created if we need to functionally distinguish the potential output complex somewhere else in the diagram. The same dashed lines used to link an EntitySet and its contained members will be used to link the individual complex A:B to Complex (Set [A, A’, A”]:EWAS B).

Use of isOrdered attribute:

Creating a Pathway

Please see *note* below if you are adding a pathway that will be a new top level pathway. Here is a description of how the outlined mini pathway "Regulation of activated PAK-2p34 by proteasome mediated degradation" is built from its component events in the curator tool. The pathway consists of 2 reactions: "Ubiquitination of PAK-2p34" and "Proteasome mediated degradation of PAK-2p34". For simplicity, the reactions have already been created (see section on creating inferred reactions for an example.)

II. Regulation of activated PAK-2p34 by proteasome mediated degradation PMID: 12853446

Summary

Stimulation of cell death by PAK-2 requires the generation and stabilization of the caspase-activated form, PAK-2p34 (Walter et al., 1998 PMID: 9786869 ;Jakobi et al., 2003 PMID: 12853446) . Levels of proteolytically activated PAK-2p34 protein are controlled by ubiquitin-mediated proteolysis. PAK-2p34 but not full-length PAK-2 is degraded by the 26 S proteasome (Jakobi et al., 2003). It is not known whether ubiquitination and degradation of PAK-2p34 occurs in the cytoplasm or in the nucleus.

IIa Ubiquitination of PAK-2p34 [cytosol]

phospho-PAK-2p34 (Thr 402) [cytosol] + Ubiquitin [cytosol] => phospho-PAK-2p34:Ubiquitin [cytosol] complex
Catalyst: unknown ubiquitin ligase
GO:0004842 Ubiquitin protein ligase activity

Inferred: Oryctolagus cuniculus, Homo sapiens PMID: 12853446
phospho-PAK-2p34 (Thr 402)(Oryctolagus cuniculus) [cytosol] + Ubiquitin (human)[cytosol] => phospho-PAK-2p34 (Oryctolagus cuniculus):Ubiquitin (human) [cytosol] complex

Summary
PAK-2p34 is ubiquitinated prior to degradation (Jakobi et al., 2003PMID: 12853446). Here, ubiquitination of PAK-2p34 is described as occurring in the cytosol. However, to date it is not known whether this occurs in the nucleus or in the cytoplasm. Evidence for this reaction comes from experiments using both human and rabbit proteins.

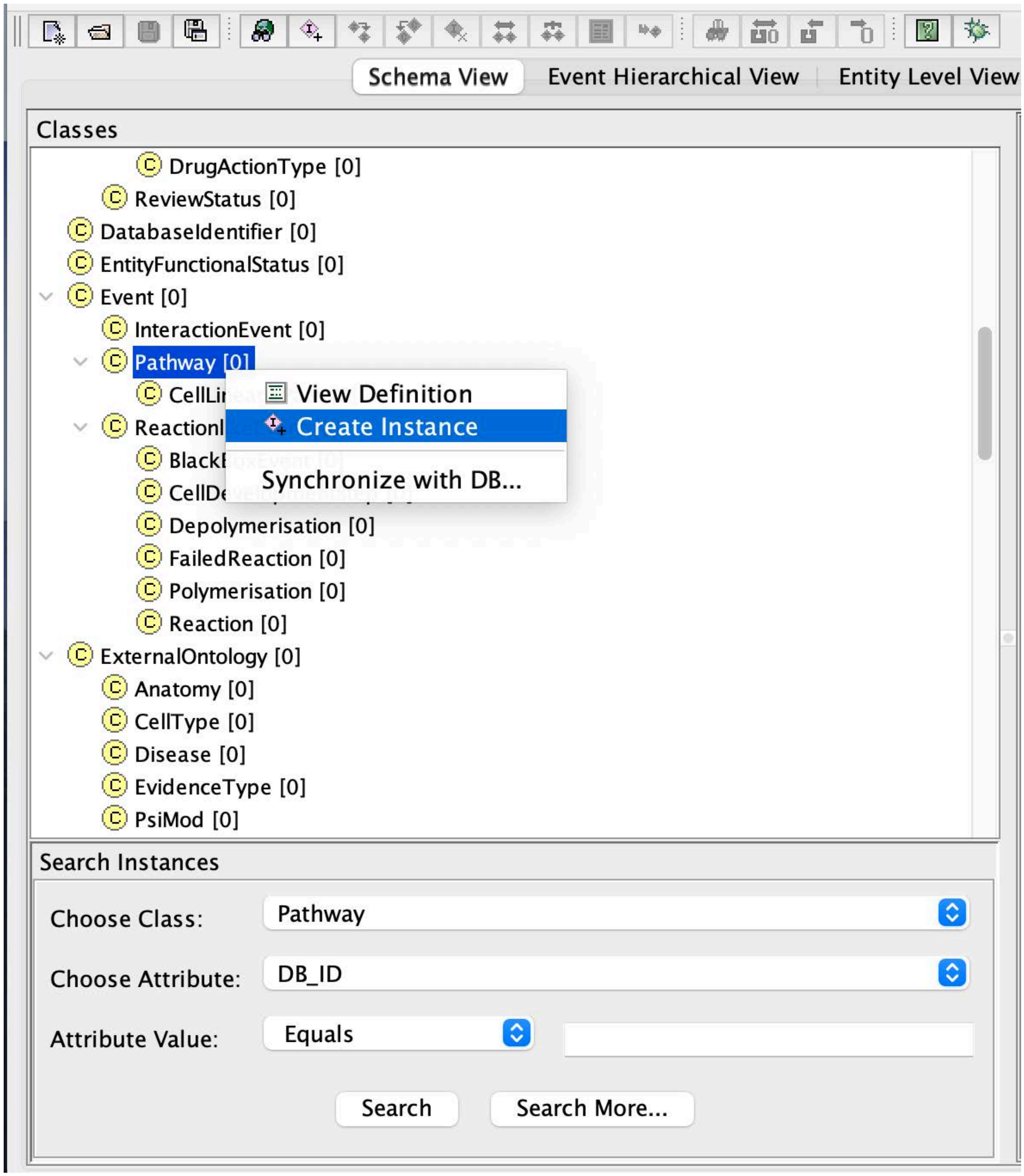
IIb Proteasome mediated degradation of PAK-2p34 [cytosol]

PAK-2p34:ubiquitin [cytosol] =>ubiquitin [cytosol]
catalyst =ubiquitin ligase [cytosol]
GO:0004175 Endopeptidase activity

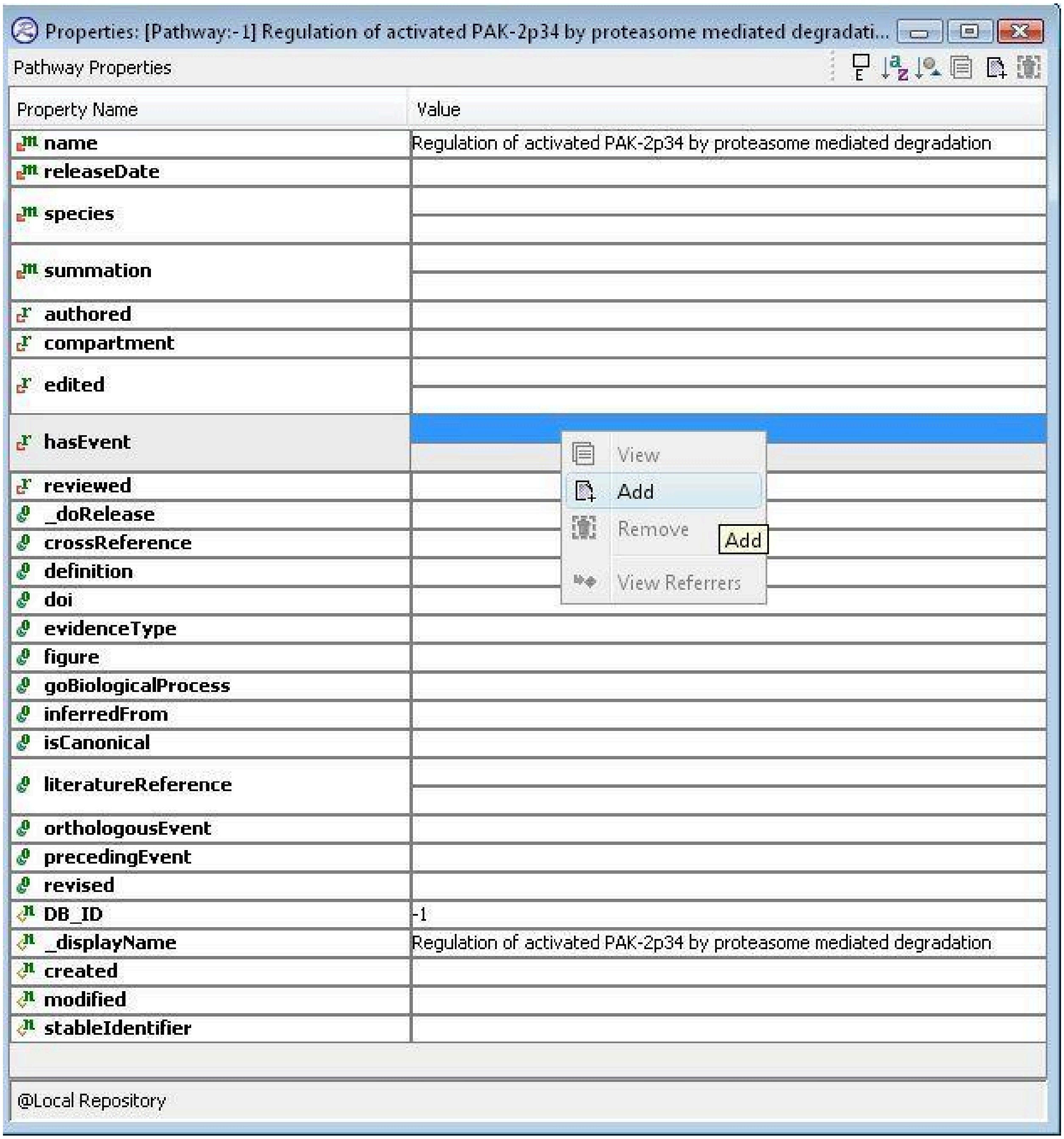
Inferred: Oryctolagus cuniculus, Homo sapiens PMID: 12853446
PAK-2p34:ubiquitin (human,Oryctolagus cuniculus) [cytosol] =>ubiquitin (human) [cytosol]
catalyst =ubiquitin ligase[cytosol]
GO:0004175 Endopeptidase activity

Summary
Proteolytically activated PAK-2p34, but not full-length PAK-2, is degraded rapidly by the proteasome (Jakobi et al., 2003PMID: 12853446). Here, degradation of PAK-2p34 is described as occurring in the cytosol. However, to date it is not known whether this occurs in the nucleus or in the cytoplasm.

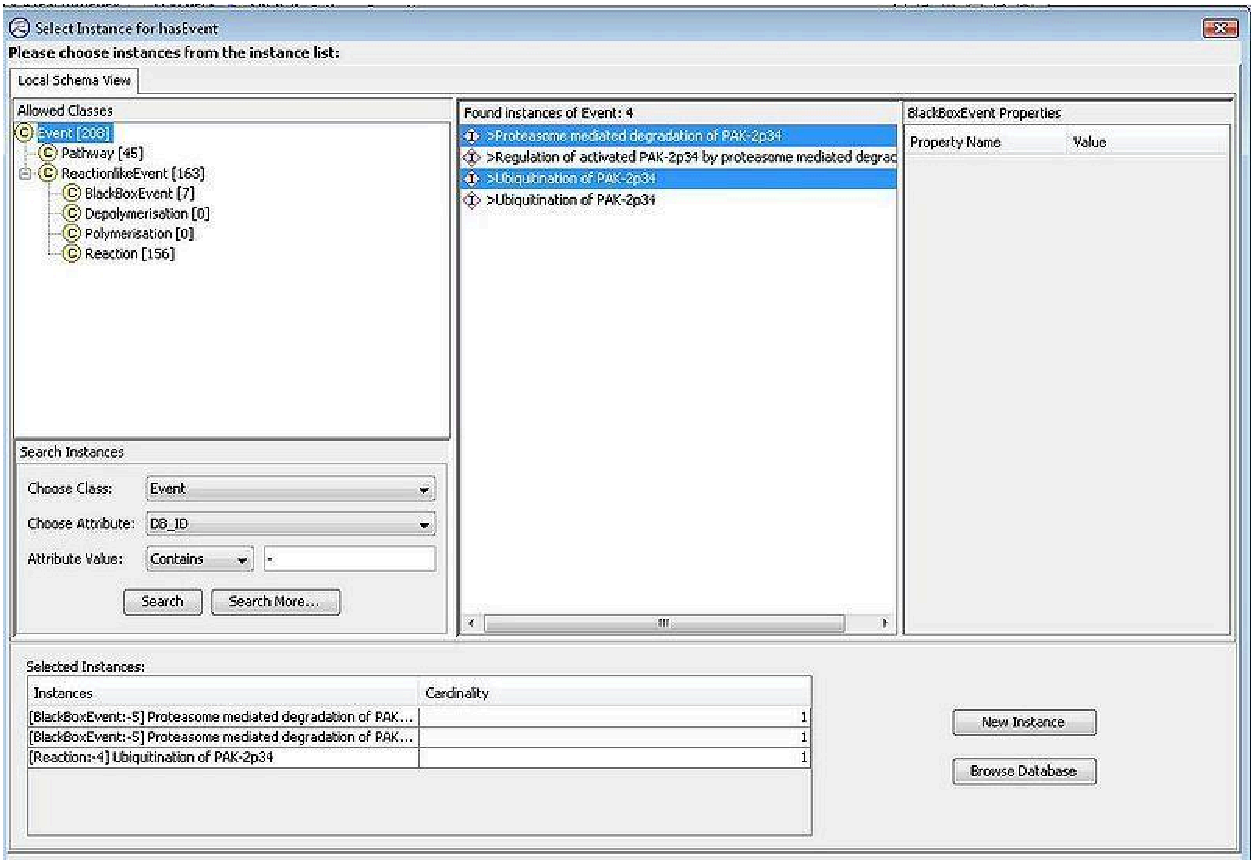
With the pathway class highlighted in the class hierarchy, right click and select Create instance, or use the create instance button in the menu bar at the top of the tool.



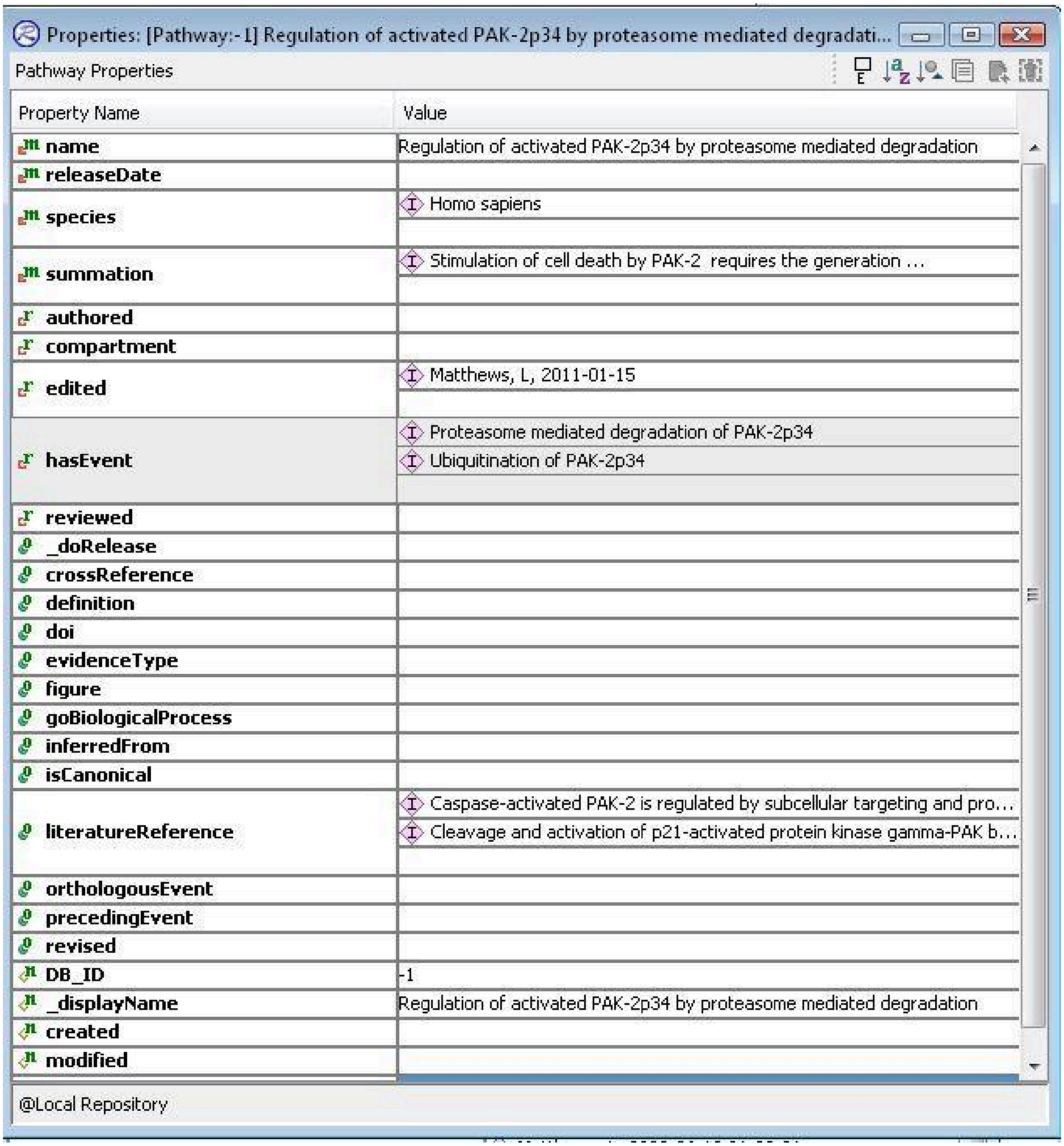
After adding the pathway title, associate reactions as component events of the pathway by right clicking on the hasEvent slot and selecting "Add"



You can then select the events that you need one at a time or, as a shortcut, you can search for your newly created events in your project if they have not yet been submitted to gk_central (they will all have DB_ID attributes that are negative). To do this, search in your project for events with DB_ID containing - . From this list, you can hold the control key and select the events of interest.



The order of the events in a pathway is described through the use of the "precedingEvent" attribute on the reactions that are components of the pathway. If/when no preceding events are specified, the order of the events displayed on the webpage reflects the order in which they are listed as components in the pathway instance that you are creating. Curators should check for precedingEvents that they may have missed within a pathway diagram by using the “Infer reaction relationship” curator tool feature (described in more detail here, Appendix C7). This feature can be accessed by right-clicking in a diagram. Here, the curator is offered a list of preceding events based on shared entities. The curator can then opt to approve a preceding event, in which case that reaction is automatically entered into the appropriate reaction’s precedingEvent attribute. If the curator decides the precedingEvent is not biologically relevant, it can be rejected with a reason for the rejection specified. This “rejection” of a preceding event is recorded in the “negativePrecedingEvent” attribute for the reaction.



Once the component events have been added, the remaining required attributes are added. Identify a GO Biological process (BP) term that corresponds to your pathway, find the corresponding GO_BiologicalProcess instance in gk_central, right click on the goBiologicalProcess slot and select set to add it. If you can't find the term of interest, double check with Peter or Lisa. If no term is available, you can use GOC GitHub Tracker to create a ticket to request a new term as described here. When suggesting a GO term, be sure to look over the guidelines described here (https://github.com/geneontology/go-ontology/blob/master/CONTRIBUTING.md) and include label (name), definition, references, and position in hierarchy. ***Never*** create a new GO term instance yourself in gk_central!

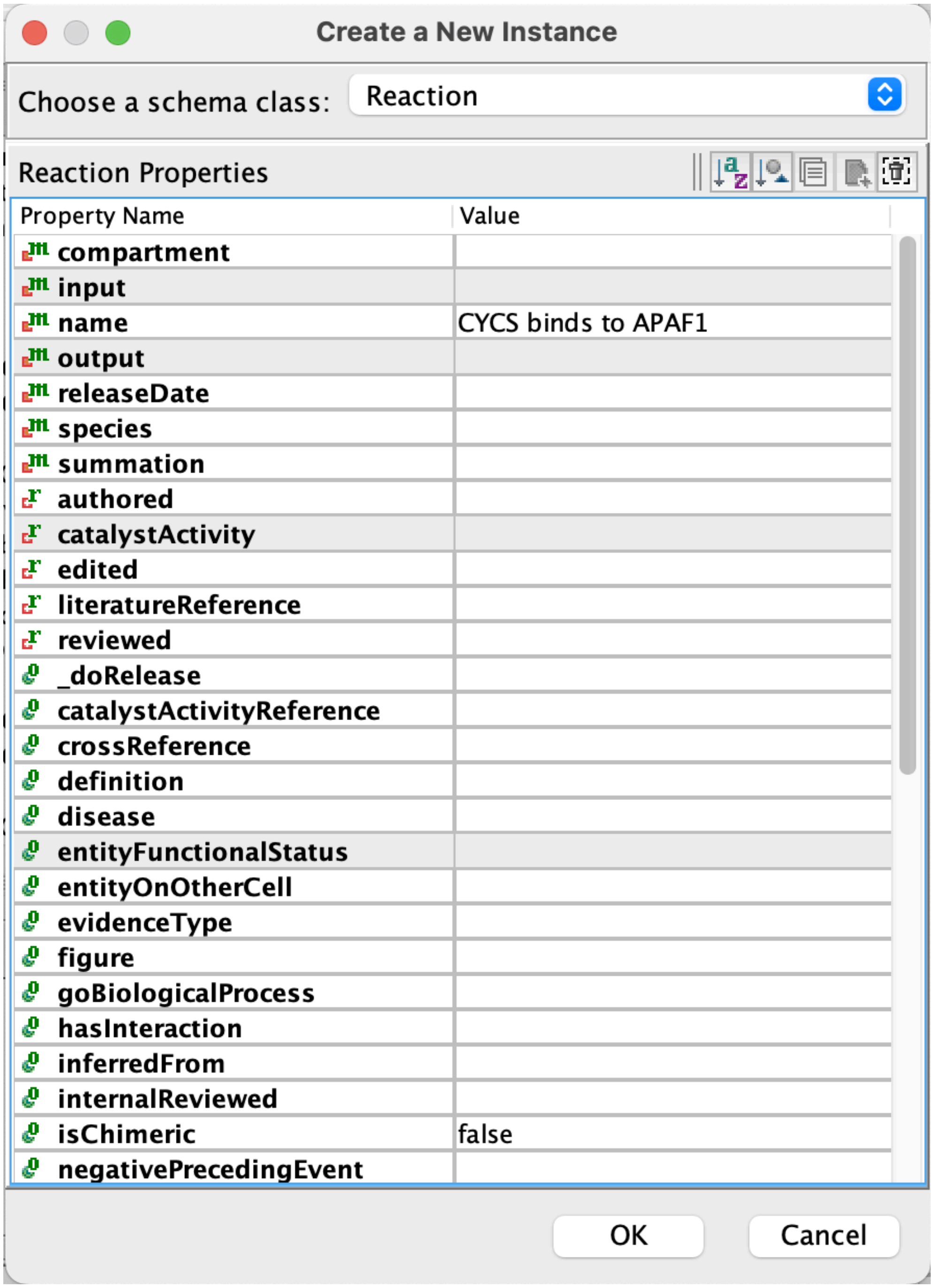
- note: If you are creating a top level pathway(check with Peter if this is appropriate), it must be listed as frontPage item. To do this, check out the frontPage instance from gk_central and add your pathways in the frontPageItem slot. Also please mark this in the editorial calendar as a front page item.

Creating a ReactionLikeEvent

There are subclasses of ReactionlikeEvent: Reaction, BlackBox, Polymerization, Depolymerization, FailedReaction, and CellDevelopmentStep. The latter two will be covered in the Disease and the Cell Lineage Path annotation sections of this guide.

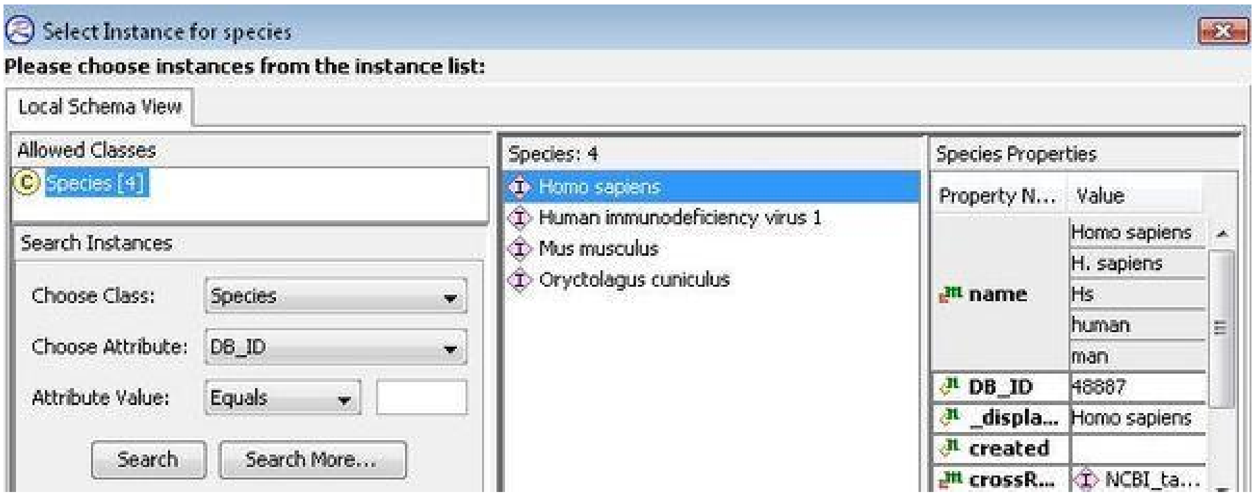
Like all other classes, you can create a new reaction by selecting the Reaction class in the Schema view, right-click and select Create Instance, but it is perhaps better practice and more intuitive to create new reactions inside a pathway. To do this, select a pathway in the Event Hierarchical View, select the hasEvent property name in the details panel on the right, right-click and select Add. This leads to a dialogue for providing details of the reaction.

When creating a reaction, first enter the name:

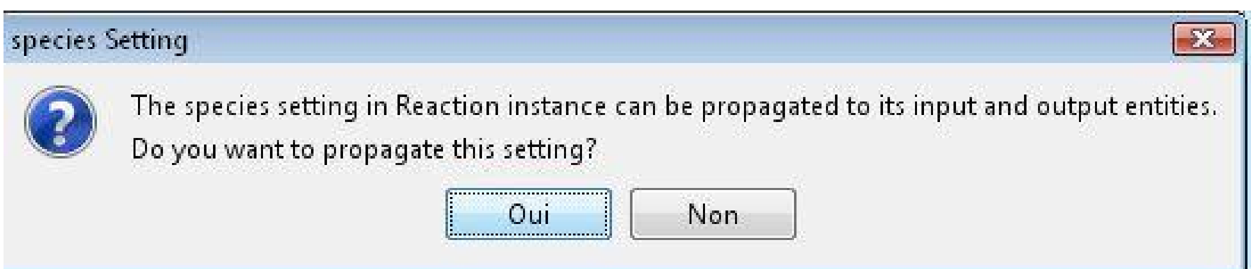


This may be enough detail for the moment, if you are simply creating a placeholder click OK.

To set the species of the reaction right click and select add in the species field.

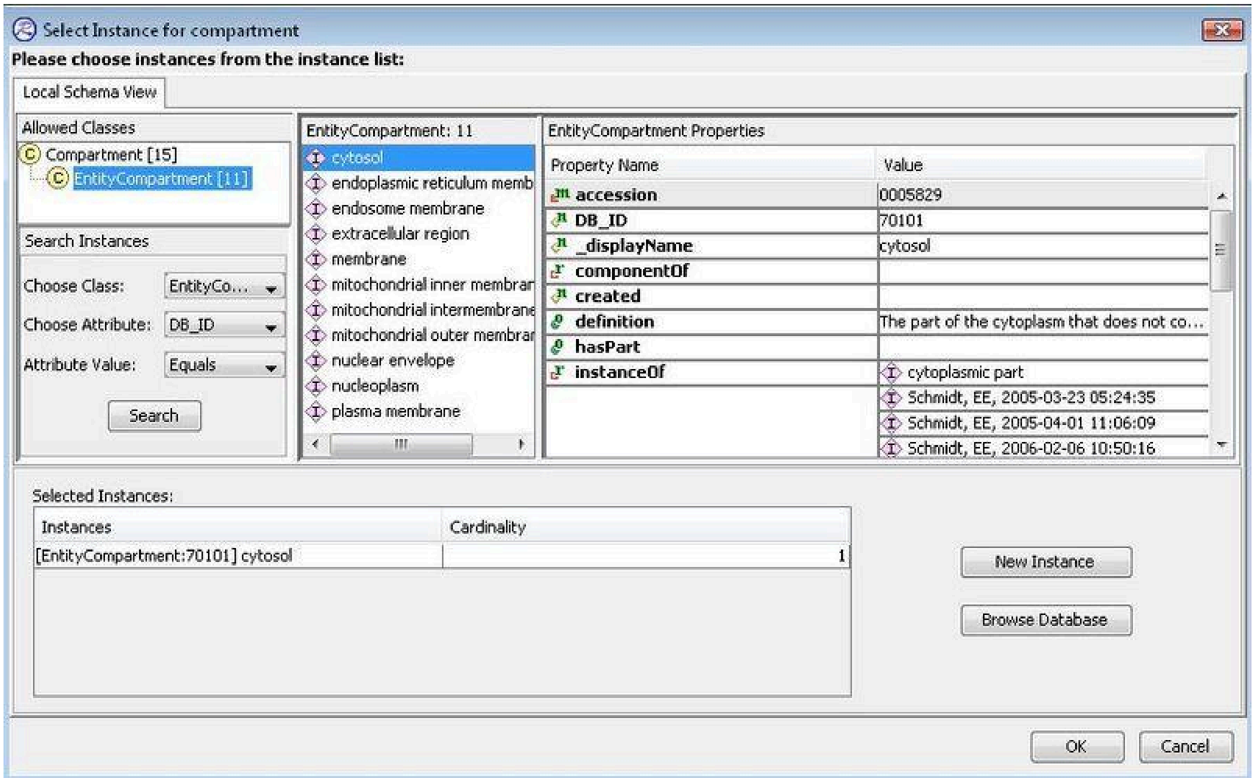


If the species that you happen to be working with is not in your local project you can opt to search for it in the database using the "Browse Database" Button in the dialog box. When you set/change the species or the compartment of a reaction, it will ask if you want to propagate the species/compartment to all of the component molecules as well.



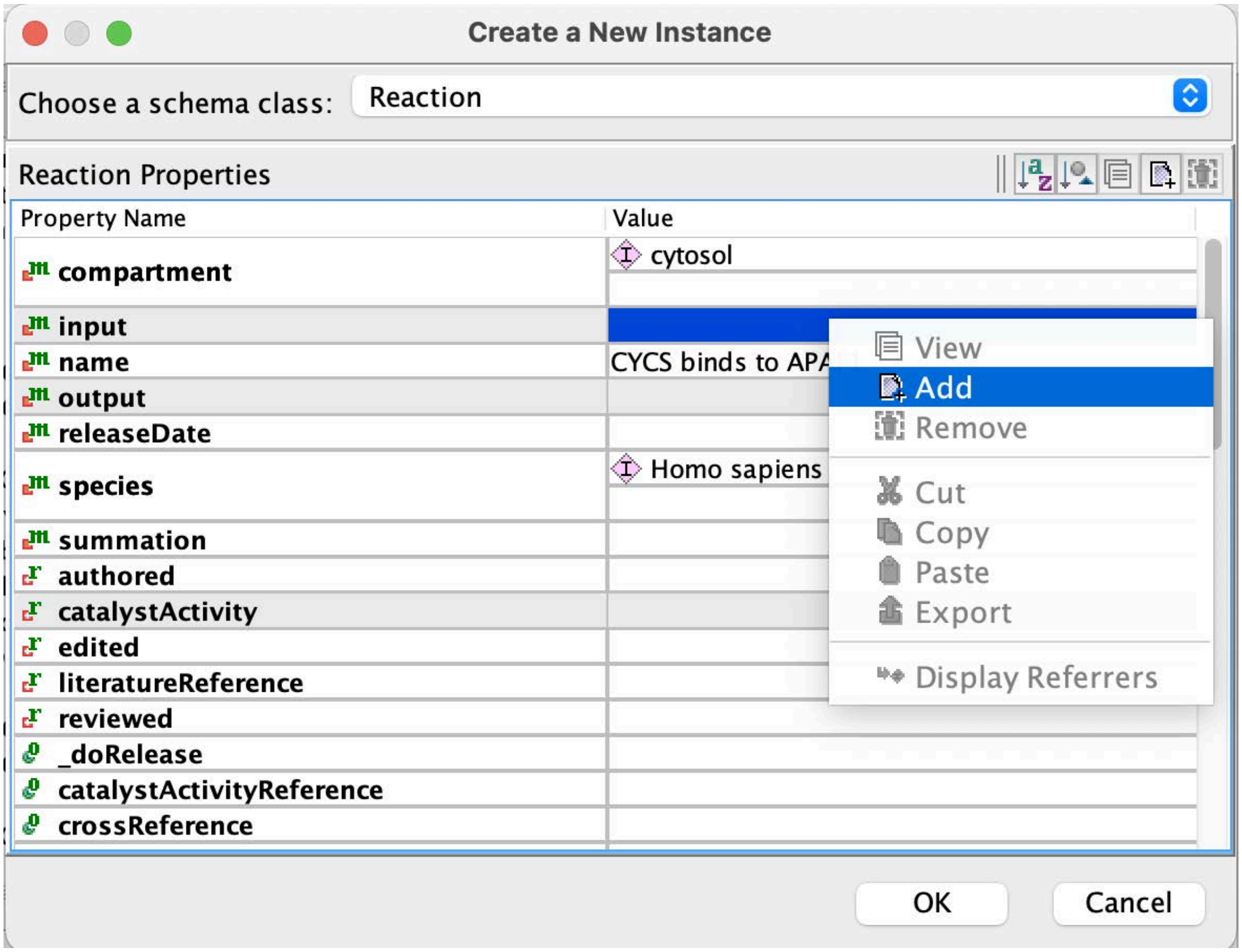
Use caution when selecting Yes. Generally, this is ONLY done when you have just created (but not committed) all of the entities used in the reaction. **ONLY** say yes here if you know that event and contained molecules have no referrers in the database that would be affected. (In other words ALL other reactions or complexes or sets in the db that make use of these now "changed" molecules would be affected).

Next add the compartment to the reaction. Right click in the compartment box and select add.

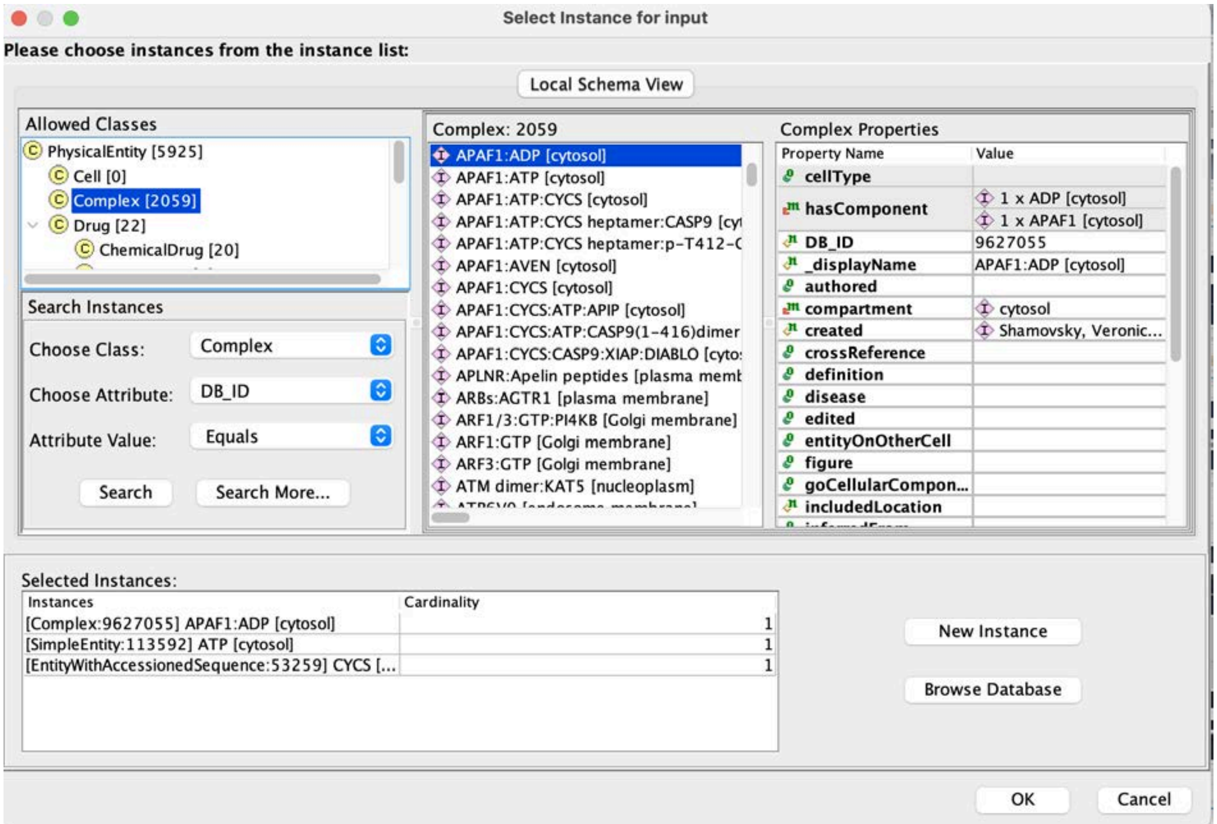


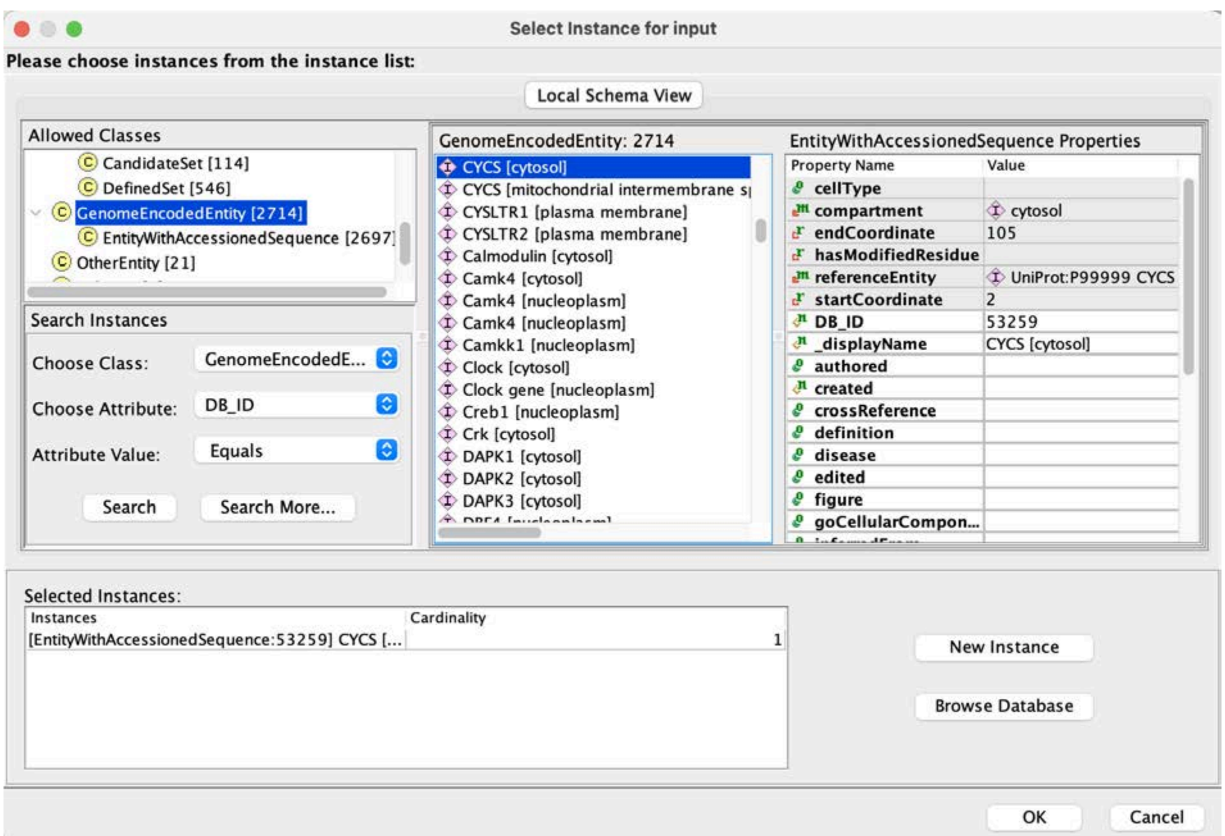
Again if you don't have the compartment you need in your project, you can Browse Database to find the one you need. ***If you still don't find the necessary compartment, DO NOT create a new one. Contact the managing editor. New compartment creation requires multiple steps as well as changes in the UI which need to be coordinated with developers.***

Now add the input and output molecules. Right click in the respective box and select "Add".



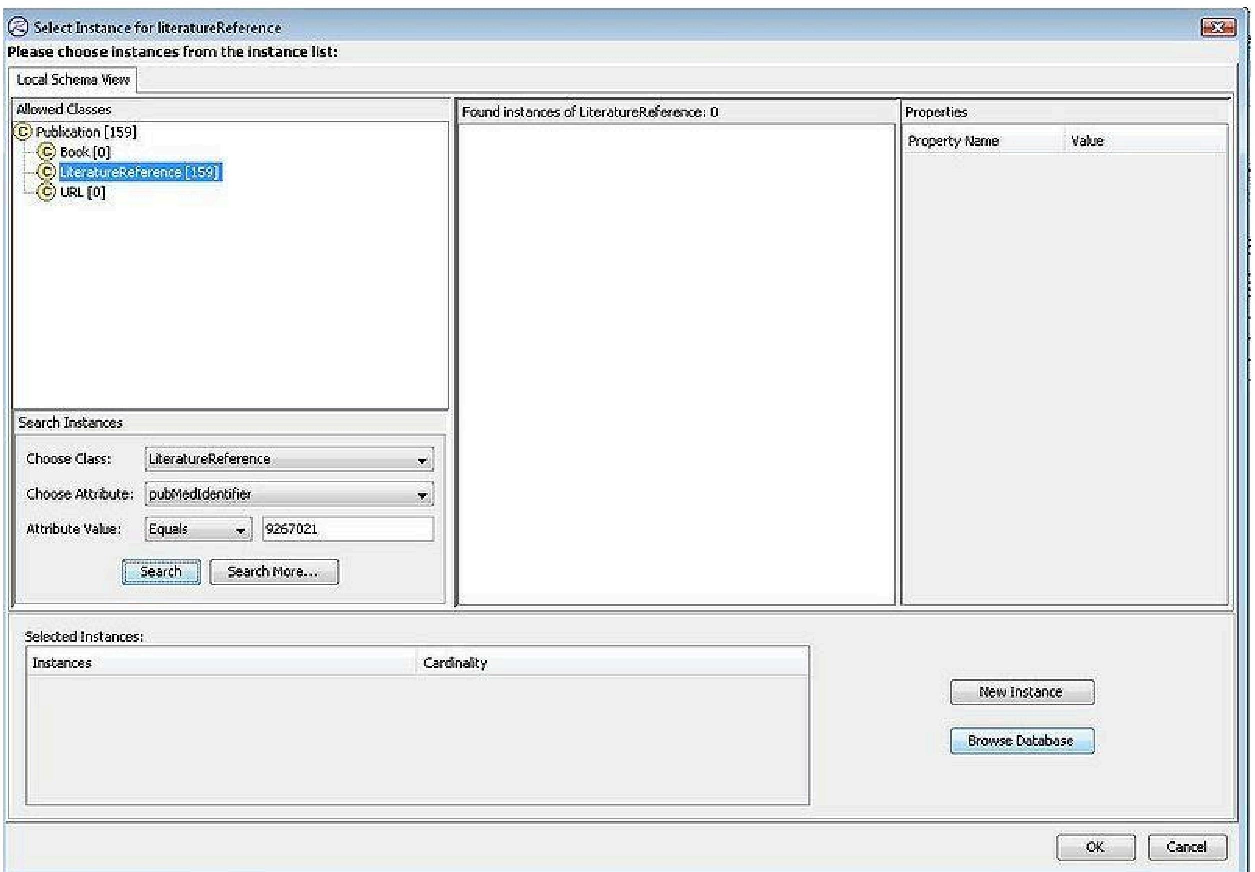
Select the molecule from the appropriate class:

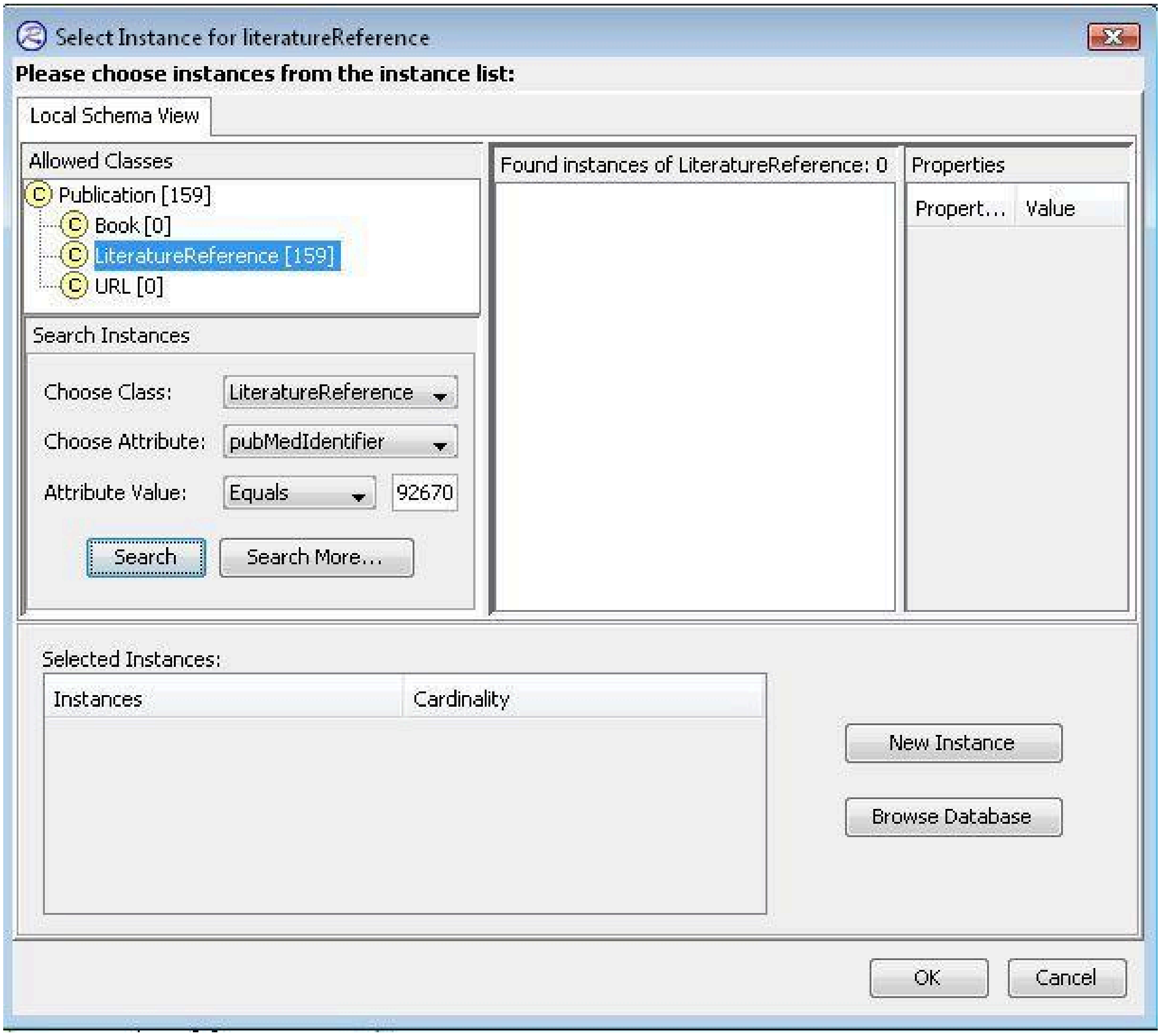

Select Instance for input
Please choose instances from the instance list:
Local Schema View
Allowed Classes
PhysicalEntity [5925]
Cell [0]
Complex [2059]
Drug [22]
ChemicalDrug [20]
Search Instances
Choose Class: Complex
Choose Attribute: DB_ID
Attribute Value: Equals
Search
Search More...
Complex: 2059
APAF1:ADP [cytosol]
APAF1:ATP [cytosol]
APAF1:ATP:CYCS [cytosol]
APAF1:ATP:CYCS heptamer:CASP9 [cy
APAF1:ATP:CYCS heptamer:p-T412-C
APAF1:AVEN [cytosol]
APAF1:CYCS [cytosol]
APAF1:CYCS:ATP:APIP [cytosol]
APAF1:CYCS:ATP:CASP9(1-416)dimer
APAF1:CYCS:CASP9:XIAP:DIABLO [cyto
APLNR:Apelin peptides [plasma memb
ARBs:AGTR1 [plasma membrane]
ARF1/3:GTP:PI4KB [Golgi membrane]
ARF1:GTP [Golgi membrane]
ARF3:GTP [Golgi membrane]
ATM dimer:KAT5 [nucleoplasm]
Complex Properties
Property Name
Value
cellType
hasComponent
1 x ADP [cytosol]
1 x APAF1 [cytosol]
DB_ID
9627055
_displayName
APAF1:ADP [cytosol]
authored
compartment
cytosol
created
Shamovsky, Veronic...
crossReference
definition
disease
edited
entityOnOtherCell
figure
goCellularCompon...
includedLocation
Selected Instances:
Instances
Cardinality
[Complex:9627055] APAF1:ADP [cytosol]
1
[SimpleEntity:113592] ATP [cytosol]
1
[EntityWithAccessionedSequence:53259] CYCS [...
1
New Instance
Browse Database
OK
Cancel


Select Instance for input
Please choose instances from the instance list:
Local Schema View
Allowed Classes
CandidateSet [114]
DefinedSet [546]
GenomeEncodedEntity [2714]
EntityWithAccessionedSequence [2697]
OtherEntity [21]
Search Instances
Choose Class: GenomeEncodedE...
Choose Attribute: DB_ID
Attribute Value: Equals
Search
Search More...
GenomeEncodedEntity: 2714
CYCS [cytosol]
CYCS [mitochondrial intermembrane s
CYSLTR1 [plasma membrane]
CYSLTR2 [plasma membrane]
Calmodulin [cytosol]
Camk4 [cytosol]
Camk4 [nucleoplasm]
Camk4 [nucleoplasm]
Camkk1 [nucleoplasm]
Clock [cytosol]
Clock gene [nucleoplasm]
Creb1 [nucleoplasm]
Crk [cytosol]
DAPK1 [cytosol]
DAPK2 [cytosol]
DAPK3 [cytosol]
EntityWithAccessionedSequence Properties
Property Name
Value
cellType
compartment
cytosol
endCoordinate
105
hasModifiedResidue
referenceEntity
UniProt:P99999 CYCS
startCoordinate
2
DB_ID
53259
_displayName
CYCS [cytosol]
authored
created
crossReference
definition
disease
edited
figure
goCellularCompon...
Selected Instances:
Instances
Cardinality
[EntityWithAccessionedSequence:53259] CYCS [...
1
New Instance
Browse Database
OK
Cancel

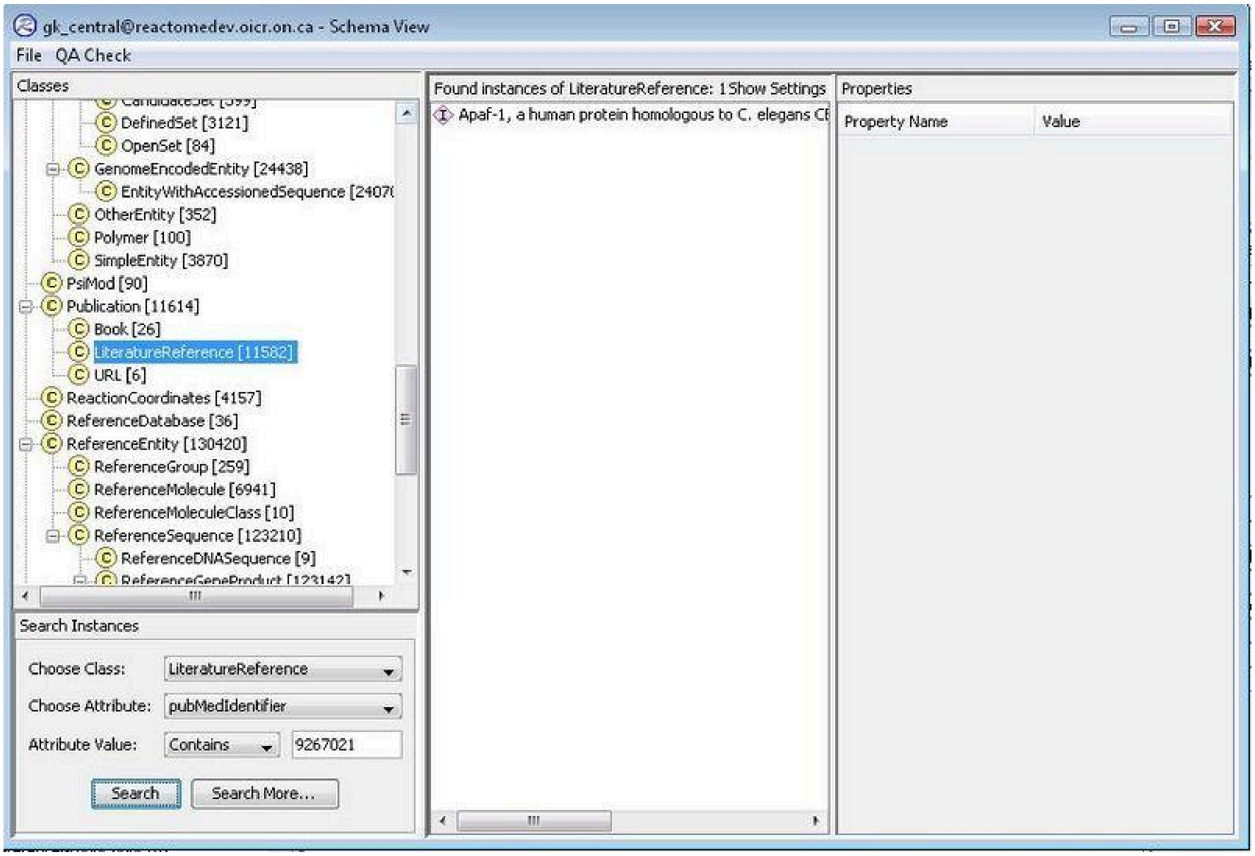
Important: After adding your input and output molecules, it is important to verify that your reaction is balanced (all molecules represented as input are also present as output). See the QA section below for a description of how to do this.

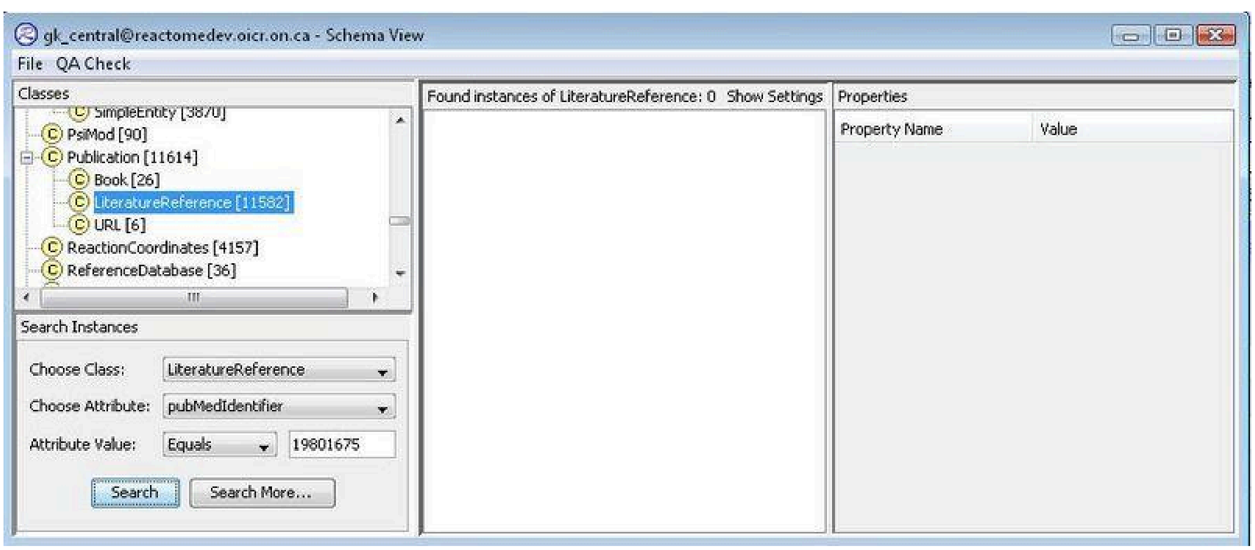
Add the literature reference(s). The references associated with a reaction MUST provide direct experimental evidence for the occurrence of that reaction in the species you are annotating (i.e. human for human Reactome). If there is no direct experimental evidence in human then you need to create an inferred human reaction as described in the section below. Enter the PMID for journal articles, and say yes when prompted to have the details filled in automatically.





If you don't see the reference in your local repository then opt to Browse Database.



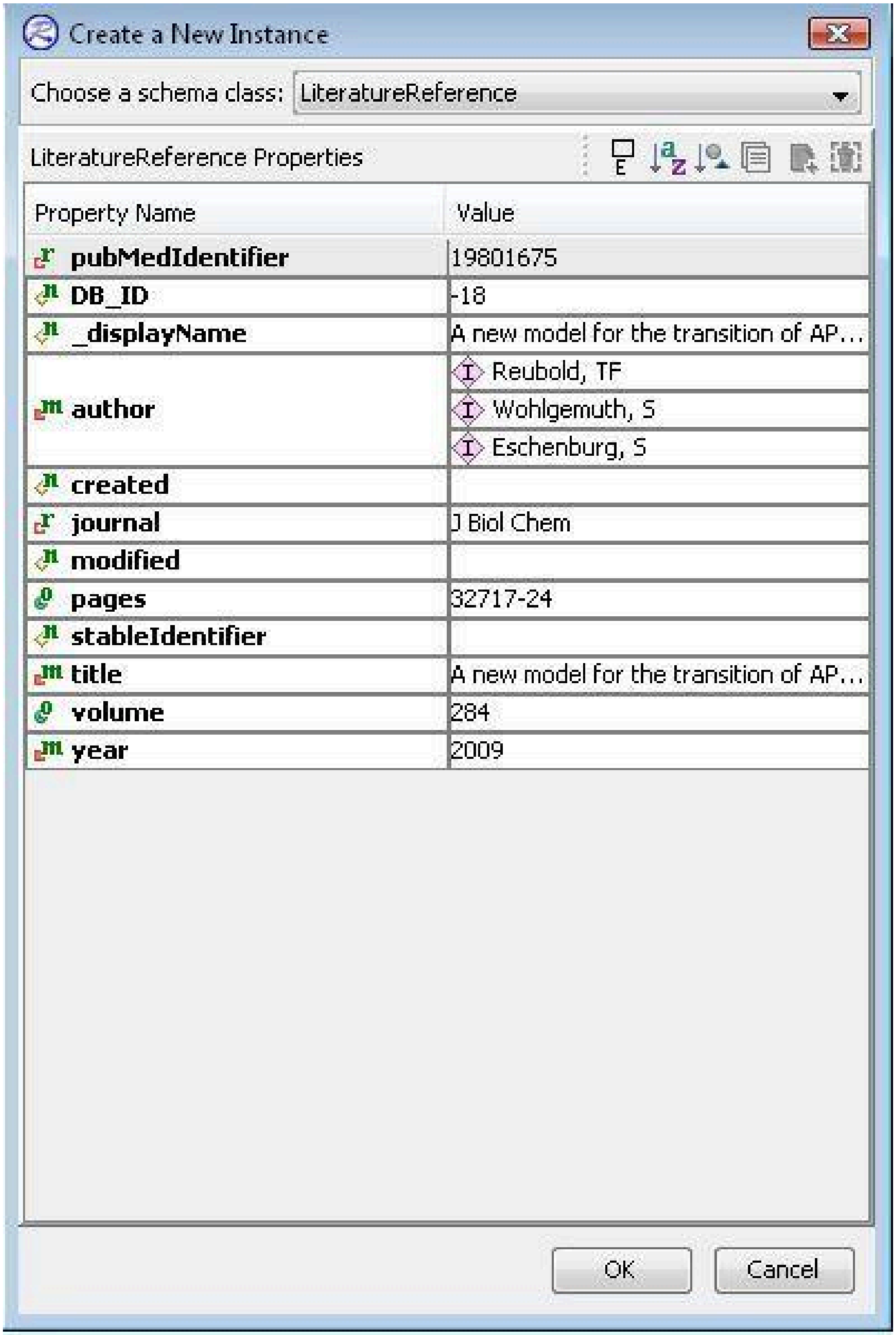


If you can't find the literature reference in the database either, then you need to create a new one:

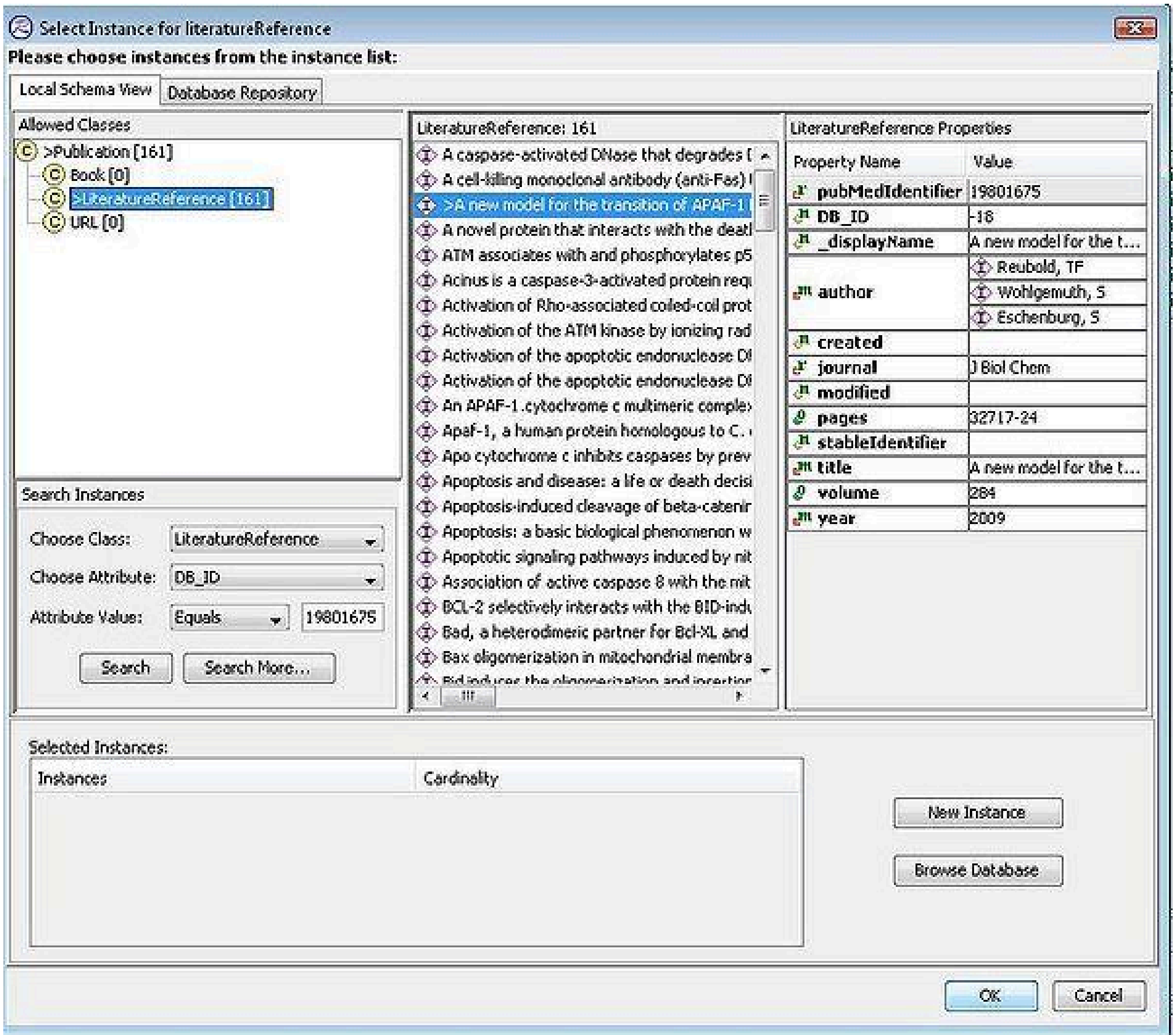
Enter the PMID (number only). Click out of the PMID box. You will be asked if you want to import the PMID record information. Say yes.


Create a New Instance
Choose a schema class:
LiteratureReference
Specify properties for new instance:
Property Name
Value
pubMedIdentifier
19801675
DB_ID
-18
_displayName
author
created
journal
modified
pages
Fetching Information?
Do you want to fetch other information for the specified PMID?
Oui
Non
OK
Cancel

You will now see the full record:

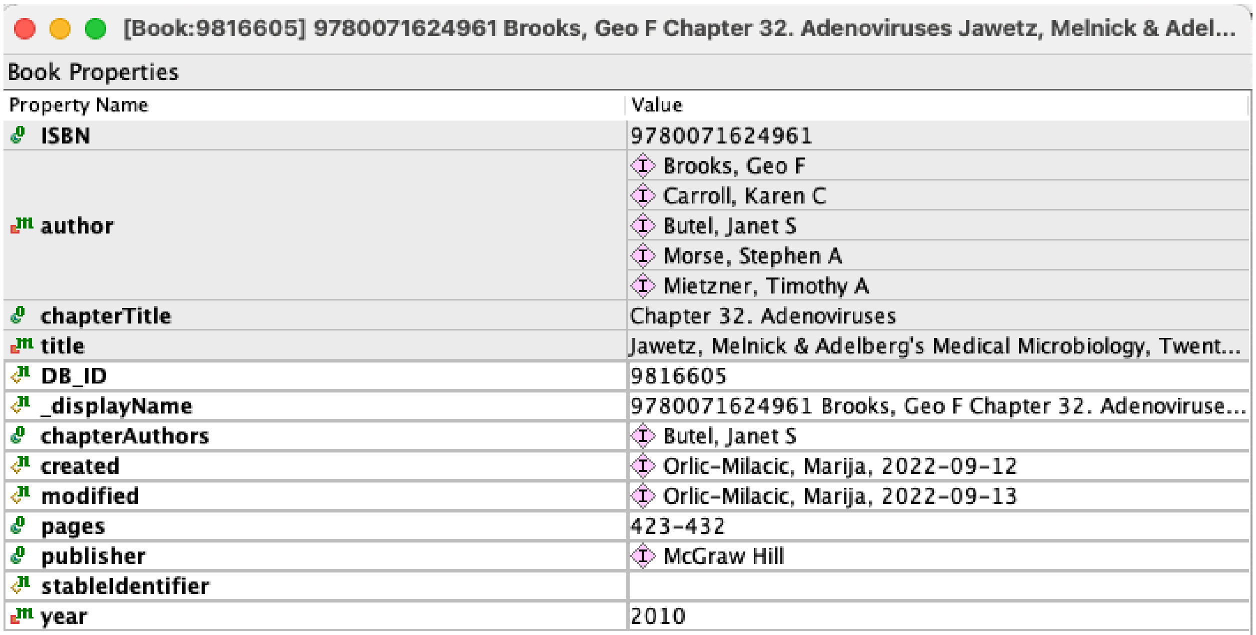

Create a New Instance
Choose a schema class: LiteratureReference
LiteratureReference Properties
Property Name
Value
pubMedIdentifier
19801675
DB_ID
-18
_displayName
A new model for the transition of AP...
author
Reubold, TF
Wohlgemuth, S
Eschenburg, S
created
journal
J Biol Chem
modified
pages
32717-24
stableIdentifier
title
A new model for the transition of AP...
volume
284
year
2009
OK
Cancel

Look back in your local repository and you will see the new literature reference. You can now add this as a reference for your reaction.



The process is similar for adding other Publication types (Books and urls) with the exception that all information must be added manually - there is no automatic import.

To make a Publication record for a book, all information must be entered manually: author(s), title, year (mandatory attributes), ISBN, title, chapterTitle, pages (optional) as shown:

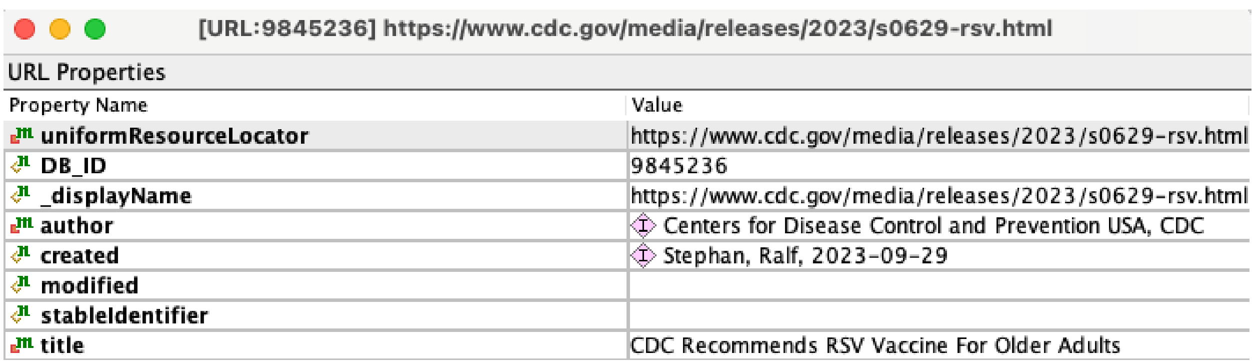


[Book:9816605] 9780071624961 Brooks, Geo F Chapter 32. Adenoviruses Jawetz, Melnick & Adel...

Book Properties

Property Name	Value
ISBN	9780071624961
author	Brooks, Geo F Carroll, Karen C Butel, Janet S Morse, Stephen A Mietzner, Timothy A
chapterTitle	Chapter 32. Adenoviruses
title	Jawetz, Melnick & Adelberg's Medical Microbiology, Twent...
DB_ID	9816605
_displayName	9780071624961 Brooks, Geo F Chapter 32. Adenoviruse...
chapterAuthors	Butel, Janet S
created	Orlic-Milacic, Marija, 2022-09-12
modified	Orlic-Milacic, Marija, 2022-09-13
pages	423-432
publisher	McGraw Hill
stableIdentifier	
year	2010

The image below shows an example of a Publication record for a url; again, information must be entered manually.

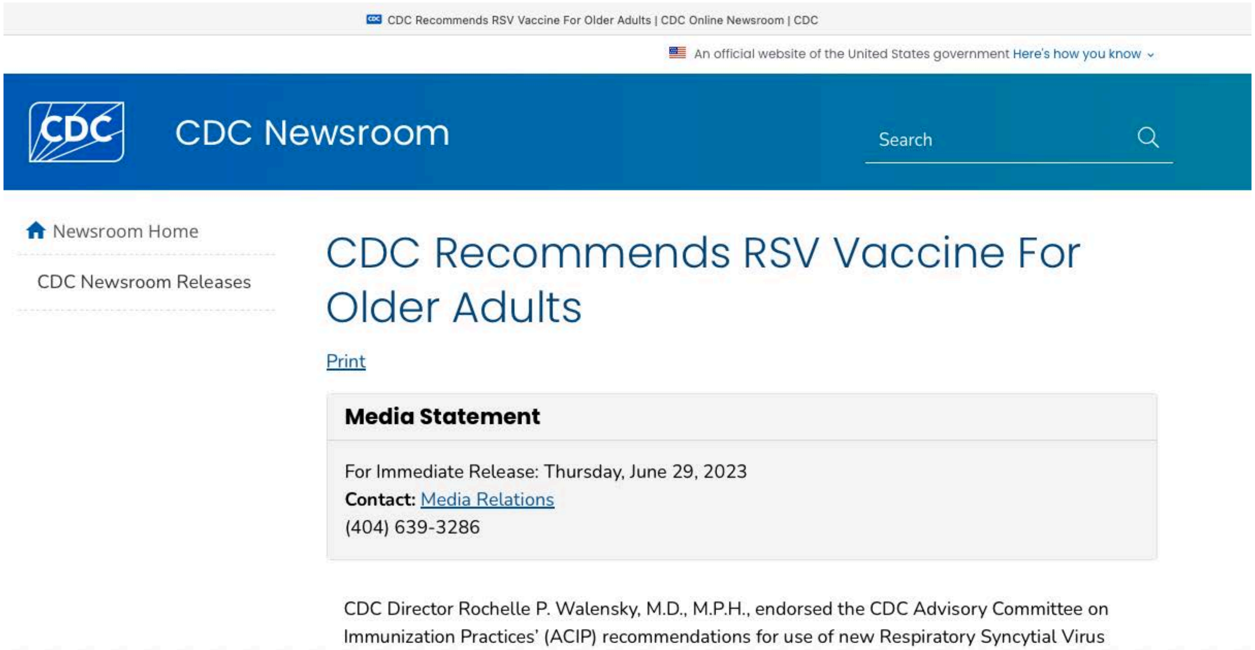


[URL:9845236] https://www.cdc.gov/media/releases/2023/s0629-rsv.html

URL Properties

Property Name	Value
uniformResourceLocator	https://www.cdc.gov/media/releases/2023/s0629-rsv.html
DB_ID	9845236
_displayName	https://www.cdc.gov/media/releases/2023/s0629-rsv.html
author	Centers for Disease Control and Prevention USA, CDC
created	Stephan, Ralf, 2023-09-29
modified	
stableIdentifier	
title	CDC Recommends RSV Vaccine For Older Adults

There is some discretionary latitude in what can be entered as the url title and author. The title may be the heading on the page itself, as in the case below:


CDC Recommends RSV Vaccine For Older Adults | CDC Online Newsroom | CDC

An official website of the United States government Here's how you know

CDC Newsroom

Search

Newsroom Home

CDC Newsroom Releases

CDC Recommends RSV Vaccine For Older Adults

Print

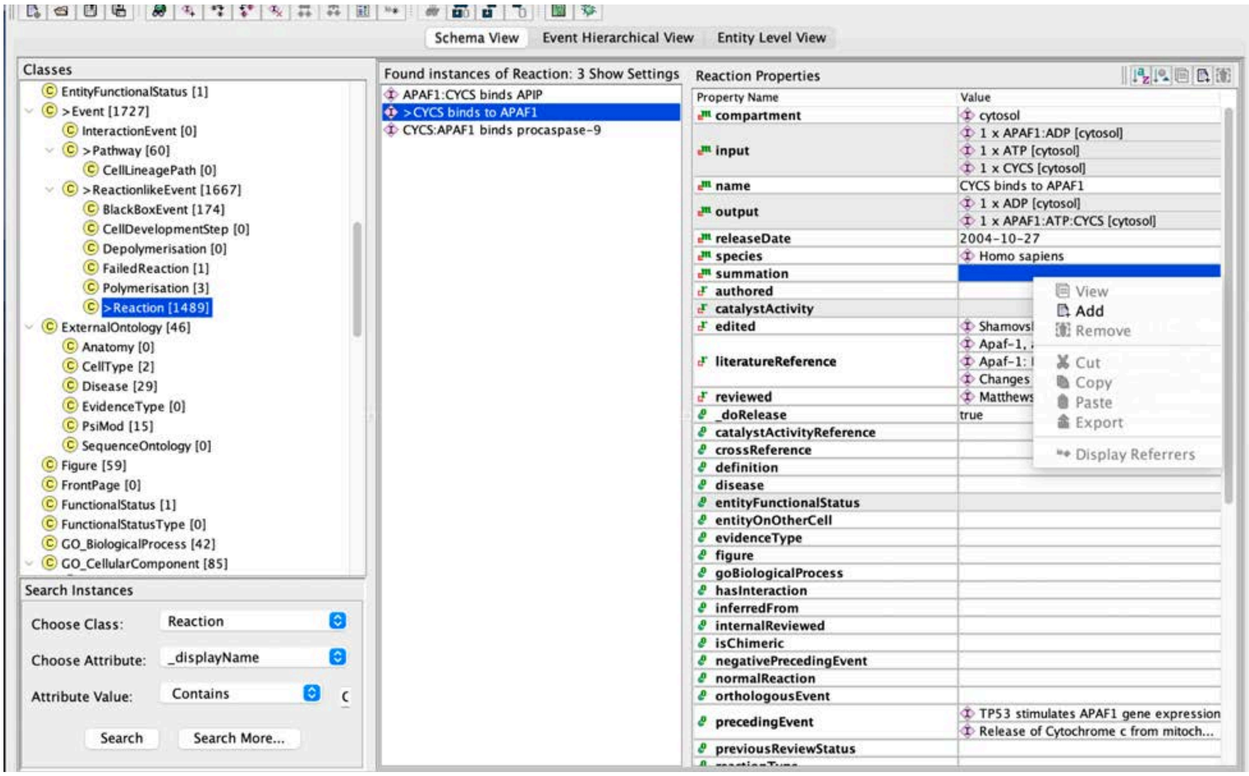
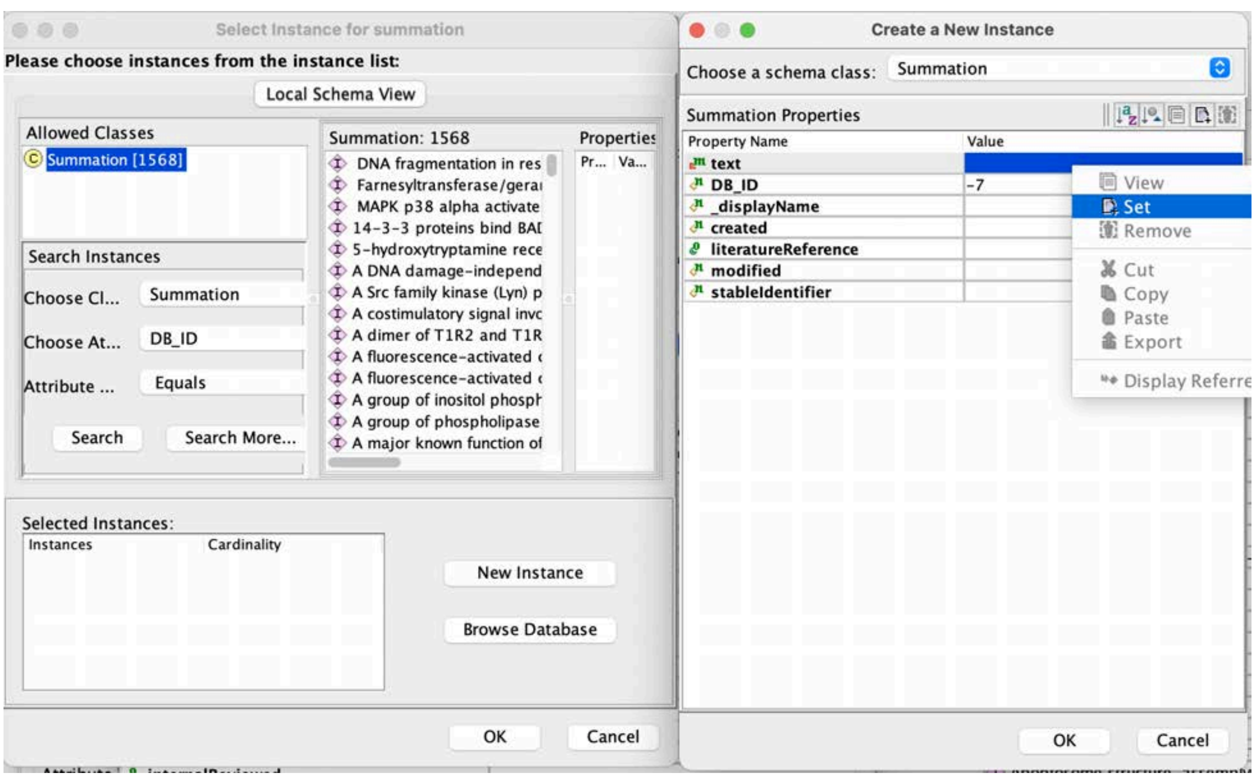
Media Statement

For Immediate Release: Thursday, June 29, 2023
Contact: Media Relations
(404) 639-3286

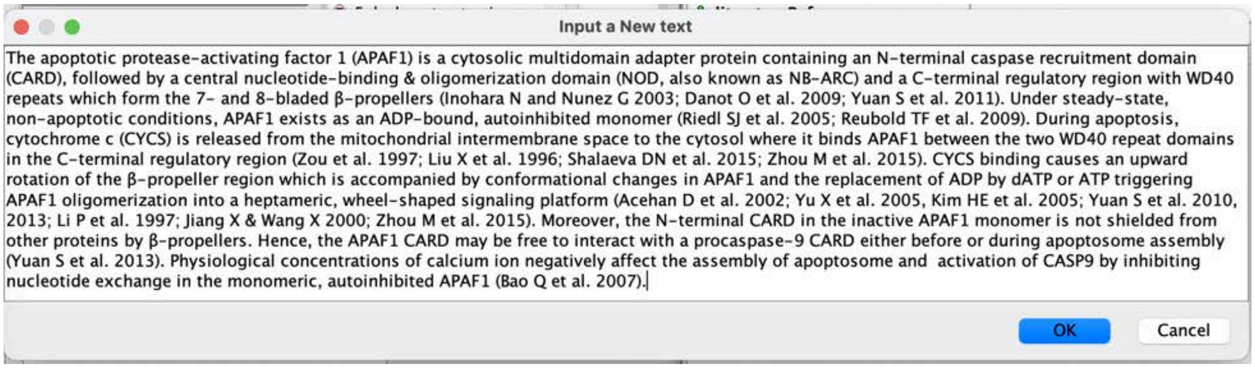
CDC Director Rochelle P. Walensky, M.D., M.P.H., endorsed the CDC Advisory Committee on Immunization Practices' (ACIP) recommendations for use of new Respiratory Syncytial Virus

If in doubt, consult gk_central for existing records for similar urls.

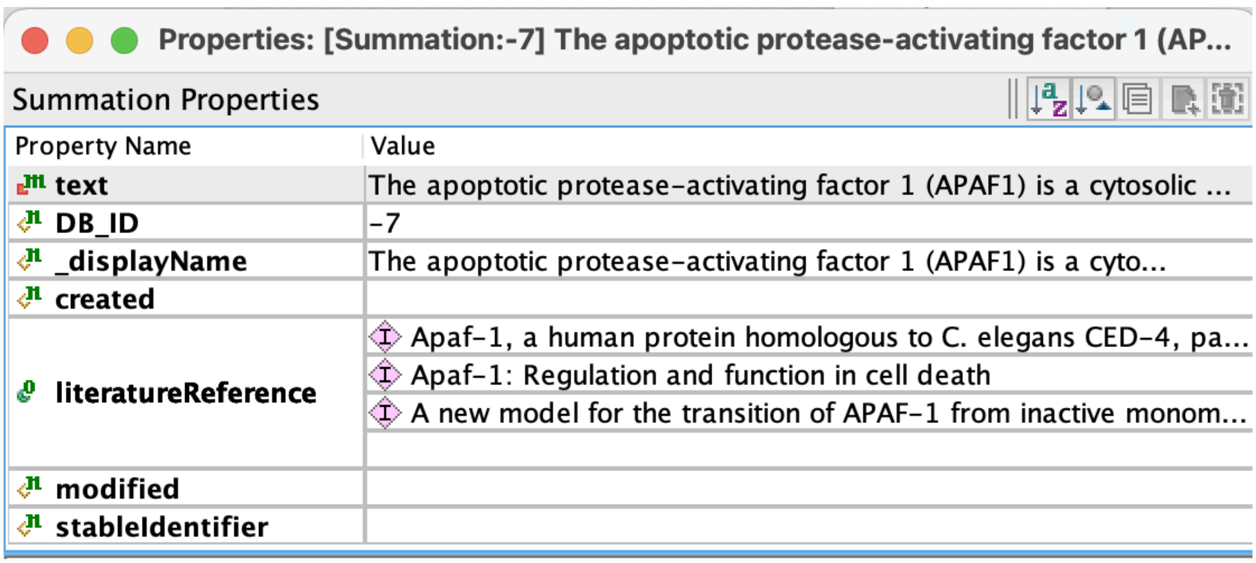
Once you have added your references, you can add a text summary for the reaction in the "summation"slot. Right click and select add.


Schema View
Event Hierarchical View
Entity Level View
Classes
EntityFunctionalStatus [1]
>Event [1727]
InteractionEvent [0]
>Pathway [60]
CellLineagePath [0]
>ReactionlikeEvent [1667]
BlackBoxEvent [174]
CellDevelopmentStep [0]
Depolymerisation [0]
FailedReaction [1]
Polymerisation [3]
>Reaction [1489]
ExternalOntology [46]
Anatomy [0]
CellType [2]
Disease [29]
EvidenceType [0]
PsiMod [15]
SequenceOntology [0]
Figure [59]
FrontPage [0]
FunctionalStatus [1]
FunctionalStatusType [0]
GO_BiologicalProcess [42]
GO_CellularComponent [85]
Search Instances
Choose Class: Reaction
Choose Attribute: _displayName
Attribute Value: Contains
Search
Search More...
Found instances of Reaction: 3 Show Settings
APAF1:CYCS binds APIP
>CYCS binds to APAF1
CYCS:APAF1 binds procaspase-9
Reaction Properties
Property Name
Value
compartment
cytosol
input
1 x APAF1:ADP [cytosol]
1 x ATP [cytosol]
1 x CYCS [cytosol]
name
CYCS binds to APAF1
output
1 x ADP [cytosol]
1 x APAF1:ATP:CYCS [cytosol]
releaseDate
2004-10-27
species
Homo sapiens
summation
authored
catalystActivity
edited
literatureReference
reviewed
_doRelease
true
catalystActivityReference
crossReference
definition
disease
entityFunctionalStatus
entityOnOtherCell
evidenceType
figure
goBiologicalProcess
hasInteraction
inferredFrom
internalReviewed
isChimeric
negativePrecedingEvent
normalReaction
orthologousEvent
precedingEvent
TP53 stimulates APAF1 gene expression
Release of Cytochrome c from mitoch...
previousReviewStatus
View
Add
Remove
Cut
Copy
Paste
Export
Display Referrers

Select Instance for summation
Please choose instances from the instance list:
Local Schema View
Allowed Classes
Summation [1568]
Summation: 1568
Properties
DNA fragmentation in res
Farnesyltransferase/gerai
MAPK p38 alpha activate
14-3-3 proteins bind BAI
5-hydroxytryptamine rece
A DNA damage-independ
A Src family kinase (Lyn) p
A costimulatory signal invc
A dimer of T1R2 and T1R
A fluorescence-activated
A fluorescence-activated
A group of inositol phosph
A group of phospholipase
A major known function of
Search Instances
Choose Cl... Summation
Choose At... DB_ID
Attribute ... Equals
Search
Search More...
Selected Instances:
Instances
Cardinality
New Instance
Browse Database
OK
Cancel
Create a New Instance
Choose a schema class: Summation
Summation Properties
Property Name
Value
text
DB_ID
-7
_displayName
created
literatureReference
modified
stableIdentifier
View
Set
Remove
Cut
Copy
Paste
Export
Display Referre
OK
Cancel

Then input your text and citations.



Associate the references associated with the citations in the text summary using the "LiteratureReference" slot for the summation as described above for reactions.



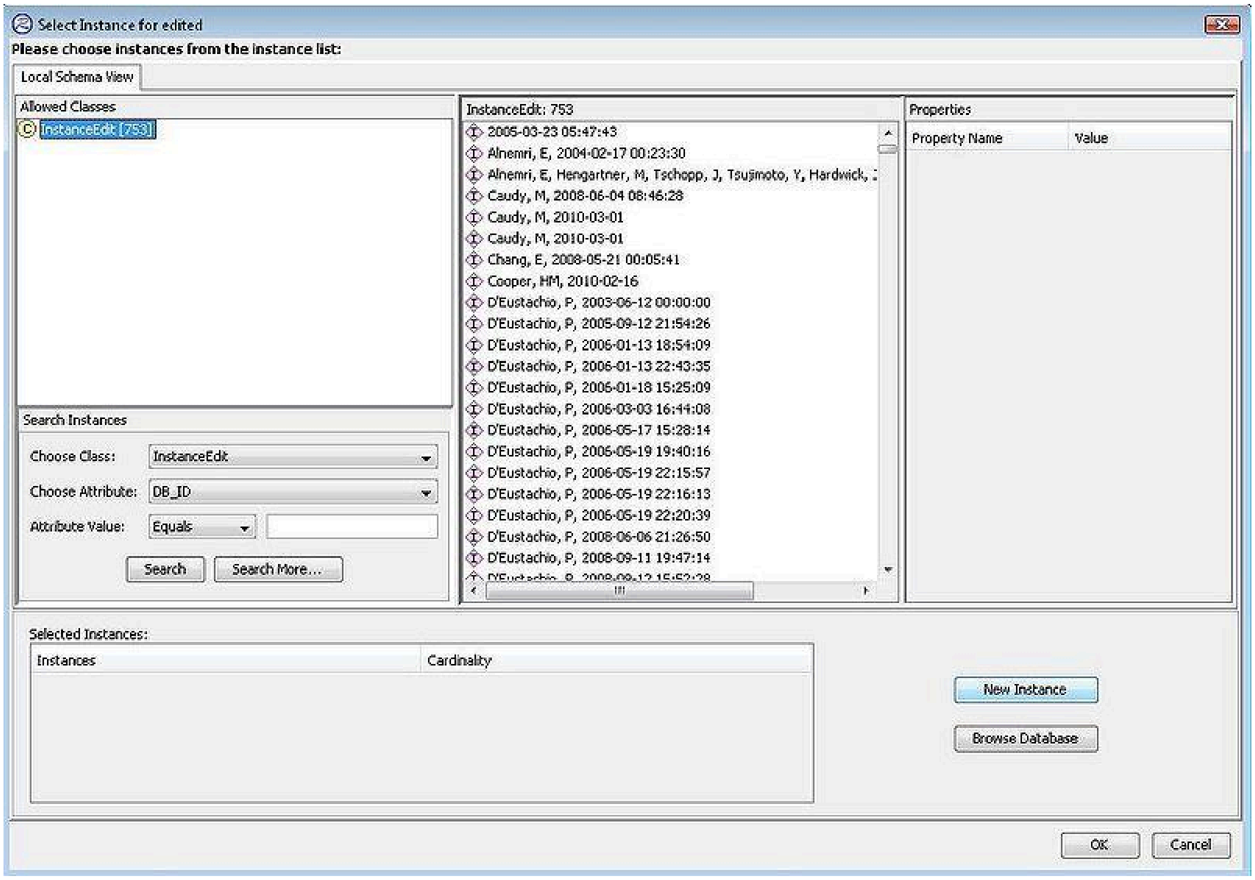
Once the mandatory attributes have been filled, enter the remaining required attributes. Right click on edited to choose an instanceEdit from your project.

Properties: [Reaction:-2] CYCS binds to APAF1

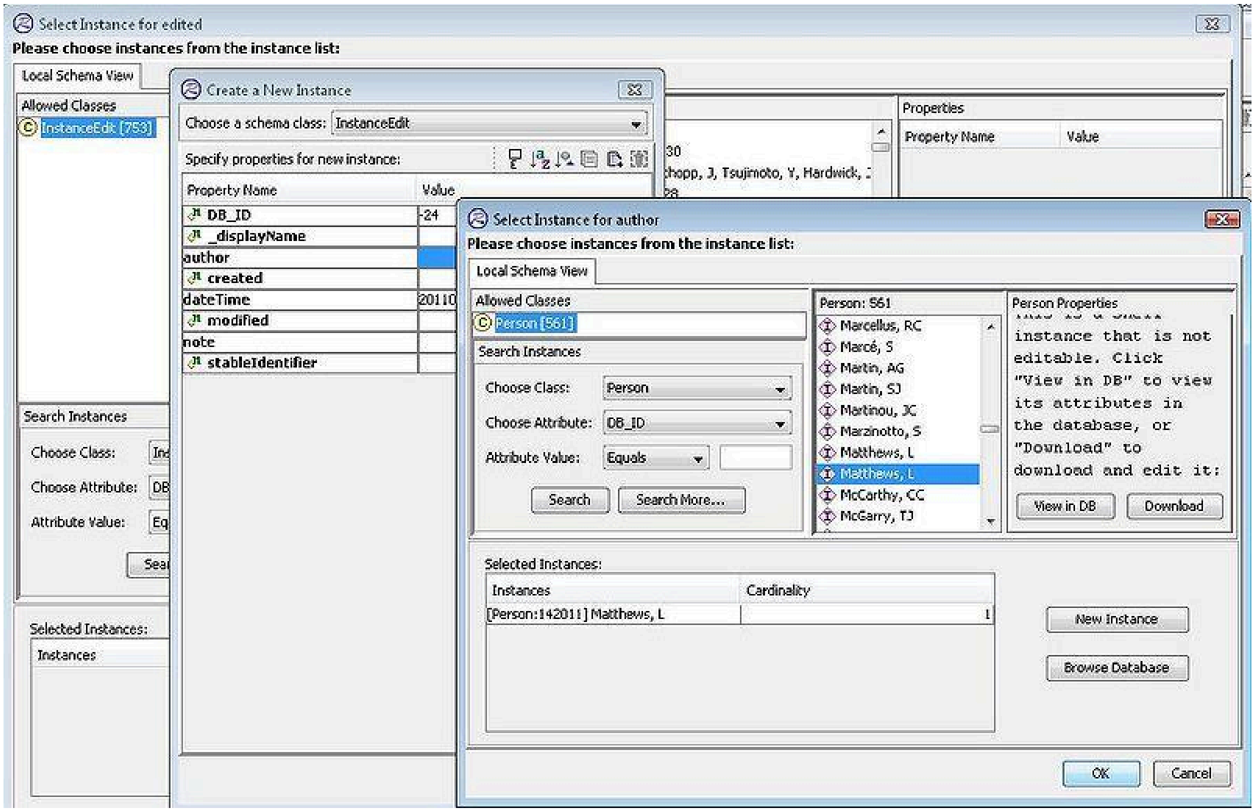
Reaction Properties

Property Name	Value
compartment	cytosol
input	1 x APAF1:ADP [cytosol] 1 x ATP [cytosol] 1 x CYCS [cytosol]
name	CYCS binds to APAF1
output	1 x ADP [cytosol] 1 x APAF1:ATP:CYCS [cytosol]
releaseDate	
species	Homo sapiens
summation	The apoptotic protease-activating factor 1 (APAF1) is ...
authored	
catalystActivity	
edited	
literatureReference	Apaf-1, a human protein homologous to C. elegans CE... Apaf-1: Regulation and function in cell death Changes in Apaf-1 conformation that drive apoptosom...
reviewed	
_doRelease	
catalystActivityReference	
crossReference	
definition	
disease	
entityFunctionalStatus	
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	
internalReviewed	
isChimeric	false
negativePrecedingEvent	
normalReaction	
orthologousEvent	
precedingEvent	
previousReviewStatus	
reactionType	

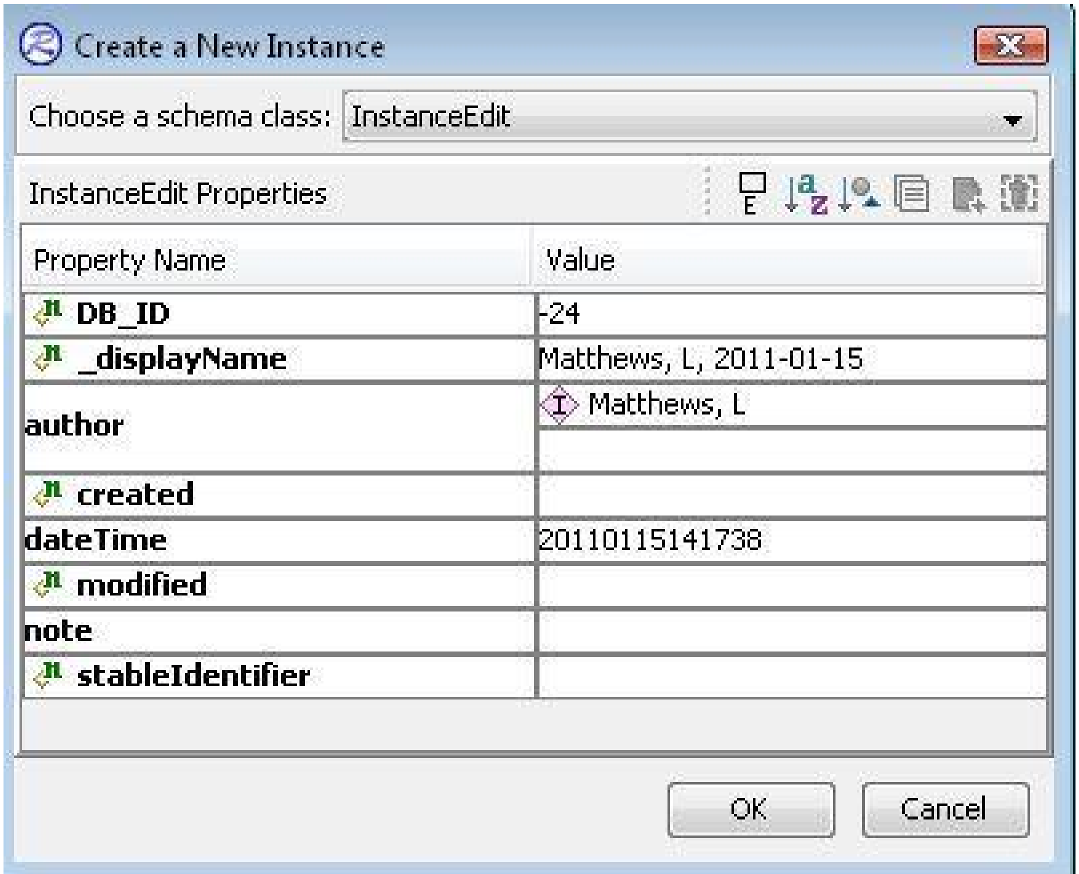
If you haven't created one previously, opt to create a new one by clicking on the New Instance button.



Then right click on author to select a person. If you don't see the one you want locally, browse the database. The edited slot holds the name of the curator that should be credited with creating and editing the event and the date it was edited. This slot is filled with an instanceedit instance that contains this information.



A date time stamp is created automatically for that instanceEdit and clicking "OK" will add this instanceEdit to the edited slot in your reaction.



The authored and reviewed slots hold the instanceEdits describing the author/reviewer and dates of authoring/reviewing respectively. If a module has an external author (and no external reviewer) the author should be listed in both the authored and reviewed attribute.(These are created as described above for the "editor" slot. When the reaction is read for release, the do_Release flag should be set to TRUE and the releaseDate slot should be filled with the appropriate release date.

Property Name	Value
compartment	cytosol
input	1 x APAF1:ADP [cytosol] 1 x ATP [cytosol] 1 x CYCS [cytosol]
name	CYCS binds to APAF1
output	1 x ADP [cytosol] 1 x APAF1:ATP:CYCS [cytosol]
releaseDate	
species	Homo sapiens
summation	The apoptotic protease-activating factor 1 (APAF1) is...
authored	
catalystActivity	
edited	Matthews, L, 2011-01-15
literatureReference	Apaf-1, a human protein homologous to C. elegans ... Apaf-1: Regulation and function in cell death Changes in Apaf-1 conformation that drive apoptoso...
reviewed	
_doRelease	
catalystActivityReference	
crossReference	
definition	
disease	
entityFunctionalStatus	
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	
internalReviewed	
isChimeric	false
negativePrecedingEvent	
normalReaction	
orthologousEvent	

Important: After adding your input and output molecules, it is important to verify that your reaction is balanced (all molecules represented as input are also present as output). See the QA section below for a description of how to do this.

When you can define preceding events for an event it should be done. One precision on that point is that a preceding event for a reaction should always be a reaction/reaction like event and not a pathway and the preceding event for a pathway should be another pathway when relevant and not a reaction or reaction like event.

Creating an inferred event

When constructing a human pathway, a curator will come across events that have no direct experimental evidence in humans, but have supporting experimental data from another/other species. If experts in the field believe that the event can in fact occur in humans, the 'other species event' can be used to infer the human event. In the case below a human reaction is inferred from in vitro experimental results using proteins from human and Oryctolagus cuniculus (rabbit).

The reaction "Proteosome mediated degradation of PAK-2p34" is created and the species Homo sapiens and Oryctolagus cuniculus are assigned

Properties: [Reaction:-2] Ubiquitination of PAK-2p34

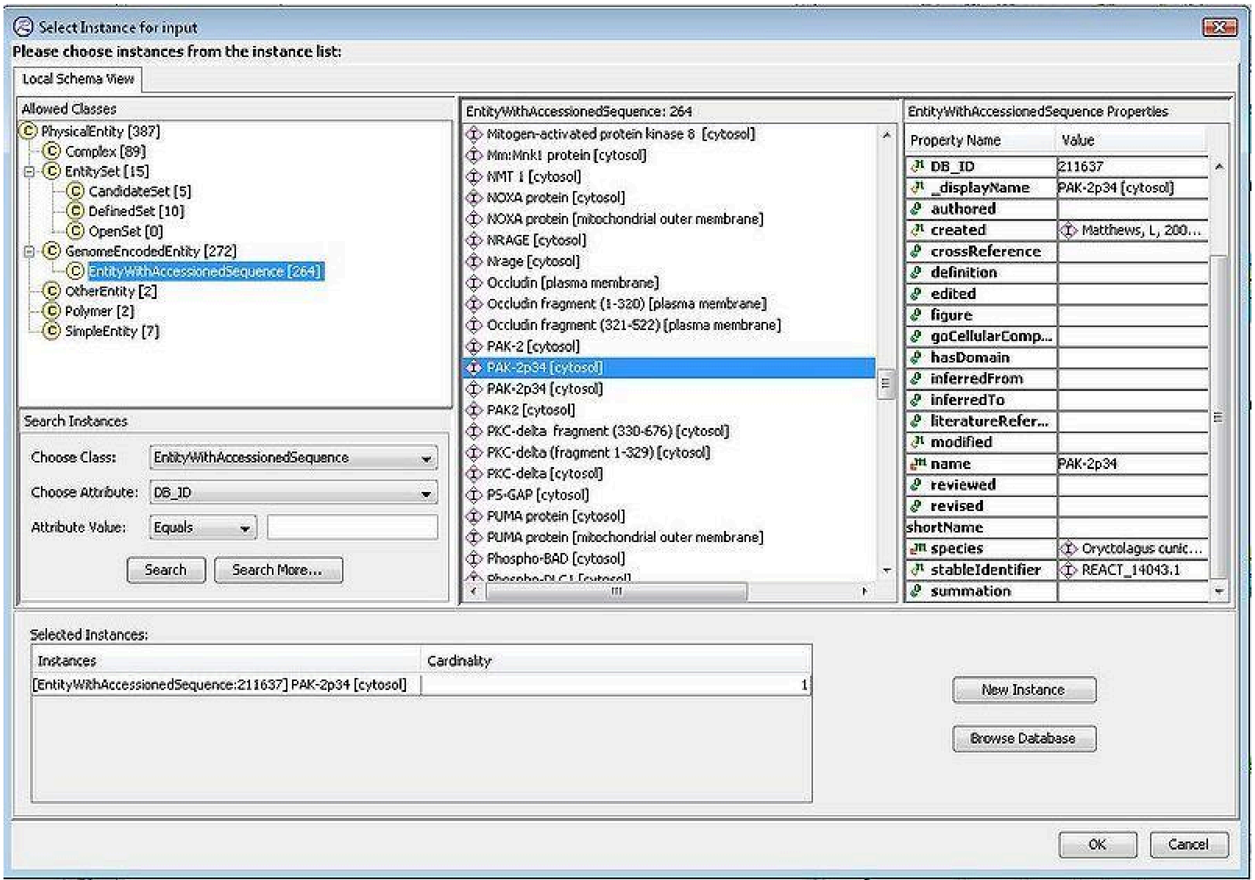
Reaction Properties

Property Name	Value
catalystActivity	
input	
output	
DB_ID	-2
_displayName	Ubiquitination of PAK-2p34
_doRelease	
authored	
compartment	cytosol
created	
crossReference	
definition	
edited	Matthews, L, 2011-01-15
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasMember	
inferredFrom	
isChimeric	true
literatureReference	Caspase-activated PAK-2 is regulated by subcellular t...
modified	
name	Ubiquitination of PAK-2p34
orthologousEvent	
precedingEvent	
releaseDate	
requiredInputComponent	
reverseReaction	
reviewed	
revised	
species	Oryctolagus cuniculus
	Homo sapiens
stableIdentifier	
summation	PAK-2p34 is ubiquitinated prior to degradation (Jakobi...

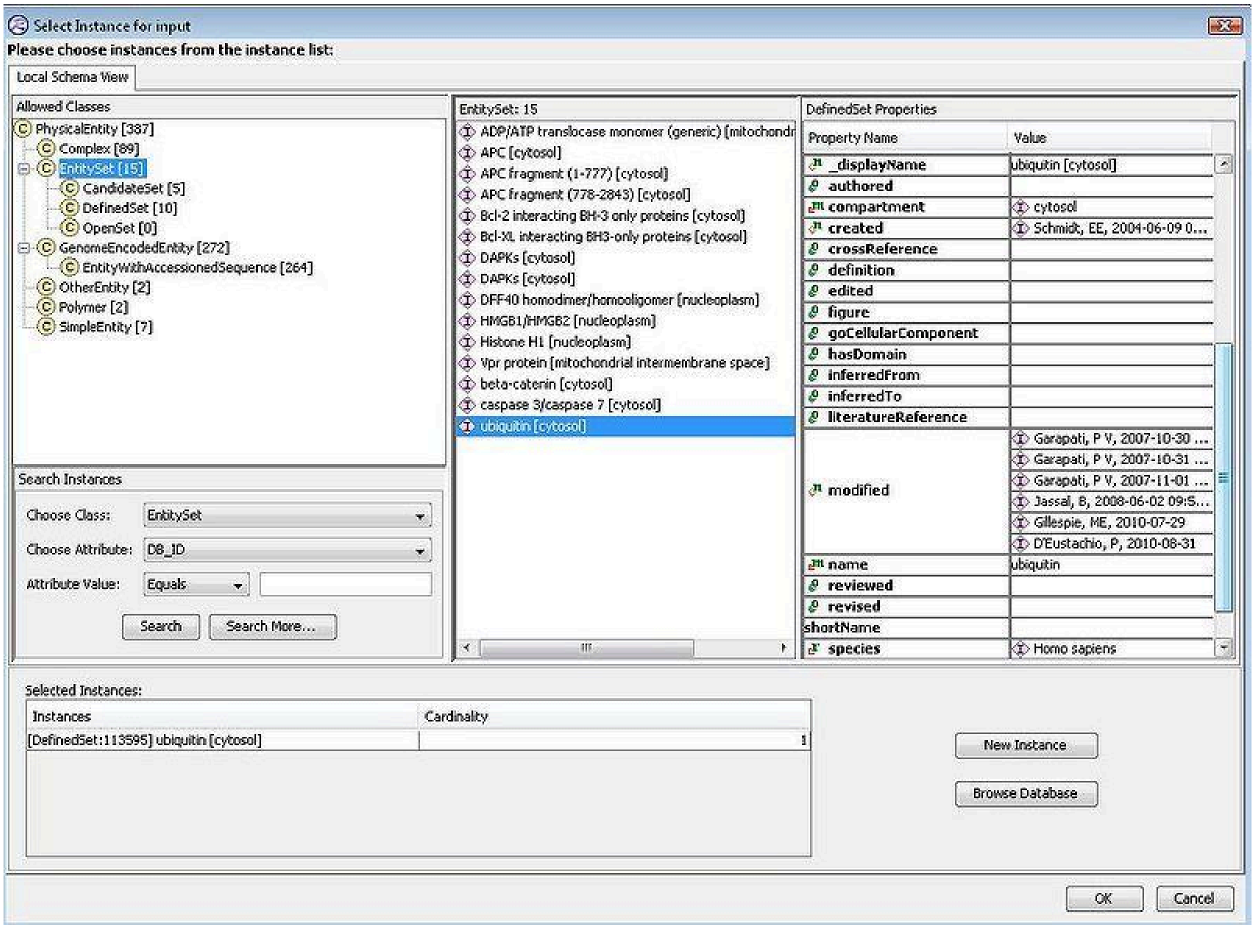
@Local Repository

To avoid having a human and a non-human reaction with identical names, it can be useful to use capitalized forms of object names for the non-human reaction and all-uppercase names for human, e.g. Jak2 and JAK2 for the non-human and human proteins respectively. A text summary and the literature reference providing evidence for this mixed species reaction is added.

The rabbit protein "PAK-2p34" is selected as an input

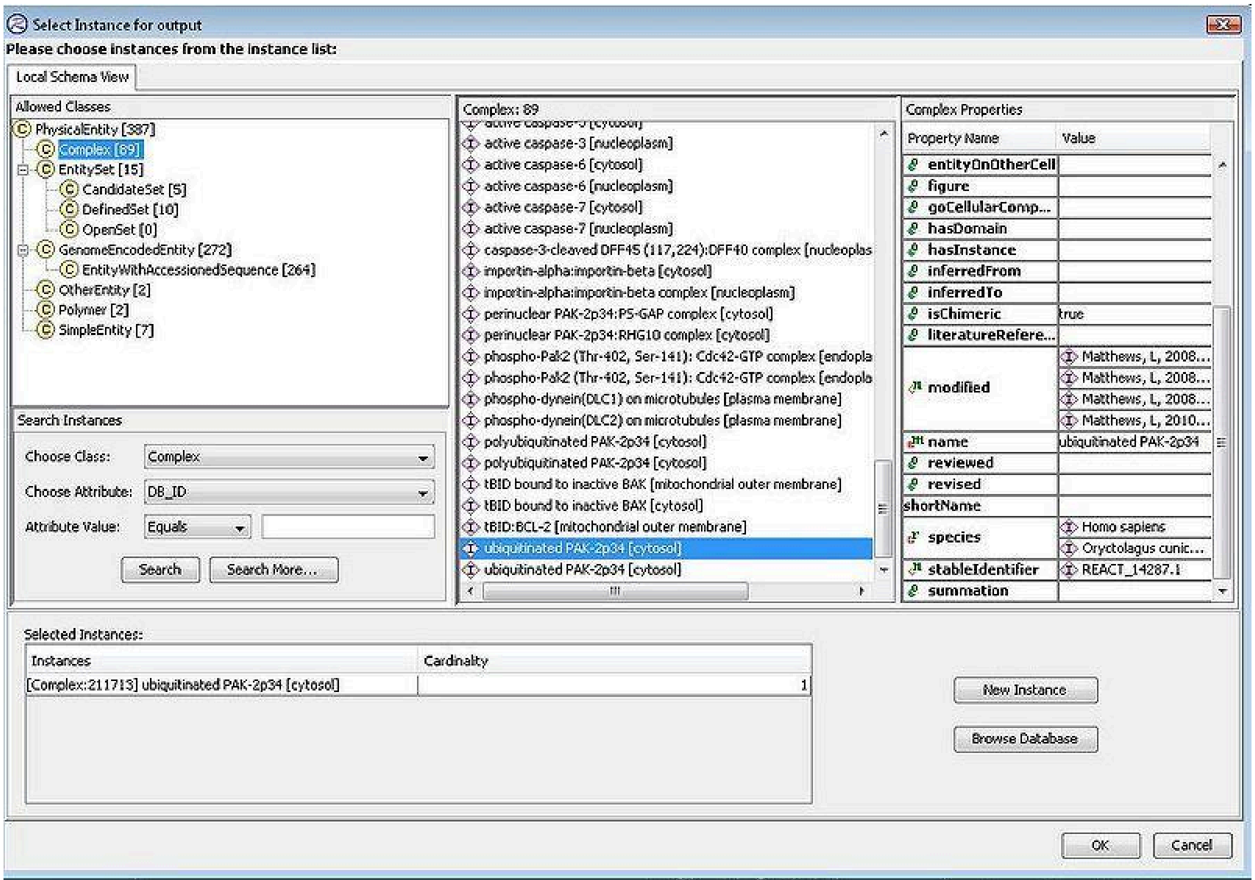


The human set "ubiquitin" is selected as an input...

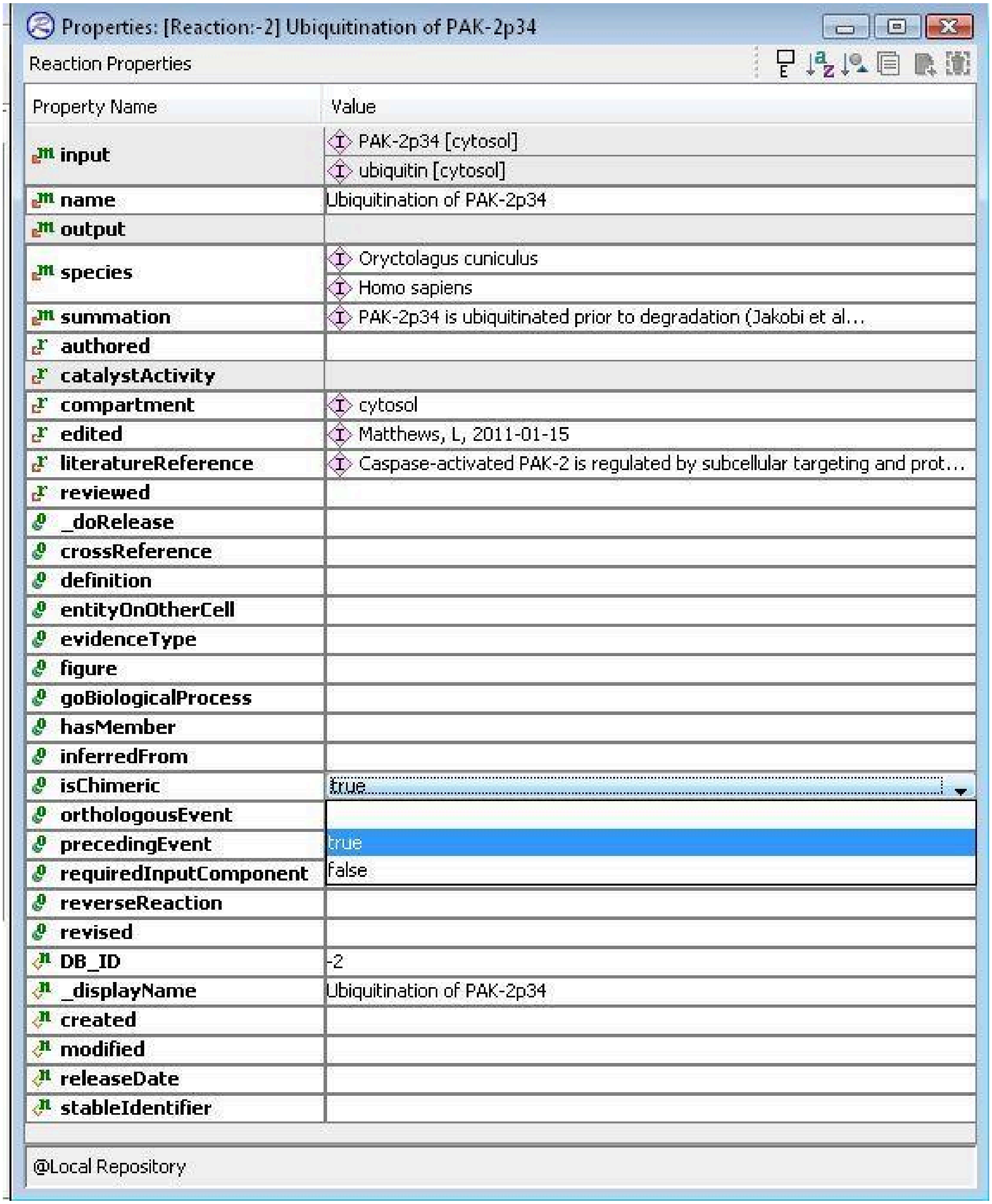


...and the mixed species output complex "PAK-2p34" is selected as the output.

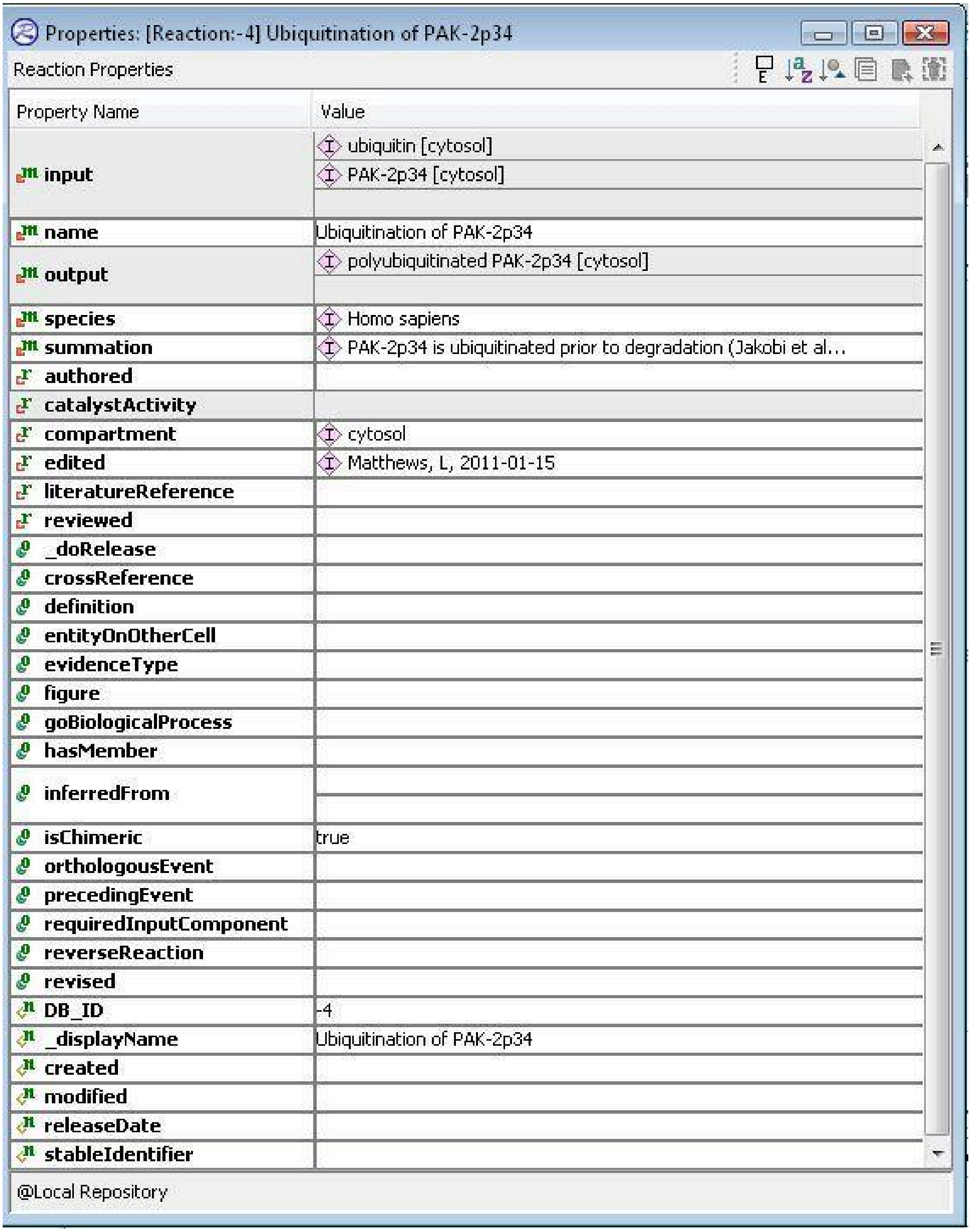
Note that this complex is not natural, it only occurs in vitro. To flag this the complex "isChimeric" attribute is set to True.



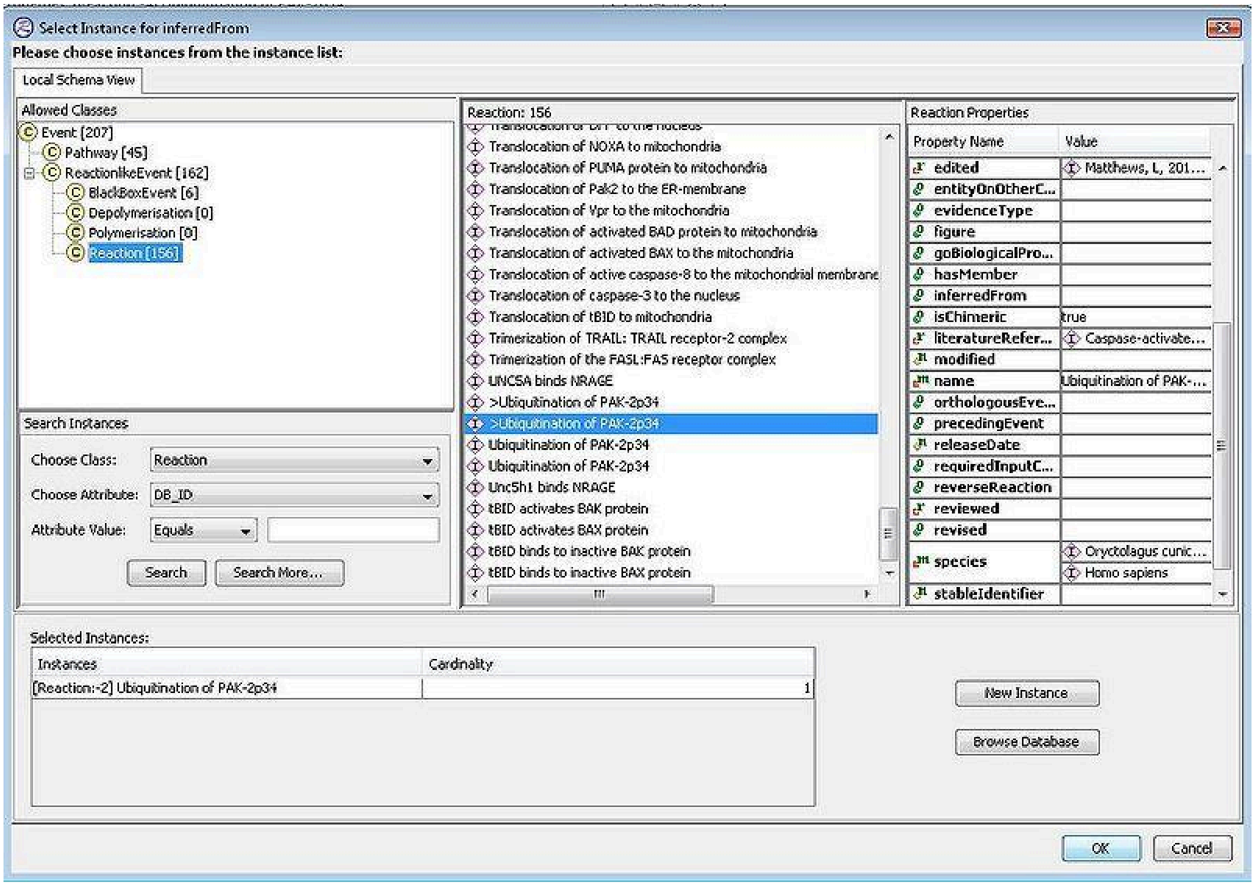
Since the reaction is also multispecies and not a natural event, the isChimera flag is also set to true for the reaction.



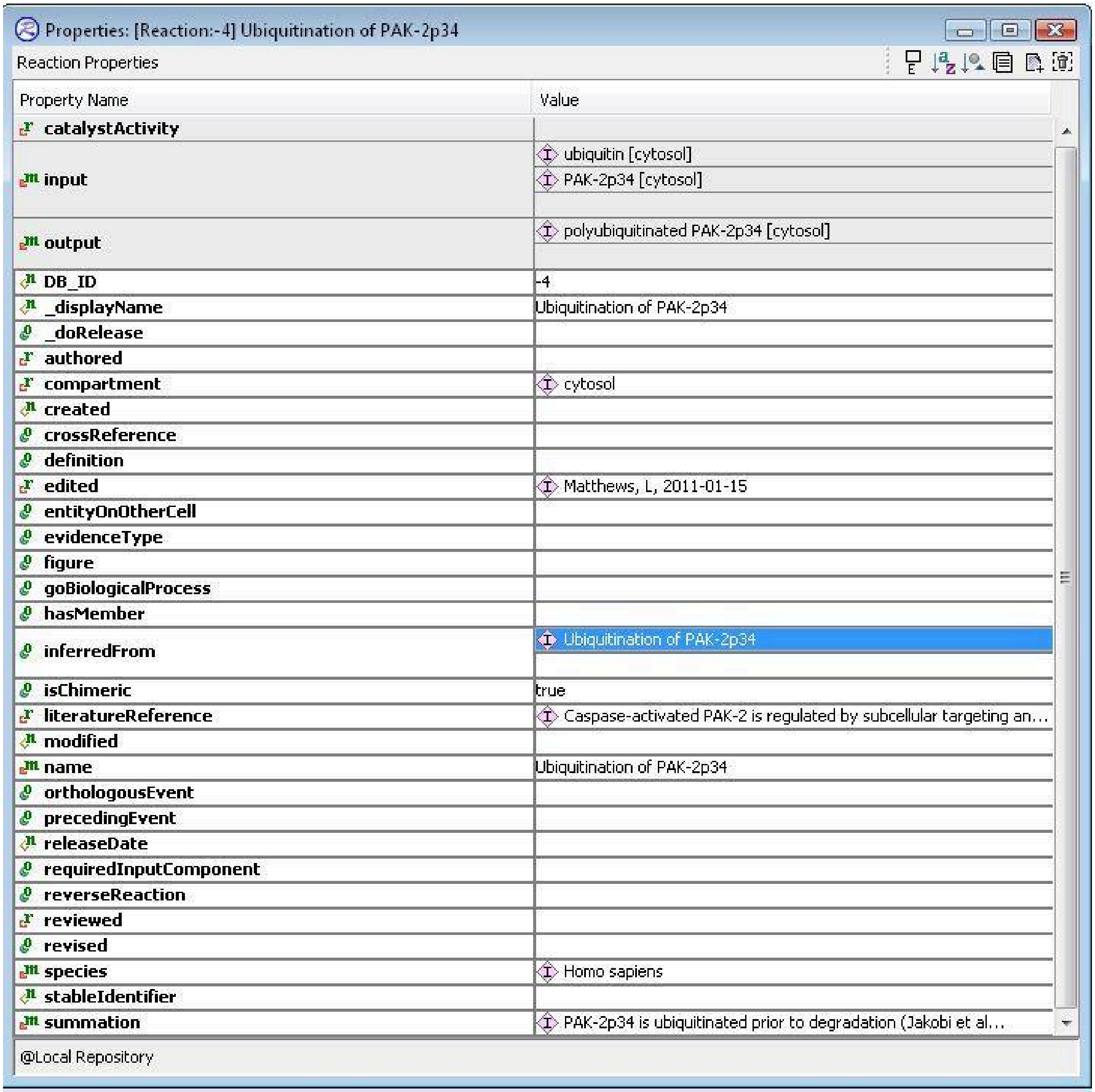
Now that the reaction to be used for inference has been created, create the same reaction for human, using human participating molecules. In the "inferredFrom" field, right click and select "Add" to enter the non-human reaction used for inference. Add literature references and summation as usual.



Selecting the mixed species reaction created previously...



...now the link to the inferred reaction has been made.



- Note also that if your annotations involve a species that is new to Reactome, please contact Peter D'Eustachio as it will need to have an abbreviation assigned by him.

Creating a Catalyst

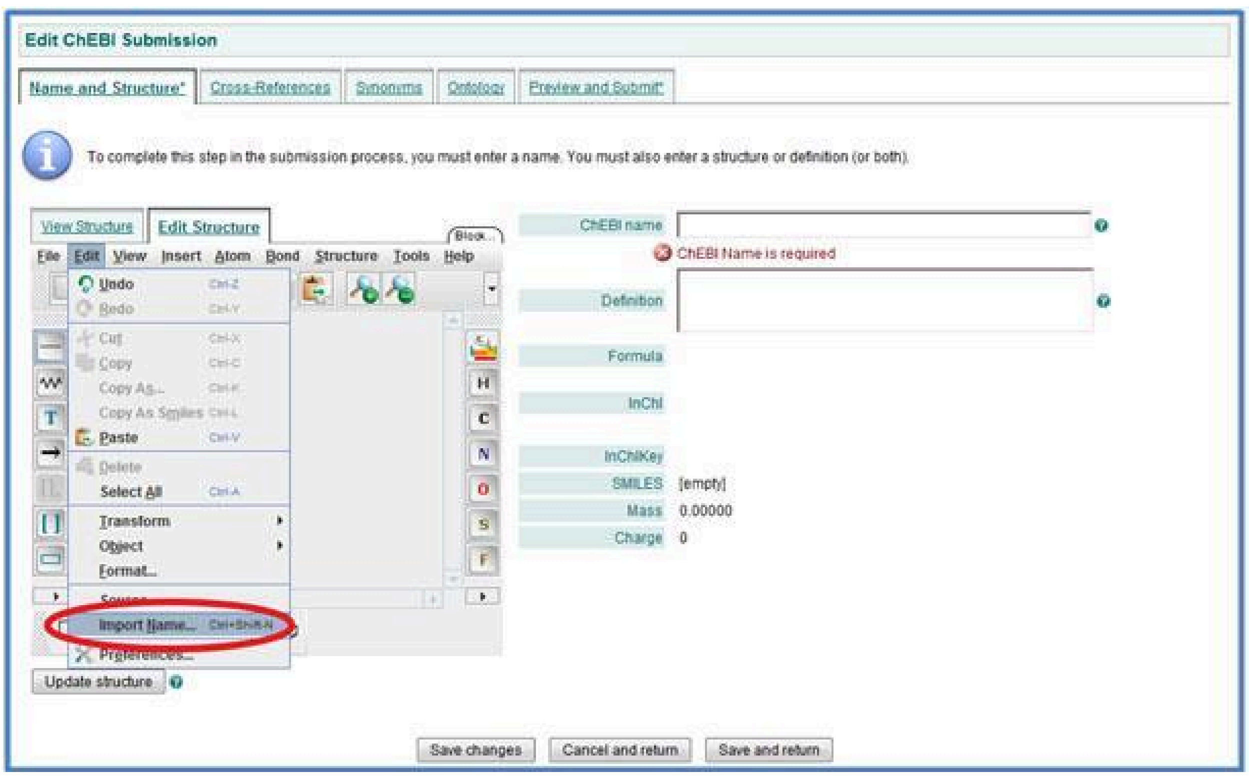
Reactions that involve a catalyst should include this information. Within the Reaction Details, the field is called catalystActivity. To complete this you need two things: the physicalEntity or object that is acting as catalyst, and the Activity of that object, defined as a GO molecular function. If the catalyst is a complex and there is a specific component of the complex that has the catalytic activity, that component must be specified in the “activeUnit” attribute. Note that if there if the whole complex acts as the catalyst or the component that is acting is not known, this attribute should be left blank. The PhysicalEntity itself should never be listed as the activeUnit. The physicalEntity will be a molecule or set or complex probably in your local project. The GO molecular function can be identified by consulting Uniprot, look at the ontologies section for GO Molecular Function. If none of the listed terms seems to be appropriate, either Browse Database for terms in gk_central, or use the OLS website at http://www.ebi.ac.uk/ontology-lookup/ to identify the correct term first. Use the most specific term possible.

Creating a new ChEBI entry

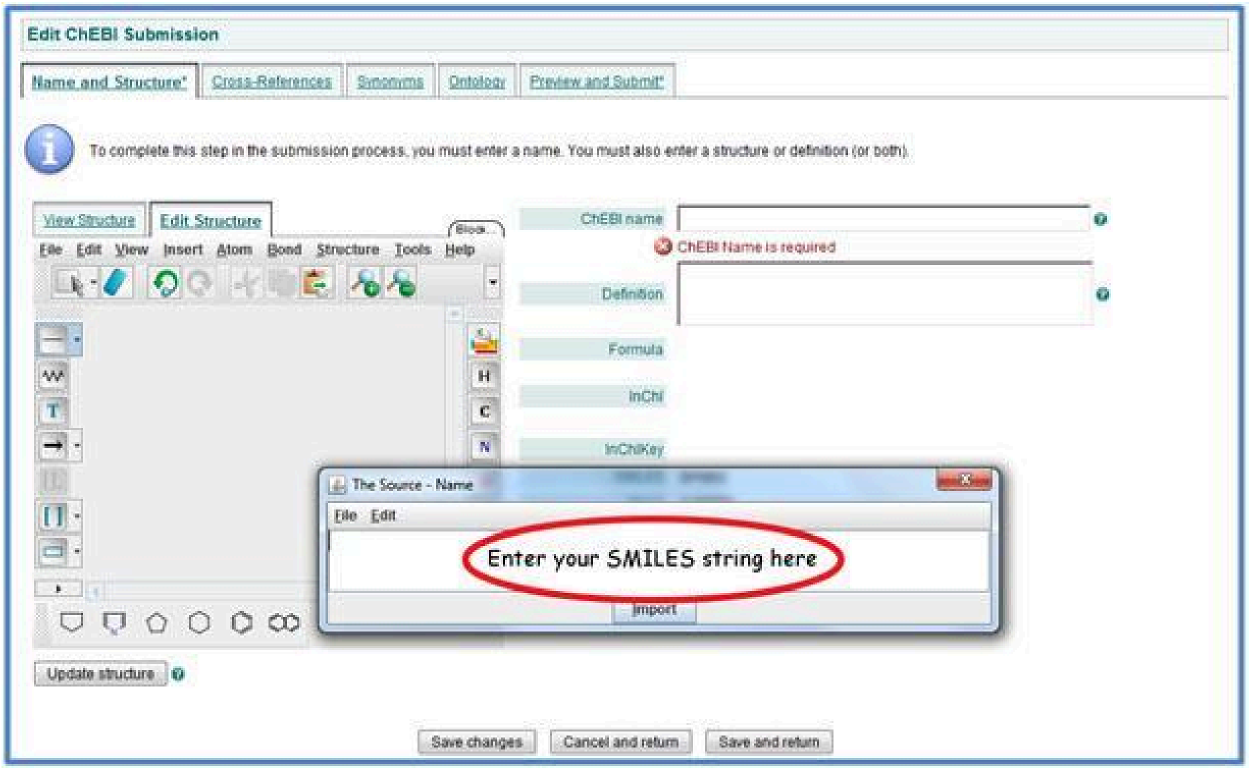
See also Best_Practices:_Using_Chemical_Compound_Databases, Appendix C8 .

Using SMILES strings as input for the ChEBI submission tool

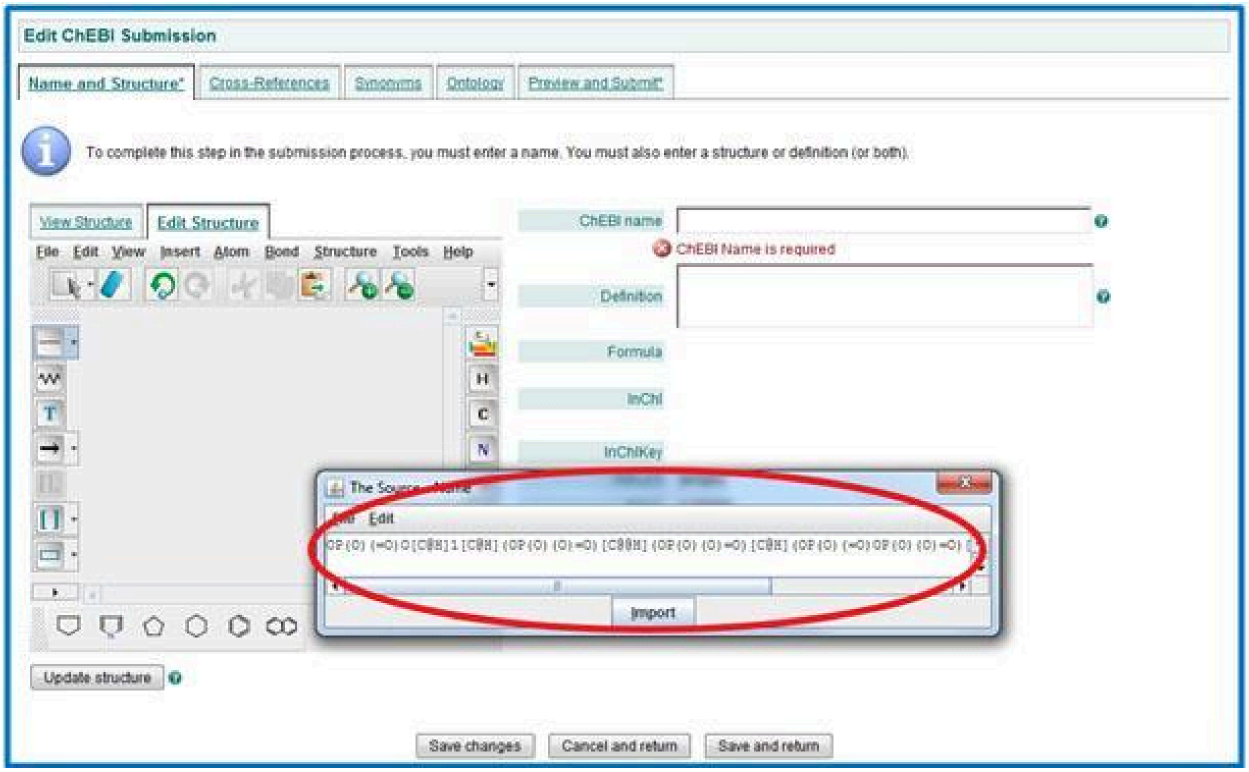
To use the submission tool, you must get a user name and password, and log in. Go here (https://www.ebi.ac.uk/chebi/submission/login) to do that. Once you have logged in, click "create a new submission" from the choices at the bottom of the page. That will cause a new line to appear in the table of "your active submissions" on that page. Click the "edit submission" option on that line to open the actual submission form.
Under the ‘Name And Structure’ section of the Submission tool, select ‘Edit Structure’. Under the ‘Edit’ Menu, select ‘Import Name’.



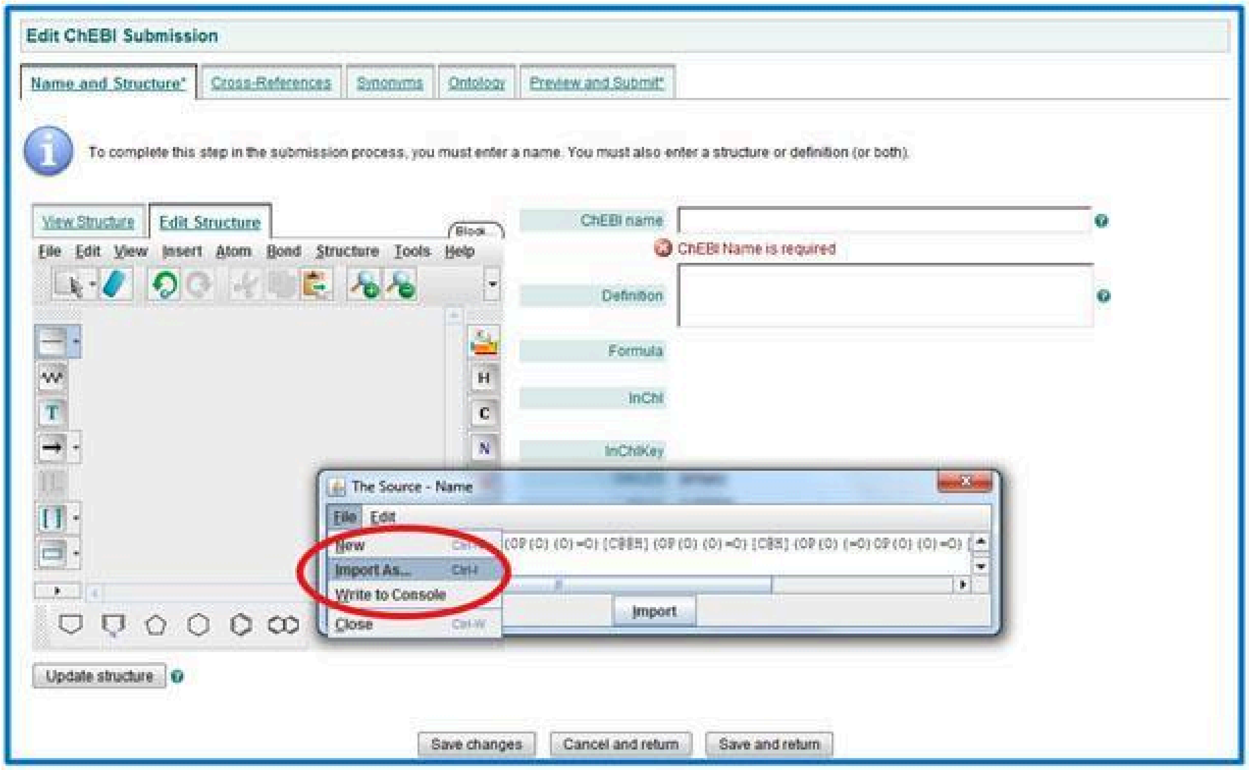
An input box named ‘The Source – Name’ appears, this is the place to paste your SMILES string.



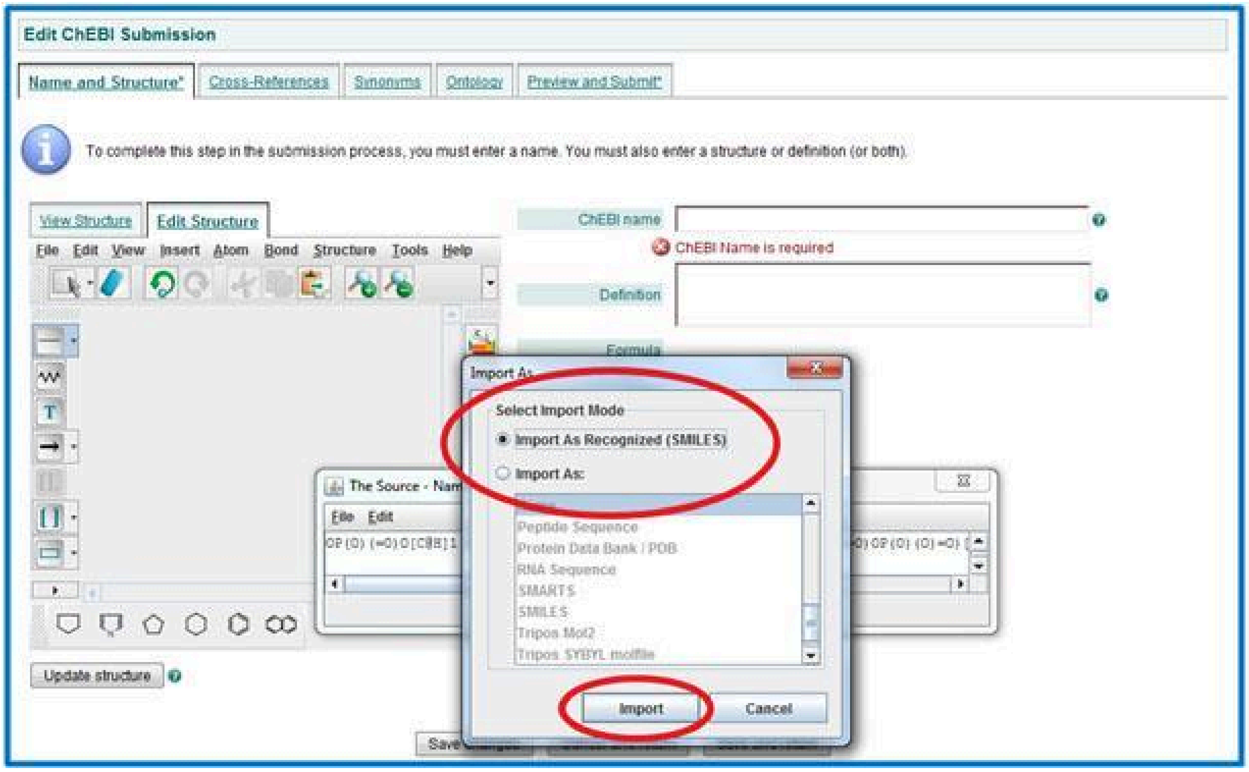
In this example, the following string for 1-PP-IP5 is used:
OP(O)(=O)O[C@H]1[C@H](OP(O)(O)=O)[C@@H](OP(O)(O)=O)[C@H](OP(O)(=O)OP(O)(O)=O)[C@H](OP(O)(O)=O)[C@@H]1OP(O)(O)=O



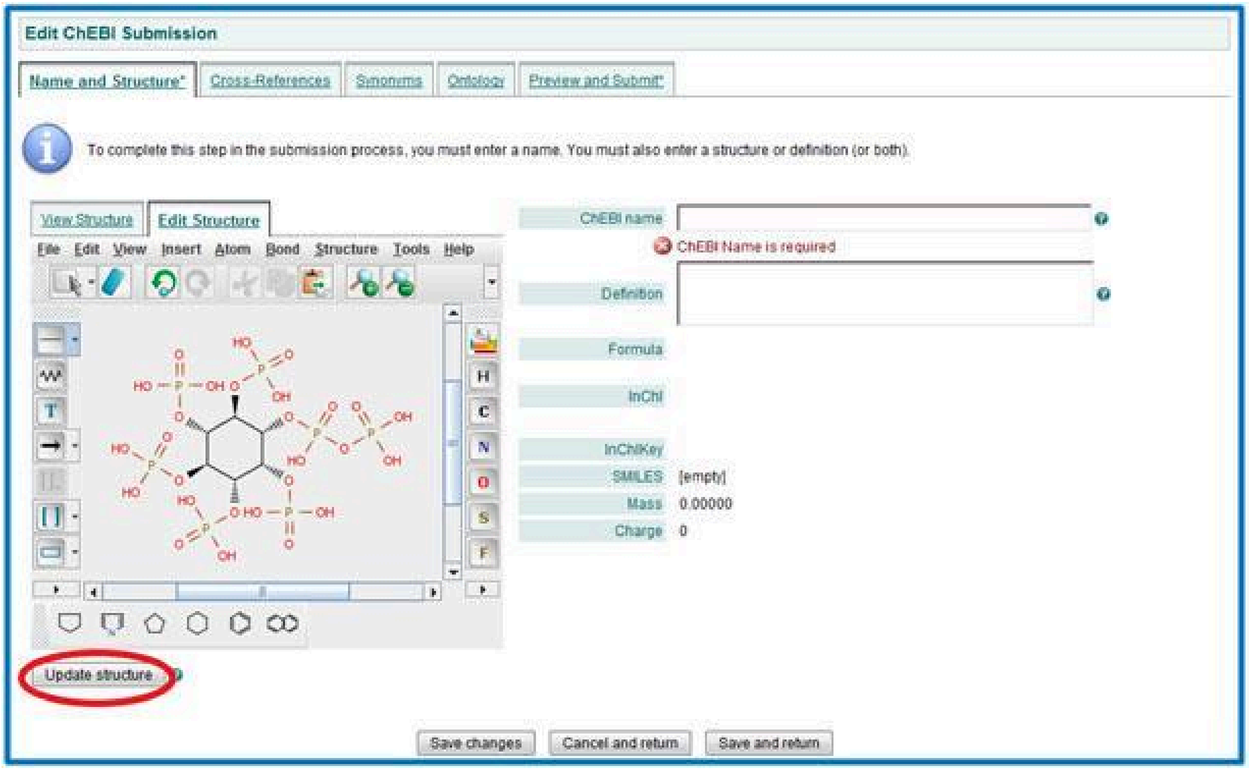
After the SMILES string has been pasted, select the 'File' menu in 'The Source – Name' input box. Select 'Import As'. Maybe displace circle or enlarge so as not to hide word "File" in menu



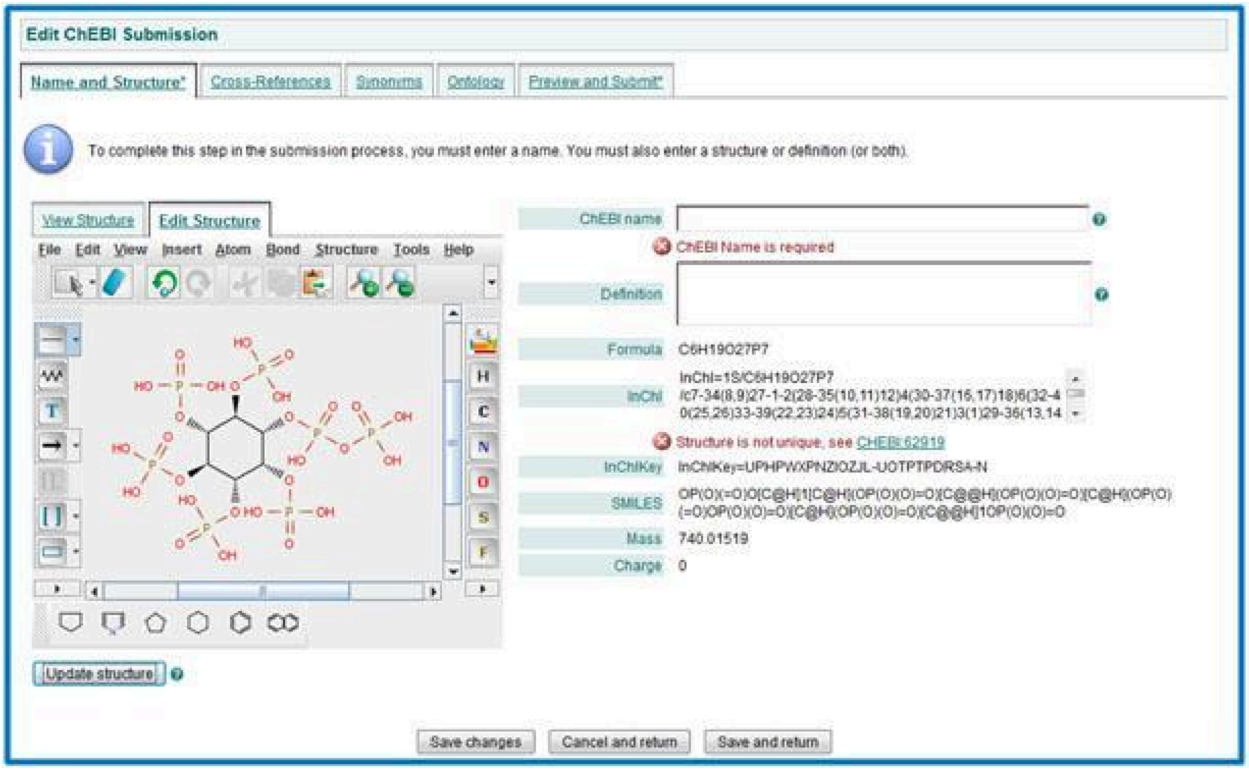
Make sure ‘Import as Recognized (SMILES)’ Import Mode has been selected and click on ‘Import’.



The chemical structure defined by the SMILES string should now be present in the box, ready for editing. Press ‘Update structure’ to obtain details of the structure.



Structure details are now displayed on the right of the page. The structure on the left hand side can now be edited as you wish.



Creating a new Species

If a curator needs to create a new "species" instance for their annotation, please contact the Editor-in-Chief, Peter D'Eustachio as it will need to have an abbreviation assigned by him. the superTaxon slot must* be filled with the "taxon" instance from gk_central **corresponding to the genus of the new species** as specified in the NCBI taxonomy entry for the species. For the bacterial species Bacillus thermoproteolyticus, that genus taxon is Bacillus, so that instance should be imported into the local project and used to fill the superTaxon slot for the new species. If the genus taxon does not exist, then the curator should create it. It too has a superTaxon slot which must be filled with the taxon corresponding to the appropriate family as given in the NCBI taxononmy. This process continues to the root of all life. Note that taxonomists use more levels than the simple kingdom - phylum - class - family - genus - species list. All of the levels shown in NCBI should be instantiated in gk_central for a species.

Annotating the regulation of a process

The following organization of regulation events works well for many kinds of processes, and it fits with our view that all parts of a process should be grouped while respecting GO's view that regulatory events should be distinguishable from the rest of the process.

All about [process] (pathway) --The steps of [process] (pathway)

[process]reaction 1

[process] reaction 2

etc.

--Regulation of [process] (pathway)

[process] regulatory reaction 1

[process regulatory reaction 2

etc.

Regulation here can include reactions that are themselves concrete molecular transformations whose effect is to modulate one of the main process reactions by activating an enzyme, or providing or sequestering an input molecule but can also include airy things like "[this regulatory event] by an unknown molecular mechanism positively or negatively regulates [process reaction #] . Note that regulation instances contain a "regulator", activity, and activeUnit. Regulation instances are associated with RLEs through the regulatedBy attribute of the RLE. We should show the RLE within the pathway that is being regulated. The reason for this is that we want to be as specific as possible about the molecular mechanism of the regulation. Annotating Gene Expression:

As far as gene expression regulation is concerned, note that:

- Even if the regulation is at the transcriptional level, show the protein as the output, since we are a protein-oriented database, and we assume that mRNA will get translated into a protein.
- Only show mRNA as the output of the transcription BBE only if this mRNA can be used as an input for another reaction – mainly regulation by microRNAs.
-

A detailed description of how to annotate regulation of gene expression is described below.

NOTE: *When considering which type of evidence to use when creating events in Reactome, note that you should NOT include events based solely on microarray, proteomics or similar high throughput data, unless either 1) an expert author has indicated that they believe the results are correct and represent the current consensus, or 2) The results are a meta-data analysis that combines several independent studies to show the overlap between them.*

Regulation where direct binding of a transcription factor to a regulatory sequence of a target gene has been demonstrated.

At least two of the following three criteria need to be met to create gene expression events and regulatory instances of this type:

1. Presence of specific transcription factor binding elements in regulatory regions of the target gene and/or demonstration of physical association of transcription factor with target gene DNA via chromatin immunoprecipitation (ChIP) or electrophoretic mobility shift assay (EMSA)
2. Evidence that the amount of a transcription factor (altered by overexpression from a recombinant vector and/or downregulation by gene deletion, targeted mutation and/or RNA interference) correlates with the level of target gene expression as measured by luciferase reporter assays and/or quantitative RT-PCR
3. Evidence that disruptive mutations of the transcription factor DNA binding sites in regulatory regions of a target gene affect target gene expression.

For this type of regulation we annotate the binding reaction between transcription factor and target gene. In the following example, the transcription factor is the MTF1 dimer bound to zinc ions (MTF1 dimer:12Zn2+). This transcription factor binds to the CSRP1 gene:

Reaction Properties

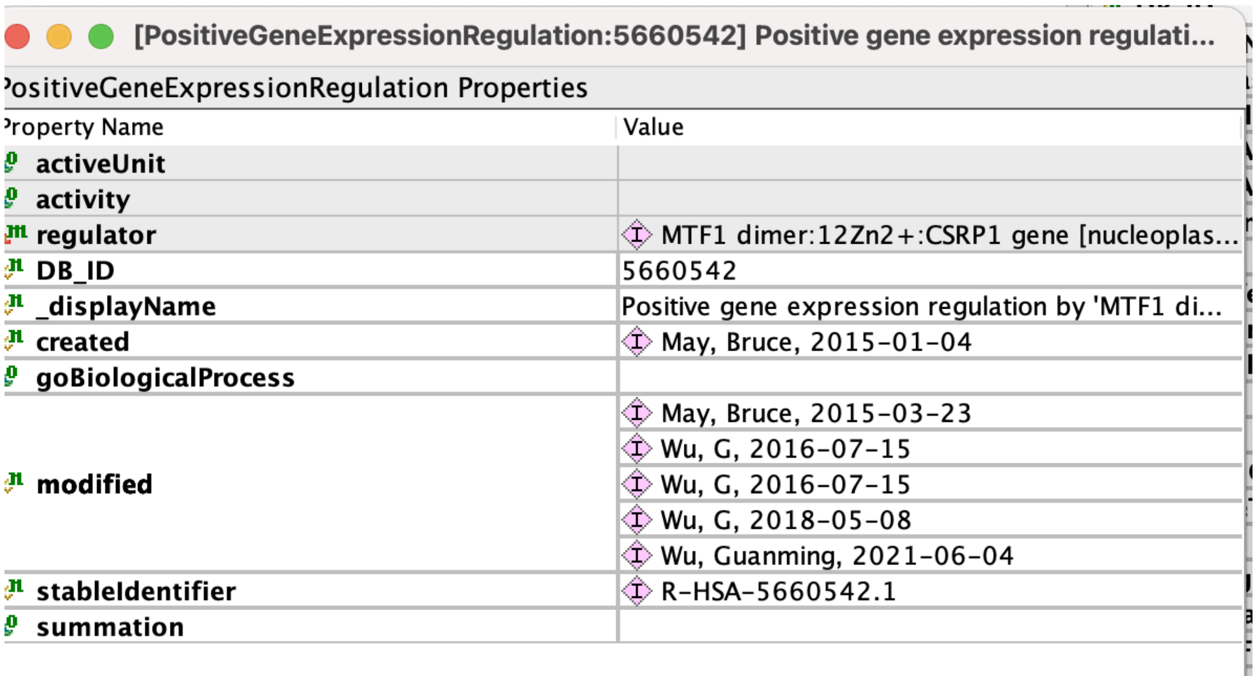
Property Name	Value
catalystActivity	
entityFunctionalStatus	
input	1 x CSRP1 gene [nucleoplasm] 1 x MTF1 dimer:12Zn2+ [nucleopl...
output	1 x MTF1 dimer:12Zn2+:CSRP1 ge...
DB_ID	5660514
_displayName	MTF1 dimer:12Zn2+ binds CSRP1 gene
_doRelease	true
authored	May, Bruce, 2014-12-27
catalystActivityReference	
compartment	nucleoplasm
created	May, Bruce, 2015-01-04
crossReference	
definition	
disease	
edited	May, Bruce, 2014-12-27
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	Mtf1 dimer:12Zn2+ binds Csrp1 g...
internalReviewed	
isChimeric	false
literatureReference	
modified	May, Bruce, 2015-03-15 Wu, G, 2016-07-15 May, Bruce, 2017-01-27 Matthews, Lisa, 2023-03-08
name	MTF1 dimer:12Zn2+ binds CSRP1 gene
negativePrecedingEvent	
normalReaction	
orthologousEvent	
precedingEvent	MTF1 dimer:12Zn2+ translocates f...
previousReviewStatus	
reactionType	
regulatedBy	
regulationReference	
relatedSpecies	
releaseDate	2017-03-27
releaseStatus	
requiredInputComponent	
reverseReaction	
reviewStatus	five stars
reviewed	Ford, Dianne, 2017-01-27
revised	
species	Homo sapiens
stableIdentifier	R-HSA-5660514.2
structureModified	
summation	As inferred from the mouse homol...
systematicName	

This transcription factor to DNA binding reaction is subsequently defined as a preceding event for the gene expression black box event, where input is a gene and the output is either a protein or an mRNA. In the following example, the input is the CSRP1 gene, and the output is protein CSRP1. Please note the preceding event:

BlackBoxEvent Properties

Property Name	Value
deletedInstanceDB_ID	
entityFunctionalStatus	
input	1 x CSRP1 gene [nucleoplasm]
output	1 x CSRP1 [nucleoplasm]
DB_ID	5660503
_displayName	Expression of CSRP1
_doRelease	true
authored	May, Bruce, 2014-12-27
catalystActivity	
catalystActivityReference	
compartment	nucleoplasm
created	May, Bruce, 2015-01-04
crossReference	
definition	
deletedStableIdentifier	
disease	
edited	May, Bruce, 2014-12-27
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	Expression of Csrp1
internalReviewed	
isChimeric	false
literatureReference	Human cysteine-rich protein. A memb... Zinc sensing by metal-responsive tran...
modified	May, Bruce, 2015-03-15 Wu, G, 2016-07-15 May, Bruce, 2017-01-07 May, Bruce, 2017-01-27 Wu, G, 2018-04-19 May, Bruce, 2018-10-31 Wu, G, 2019-04-04 Wu, Guanming, 2021-06-04 Matthews, Lisa, 2023-03-08
name	Expression of CSRP1
negativePrecedingEvent	
normalReaction	
orthologousEvent	
precedingEvent	MTF1 dimer:12Zn2+ binds CSRP1 gene
previousReviewStatus	
reactionType	
regulatedBy	Positive gene expression regulation by...
regulationReference	Positive gene expression regulation by...
relatedSpecies	

Regulation, in cases where regulator is a complex of transcription factor and target gene, is annotated as a **PositiveGeneExpressionRegulation'** ***or a*** **NegativeGeneExpressionRegulation**:

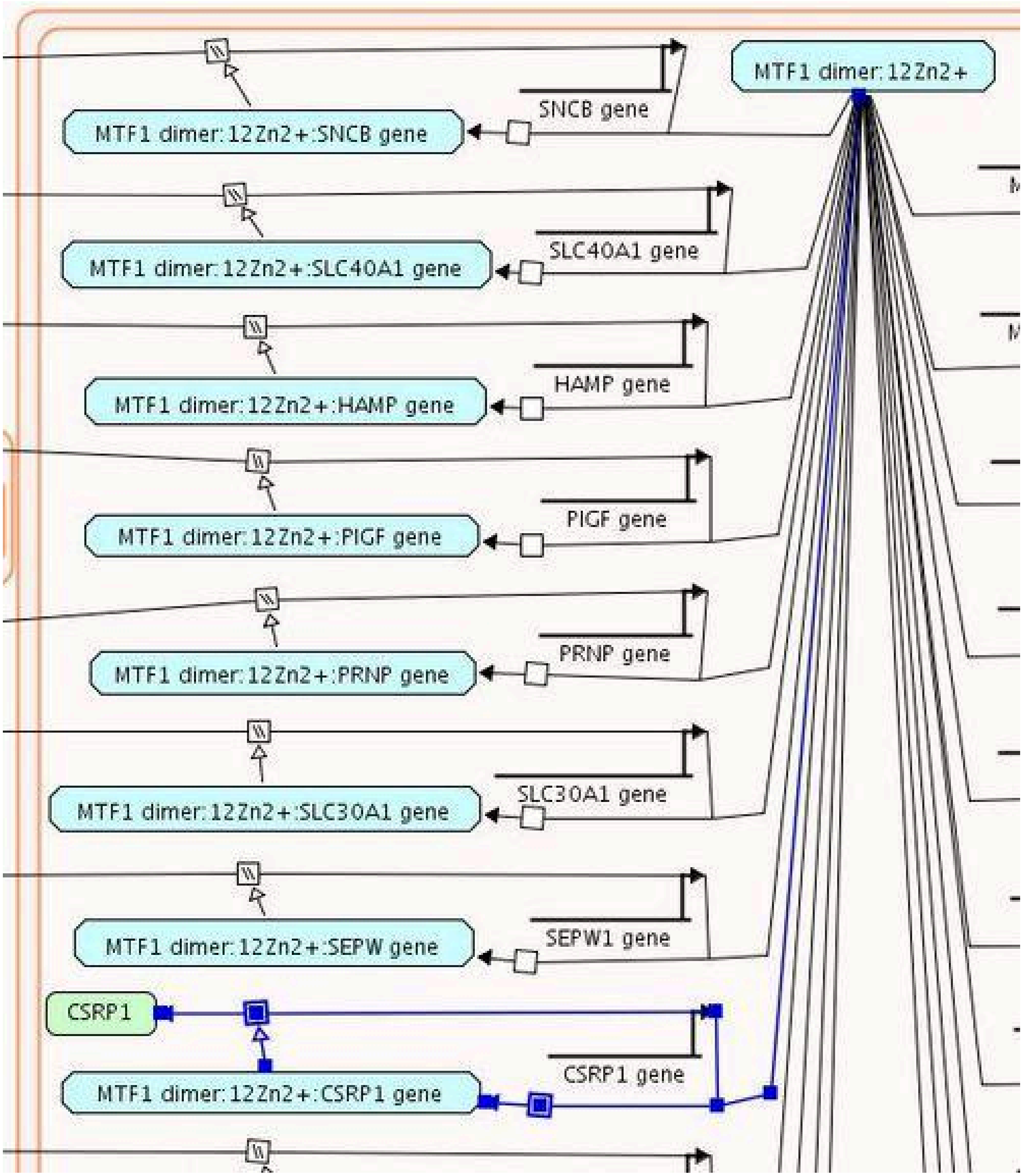


[PositiveGeneExpressionRegulation:5660542] Positive gene expression regulati...

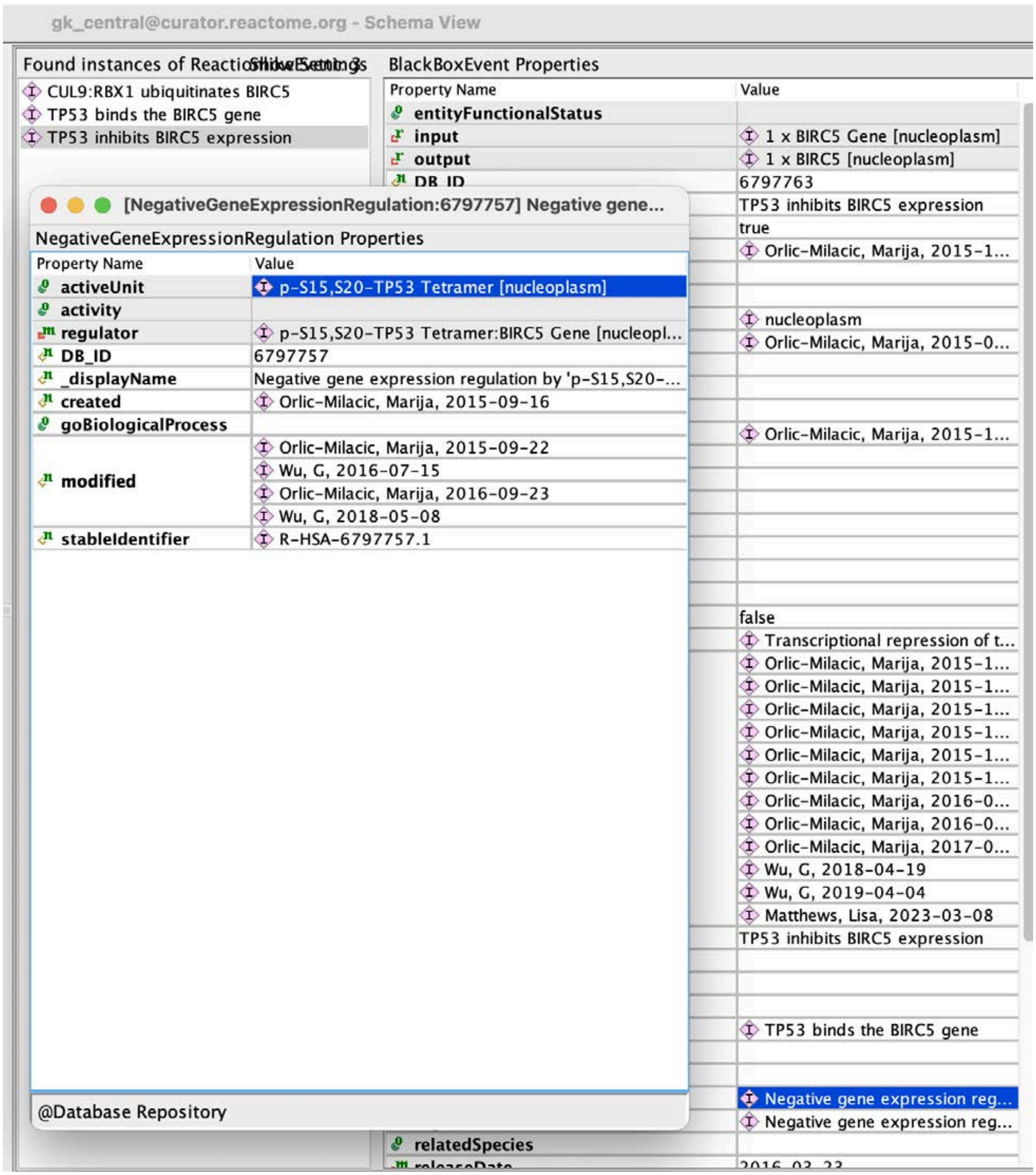
PositiveGeneExpressionRegulation Properties

Property Name	Value
activeUnit	
activity	
regulator	MTF1 dimer:12Zn2+:CSRP1 gene [nucleoplas...
DB_ID	5660542
_displayName	Positive gene expression regulation by 'MTF1 di...
created	May, Bruce, 2015-01-04
goBiologicalProcess	
modified	May, Bruce, 2015-03-23
	Wu, G, 2016-07-15
	Wu, G, 2016-07-15
	Wu, G, 2018-05-08
	Wu, Guanming, 2021-06-04
stableIdentifier	R-HSA-5660542.1
summation	

This is how this two-step CSRP1 gene expression regulation (transcription factor binding to DNA, followed by gene expression black box event) looks like in a pathway diagram:



From a computational modelling standpoint, when a regulator is a complex of a transcription factor and its target gene, it is important to define the **activeUnit of the regulation instance**. For example, TP53 negatively regulates BIRC5 gene expression by binding to the BIRC5 gene promoter. The regulator of the NegativeGeneExpressionRegulation instance is the complex of TP53 and BIRC5 gene (p-S15,S20-TP53 Tetramer:BIRC5 gene). Since the BIRC5 gene is both an input and a component of the negative regulator of the gene expression black box event, computational modelling tools may predict that increased copy number of the BIRC5 gene results in a decreased amount of BIRC5 protein. If TP53 is annotated as the activeUnit of the negativeGeneExpressionRegulation instance, this computational problem is prevented:



*Note that for regulatory events in which transcription factor(s) actually bind to an enhancer outside of the regulated gene, the annotation should still show binding of the transcription factor to the gene EWAS (and the gene coordinates should be set -1,1 regardless of gene size and how much of the regulatory region we are trying to capture).

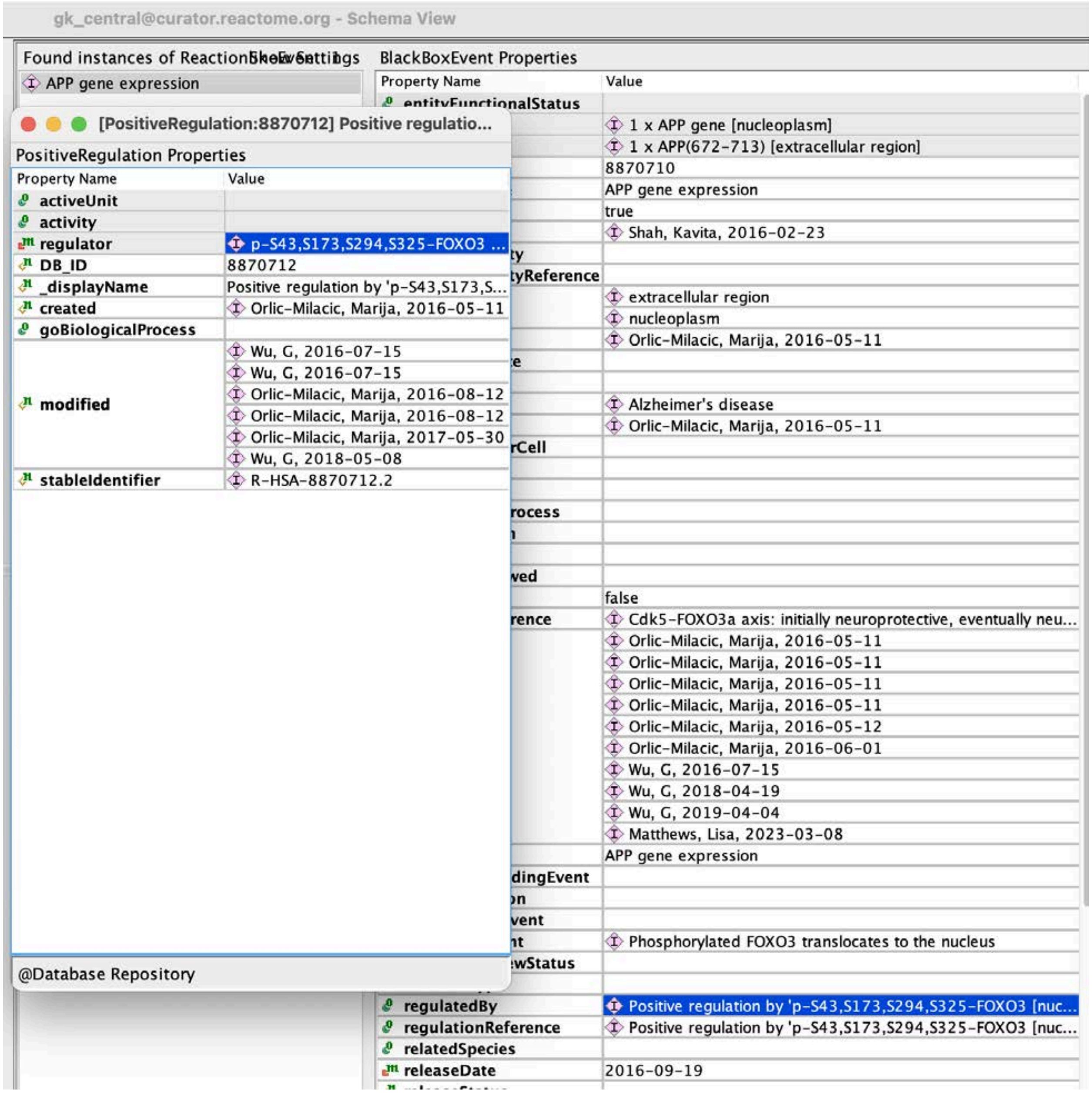
If a translational regulation will be shown, e.g. via noncoding RNAs, then mRNA is annotated as the output of a gene expression black box event. If it is anticipated that a translational regulation will not be made presently but may be added in future releases, then a protein should be annotated as the output of a gene expression black box event that will act as a releasable "placeholder". In a later release, separate transcription and translation black box events will be made and substituted for the "placeholder" black box event. The "placeholder" black box event should be deleted with a deletion instance listing the new transcription and translation black box events as replacements.

Regulation where no direct binding of a regulator to a regulatory DNA sequence has been demonstrated.

For this type of annotation, one criterion needs to be met:

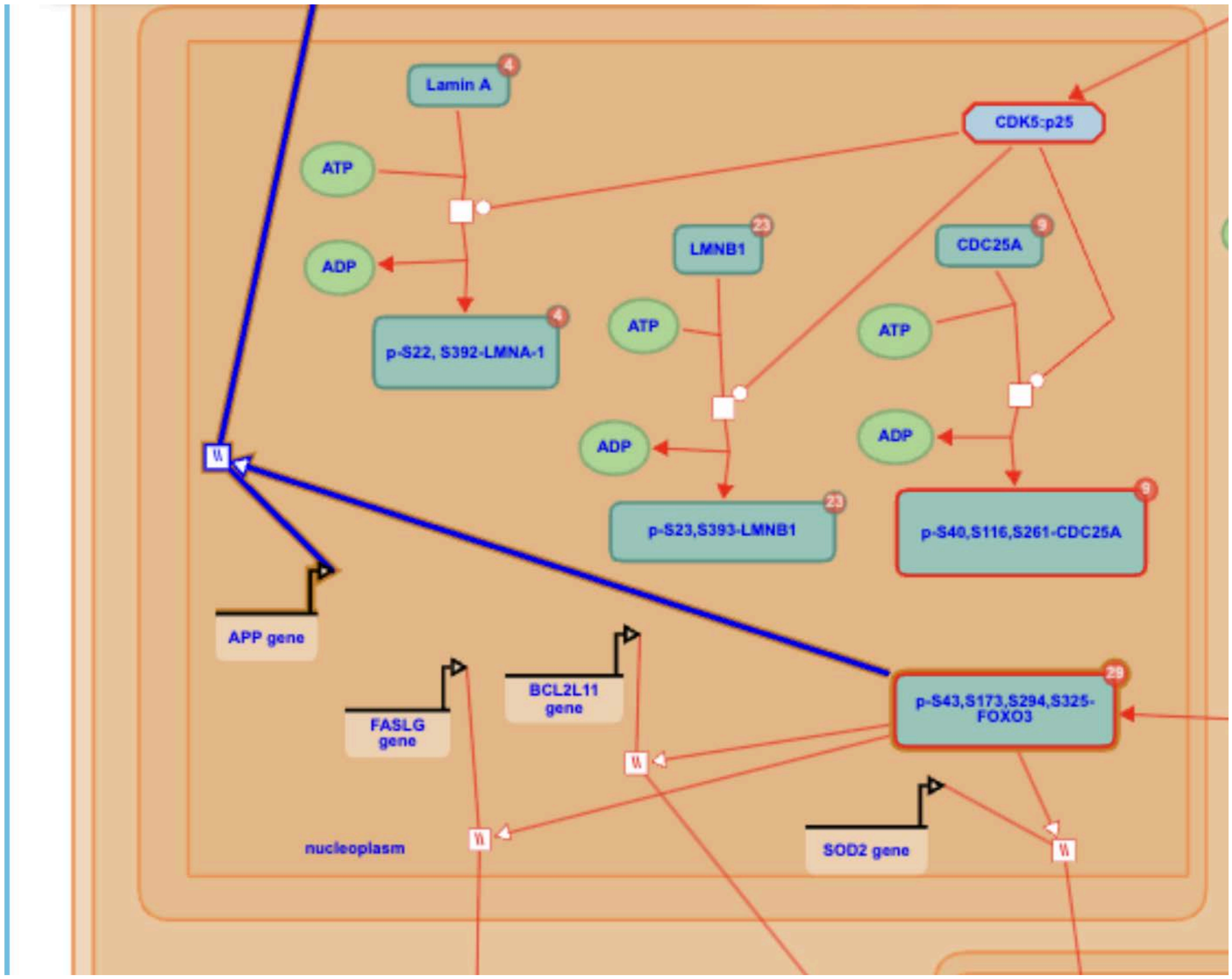
4. Evidence that the amount and/or the activity of a regulator (altered by overexpression from a recombinant vector and/or downregulation by gene deletion, targeted mutation, RNA interference and/or agonists/antagonists) correlates with the level of target gene expression measured by luciferase reporter assays and/or quantitative RT-PCR.

In this case, just a black box gene expression event is created. For example, FOXO3 positively regulates transcription of APP, but no direct binding of FOXO3 to the APP gene locus has been demonstrated:



Regulation instance, when no direct transcription factor binding has been demonstrated, is captured as a general **positiveRegulation** or **negativeRegulation**:

This is how the one-step APP gene expression regulation looks in a pathway diagram:



Regulation of gene expression at the level of translation

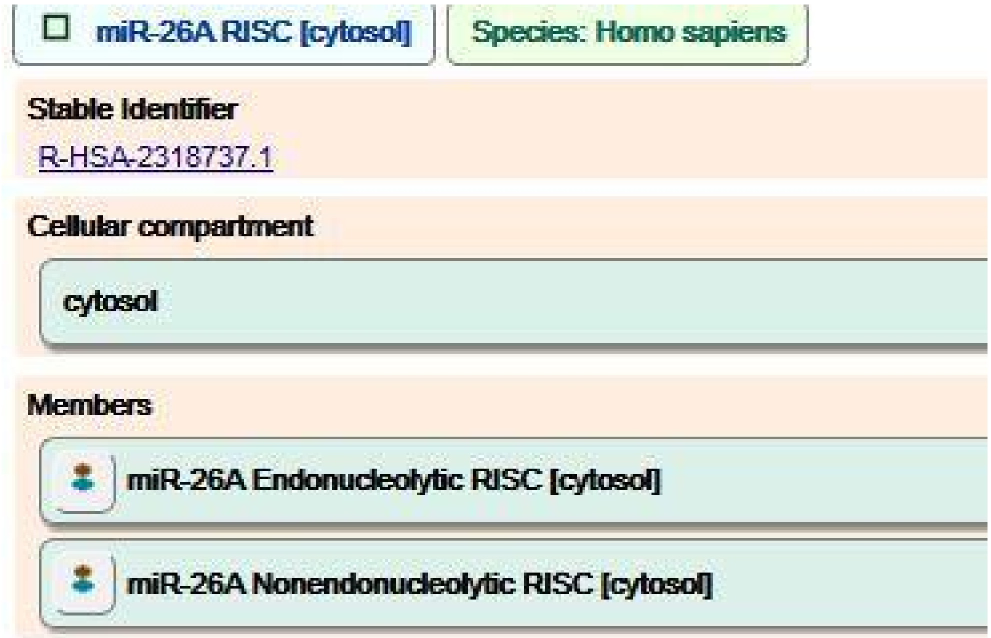
mRNA translation into protein can be regulated by non-coding RNAs, such as microRNAs (miRNAs). We only annotate miRNAs for which direct binding to target mRNAs and regulation of protein expression has been experimentally validated by either:

5. A reporter assay, where the miRNA is combined with a reporter-fused 3'UTR of the mRNA and observed for altered levels of reporter expression;
6. Co-immunoprecipitation of the miRNA:mRNA complex, together with an assay demonstrating that the miRNA alters the levels of either the mRNA (qRT-PCR) or protein (western blot).

MicroRNAs are always shown to bind to target mRNAs in the context of the RISC complex. Two types of RISC complex exist: nonendonucleolytic RISC (interferes with translation without degrading target mRNA) and endonucleolytic RISC (degrades target mRNA to prevent translation). Composition of different RISCs and their interaction with microRNAs is described in detail in the pathway [Gene Silencing by RNA] (R-HSA-211000):

- If an miRNA causes reduction of protein levels, but not mRNA levels of a target gene, it is annotated as a component of a nonendonucleolytic RISC.

- If an miRNA causes reduction of both mRNA and protein levels of a target gene, we usually annotate a RISC set where endonucleolytic RISC is a member and non-endonucleolytic RISC, depending on the experimental evidence, is either a candidate or a member.



If a non-endonucleolytic RISC for an miRNA of interest was definitely experimentally excluded, we would annotate the miRNA as a component of an endonucleolytic RISC only:

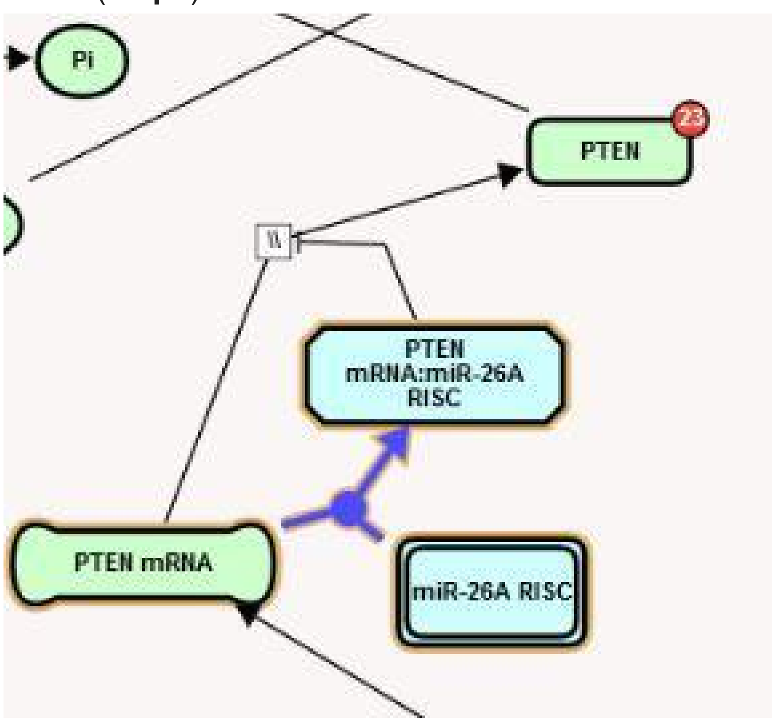
The easiest way to create endonucleolytic RISC and non-endonucleolytic RISC is by cloning an already annotated RISC complex and replacing the miRNA component of the clone by the miRNA of interest:

1. As there are over 100 miRNA RISC complexes in gk_central, please make sure that the specific RISC that you are trying to create does not already exist in the central repository.

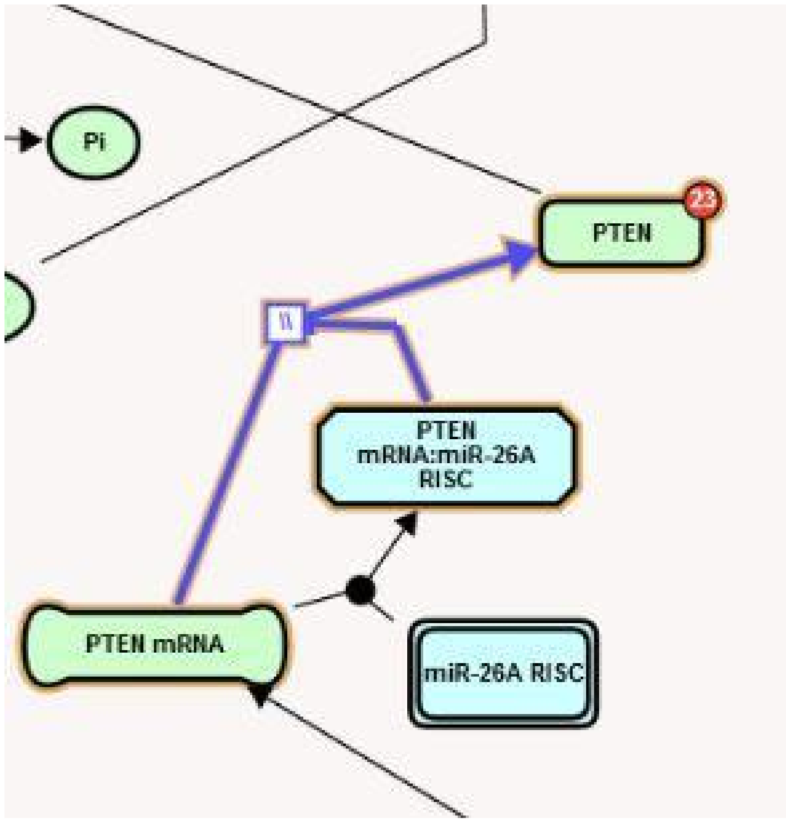
2. Check out an existing human endonucleolytic and/or non-endonucleolytic RISC to your local project.
3. Note that both endonucleolytic RISC and non-endonucleolytic RISC are nested complexes, in which miRNA is a component of a subcomplex named “minimal endonucleolytic RISC” and “minimal non-endonucleolytic RISC”, respectively. Create separate clones of the minimal endonucleolytic RISC/minimal non-endonucleolytic RISC, and endonucleolytic RISC/non-endonucleolytic RISC, **declining the offer to clone contain subunits recursively.**
4. In the clone(s) of minimal endonucleolytic RISC/minimal non-endonucleolytic RISC, replace the miRNA component with your miRNA of interest. Rename the clone to reflect its new miRNA component.
5. In the clone(s) of endonucleolytic RISC/non-endonucleolytic RISC, replace the minimal endonucleolytic RISC/minimal non-endonucleolytic RISC components with cloned and edited minimal endonucleolytic RISC/minimal non-endonucleolytic RISC. Rename the clone to reflect its new minimal endonucleolytic RISC/minimal non-endonucleolytic RISC component.

Translational regulation of gene expression is annotated in two steps, with step1 defined as a preceding event of step2:

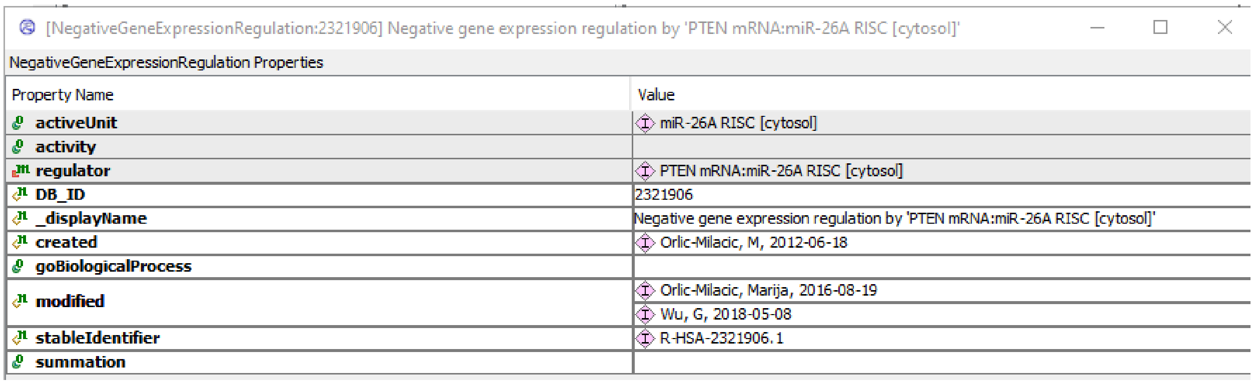
1. Binding of an inhibitor (e.g. miRNA-containing RISC complex) to a target mRNA (**step1**).



2. Gene expression black box event where mRNA is an input and protein is an output (**step2**).



Step2 BBE has regulatedBy attribute populated by a **negativeGeneExpressionRegulation** instance. The output of step1 is an inhibitor:mRNA complex, which is annotated as a negative regulator of step2. Again, from a computational modeling standpoint, when a regulator is a complex of an inhibitor and its target mRNA, it is important to define the **activeUnit of the regulation instance**. For example, miR-26A negatively regulates PTEN mRNA translation by binding to the 3'UTR of PTEN mRNA. The regulator of the NegativeGeneExpressionRegulation instance is the complex of PTEN mRNA and miR-26A RISC (PTEN mRNA:miR-26A RISC). Since PTEN mRNA is both an input and a component of the negative regulator of the translation black box event, computational modeling tools may predict that the increased level of PTEN mRNA results in a decreased amount of PTEN protein. If miR-26A RISC is annotated as the activeUnit of the negativeGeneExpressionRegulation instance, this computational problem is prevented:



[NegativeGeneExpressionRegulation:2321906] Negative gene expression regulation by 'PTEN mRNA:miR-26A RISC [cytosol]'

NegativeGeneExpressionRegulation Properties

Property Name	Value
activeUnit	miR-26A RISC [cytosol]
activity	
regulator	PTEN mRNA:miR-26A RISC [cytosol]
DB_ID	2321906
_displayName	Negative gene expression regulation by 'PTEN mRNA:miR-26A RISC [cytosol]'
created	Orlic-Milacic, M, 2012-06-18
goBiologicalProcess	
modified	Orlic-Milacic, Marija, 2016-08-19 Wu, G, 2018-05-08
stableIdentifier	R-HSA-2321906.1
summation	

Drawing a pathway diagram

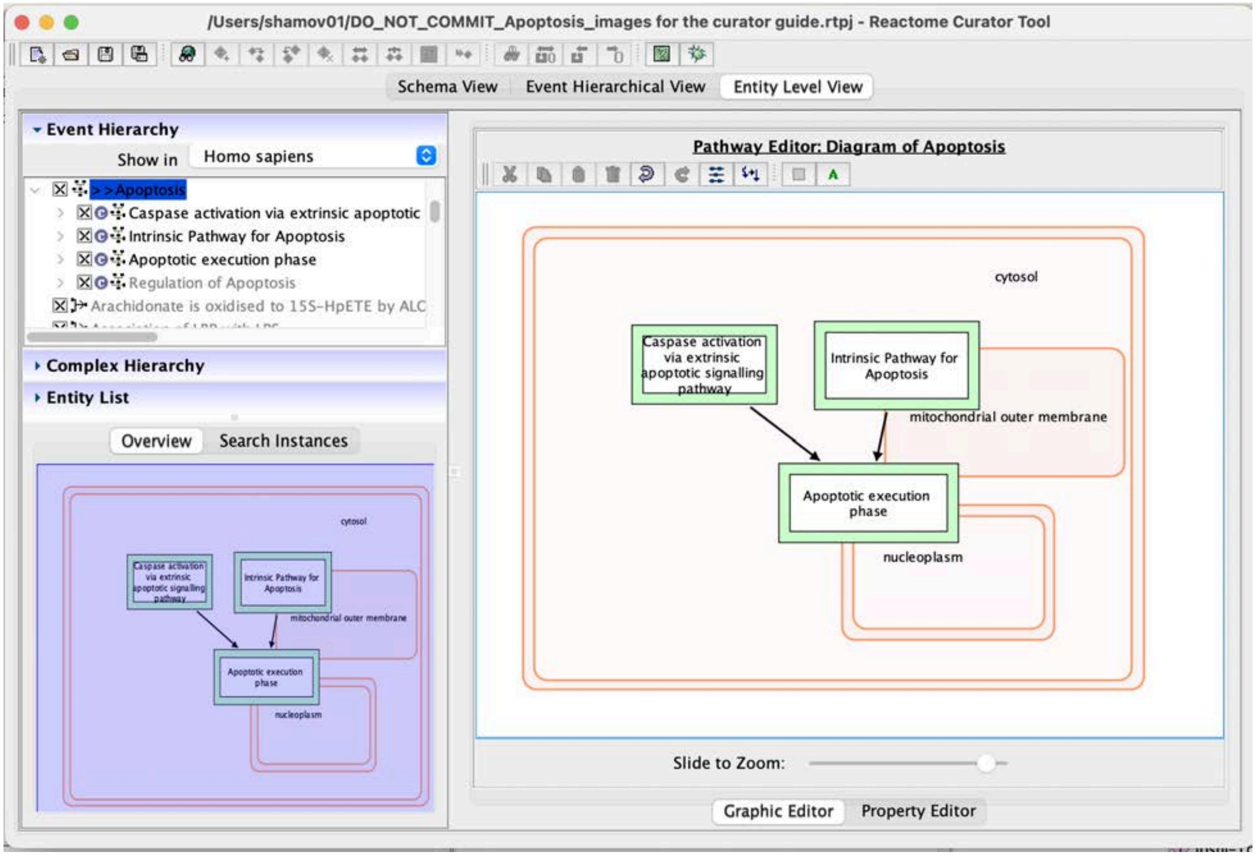
Reactome pathway diagrams are drawn and viewed in the curator tool using the ELV pane of the tool.

Below is an example to show how the superpathway "Regulation of Apoptosis" is diagrammed and how it’s parent pathway must also be modified in the process to reflect that a new navigable subpathway diagram now exists. These instructions are followed by important hints about revising existing diagrams.

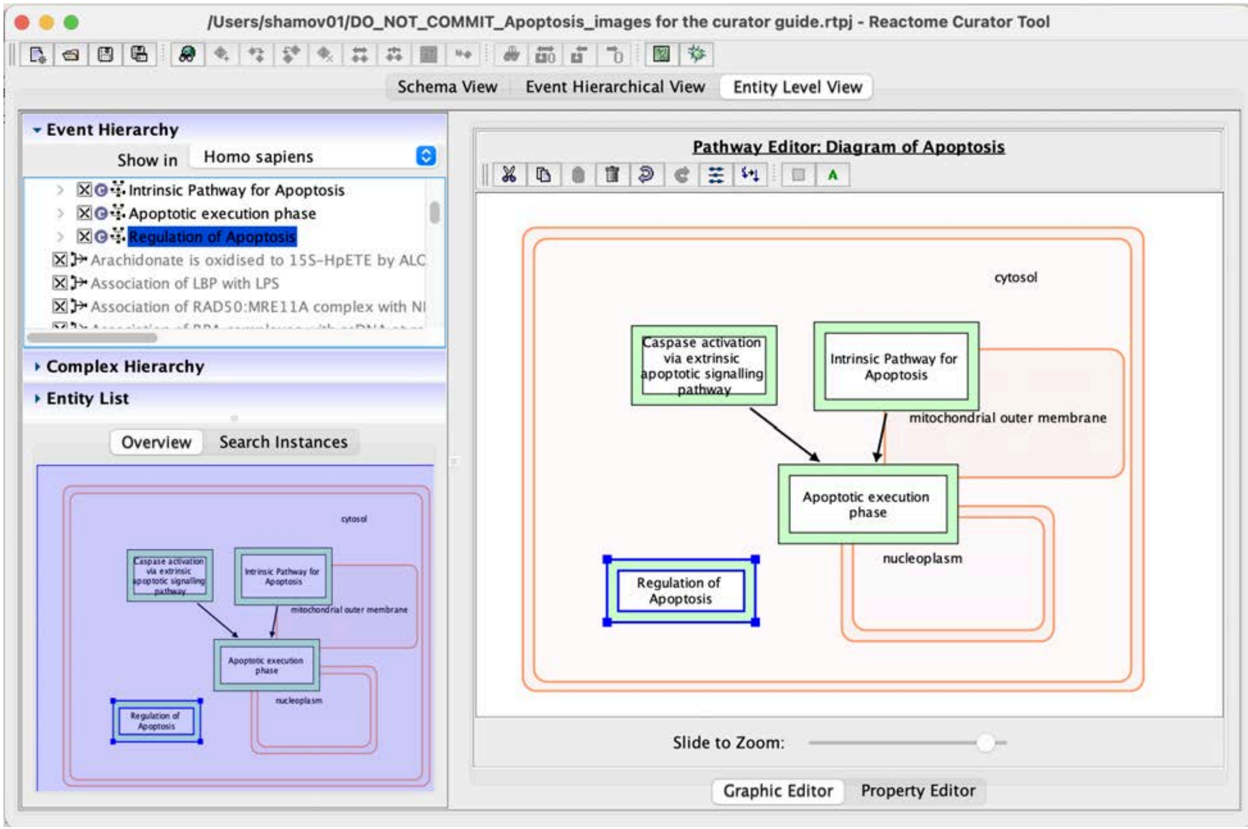
Example of updating the parent of a pathway that has a new diagram

The pathway Regulation of Apoptosis is part of the supercanonical Apoptosis pathway. First, go to gk_central in the Event browser, find Apoptosis, right click and opt to "build project". In the curator tool entitylevel view, unfurl Apoptosis. In this example, you can tell that Regulation of Apoptosis has been annotated but not yet incorporated in the Apoptosis diagram because the pathway is greyed out in the event hierarchy.

Click on apoptosis and open the diagram.



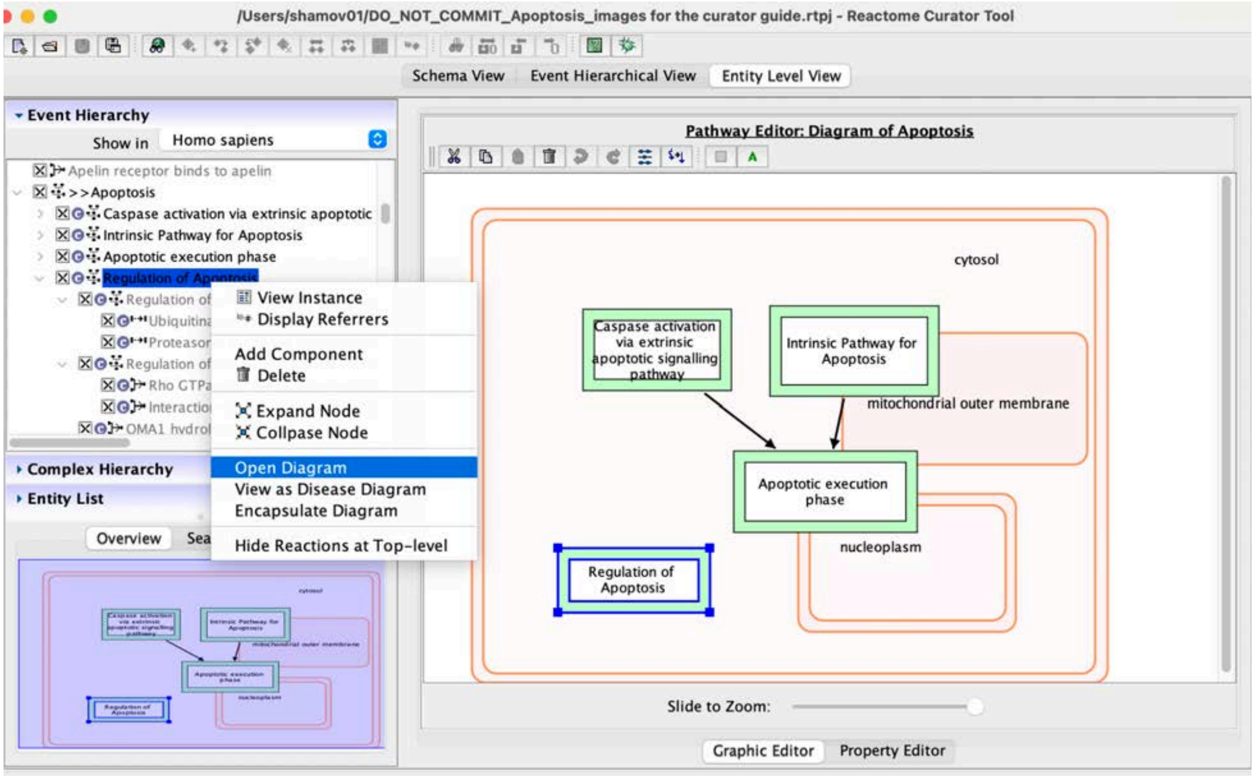
To incorporate the Regulation of Apoptosis pathway into the Apoptosis diagram, simply click on, hold, and drag the pathway from the hierarchy to the diagram.



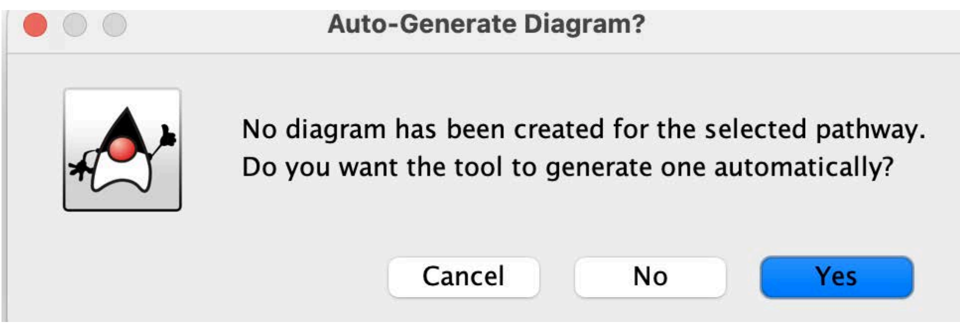
Example: Create a diagram for Regulation of Apoptosis

Right click on pathway in the hierarchy or in the diagram and opt to "open diagram"

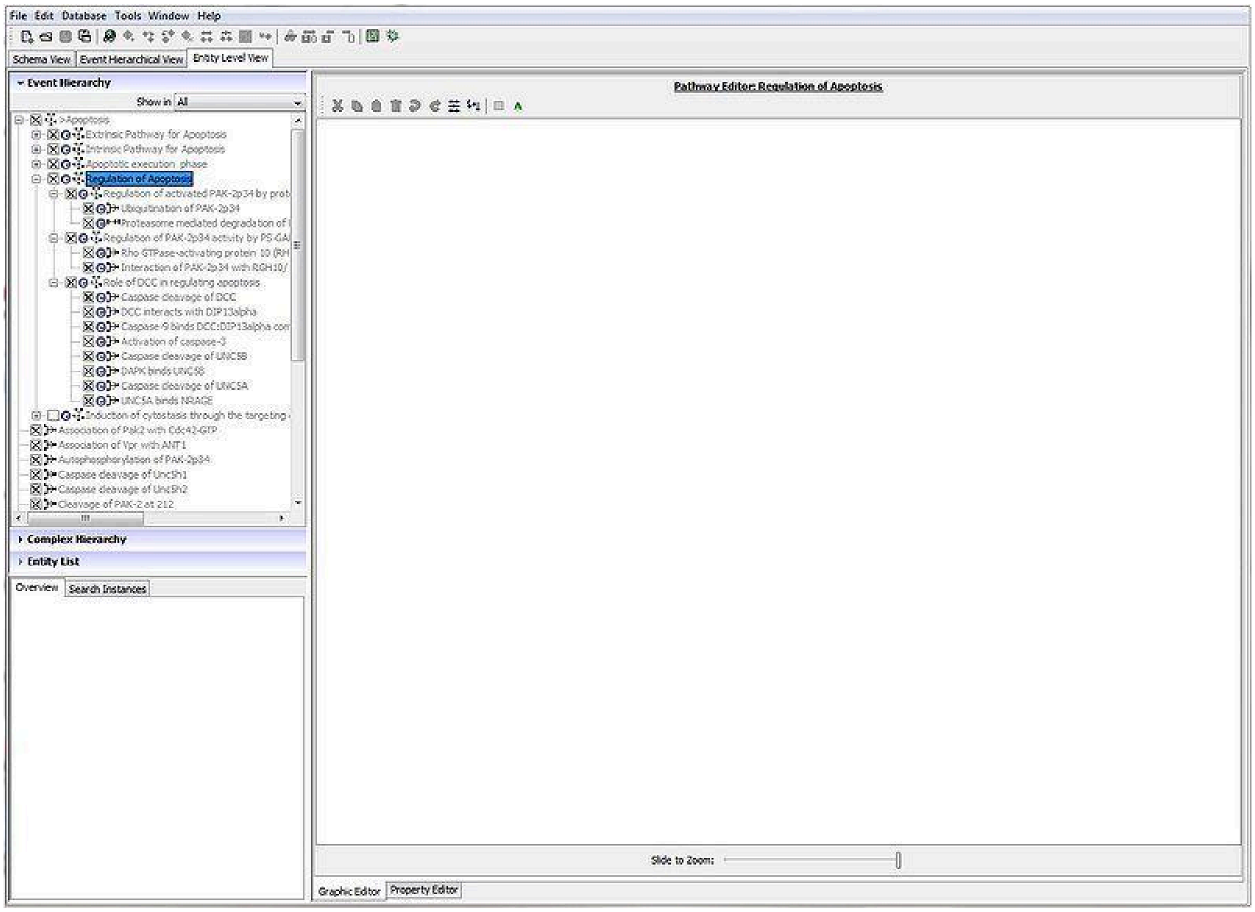
If one already exists, it will open.



If it has not, as in the case of Regulation of Apoptosis, you get the below message.

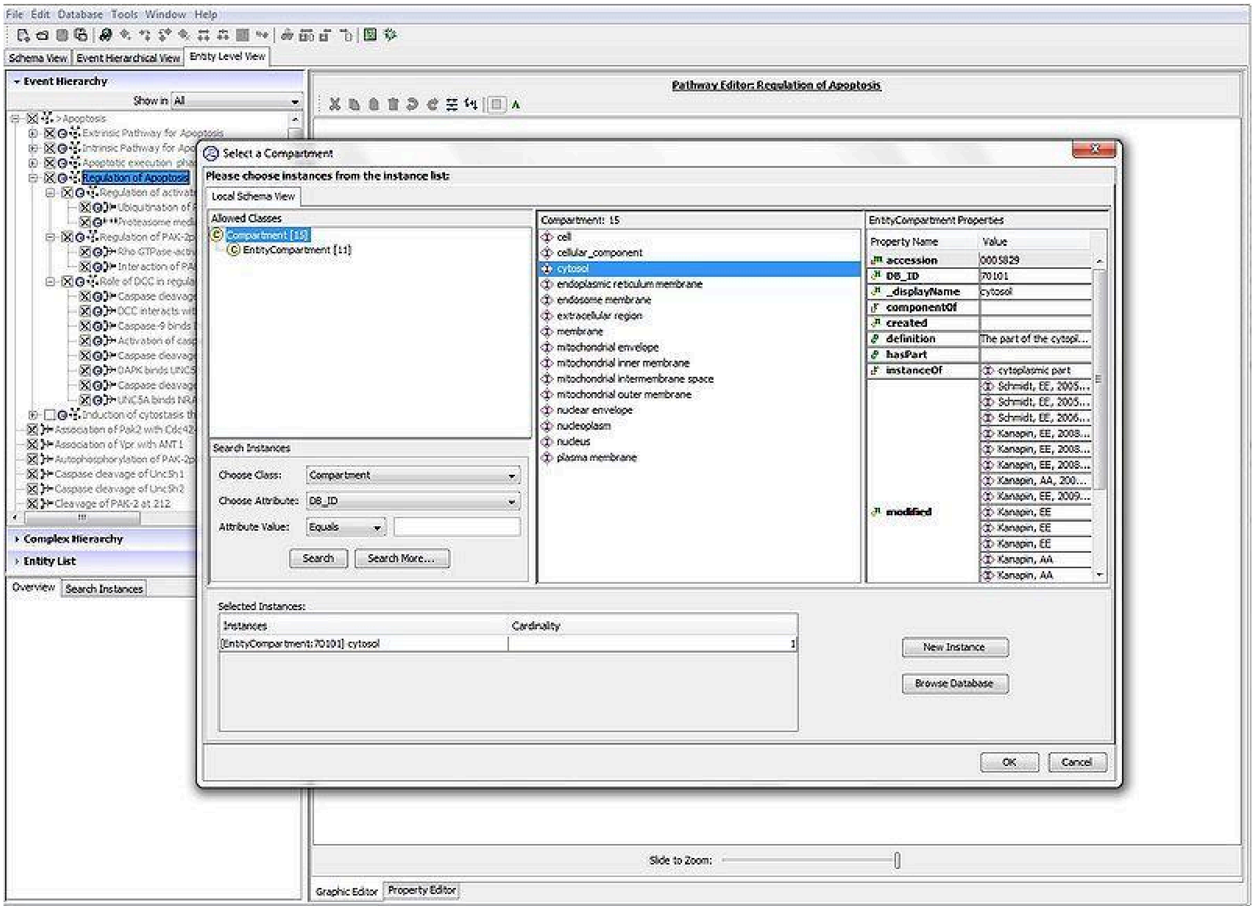


Answering “Yes” may be helpful if there are just a few reactions but can create a tangle of reactions that are difficult/time consuming to readjust. If you do want to create a diagram but want to lay it out yourself, hit “NO”and an empty diagram will be opened in the pathway editor pane.

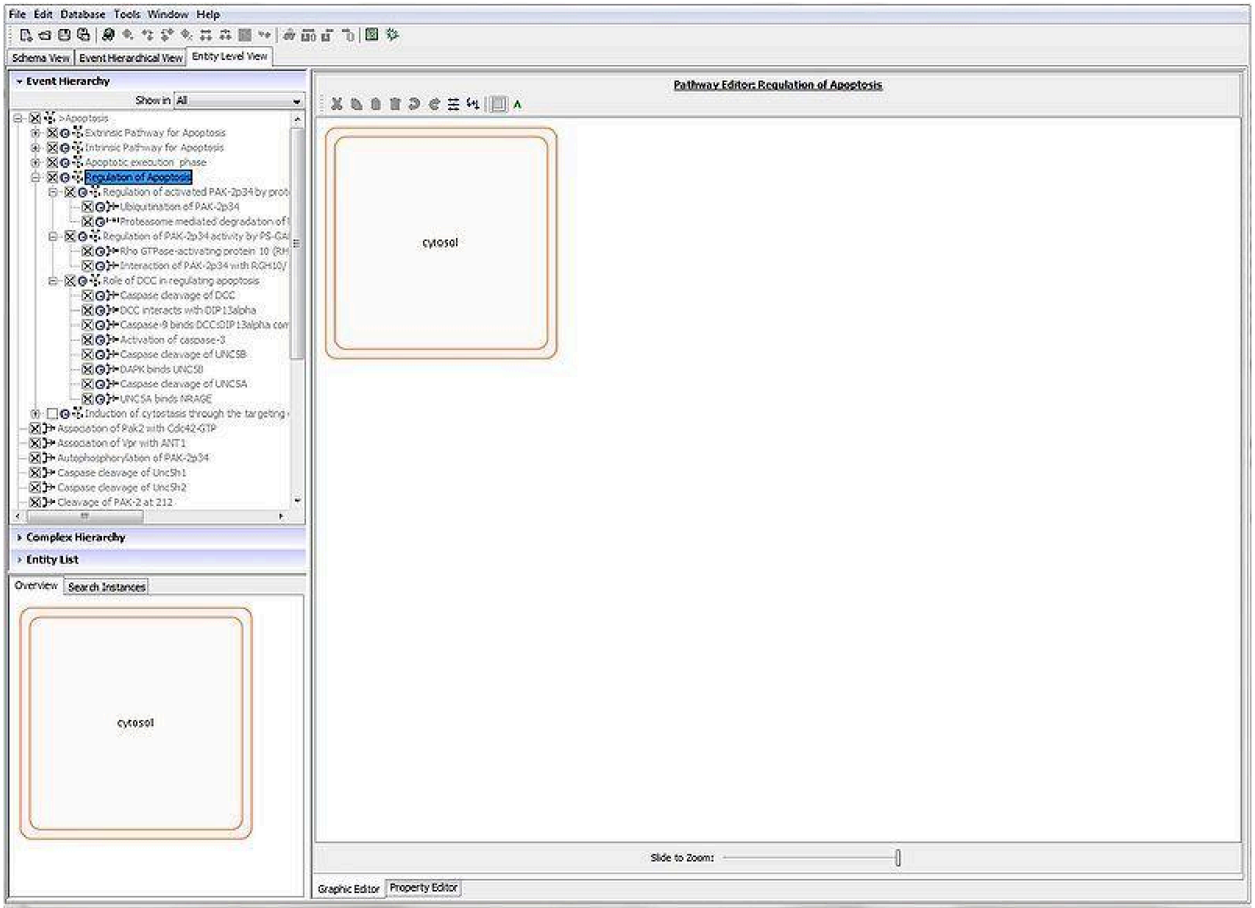


Adding cellular compartments to diagram

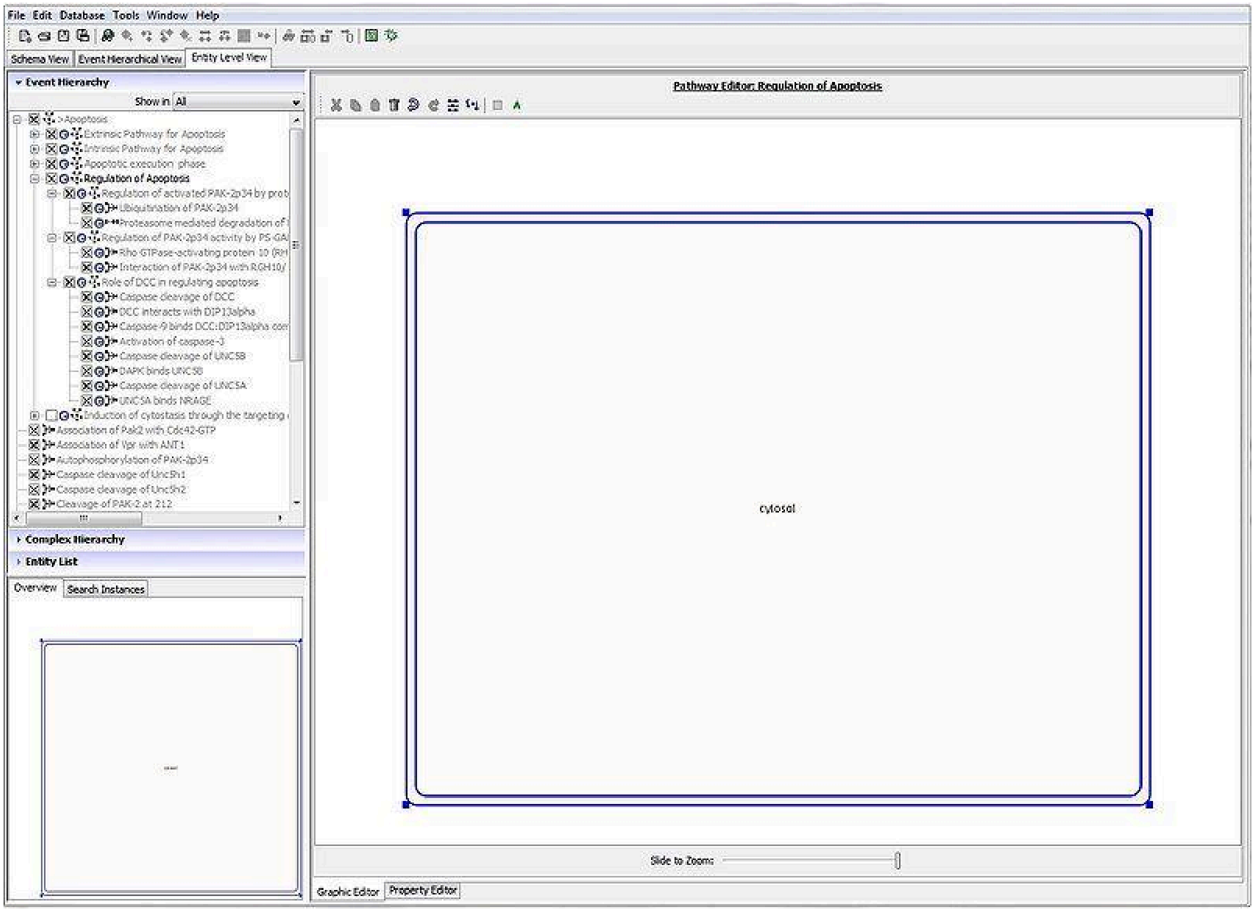
Click on the shaded square in the menu bar for the Pathway editor. It is best to add all the compartments that you will need before you start to lay out reactions. Here cytosol is added.



To reposition the compartment, click on it and drag.

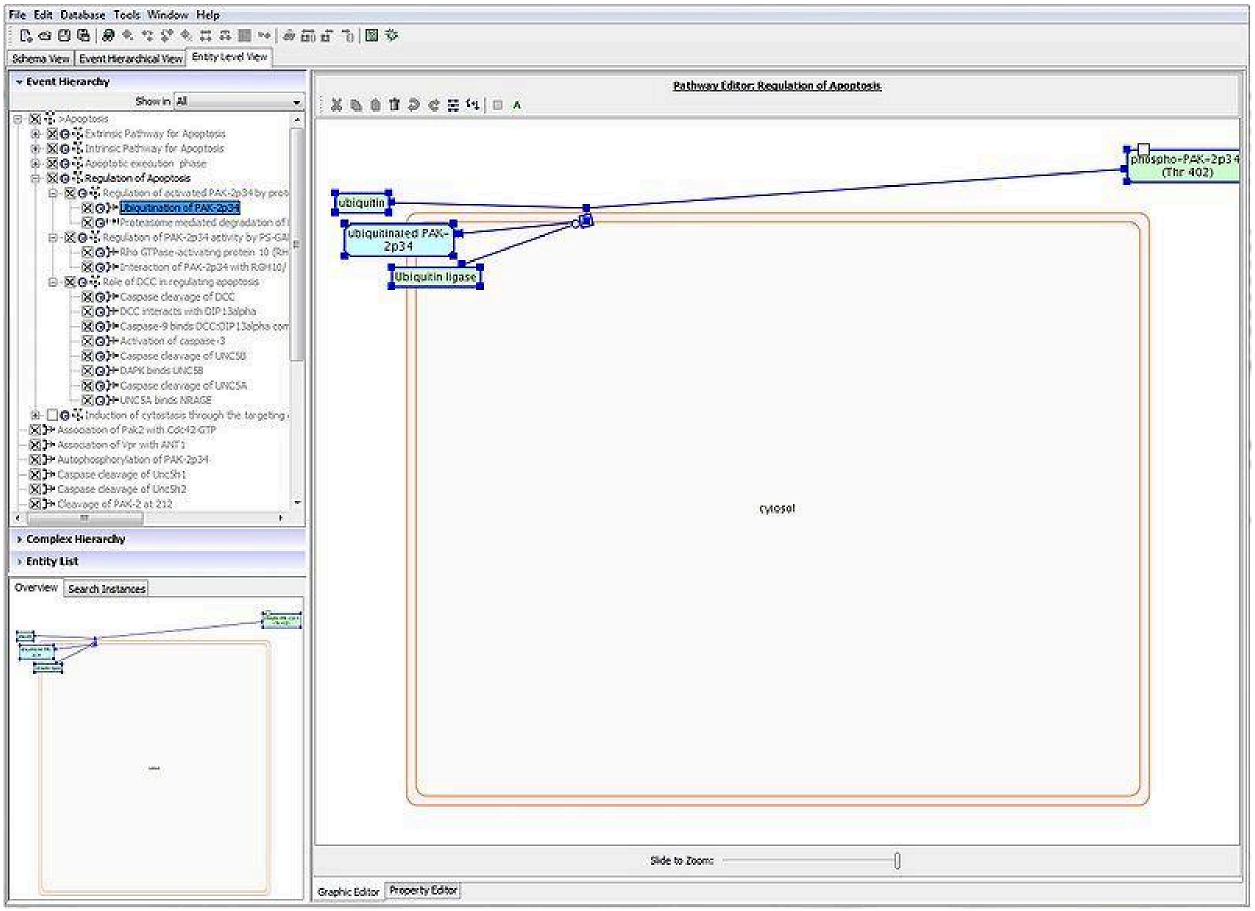


To enlarge the compartment, click on it to select it and then grab the compartment at one of its nodes in the corners. Then drag outward.

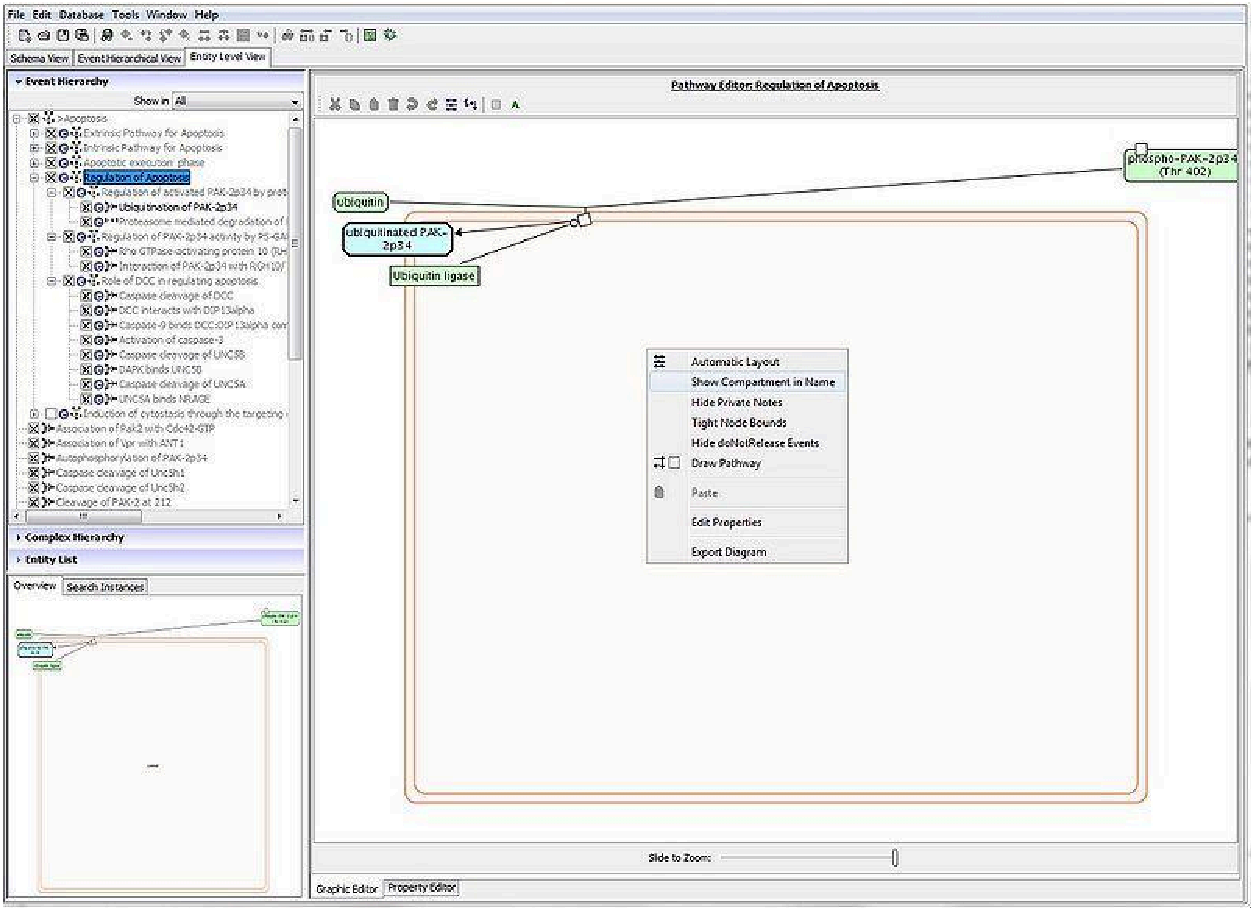


Drawing the pathway

To begin drawing, select a reaction from the event hierarchy and drag it onto the diagram.



To see the names of the compartments of the reaction participating molecules, right click anywhere on the diagram and opt to show compartment names. This will make it easier to see that the molecules have been positioned in the correct compartment.


File Edit Database Tools Window Help
Schema View
Event Hierarchical View
Entity Level View
Event Hierarchy
Show in All
Apoptosis
Extrinsic Pathway for Apoptosis
Intrinsic Pathway for Apoptosis
Apoptotic execution phase
Regulation of Apoptosis
Ubiquitination of PAK-2p34
Role of DCC in regulating apoptosis
Caspase cleavage of DCC
DCC interacts with DIP13alpha
Activation of caspase-3
Caspase cleavage of UNC5B
DAPK binds UNC5B
Caspase cleavage of UNC5A
UNC5A binds NRAGE
Association of Pak2 with Cdc42-GTP
Association of Vpr with ANT1
Autophosphorylation of PAK-2p34
Caspase cleavage of Unc5h1
Caspase cleavage of Unc5h2
Cleavage of PAK-2 at 212
Complex Hierarchy
Entity List
Overview
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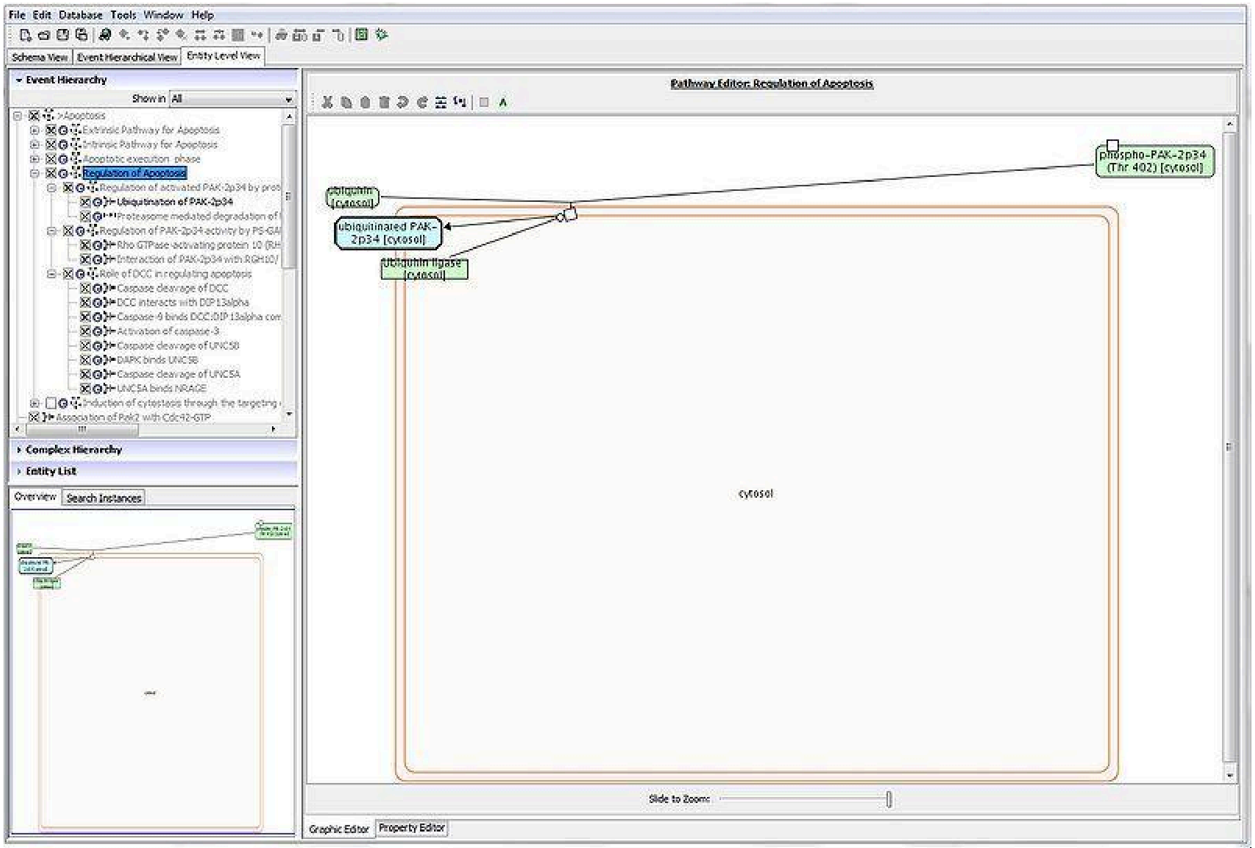
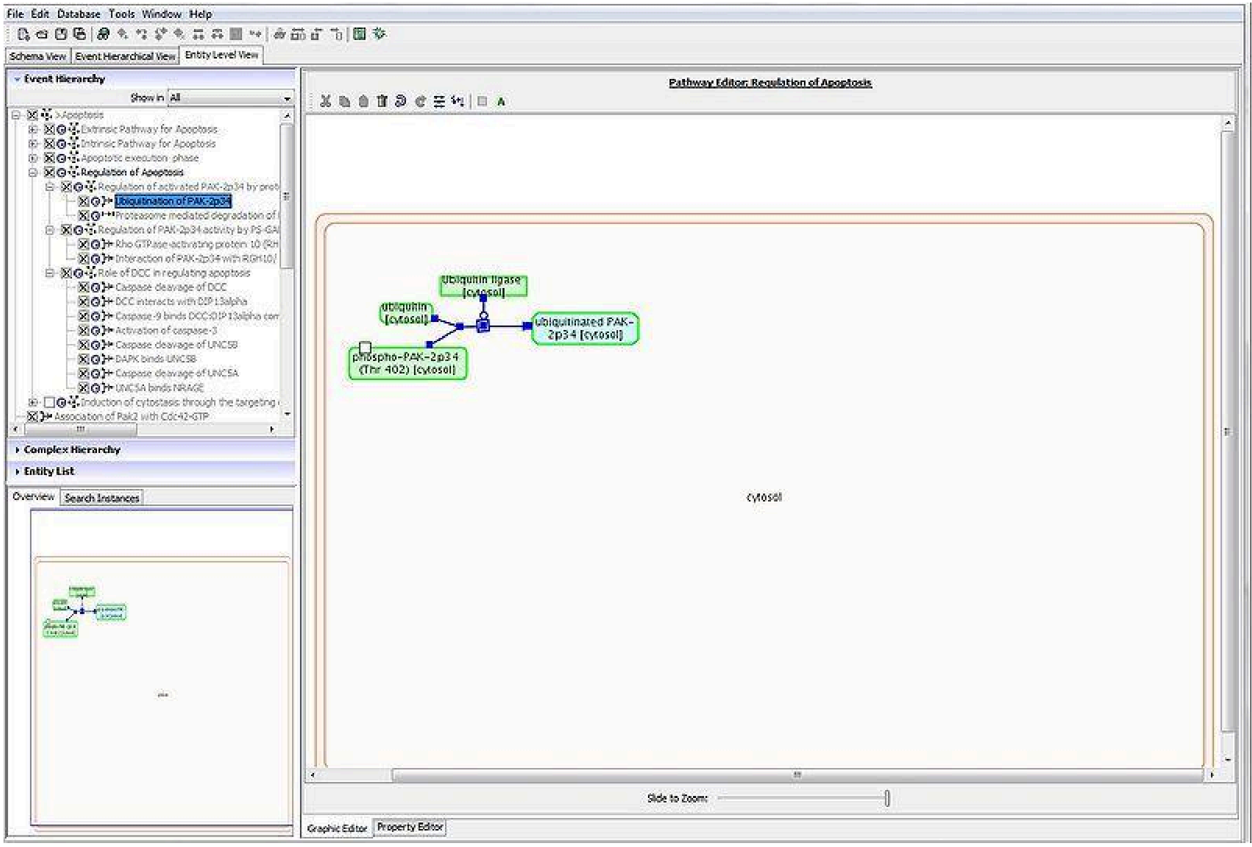
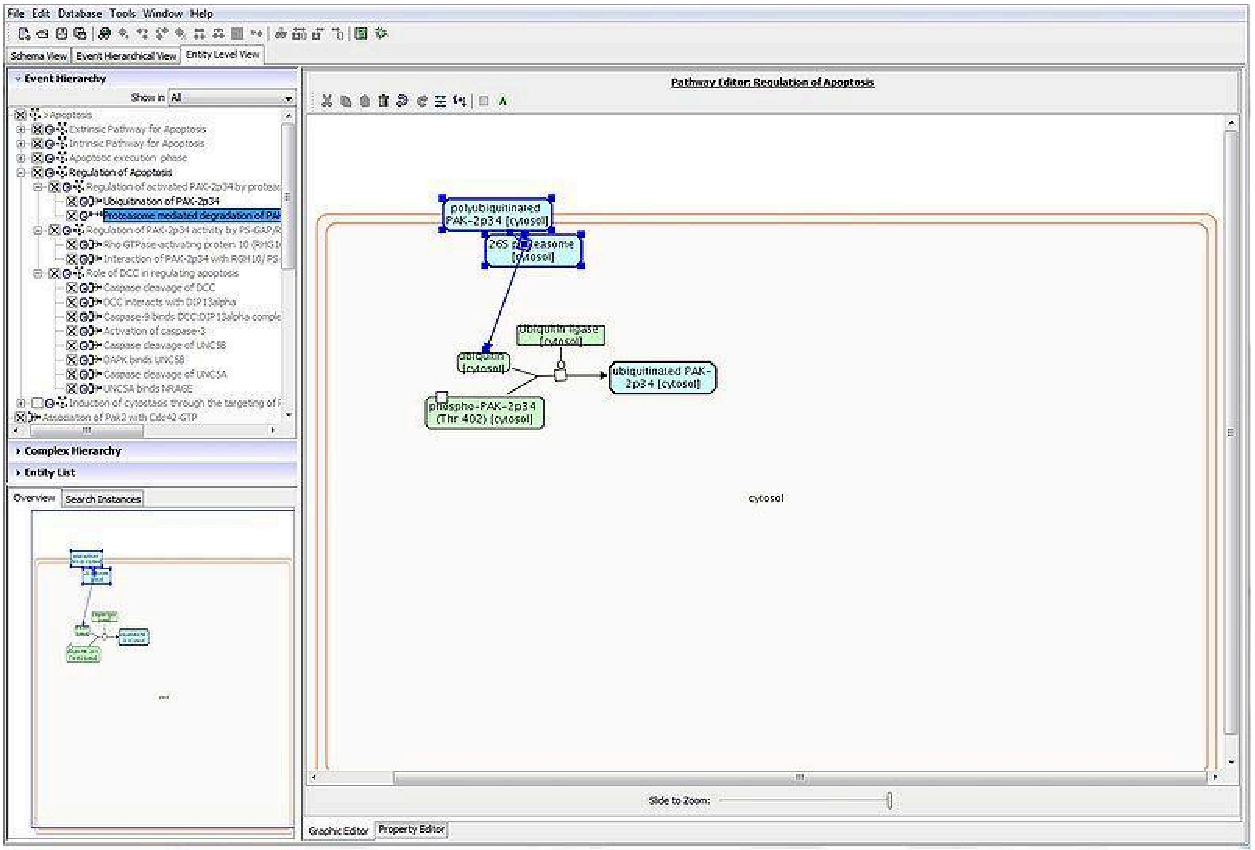
Pathway Editor: Regulation of Apoptosis
phospho-PAK-2p34 (Thr 402)
ubiquitin
ubiquitinated PAK-2p34
Ubiquitin ligase
Automatic Layout
Show Compartment in Name
Hide Private Notes
Tight Node Bounds
Hide doNotRelease Events
Draw Pathway
Paste
Edit Properties
Export Diagram
Slide to Zoom:
Graphic Editor
Property Editor



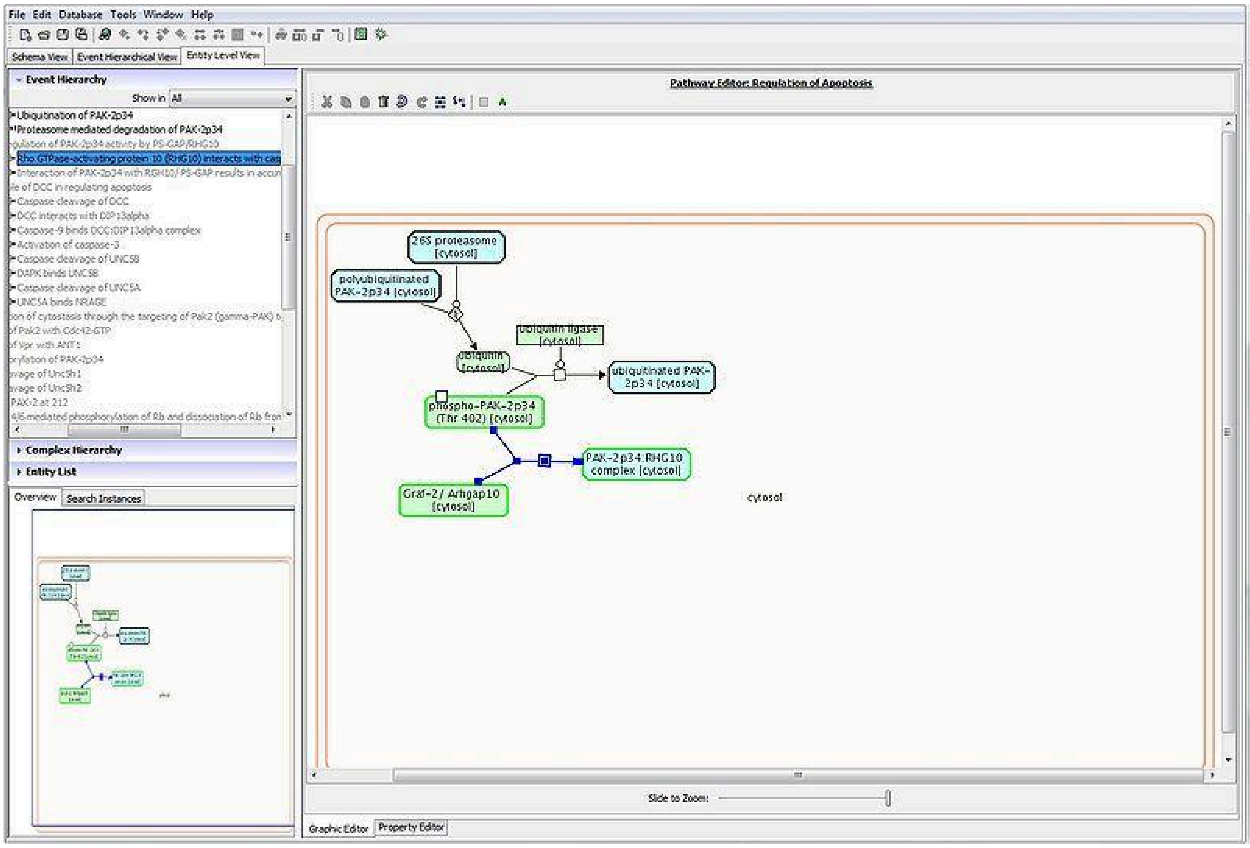
To reposition the reaction you can click and drag different components or you can select the entire reaction by clicking and dragging a selection box over it. Then the reaction and all its component molecules can be moved as a unit.



Additional reactions are dragged out and positioned one at a time.

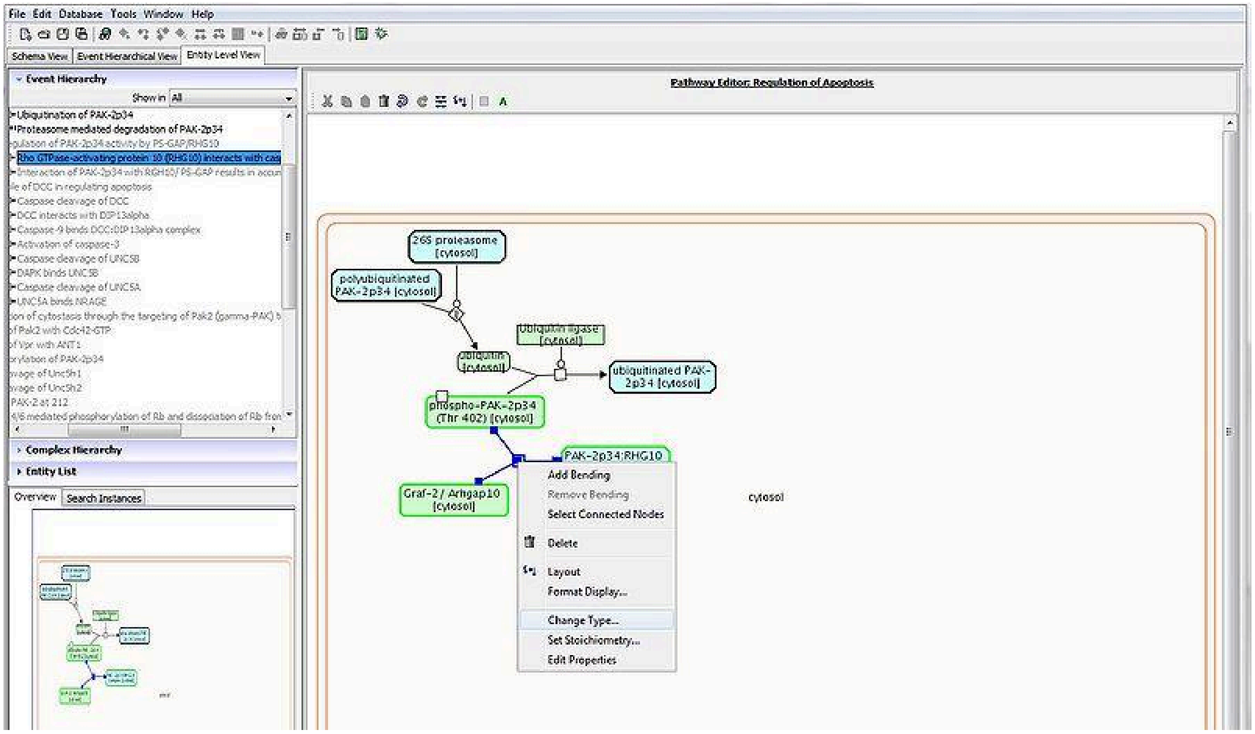


File Edit Database Tools Window Help
Schema View
Event Hierarchical View
Entity Level View
Event Hierarchy
Pathway Editor: Regulation of Apoptosis
Apoptosis
Extrinsic Pathway for Apoptosis
Intrinsic Pathway for Apoptosis
Apoptotic execution phase
Regulation of Apoptosis
Ubiquitination of PAK-2p34
Caspase cleavage of DCC
DCC interacts with DIP13alpha
Activation of caspase-3
Caspase cleavage of UNC5B
DAPK binds UNC5B
Caspase cleavage of UNC5A
UNC5A binds NRAGE
Association of Pak2 with Cdc42-GTP
Complex Hierarchy
Entity List
Overview
Search Instances
polyubiquitinated PAK-2p34 [cytosol]
ubiquitinated PAK-2p34 [cytosol]
phospho-PAK-2p34 (Thr 402) [cytosol]
cytosol
Slide to Zoom:
Graphic Editor
Property Editor

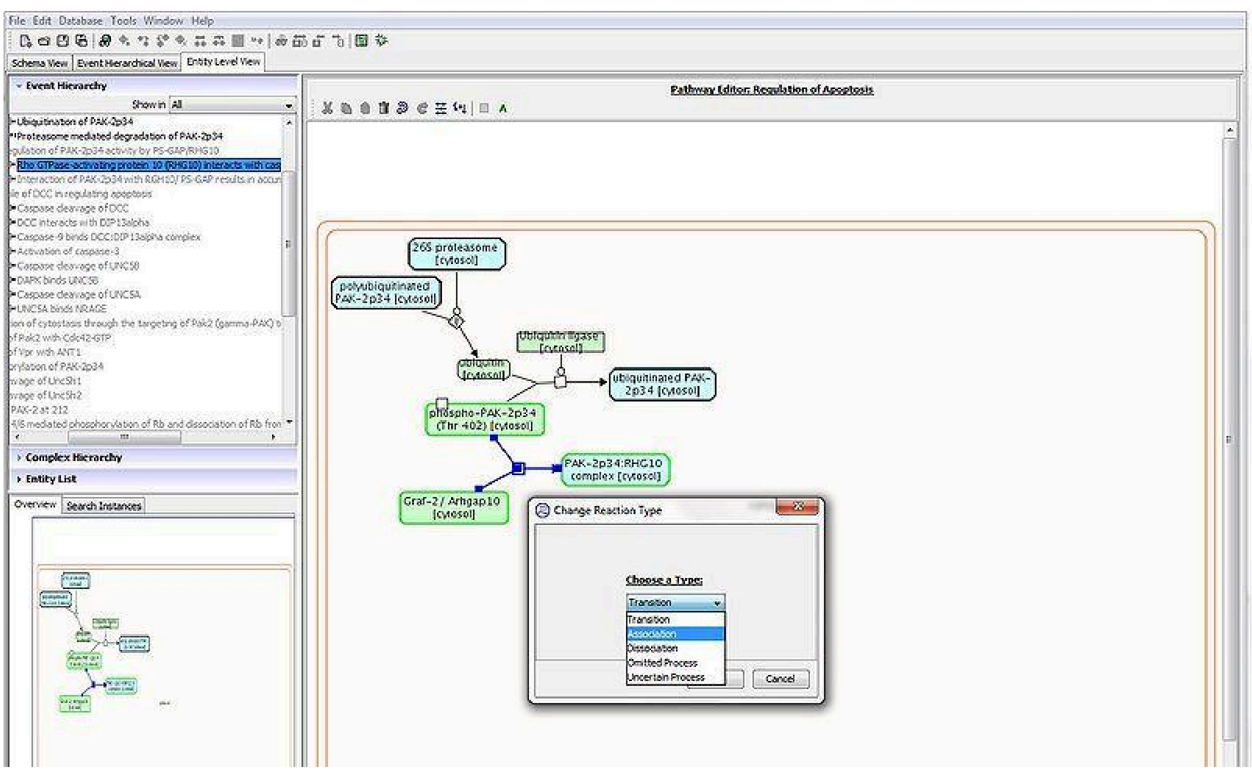


Reaction types

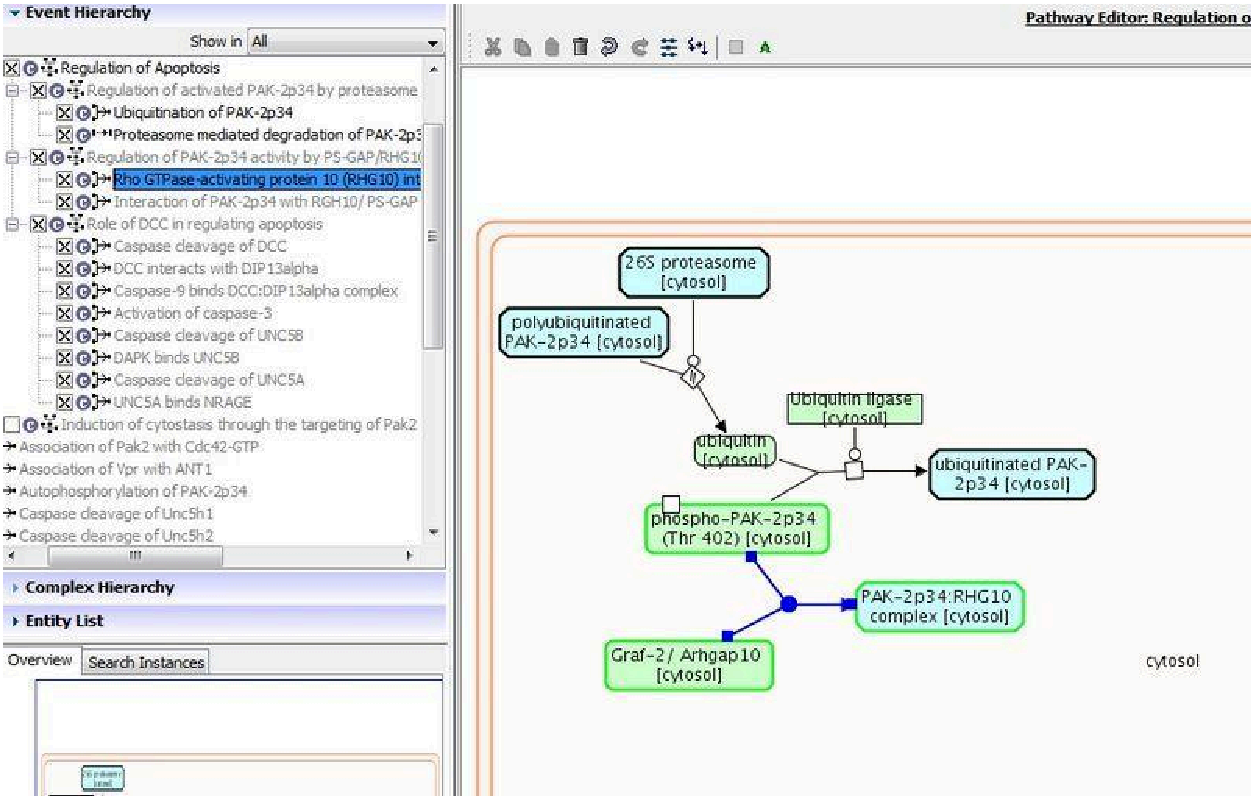
Reactions are of one of 5 types: Transition, Association, Dissociation, Omitted process, and Uncertain process. Transitions involve the molecules changing state, Association is a binding reaction, Dissociation is the Dissociation of a complex. Omitted process, and Uncertain process can be used to describe Black Box Events (BBEs). Omitted process for a BBE describes a process where steps have been deliberately consolidated eg. a gene producing a protein. Uncertain process for a BBE describes a process where the exact mechanism is unknown. To apply a reaction type to a reaction, right click on reaction, select change type .

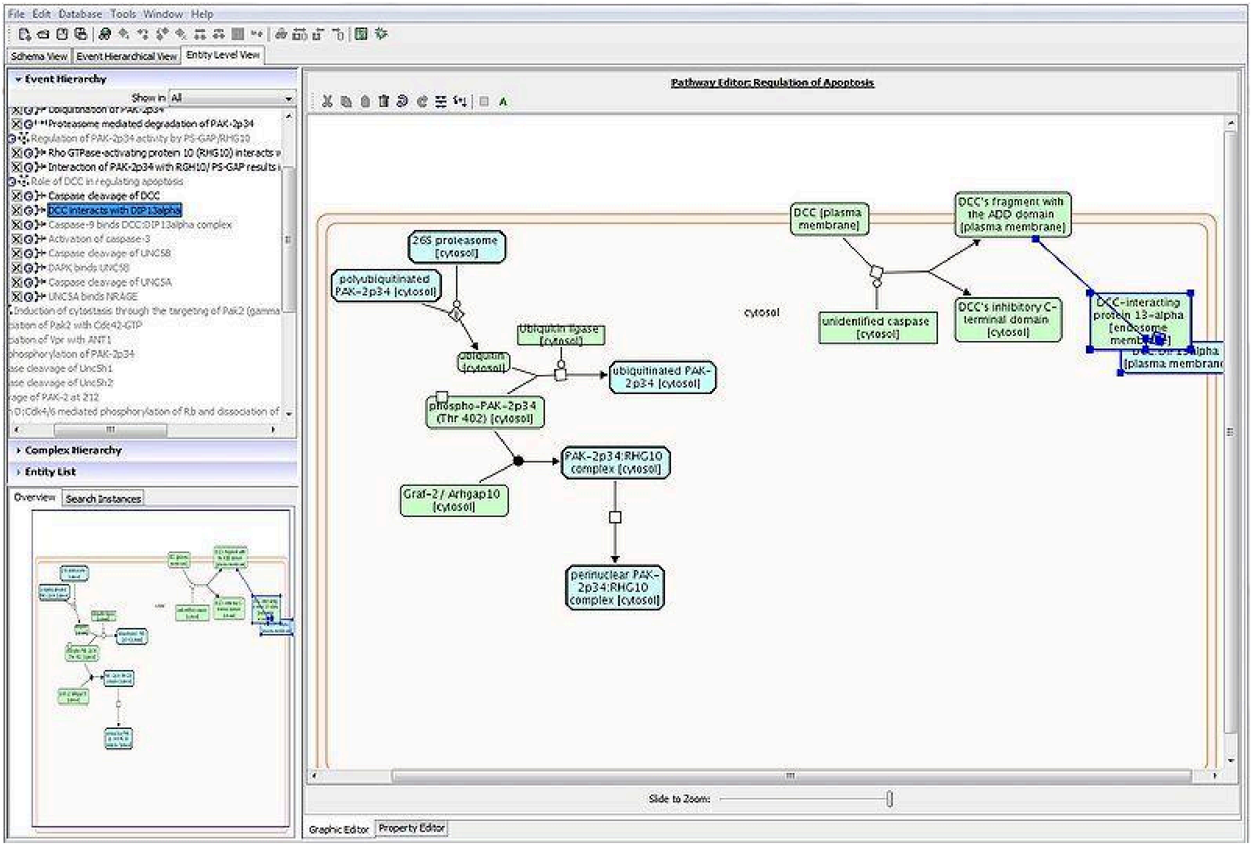


Select type of interest. Here it is an association.



Continue to drag out, and map, and assign reaction type to the remaining reactions.





File Edit Database Tools Window Help
Schema View
Event Hierarchical View
Entity Level View
Event Hierarchy
Show in All
Proteasome mediated degradation of PAK-2p34
Regulation of PAK-2p34 activity by PS-GAP/RHG10
Rho GTPase-activating protein 10 (RHG10) interacts w
Interaction of PAK-2p34 with RGH10/ PS-GAP results i
Role of DCC in regulating apoptosis
Caspase cleavage of DCC
DCC interacts with DIP13alpha
Caspase-9 binds DCC:DIP13alpha complex
Activation of caspase-3
Caspase cleavage of UNC5B
DAPK binds UNC5B
Caspase cleavage of UNC5A
UNC5A binds NRAGE
Complex Hierarchy
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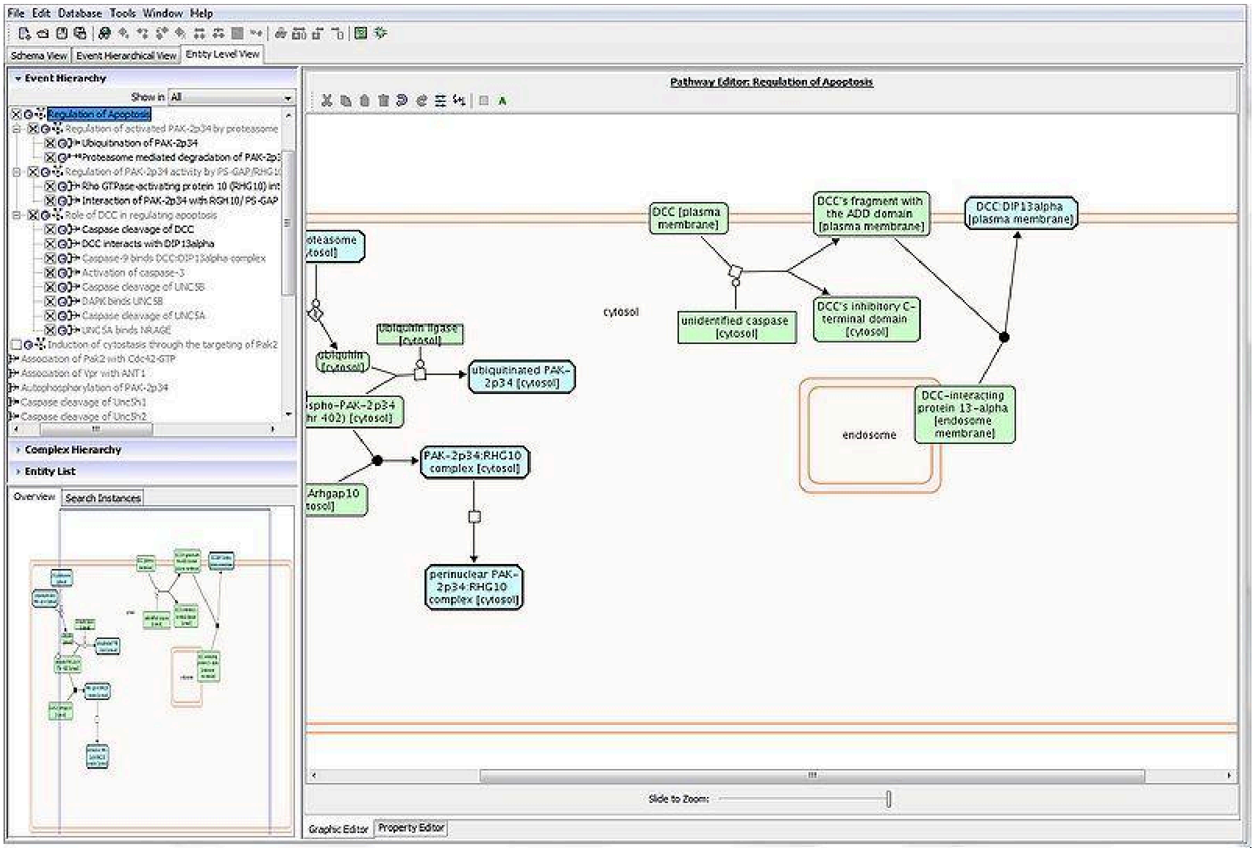
Overview
Search Instances
Pathway Editor: Regulation of Apoptosis
26S proteasome [cytosol]
polyubiquitinated PAK-2p34 [cytosol]
Ubiquitin ligase [cytosol]
ubiquitinated PAK-2p34 [cytosol]
phospho-PAK-2p34 (Thr 402) [cytosol]
PAK-2p34:RHG10 complex [cytosol]
Graf-2 / Arhgap10 [cytosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
cytosol
DCC [plasma membrane]
DCC's fragment with the ADD domain [plasma membrane]
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
DCC-interacting protein 13-alpha
Slide to Zoom:
Graphic Editor
Property Editor


File Edit Database Tools Window Help
Schema View
Event Hierarchical View
Entity Level View
Event Hierarchy
Show in All
Regulation of Apoptosis
Regulation of activated PAK-2p34 by proteasome
Ubiquitination of PAK-2p34
Proteasome mediated degradation of PAK-2p3
Regulation of PAK-2p34 activity by PS-GAP/RHG1
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Caspase cleavage of Unc5h2
Complex Hierarchy
Entity List
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Search Instances
Pathway Editor: Regulation of Apoptosis
DCC [plasma membrane]
DCC's fragment with the ADD domain [plasma membrane]
DCC:DIP13alpha [plasma membrane]
oteasome tosol]
cytosol
unidentified caspase [cytosol]
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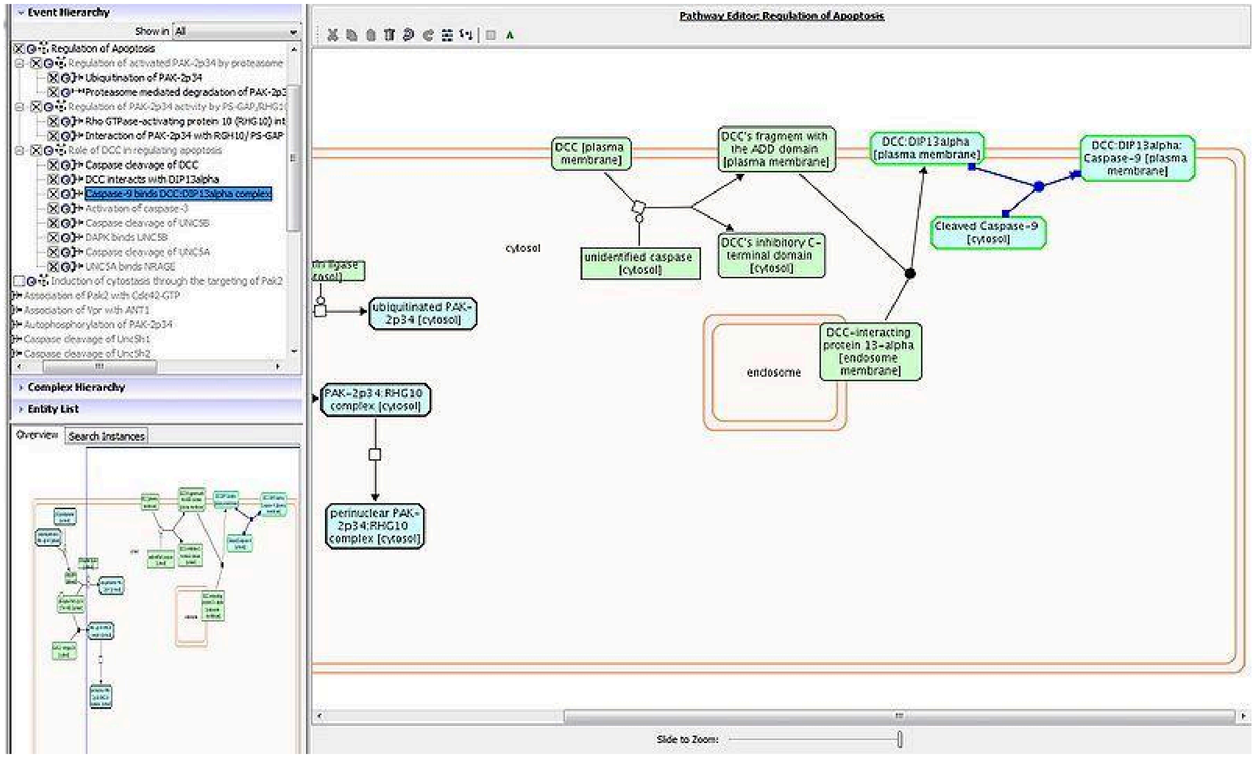
ubiquitin [cytosol]
ubiquitinated PAK-2p34 [cytosol]
spho-PAK-2p34 hr 402) [cytosol]
DCC-interacting protein 13-alpha [endosome membrane]
endosome
PAK-2p34:RHG10 complex [cytosol]
Arhgap10 tosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
Slide to Zoom:
Graphic Editor
Property Editor


Event Hierarchy
Show in All
Regulation of Apoptosis
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Ubiquitination of PAK-2p34
Proteasome mediated degradation of PAK-2p3
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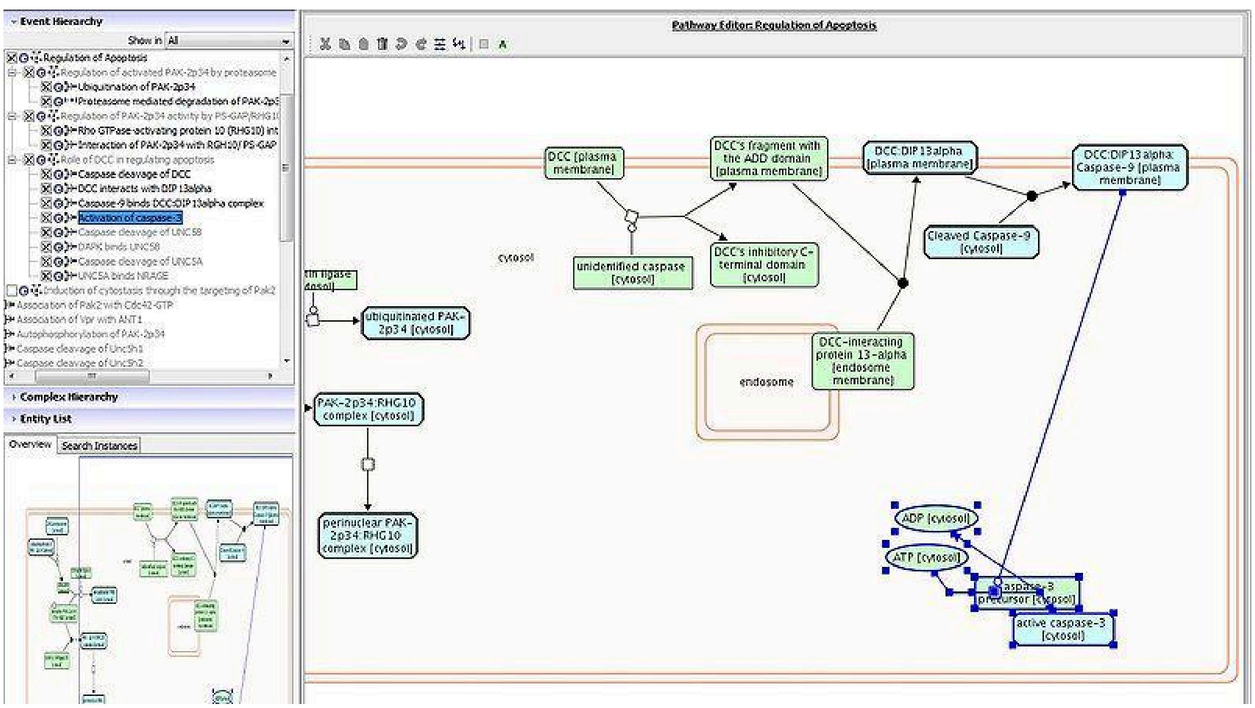
Pathway Editor: Regulation of Apoptosis
DCC [plasma membrane]
DCC's fragment with the ADD domain [plasma membrane]
DCC:DIP13alpha [plasma membrane]
DCC:DIP13alpha: Caspase-9 [plasma membrane]
Cleaved Caspase-9 [cytosol]
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
cytosol
ubiquitinated PAK-2p34 [cytosol]
endosome
DCC-interacting protein 13-alpha [endosome membrane]
PAK-2p34:RHG10 complex [cytosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
Slide to Zoom: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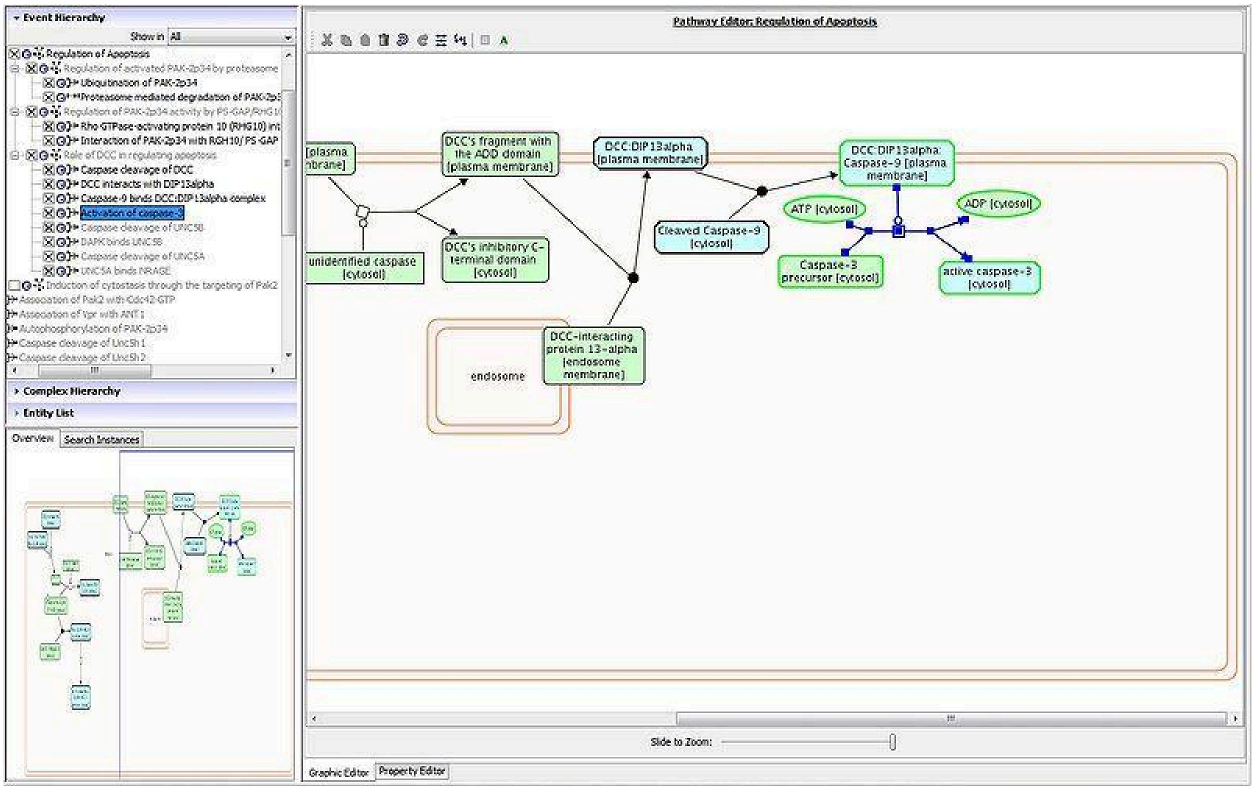
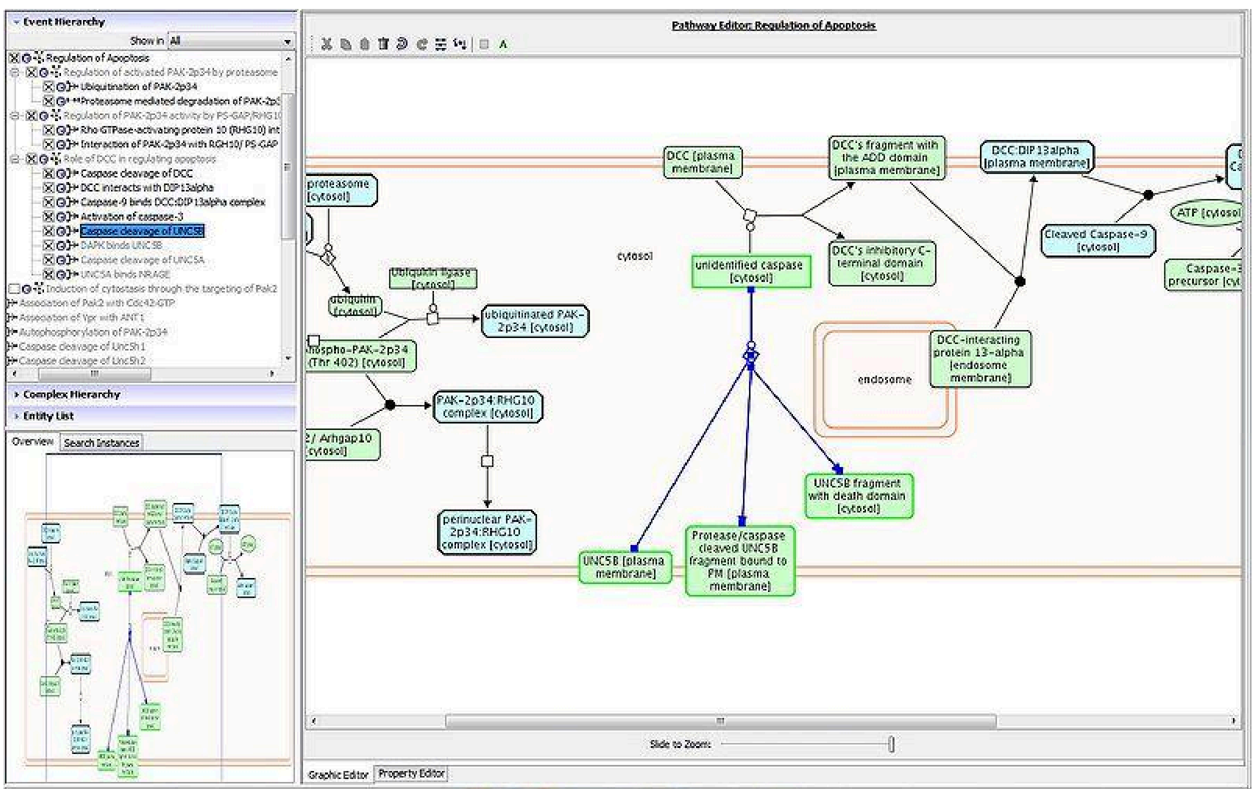
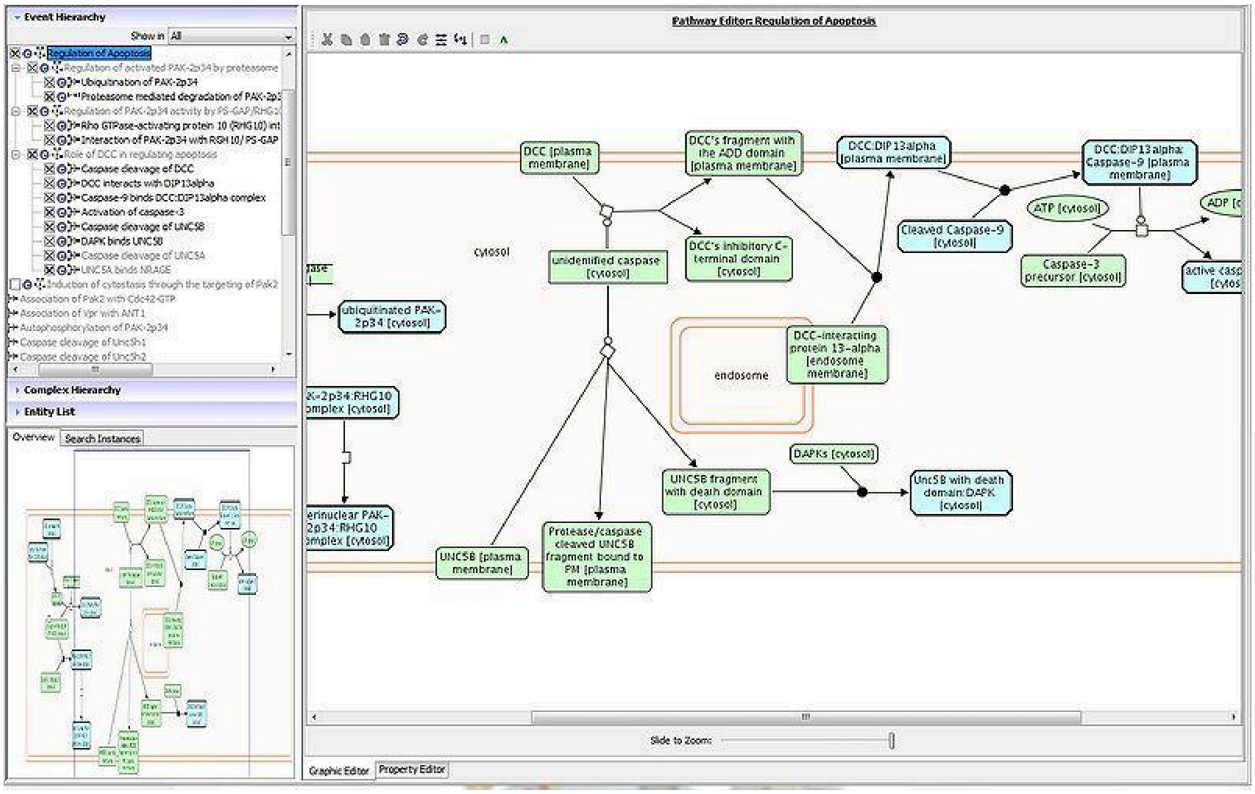
DCC:DIP13alpha [plasma membrane]
DCC:DIP13alpha: Caspase-9 [plasma membrane]
Cleaved Caspase-9 [cytosol]
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
cytosol
ubiquitinated PAK-2p34 [cytosol]
endosome
DCC-interacting protein 13-alpha [endosome membrane]
PAK-2p34:RHG10 complex [cytosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
ADP [cytosol]
ATP [cytosol]
Caspase-3 precursor [cytosol]
active caspase-3 [cytosol]



Event Hierarchy
Show in All
Regulation of Apoptosis
Regulation of activated PAK-2p34 by proteasome
Ubiquitination of PAK-2p34
Proteasome mediated degradation of PAK-2p3
Regulation of PAK-2p34 activity by PS-GAP/RHG1
Rho GTPase-activating protein 10 (RHG10) int
Interaction of PAK-2p34 with RGH10/ PS-GAP
Role of DCC in regulating apoptosis
Caspase cleavage of DCC
DCC interacts with DIP13alpha
Caspase-9 binds DCC:DIP13alpha complex
Activation of caspase-3
Caspase cleavage of UNC5B
DAPK binds UNC5B
Caspase cleavage of UNC5A
UNC5A binds NRAGE
Induction of cytostasis through the targeting of Pak2
Association of Pak2 with Cdc42-GTP
Association of Vpr with ANT1
Autophosphorylation of PAK-2p34
Caspase cleavage of UncSh1
Caspase cleavage of UncSh2
Complex Hierarchy
Entity List
Overview
Search Instances
Pathway Editor: Regulation of Apoptosis
DCC's fragment with the ADD domain [plasma membrane]
DCC:DIP13alpha [plasma membrane]
DCC:DIP13alpha: Caspase-9 [plasma membrane]
ATP [cytosol]
ADP [cytosol]
Cleaved Caspase-9 [cytosol]
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
Caspase-3 precursor [cytosol]
active caspase-3 [cytosol]
DCC-interacting protein 13-alpha [endosome membrane]
endosome
Slide to Zoom:
Graphic Editor
Property Editor



Event Hierarchy
Show in All
Regulation of Apoptosis
Caspase cleavage of UNC5B
Complex Hierarchy
Entity List
Overview
Search Instances
Pathway Editor: Regulation of Apoptosis
proteasome [cytosol]
DCC [plasma membrane]
DCC's fragment with the ADD domain [plasma membrane]
DCC:DIP13alpha [plasma membrane]
Cleaved Caspase-9 [cytosol]
ATP [cytosol]
cytosol
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
Ubiquitin ligase [cytosol]
ubiquitin [cytosol]
ubiquitinated PAK-2p34 [cytosol]
Thr 402) [cytosol]
PAK-2p34:RHG10 complex [cytosol]
2/ Arhgap10 [cytosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
endosome
DCC-interacting protein 13-alpha [endosome membrane]
UNC5B fragment with death domain [cytosol]
UNC5B [plasma membrane]
Protease/caspase cleaved UNC5B fragment bound to PM [plasma membrane]
Slide to Zoom:
Graphic Editor
Property Editor


Event Hierarchy
Pathway Editor: Regulation of Apoptosis
Regulation of Apoptosis
DCC [plasma membrane]
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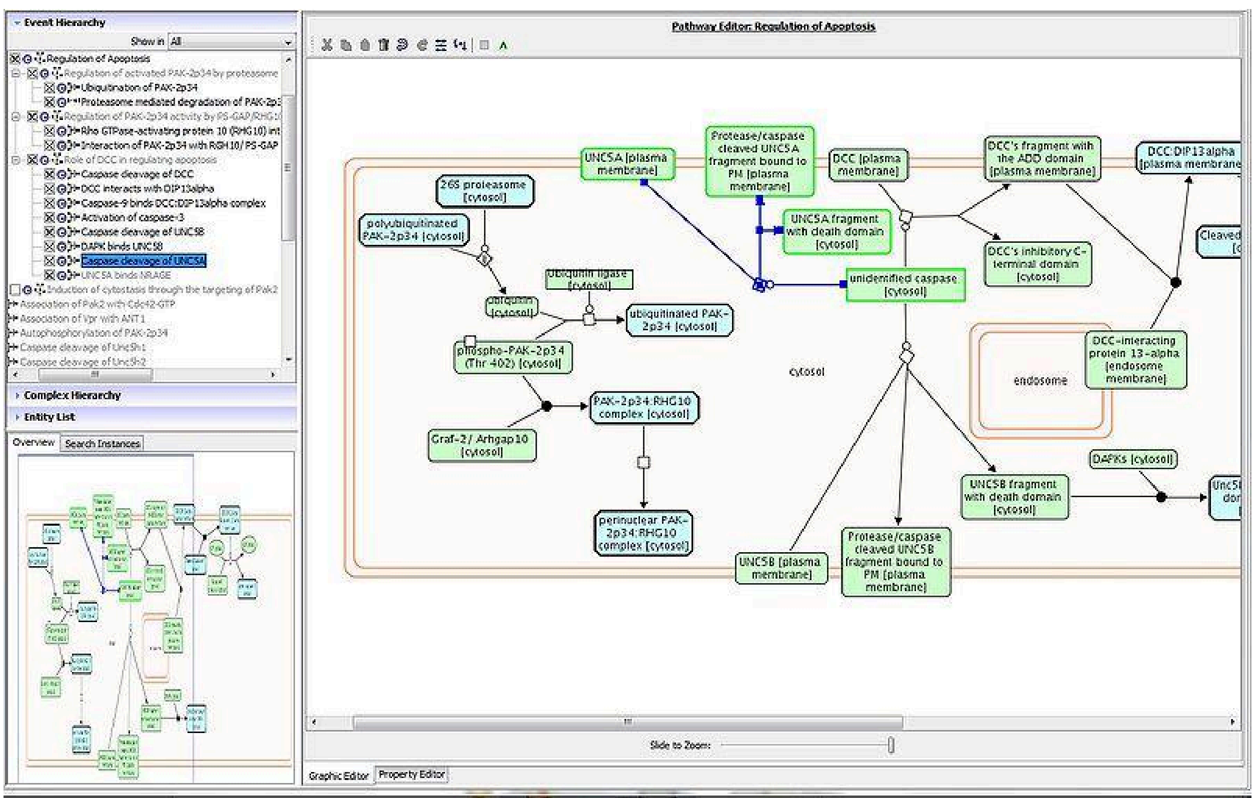
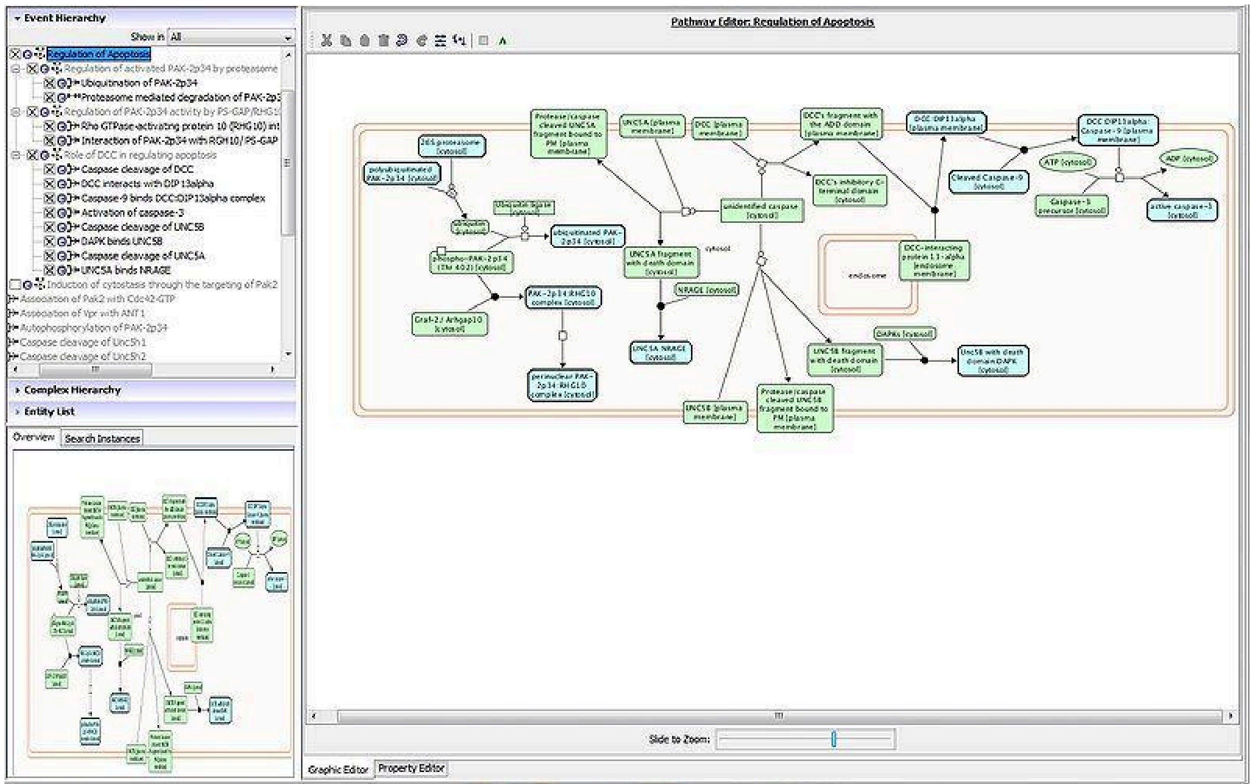
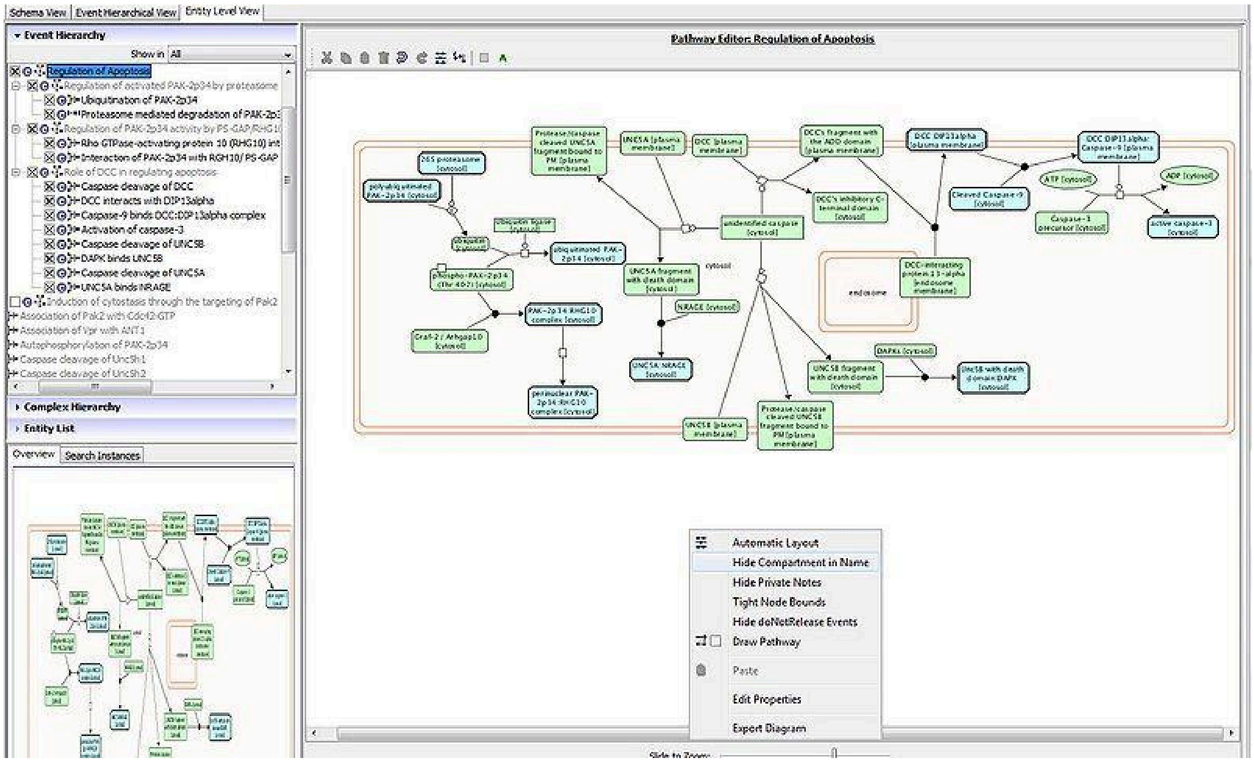
DCC:DIP13alpha [plasma membrane]
DCC:DIP13alpha:Caspase-9 [plasma membrane]
Cleaved Caspase-9 [cytosol]
ATP [cytosol]
Caspase-3 precursor [cytosol]
unidentified caspase [cytosol]
DCC's inhibitory C-terminal domain [cytosol]
cytosol
endosome
DCC-interacting protein 13-alpha [endosome membrane]
DAPKs [cytosol]
UNC5B fragment with death domain [cytosol]
Unc5B with death domain:DAPK [cytosol]
Protease/caspase cleaved UNC5B fragment bound to PM [plasma membrane]
UNC5B [plasma membrane]
ubiquitinated PAK-2p34 [cytosol]
Complex Hierarchy
Entity List
Overview
Search Instances
Slide to Zoom:
Graphic Editor
Property Editor

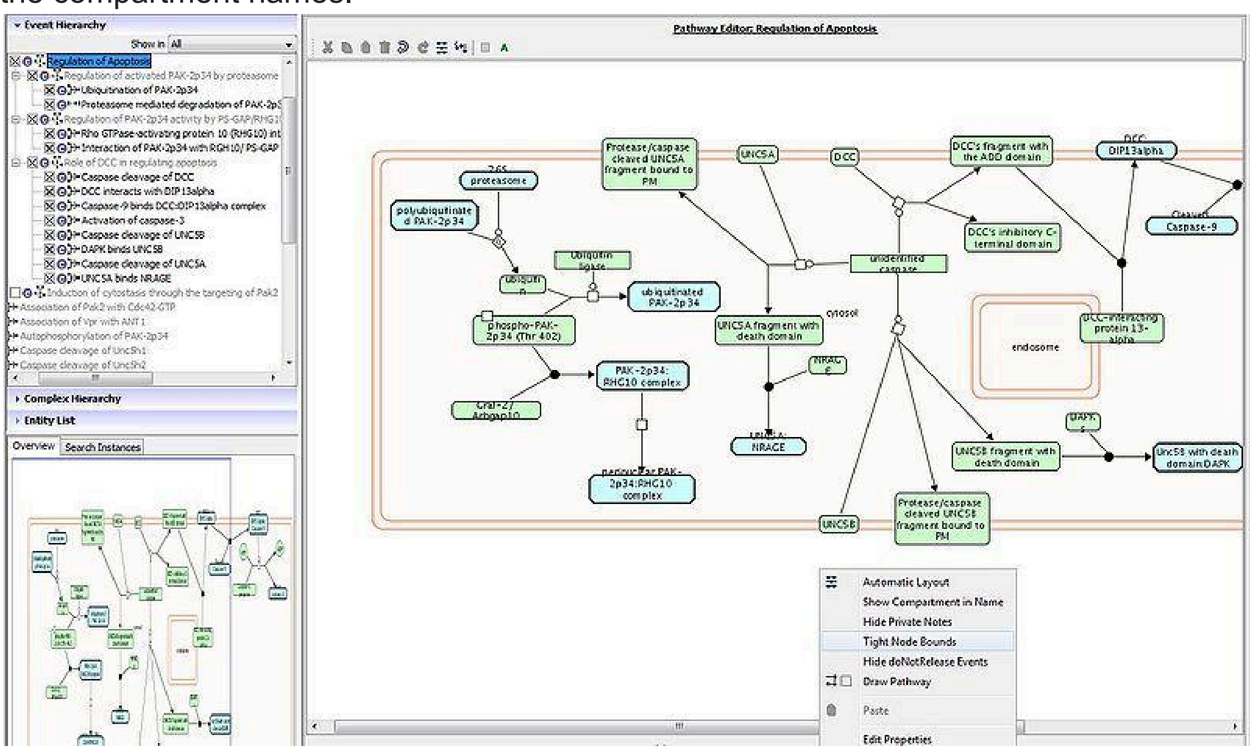

Event Hierarchy
Pathway Editor: Regulation of Apoptosis
Caspase cleavage of UNC5A
UNC5A [plasma membrane]
Protease/caspase cleaved UNC5A fragment bound to PM [plasma membrane]
UNC5A fragment with death domain [cytosol]
unidentified caspase [cytosol]
26S proteasome [cytosol]
polyubiquitinated PAK-2p34 [cytosol]
Ubiquitin ligase [cytosol]
ubiquitinated PAK-2p34 [cytosol]
phospho-PAK-2p34 (Thr 402) [cytosol]
Graf-2/ Arhgap10 [cytosol]
PAK-2p34:RHG10 complex [cytosol]
perinuclear PAK-2p34:RHG10 complex [cytosol]
DCC [plasma membrane]
DCC's fragment with the ADD domain [plasma membrane]
DCC:DIP13alpha [plasma membrane]
DCC's inhibitory C-terminal domain [cytosol]
cytosol
endosome
DCC-interacting protein 13-alpha [endosome membrane]
DAPKs [cytosol]
UNC5B fragment with death domain [cytosol]
UNC5B [plasma membrane]
Protease/caspase cleaved UNC5B fragment bound to PM [plasma membrane]
Slide to Zoom:
Graphic Editor
Property Editor



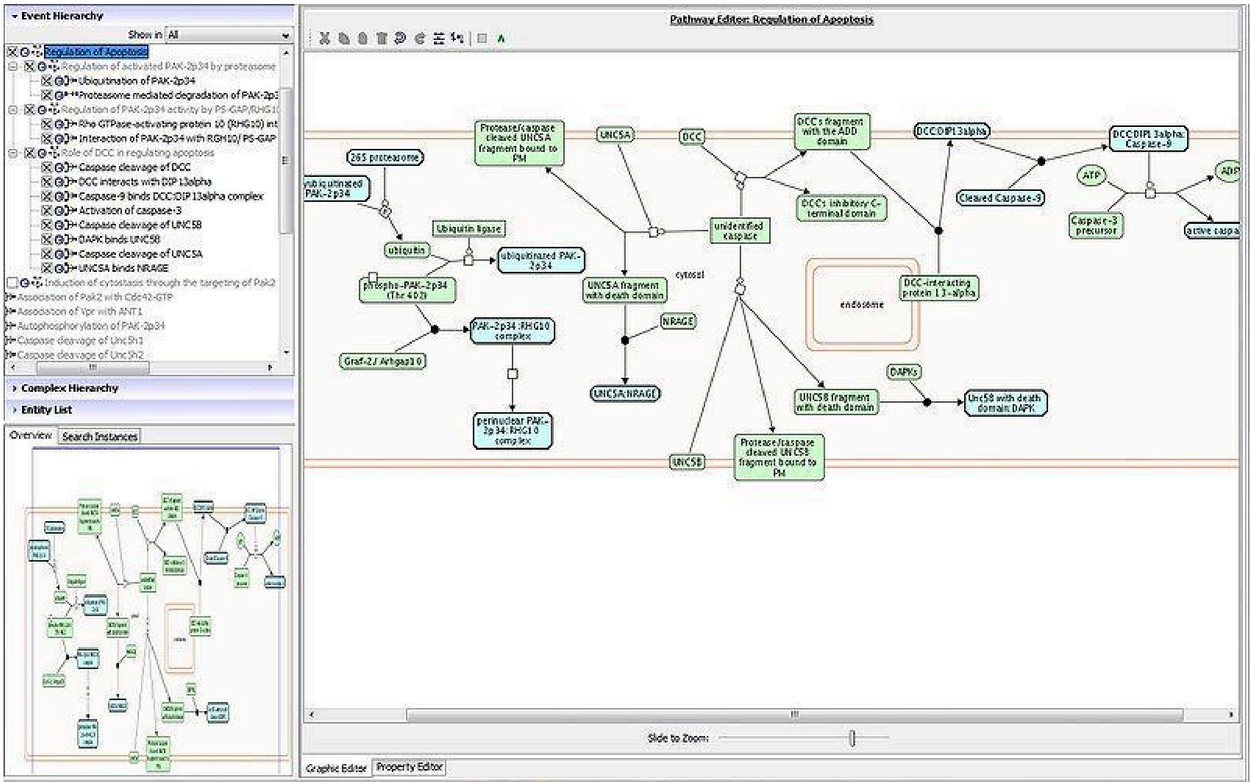
Once all of the reactions have been laid out, the compartment names on molecules can be hidden by right clicking on the diagram anywhere and selecting "Hide compartment in names".



Right click and select "Tight Node Bounds. This will reduce the space left by removing the compartment names.



Here is the completed diagram.



Verify that all entity names are contained in the icons, resizing manually if necessary.

Connecting sets with their members

In some cases sets of entities are used in a pathway, and also the entities themselves. In the diagram you may want to indicate that an entity belongs to a set. To make the connection, click on the set and one of the entities you want to connect to (done by holding shift as you choose both), right-click with both highlighted and choose “Add EntitySet and Member Link”. This will link the set to the chosen entity with a dotted line.

Use of pathways icons as links to related "pathways" in diagrams

If you want to include a link to a pathway that is not actually "part of" the pathway that you are diagramming, you can do this by checking out both the pathway you are diagramming (pathway A) as well as the pathway you'd like to include as a icon (pathway B) using the Event view. Open the diagram of pathway A in the ELV view. Then, drag the icon of pathway B into the ELV. **Note that you must indicate the reaction that links the diagrams via a flowline between the pathway icon and the output of the connecting reaction. Save and commit the changes. Note that the pathway whose icon you would like to include MUST have a pathway diagram associated with it.

To make an association between a reaction in one diagram and another pathway diagram in which the reaction (entity) acts as a preceding event (participant), you can draw a flow line from the entity to the pathway icon. Note that the pathway displayed as an icon in such a case MUST, itself, have an ELV in order to link this way.

To make this connection between events (on a mac) go to preferences under Reactome curator tool on top bar and unclick 'Use ELV as drawing tool'.

Then if you go into a diagram and hover over entity you want to connect, you get the shadow arrows that you can extend to your green box icon, with options for either activate or inhibit.
On a PC I go to the Tools tab, select Options, and then uncheck "Use ELV as a drawing tool".

Drawing a new Top level pathway diagram

If you are creating a diagram for a new top level pathway (check with editorial managers if this is appropriate), remember that the pathway itself must be listed as frontPage item in order to see the diagram. To do this, check out the frontPage instance from gk_central and add your pathways in the frontPageItem slot. Also, please mark this in the editorial calendar as a front page item and inform.

Editing existing pathways and corresponding diagrams

Any time a change is made to events in a pathway, the corresponding reactions/RLEs in a diagram must be adjusted accordingly. To start on revisions, be sure to "Build" project into a local CT project. Make sure you build the pathway that has the diagram associated with it, as opposed to one of its children.

Note that if you build a project (or try to open a pathway within a built project) that does not have its own pathway diagram, and you opt to open it in the ELV panel of the curator tool, you will get the message "The pathway has no diagram, do you want to automatically generate one?" The first thing you should decide is whether you actually need a new diagram (i.e the pathway is <u>not</u> already part of another parent pathway's diagram). If the pathway is already drawn in the diagram of a parent pathway, hit "Cancel" and rebuild that parent pathway) or, if the parent is already in your built project, hit "Cancel" and open the diagram for the parent. If you <u>do</u> want to make a new diagram for a pathway, Answering "Yes" may be helpful if there are just a few reactions but can create a tangle of reactions that are difficult/time consuming to adjust if there are many reactions. If you <u>do</u> want to create a diagram but want to lay it out yourself, hit "NO"

ENCAPSULATING A PATHWAY DIAGRAM

Encapsulating a sub-pathway

Encapsulate Diagram: this feature is used to refactor a section of a diagram for a sub-pathway contained by a displayed pathway diagram into a new diagram in the Entity Level View. It works as follows using the diagram for "Metabolism of steroid hormones and vitamins A and D" created by Peter and Bijay as an example:

1. Use "Build Project" or "Check Out" to create a local copy of project for Event 196071 (Pathway: Metabolism of steroid hormones and vitamins A and D)

2. In the Entity Level View, "Open Diagram" to open the diagram for the top-level pathway (Metabolism of steroid hormones and vitamins A and D).

3. Select "Vitamin D (calciferol) metabolism" in the "Event Hierarchy" at the left panel. Right click it to get a popup menu called "Encapsulate Diagram".

4. Select this action to create a diagram for this selected sub- pathway. A new pathway node for this sub-pathway will be added to the displayed diagram for pathway "Metabolism of steroid hormones and vitamins A and D". All reactions, compartments and reactions used by this sub-pathway only will be deleted from this displayed diagram.

5. To view the diagram for sub-pathway "Vitamin D (calciferol) metabolism", use "Open Diagram" after selecting this sub-pathway. 6). The user may need some fine adjustments: e.g. resize compartments in both old and new diagrams, add links to the new pathway node.

Encapsulating a connected pathway

Pathway diagrams may not be in a hierarchical relationship but still may have connections via input/output of their reactions, e.g. drug ADME pathways provide the input of drug-target reactions in metabolic pathways. To show a connected pathway B in the diagram of pathway A:

1. Build both projects together from the Database Browser --> Event View by selecting A and Ctrl-Click selecting B, then right-click Build Project

2. Open the Entity Level View, in Event Hierarchy right-click B with Open Diagram, then right-click A with Open Diagram

3. With Diagram A open select pathway B in Event Hierarchy and move it into the diagram. The encapsulated B should appear.

4. Select both the encapsulated B and the entity that is connected with B (e.g. the drug), right-click Create Flow Link.

Guidelines for encapsulating events and reusing events in other diagrams

1. If a pathway (pathway X) that has been drawn out in one pathway diagram is found to be participating in other pathways, consider encapsulating the pathway into its own diagram as described above. In this case, pathway X will be represented as an icon in the pathway diagram(s) in which pathway X participates. **Note that you can ONLY represent a pathway as a pathway icon in a diagram IF that pathway has a pathway diagram of its own.

2. It follows from the above rule, that IF pathway X has a diagram of its own, you cannot draw out pathway X in another pathway diagram. It MUST be represented as a pathway icon. You CAN, however, draw out subevents of pathway X (reactions and subpathways) in another pathway (Y) diagram, as long as the events are represented in the hierarchy of pathway Y AND as long as the subpathway you wish to represent does not have a diagram of its own. If it does you'd use the pathway icon of the subpathway.

Using the precedingEvent suggestion feature in the curator tool

See slides here (Appendix C7).

Cell Lineage Path Curation Guide

Introduction

Our bodies are built of >30 trillion cells specialized to fulfill diverse roles within our tissues, organs and organ systems. All these cells originate from a single cell, a zygote formed at conception. From zygote to fetus, and throughout childhood, adolescence and adulthood, cells divide and commit to different fates in order for the organism to develop, sustain and regenerate. The series of steps that lead from an undifferentiated progenitor cell, such as a stem cell, to one of its several possible specialized descendants constitutes a cell lineage path. Recent technological advances have allowed researchers to harvest high-throughput omics data from single cells of multicellular organisms and use it to track and manipulate cell fates (Burgess 2018; Saelens et al. 2019). This opens the door to the possibility of deciphering cell lineage paths at single cell resolution, a critical requirement for the advancement of regenerative medicine and cancer medicine.

Researchers who conduct experiments that involve cell lineage tracing through the study of genomic, epigenomic, transcriptomic and proteomic features of single cells currently expend significant effort extracting information from published sources to create short-lived, limited-scope local annotations of biological markers that are needed to interpret and analyze their high-throughput experimental data. This creates the urgent need for an open source, comprehensive, continuously maintained reference repository of cell lineage paths framed on the pre-existing knowledge of tissue morphogenesis, as well as a cell type-specific pathway resource, which can serve as a framework for interpretation and integration of latest experimental findings. Cytomics Reactome, a database of cell lineage paths and a toolset for analysis of single cell omics data, aims to be an integrative systems biology resource that fulfills this need.

Cytomics Reactome will systematically curate published studies and datasets to create a lasting, up-to-date, broad-scope and peer-reviewed online repository of cell type-specific biological markers, cell lineages, and relevant regulatory networks. The resource will be freely available to the research community, and researchers will be able to take part in the growth and maintenance of this repository by participating in community curation efforts, acting as peer reviewers, and by providing user feedback. To analyze single cell omics data, researchers will be able to download the repository and incorporate it into their own analysis pipelines, or to use online Cytomics Reactome analytic tools to apply cell lineage-oriented or pathway-oriented cell type-tailored analyses to their datasets.

By providing a peer-reviewed, standardized reference of cell lineage paths and milestones, Cytomics Reactome will increase the efficiency and reproducibility of the single cell-based research, and assist in addressing long-standing fundamental questions concerning regulation of progenitor cell division and differentiation by highlighting scientific gaps and acting as a catalyst for hypothesis generation.

Data Model

The Reactome data model provides a robust framework for annotation of biological entities and events (Joshi-Tope et al. 2005) and was easily expanded to accommodate new classes needed for Cytomics Reactome. Reactome web application and pathway visualization tools (Croft et al. 2011) have been extended to enable Cytomics Reactome visual display on the curator website, and we are working on enabling the display and search functions on the live website.

The basic unit of the Reactome data model is a reaction, an event that converts input to output physical entities. Physical entities, which can be simple or complex molecules, or molecular complexes, can also act as reaction catalysts and regulators. Reactions are grouped into causal chains to form pathways (Joshi-Tope et al. 2005).

The Event and PhysicalEntity classes were expanded to accommodate cell lineage path curation. A new Cell class, added as a subclass of the PhysicalEntity class (Figure 1A), has CellType, TissueLayer, Tissue and Organ as single-valued attributes. Multi-valued attributes of the Cell are ProteinMarker, RNAMarker, Proteome, Transcriptome and Epigenome (Figure 1C).

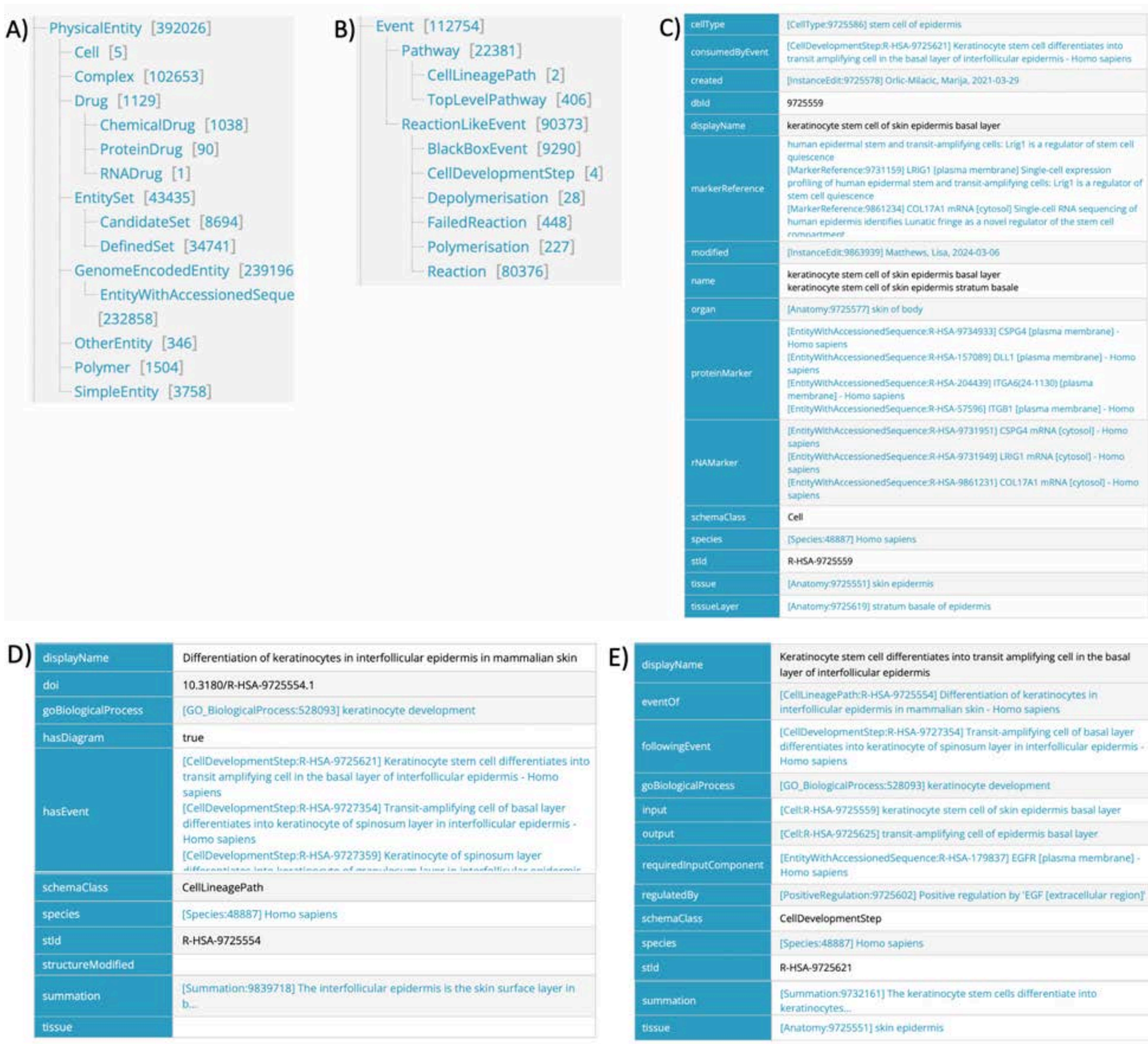


Figure 1. Changes to the Reactome data model to enable Cytomics Reactome annotations. A) Addition of the Cell subclass to the Physical Entity class. B) Addition of Cell Development Step and Cell Lineage Path subclasses to the Event class. C) Attributes of the Cell subclass. D) Attributes of the Cell Lineage Path subclass. E) Attributes of the Cell Development Step subclass.

Developmental relationships among Cells are described by two new event subclasses: Cell Development Step and Cell Lineage Path (Figure 1B), each tied to a Gene

Ontology (GO) biological process (The Gene Ontology Consortium 2019). The Cell Development Step, similar to the Reaction subclass, has inputs and outputs defined, which are instances of the Cell class (an input representing the cell of origin, and an output representing the destination cell type). Cell Development Step attributes include regulators (molecules promoting or inhibiting the step) and required input components (input cell biomarkers required for the action of regulators) (Figure 1E). The Cell Lineage Path, resembling the existing Pathway subclass, consists of Cell Development Steps connected through shared input/output Cells (Figure 1D).

Pathway Browser

Cytomics Reactome webpages will enable users to search and browse the curated lineages through tissue-level view (Figure 2A) or lineage path view (Figure 2B). Based on annotated regulators and required input components of Cell Development Steps that constitute a Cell Lineage Path, and on annotated biomarkers of participating Cells, we will generate a sub-network of Reactome's gene functional-interaction (FI) network (Wu and Stein 2012; Wu and Haw 2017) for each Path. These networks will first be extracted programmatically from annotation and then manually checked to ensure quality. This lineage path gene regulatory network, available to users as a diagram and as an FI table, will be free of cell boundaries and arranged in an approximate time progression order from left to right (Figure 2C).

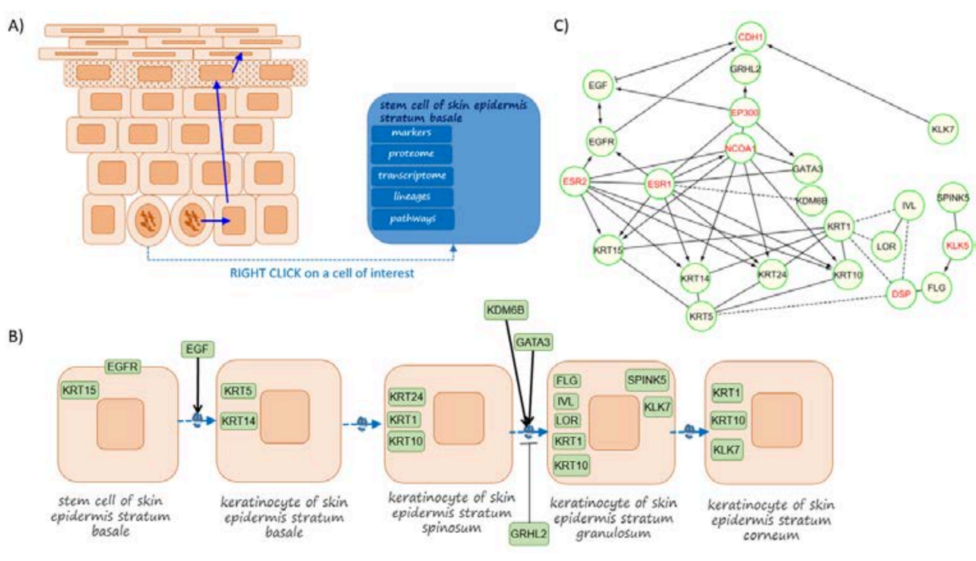


Figure 2. Development of keratinocytes in skin epidermis. A) Tissue-level view, where users can select a tissue's cell lineages and individual cells to access lists of annotated biomarkers, omics sets, and cell type-specific Reactome pathway viewer. B) Individual cell lineage path view where users can visualize and access detailed information on cell types and development steps that constitute a lineage, including biomarkers and development step regulators. C) Gene regulatory network view of the cell lineage path. Black labeled genes are path-derived biomarkers and regulators. Red labeled genes are linker genes.

Cell lineage paths are grouped into an expanded Developmental Biology chapter and are available among the browsable pathways on Reactome's website (currently available only on www.curator.reactome.org).

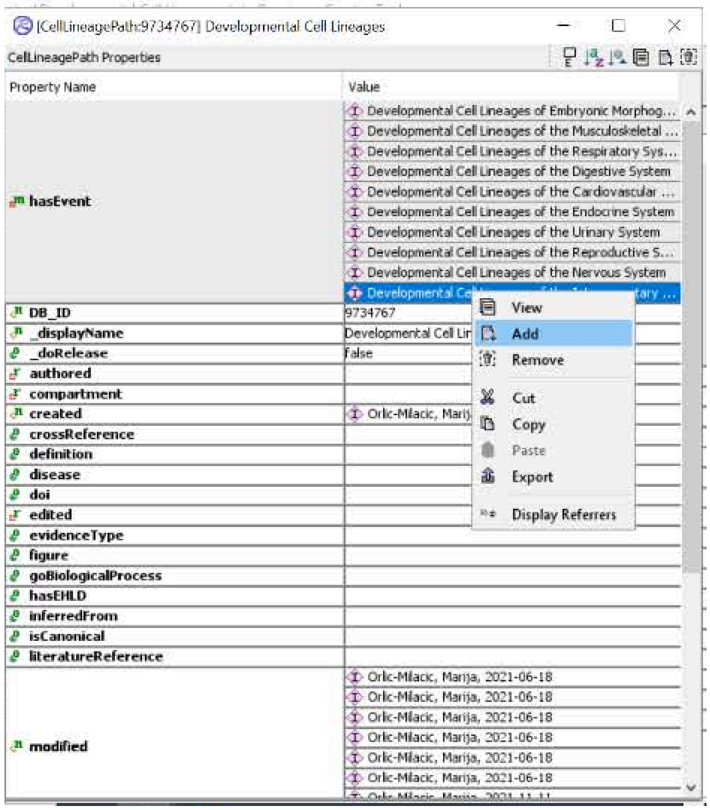
A detailed description of visual display and new website features is available in Cytomics Documentation (Appendix C9) written by Joel Weiser who worked on the new visual features with Guanming Wu.

Step-by-Step Annotation

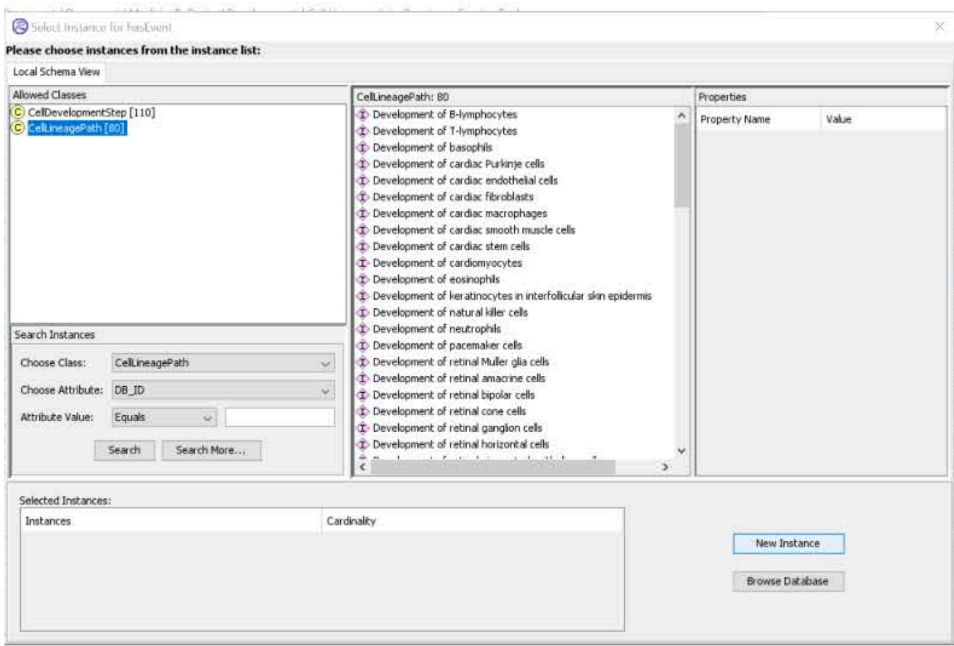
TissueLayer, Tissue and Organ attributes of Cells are populated using terms from public open source databases, the Cell Ontology (Sarntivijai et al. 2014; Osumi-Sutherland 2017) and UBERON (Haendel et al. 2014), and cross-referenced to these databases. ProteinMarker and RNAMarker attributes were intended to be semi-automatically populated from published immunohistochemistry and RNA in situ hybridization data, using the Reach tool natural language processing system (Valenzuela-Escárcega et al. 2018) and manual curation. Currently, only manual annotation of protein and RNA markers is functional. A good reference resource is the CellMarker database (Zhang et al. 2019; http://biocc.hrbmu.edu.cn/CellMarker/).

Annotation of Cell Lineage Paths

The first step in the creation of a new cell lineage path is to check out a parent cell lineage path, either Developmental Cell Lineages (DB_ID = 9734767) or any of its children into your local Curator Tool project. Open the parent cell lineage path record, select the hasEvent attribute field, right click and choose Add.

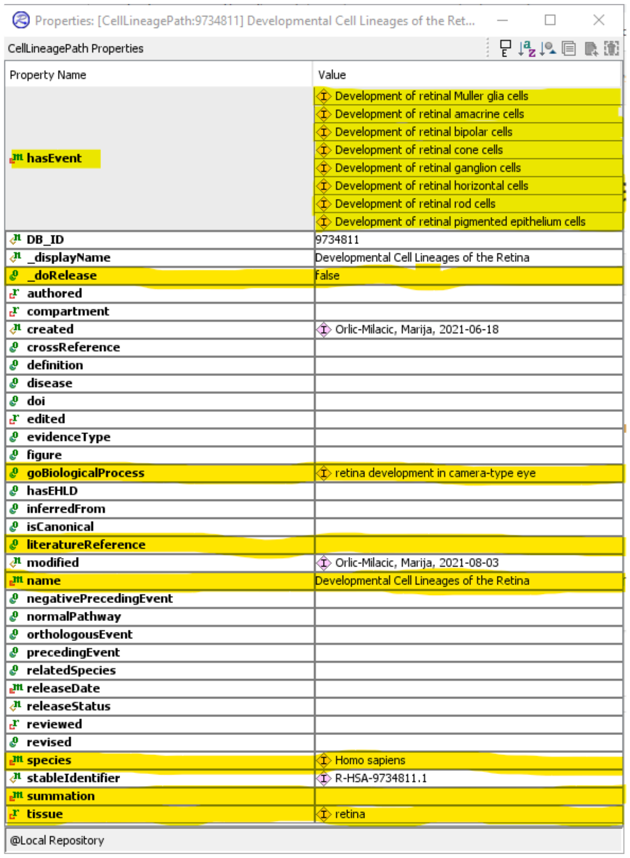


In the window that pops up, select CellLineagePath and click on New Instance.



In the new window that appears, fill in the attributes of the cell lineage path, such as **name**, **species**, **summation**, **literatureReference**, **_doRelease**,

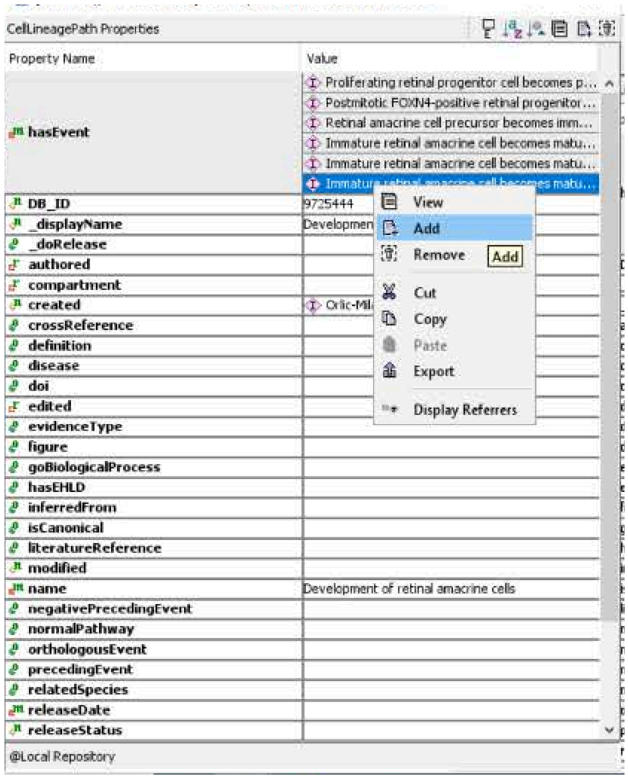
goBiologicalProcess, **tissue**, and **hasEvent**. The hasEvent attribute(s) can be cellLineagePath(s) and cellDevelopmentStep(s):



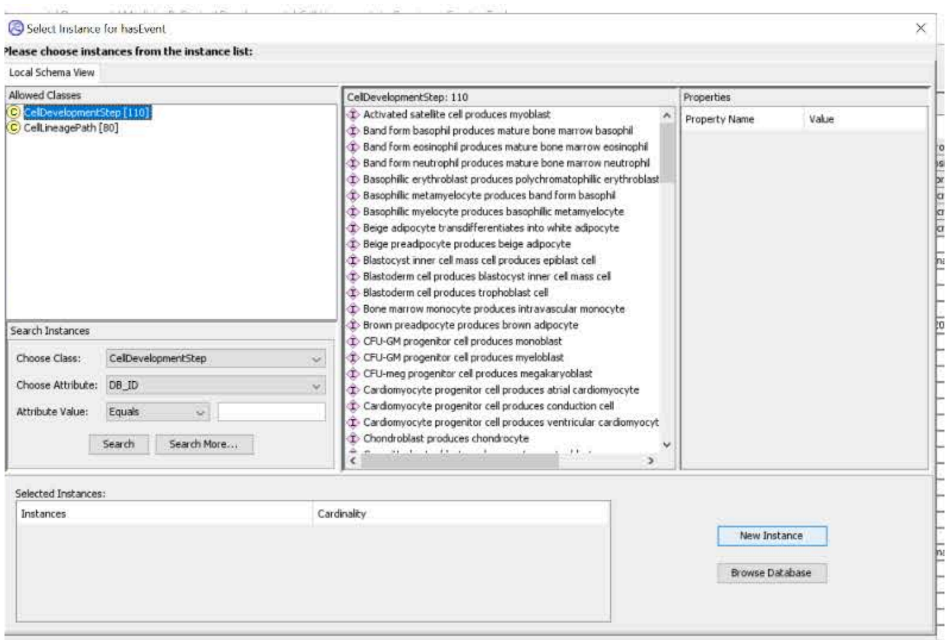
Annotation of Cell Development Steps

The first step in the annotation of a new cell development step is to identify, from published literature, the input cell type, the output cell type, regulators of the development step (such as growth factors), and required components of the input cell (molecules that regulators act on, such as growth factor receptors).

Open the parent cell lineage path record that a new cell development step belongs to, select the hasEvent attribute field, right click and choose Add.

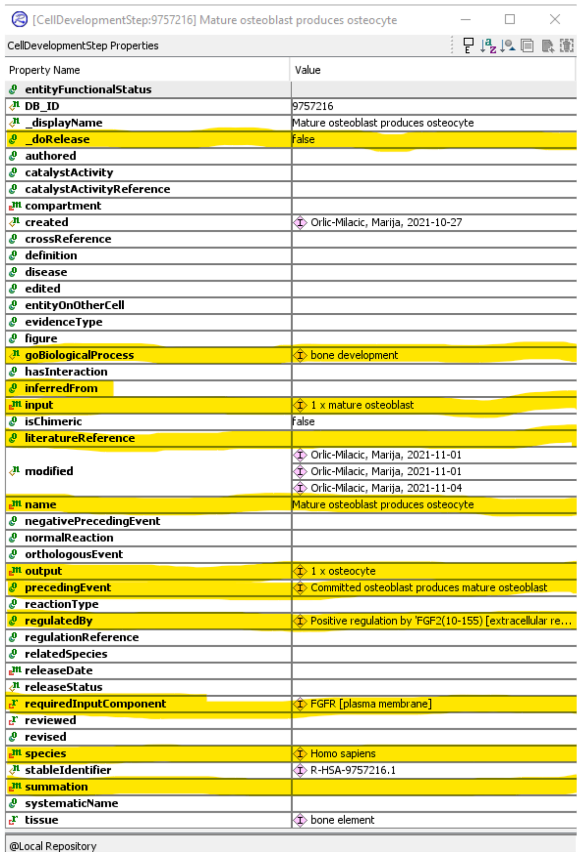


In the window that pops up, select CellDevelopmentStep and click on New Instance.



In the new window that appears, fill in the attributes of the cell development step, such as **name**, **species**, **summation**, **literatureReference**, **_doRelease**,

goBiologicalProcess, **tissue**, **input**, **output**, **precedingEvent**, **regulatedBy**, **regulationReference**, and **requiredInputComponent**.

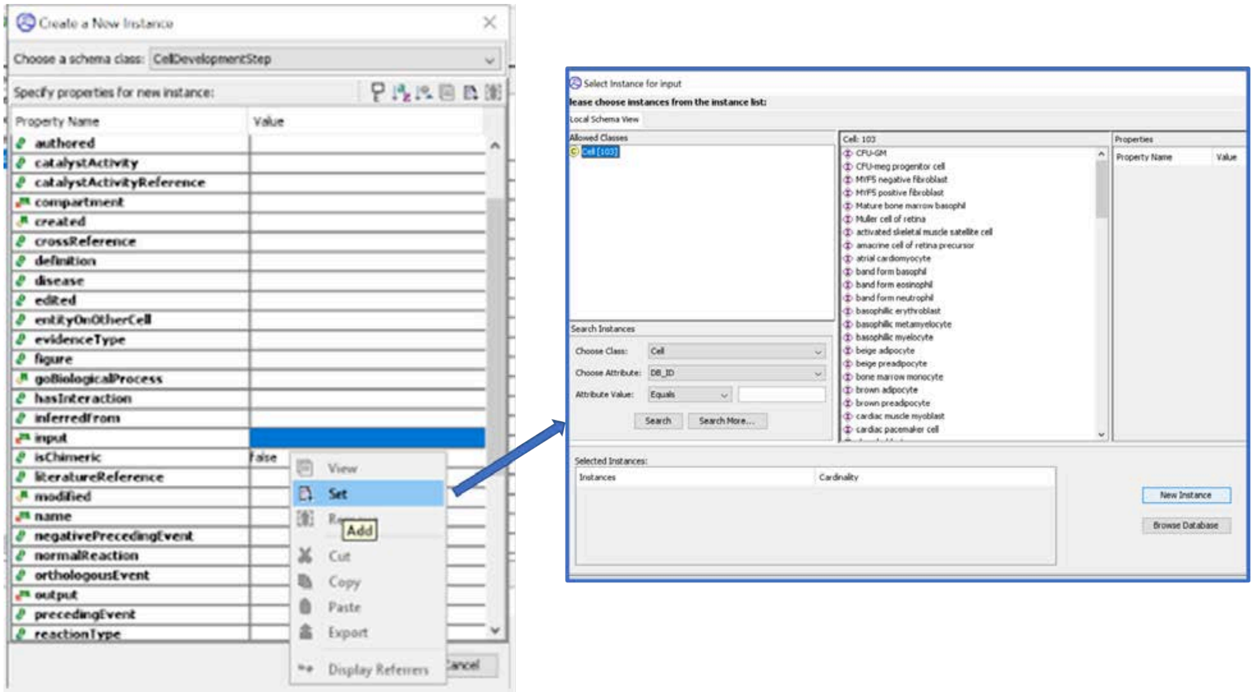


If there is no sufficient human-based experimental evidence available, a cell development step can be inferred from a model organism, similar to a reactionLikeEvent.

i. Inputs

A single input of the Cell class is to be defined for each cell development step.

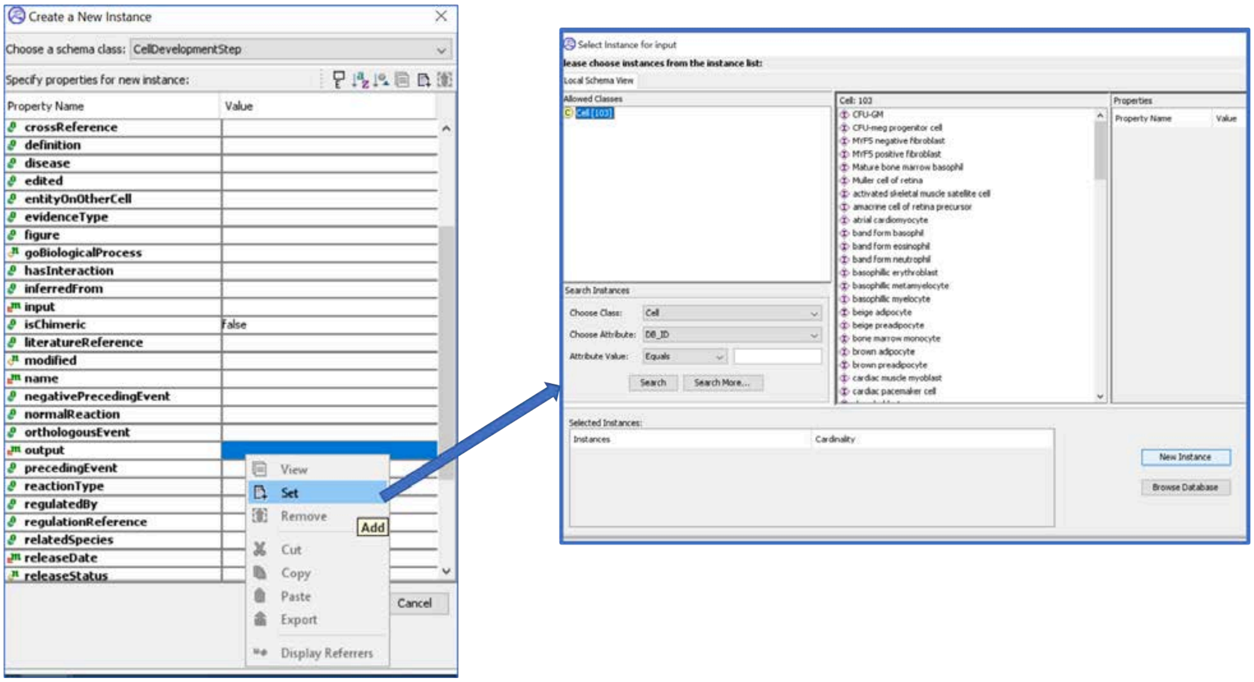
In a cellDevelopmentStep record, right click on the **input** field and select “Set”. From the window that pops up, you can choose from already annotated Cells or you can create a new cell instance.



ii. Outputs

A single output of the Cell class is to be defined for each cell development step. If an input cell can produce more than one output cell type, each input-output cell relationship is to be defined in a separate cell development step.

In a cellDevelopmentStep record, right click on the **output** field and select “Set”. From the window that pops up, you can choose from already annotated Cells or you can create a new cell instance.



iii. Regulators

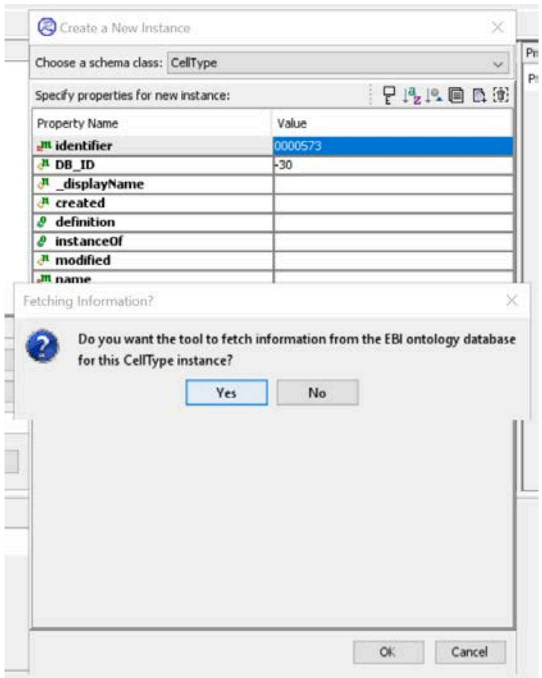
Any number of positive and negative regulators can be annotated for a cell development step using the **Regulation** subclasses - either **NegativeRegulation** or **PositiveRegulation**. Create the Regulation instances just like you would create them for a ReactionLikeEvent or use existing ones.

iv. Required Input Components

A required input component can be any PhysicalEntity present in the input cell that is needed for the action of the regulator or that was experimentally shown to be essential for the development step (e.g. a product of a gene whose conditional knockout abrogates the development step).

Annotation of Cells

A cell as a PhysicalEntity subclass is annotated by using a curator-selected Cell Ontology (https://www.ebi.ac.uk/ols4/ontologies/cl; CL) entry (identifiers starting with “CL_” prefix) as a reference **cellType**. When creating a new CellType instance, a curator needs to enter only the numerical part of the CL identifier, and the CellType record will be automatically populated:



One cellType can serve as a reference for multiple Cells, just like a referenceGeneProduct can serve as referenceEntity for multiple protein EWASs. The reason for this can be:

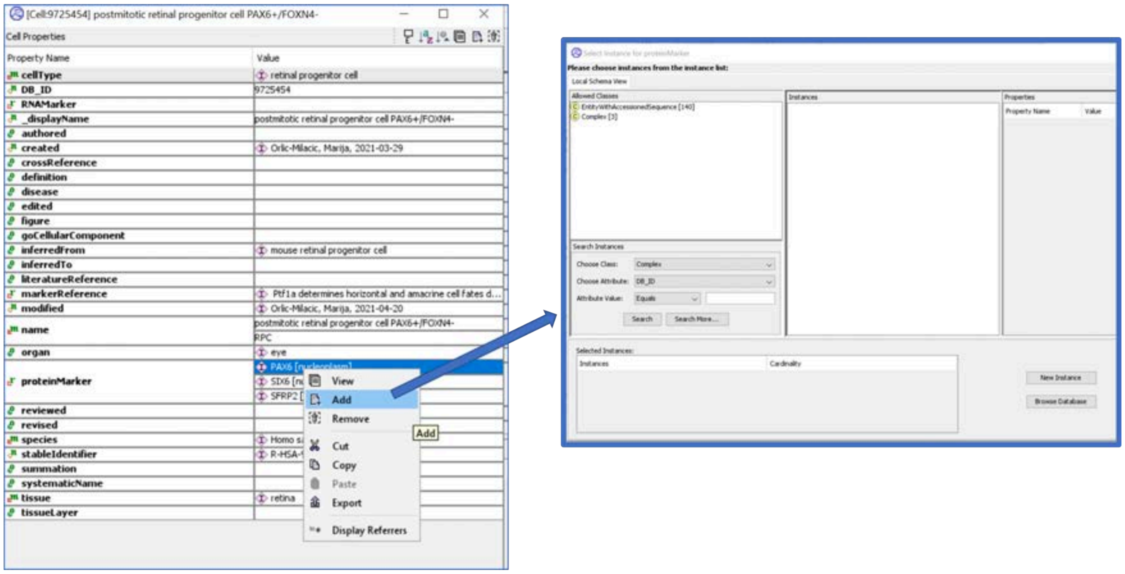
a) Cell Ontology does not have a more specific cellType term, so the curator has to use the closest match. For example, retinal horizontal cell (CL_0000745) is a reference cellType for three Cell instances: horizontal cell of retina precursor (DB_ID = 9725431), immature horizontal cell of retina (DB_ID = 9725440), and mature horizontal cell of retina (DB_ID = 9725415)
b) Two cells are closely related, but differ in their state, reflected in differential expression of key protein or RNA marker(s). The difference, although biologically relevant, is too miniscule to be captured by the CL. For example, retinal progenitor cell (CL_0002672) is a reference cellType for two Cell instances: postmitotic retinal progenitor cell PAX6+/FOXN4+ (DB_ID = 9725412) and postmitotic retinal progenitor cell PAX6+/FOXN4- (DB_ID = 9725454).

v. Protein Markers

Protein markers are proteins that are specific to the cell type and/or the cell state of the Cell or are differentially upregulated in the Cell. In immunohistochemical studies, protein markers can be used to distinguish the Cell from the neighboring cells. In fluorescence-associated cell sorting (FACS), protein markers localizing to the cell surface can be used to isolate the cells of a desired type.

A useful search term for finding literature references with information on cell type-specific protein markers is [Cell / cellType _displayName] + marker; additional terms, such as “staining” or “sorting”, can be added to narrow down the search. The CL database rarely lists protein markers in a cell type description. The CellMarker database contains 13605 human cell markers (protein and RNA markers combined) across 467 human cell types, but markers are almost exclusively derived from high throughput studies (mostly scRNAseq) and are not reconciled between different studies for the same cell type (markers derived from each study are listed separately). CellMarker entries are, therefore, a good starting point to identify the most prominent markers, which overlap between studies (and, if possible, between human and mouse), and then use these markers + cellType _displayName as search terms. Please note that CellMarker also does not standardize gene names - they derive names directly from the study, and therefore an identical marker can be named several different ways on the cell type webpage (that is, the overlap between the studies may not be obvious - UniProt has to be consulted to retrieve the synonyms). Finally, the content of the CellMarker database has not been updated since it first went live in November 2018. It is uncertain if the database is actively curated or just maintained on the programmatic side (there have been some updates in terms of cross-linking to the CL database, which was not mentioned in their manuscript from January 2019).

To annotate a protein marker, open the Cell record. Right click on the proteinMarker field and select Add. You can currently choose a protein marker from the EWAS class or from the Complex class.

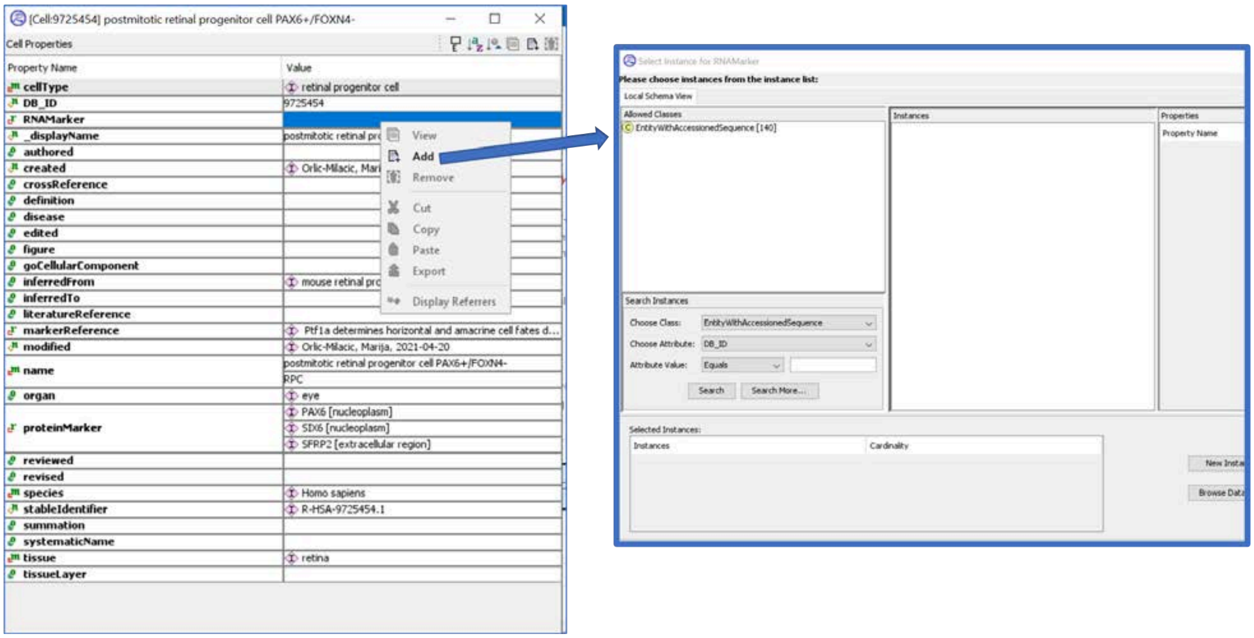


vi. RNA Markers

RNA markers are transcripts that are specific to the cell type and/or the cell state of the Cell or are differentially upregulated in the Cell. In RNA in situ hybridization studies, RNA markers can be used to distinguish the Cell from the neighboring cells. Increasingly, RNA markers are used to identify cells in scRNAseq experiments and to identify novel markers. Curators should evaluate if a novel RNA marker identified via scRNAseq or microarray expression analysis is robust enough to be annotated as an RNAMarker attribute of a Cell.

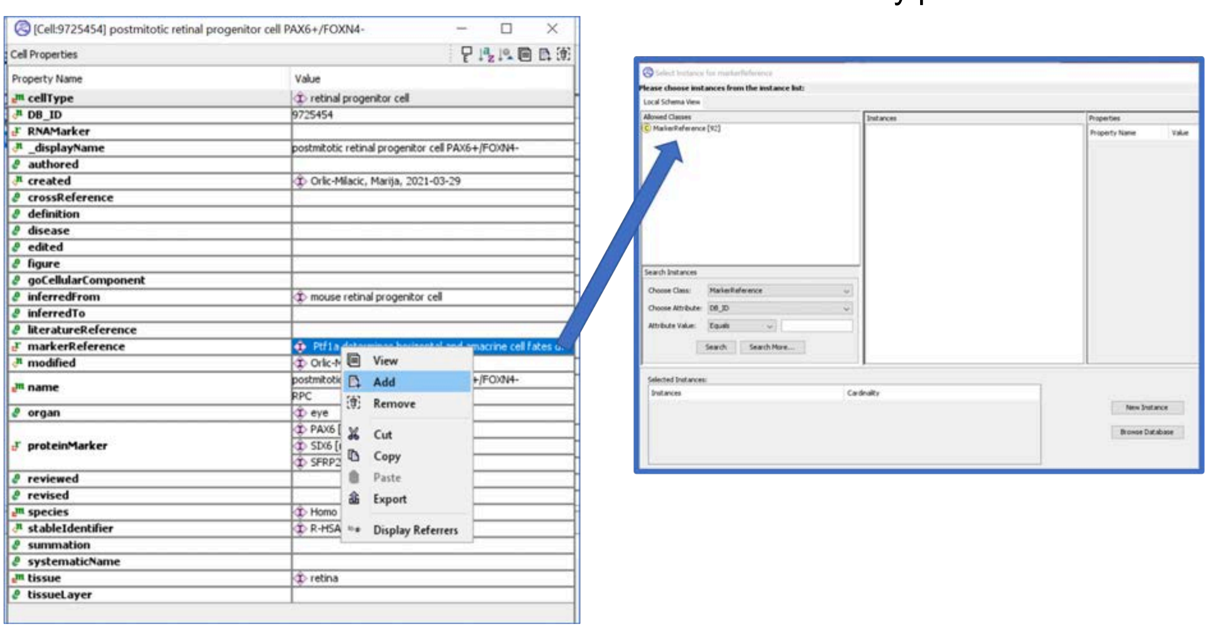
A useful search term for finding literature references with information on cell type-specific RNA markers is [Cell / cellType _displayName] + marker; additional terms, such as “hybridization” can be added to narrow down the search. The CellMarker database can be used as a starting point, as described above.

To annotate an RNA marker, open the Cell record. Right click on the RNAMarker field and select Add. RNA markers are restricted to the EWAS class.



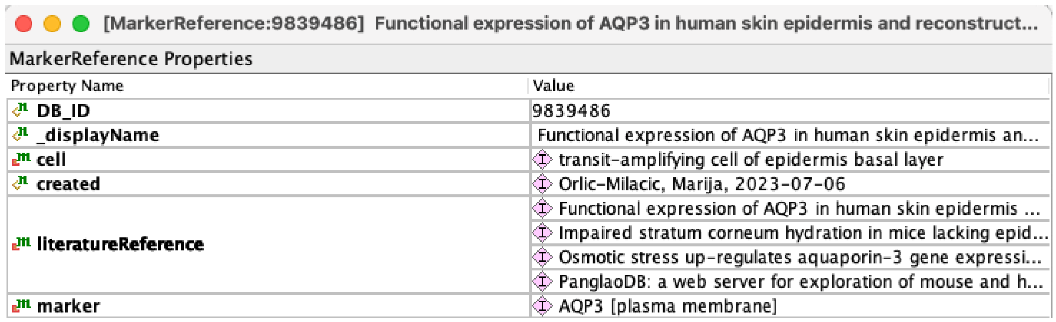
vii. Marker References

To associate literature references with individual protein or RNA markers, use the markerReference attribute of the Cell. Right clicking on the markerReference and selecting Add opens a new window in which MarkerReference is the only permitted class:



Select MarkerReference and click on New Instance. A new window will appear in which you can specify a **marker** (an RNA or a protein

EWAS), the **cell** it is a marker for and the **literatureReference(s)** supporting the assertion.. Note that a separate MarkerReference record is required for each marker annotated. Currently, marker in the MarkerReference record cannot be a protein complex, which is inconsistent with the option of specifying a complex as a proteinMarker in the Cell instance record.

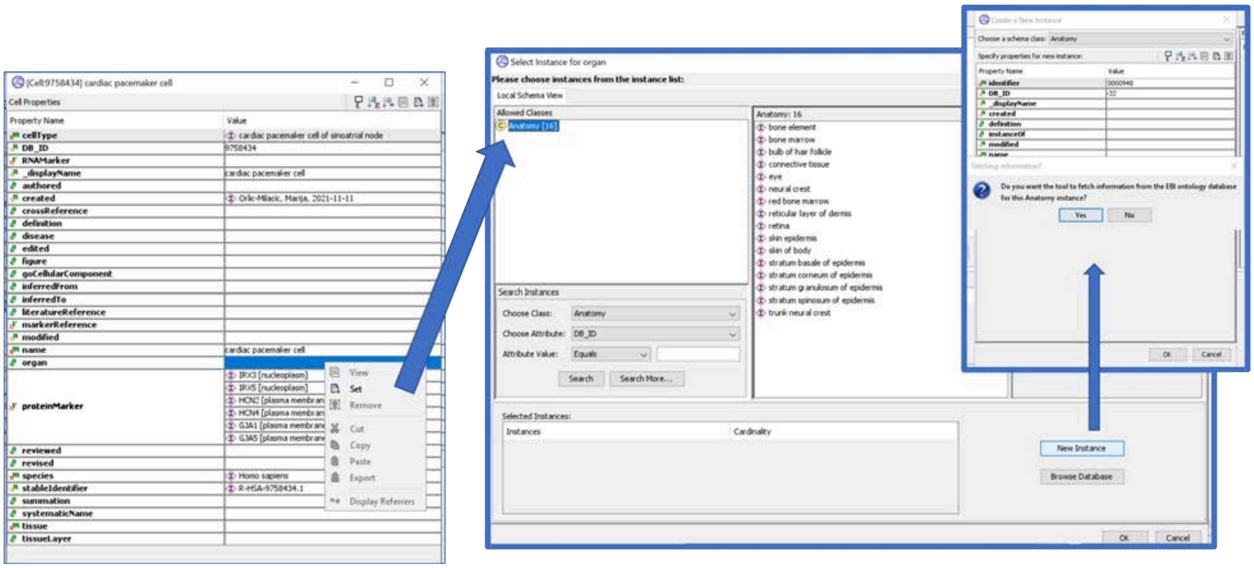


MarkerReference _displayName needs to be improved in the future editions of the Curator Tool. At the moment, the _displayName corresponds to the first literatureReference listed in the record, which makes it impossible to find relevant MarkerReference instances for a given marker by scanning the list of MarkerReferences. The optimal _displayName would be: marker._displayName: literatureReference.pubMedIdentifiers (separated by "," or "|" in case of multiple literatureReferences).

viii. Anatomical Localization

The [UBERON Ontology](https://www.ebi.ac.uk/ols4/ontologies/uberon) (https://www.ebi.ac.uk/ols4/ontologies/uberon) is used as a reference database for anatomical localization. Curators can browse UBERON or use the search engine to find anatomical terms they are interested in. UBERON identifiers have a prefix in the form of either "UBERON_" or "UBERON:".

To annotate any anatomical attribute (tissue, tissueLayer or organ), a curator has to right click on the attribute in the instance record (e.g. Cell) and select Set (each anatomical attribute is single-valued). In the new window that appears, a curator can create a new Anatomy class instance if a relevant instance is not available. Paste the numerical part of the UBERON identifier to the identifier field of the Anatomy record. The Curator Tool will offer to automatically populate the record.



1. *Tissue*
   Tissue is the mandatory attribute of Cell instances, and a required attribute of CellDevelopmentStep and CellLineagePath instances. When looking for a reference Anatomy term in UBERON, curators should make sure that the Anatomy term is defined as a tissue by UBERON, and not an organ or a tissue layer. For a reference, curators can also consult histology textbooks.
2. *Tissue Layer*
   TissueLayer is an optional attribute of Cell instances. For some highly organized tissues, such as skin epidermis or retina, it makes sense to fill in this attribute. For some other tissues, this attribute may be irrelevant. When looking for a reference Anatomy term in UBERON, curators should make sure that the Anatomy term is defined as a tissue layer by UBERON, and not a tissue or an organ For a reference, curators can also consult histology textbooks, as mentioned above.
3. *Organ*

Organ is an optional attribute of Cell instances. When looking for a reference Anatomy term in UBERON, curators should make sure that the Anatomy term is defined as an organ by UBERON, and not a tissue or a tissue layer. For a reference, curators can also consult histology textbooks, as mentioned above.

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Disease Pathway Curation Guide

Introduction

Reactome disease pathways are contained in a separate top-level pathway called “Disease”. Consistent with Reactome’s pathway-centric view, disease pathways are not comprehensive descriptions of given diseases (bladder cancer, diabetes etc), but rather show the perturbations of a given normal pathway that have been characterized, and may include tags for multiple different diseases within one pathway.

Reactome’s disease annotations are grouped by general biological processes (“Diseases of signal transduction by growth factor receptors and second messengers”, “Diseases of metabolism”, “Infectious diseases”, etc.) rather than by specific diseases. The groupings generally mirror the pathway groupings of the normal event hierarchy.

At a high level, there are two main classes of Reactome disease pathway.

The first type of disease pathway has a “normal” counterpart and shows perturbations to a normal biological process that result from disease entities with altered behavior relative to the WT entities. This type includes, for instance, cancer pathways that arise as the result of gain- or loss-of-function mutations in oncogenes or tumor suppressor genes.

Both gain- and loss-of-function reactions with a corresponding normal biological pathway will be displayed (automatically highlighted in red by virtue of the disease tag, see below) in the context of the greyed-out WT pathway. Loss-of-function and some gain-of-function disease events are automatically overlaid on the corresponding normal reactions by virtue of their “normal reaction” attribute (see below). Other gain-of-function disease events must be manually laid out in the disease pathway, also described below.

The second type of disease pathway has no “normal” counterpart. This type includes, for instance, events that occur after the introduction and expression of foreign proteins encoded by genomes of infectious agents like viruses and intracellular parasites. These novel events must be manually laid out in the curator tool. Foreign/disease entities are automatically highlighted in red by virtue of the added disease tag (see below).

Because there is no normal WT pathway for these events, the ELV is unique for the disease pathway and does not have the greyed-out background.

While the bulk of disease annotation is the same as normal curation, there are a couple of disease-specific classes and fields that need to be filled out on entity and/or event records. These include:

- FailedReaction (class)
- Disease (field needed for disease entities and events)
- Functional status (field needed for disease RLEs involving genetic disease variants)
- Normal pathway (field needed for non-infectious disease pathways that have a normal pathway counterpart)
- Normal reaction (field needed for non-infectious disease RLEs that will be displayed as an automatic overlay on the corresponding normal reaction)

Relative to normal curation, disease curation also makes heavy use of sets.

Disease curation also requires extra layers of subpathways to make the display work properly, and a corresponding increase in the number of EHLDs. This is discussed more fully below.

This disease chapter has three main sections. The first, “Researching a disease”, briefly describes some useful resources for accessing disease information for curation. The second “Structuring the disease hierarchy and disease display” describes how to structure a disease pathway and how this shows up on the live site. The third section “annotating disease events and entities” has the nuts and bolts of disease event and entity curation.

If you are starting with disease curation for the first time, it is important to work closely with a curator that has experience in this area as you get started. It is also important to preserve the normal biology that has been curated and not try to adjust it to fit a disease process. This is because the curation process to date has gone through many changes to standardise it to be done in a consistent way. Any changes to the consensus could result in QA errors and/or affect scripts, download files and analyses based on curation. In the vast majority of cases, the normal biology curated is correct for basing its disease counterpart on it. However, the data model does not support annotation of mutations at the nucleic acid level so, for example, it is not possible to model a failedReaction on a normal gene expression event. Instead, the failedReaction should be modelled on the most proximal normal RLE that uses the gene product (protein) from the output of the gene expression event. Consult with a curator experienced in disease curation before undertaking such curation as it may avoid errors in this complicated process.

A detailed introductory video describing the annotation of disease variants can be found here (https://www.youtube.com/watch?v=0irXpoqiDp4).

Researching a disease

- Focus on biology you know well
- Make sure the biology is well characterized at the molecular level with good human biochemical evidence.
- It is generally useful to start off with a few good, high-level reviews to get an overview.
- For non-infectious disease curation, make sure that the normal biological process is adequately covered in its own non-disease pathway before starting the disease annotations. This is because disease pathways and events are overlaid on the normal events for display in the curator tool and the website. It is possible to add a few normal events to the normal pathway if you discover they are needed to support the disease curation, but simultaneously curating both the normal and the disease events is a big job and should be avoided.
- Infectious diseases are novel events that have their own diagrams, independent of underlying normal biology and curated pathways. If you plan to annotate host-pathogen interactions, it can be useful to check that the impacted normal human pathways show the biology you need to reference.

Resources

OMIM

OMIM is the classic resource. The entry for SERPINC1 (antithrombin III) is a good example. The text rambles (it’s a cumulative historical narrative of relevant published results) but as an annotated bibliography it is generally quite complete, with very good coverage of papers describing molecular and cellular biology of the gene and molecular bases of associated disease. The entry for each gene includes a list of mutations with phenotypic effects (under the “table of contents” link on the right of the page, choose “allelic variants”) and links to locus-specific databases which sometimes have much more extensive catalogues. (For a while, the catalogue that had the most extensive coverage for the most human genes was HGMD, accessible from OMIM “table of contents” -> “external links” -> “variation”. It may still, but it’s hard to tell because a license is needed to access it now.) It looks like searches in OMIM for a UniProt ID fail (e.g., searching for P01008 in UniProt does not return SERPINC1) but the UniProt record for a protein includes links to its OMIM record.

COSMIC

For cancer mutations, COSMIC is useful, since it provides information on mutations detected in individual tumors (far from ideal, not all types of mutations are equally covered). In addition, it is good to start with one or two good reviews just to get a bird's eye perspective of the landscape, and then to focus on 1-3 research publications where lots of patients were analyzed, which gives you an idea on the frequency of different mutations. This helps you create your mutant top list, since we aren't capturing all of the mutants, of course. After that, you follow your favourites through PubMed. If you're lucky, you may find a group that maintains a database of mutations for a specific gene.

Gene Reviews

Gene Reviews provides concise “reviews” of the genetics and clinical biology of many human diseases, as well as links to clinical testing laboratories, support groups, and other clinical resources that are not of much use to us. Its coverage is less comprehensive than OMIM’s and much more narrowly focused on material useful to the working doctor who has just encountered an affected patient and is wondering how to proceed.

IUPHAR/Guide to Pharmacology

Citing other resources

In certain cases, there are external databases that include information that is relevant to an annotated pathway.
One example is rare disease database such as RettBase, which describes Rett syndrome-related mutations in MECP2, FOXG1 and CDKL5: http://mecp2.chw.edu.au/.
This database provides information about variants described in the disease pathway: Loss of function of MECP2 in Rett syndrome.
We are not able to cross-reference RettBase for individual MECP2, variants in the way that we cross-reference COSMIC for cancer variants because RettBase is not organized in a way that allows us to do this. However, we can mention RettBase in the pathway summation and cite their most recent publication (https://www.ncbi.nlm.nih.gov/pubmed/28544139) as a literatureReference for the Loss of function of MECP2 in Rett syndrome pathway. In addition, we can also create a second Publication instance of class "URL" and use this to cite database itself.

**NOTE that you should NOT not cite disease databases at the reaction level unless these disease databases capture biochemical/functional information.

Building a rough outline

After some preliminary research and reading, you will have a sense of what your disease pathway will look like:

- what normal pathway it is based on (in the case of non-infectious disease annotations)
- what kinds of mutations have been characterized (gain-of-function mutations, loss-of-function mutations)

- what normal functions are perturbed as a result of these genetic changes (ie, kinase-domain mutants in a receptor tyrosine kinase (RTK) don't trans-autophosphorylate and therefore downstream signaling is abrogated; constitutively active kinases phosphorylate their downstream substrates in an unregulated manner; binding domain mutants of a protein don't interact with their protein partners etc)
- drug:target interactions

Note that Reactome does not attempt to catalog every variant that has been identified for a given gene/protein. The focus is on capturing a few prevalent, well studied examples of each of the ways that the biological function of a protein can be altered in disease. Your disease pathway will likely involve sets of variants, each of which has been shown to act in a similar manner. Note that all variants with the same molecular consequence should be grouped together in a set, even if they occur in different diseases. See below for a more thorough description.

After this groundwork, you should be able to mock-up your disease pathway, grouping major classes of mutants under sub-headings/subpathways in your primary disease pathway.

As a hypothetical example, if you are annotating disease events in a RTK pathway, you might have in your outline:

SIGNALING BY RTK IN CANCER (parent pathway)

- constitutive signaling by overexpressed RTK (gain-of-function subpathway)
- constitutive signaling by RTK kinase domain mutants (gain-of-function subpathway with set of RTK mutants that have ligand-independent dimerization and activation)
- defective signaling by RTK kinase domain mutants (loss-of-function subpathway with set of RTK mutants that have lost kinase activity)
- signaling by RTK fusions (gain-of-function subpathway with set of fusion proteins that have ligand-independent dimerization and activation)
- RTK mutants bind tyrosine kinase inhibitors (essentially normal pathway showing binding of sensitive RTK mutants to known inhibitors)
- Drug resistance of RTK mutants (loss-of-function subpathway showing inability of RTK mutants to bind and be inhibited by specified TKIs)

Once you have a rough outline, you are ready to start annotating in the curator tool.

Assigning literature references for disease reactions

As with normal reactions, the references associated with a reaction MUST provide direct experimental evidence for the occurrence of that reaction in the species you are annotating (i.e. human for human Reactome). If there is no direct experimental evidence orderedSet in humans then you need to create an inferred human reaction as described in the section below. See Guidelines on evidence confidence here (Appendix C3).

Structuring the disease pathway

Ultimately, all disease pathways will end up under the “Disease” chapter on the website. You may have a general idea of where in the “Disease” hierarchy your pathway will end up, based on the process that you are annotating (“Diseases of Signal Transduction by growth factor receptors and second messengers” or “Diseases of metabolism”, for instance), or this may become more evident as your curation gets underway. This does not need to be decided before you start your curation.

To begin your disease curation, start by making your parent disease pathway. This parent pathway groups all the disease events related to a particular disease(s) that occur in the context of a given normal pathway (except in cases of infectious diseases, which are by-and-large independent of normal pathways).

For disease pathways without a corresponding “normal” pathway in Reactome

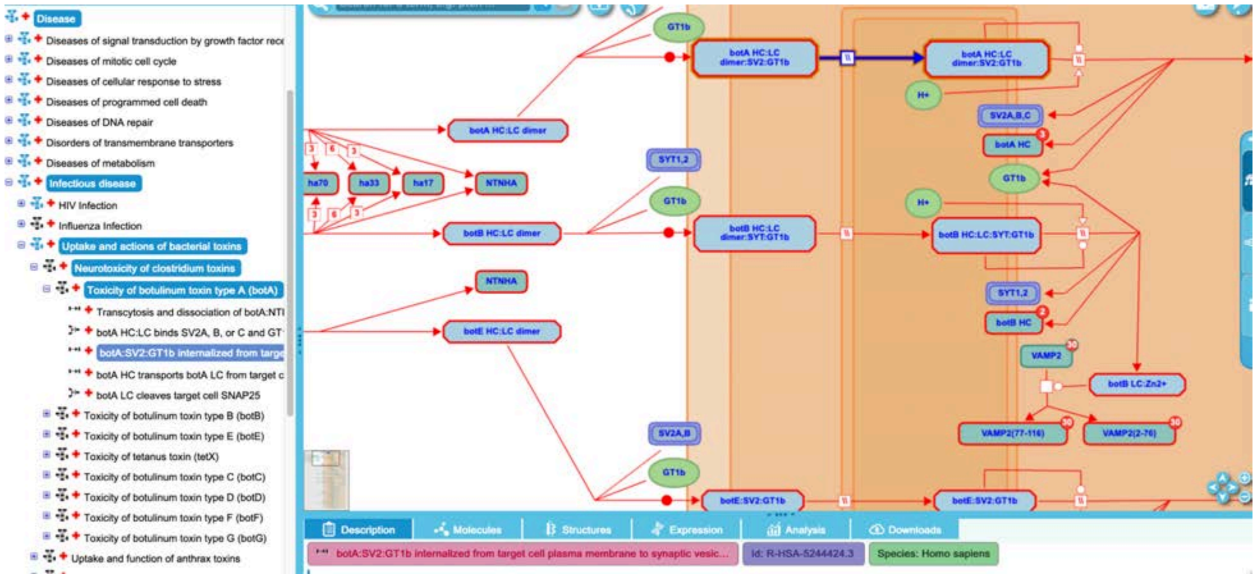
If the disease pathway will not have a corresponding "normal pathway" in Reactome, the pathway should still be placed under the disease chapter. Since these pathways are generally infectious processes, they will likely belong under the Disease child subpathway “Infectious disease”.

These pathways are labeled with an appropriate disease term (see below) and all species that contribute to the reactions and events (ie, generally, the human host and the infecting species, which is entered in the “related species slot, see below).

Pathway structure is much like that of normal curation. RLEs may be grouped into subpathways by the particular biology being described to facilitate comprehension and to manage diagram size, but because infectious disease pathways have their own diagrams and are not overlaid on the WT pathway, there are no particular requirements for how the pathways and subpathways are arranged.

Below is a screenshot of part of the “Toxicity of botulinum toxin type A” pathway, showing disease entities and disease reaction lines highlighted in red by virtue of the applied disease tag (described below). Host (human) proteins have no disease tags, are displayed bordered in black, but complexes of host and viral entities should be disease tagged and will therefore also appear bordered in red.

Because this is a disease-specific ELV, there is no greying out of a “normal pathway” background.



Below is a screenshot of the pathway record, showing the structure of the disease pathway. The parent pathway is labeled with a disease term (see below), and contains 5 disease events. The pathway is labeled with the two species that contribute to the reactions: Homo sapiens in the main species slot, and Clostridium botulinum in the "RelatedSpecies" slot. This class was introduced to prevent inappropriate inferring of pathways to the other species during website release. See below for more detailed description of the annotation of host:pathogen interactions.

Pathway Properties

Property Name	Value
deletedInstanceDB_ID	
hasEvent	Transcytosis and dissociation of botA:NTNHA:HA botA HC:LC binds SV2A, B, or C and GT1b on the target cell surface botA:SV2:GT1b internalized from target cell plasma membrane to synap... botA HC transports botA LC from target cell synaptic vesicle membrane i... botA LC cleaves target cell SNAP25
DB_ID	5250968
_displayName	Toxicity of botulinum toxin type A (botA)
_doRelease	true
authored	Krupa, S, Gopinathrao, G, 2006-06-15 10:36:11
compartment	
created	D'Eustachio, Peter, 2014-01-31
crossReference	
definition	
deletedStableIdentifier	
disease	botulism
name	Toxicity of botulinum toxin type A (botA)
negativePrecedingEvent	
normalPathway	
orthologousEvent	
precedingEvent	
previousReviewStatus	
relatedSpecies	Clostridium botulinum
releaseDate	2014-03-12
releaseStatus	
reviewStatus	five stars
reviewed	Ichtchenko, K, 2007-08-03 18:17:25 Thirunavukkarasu, Nagarajan, Sharma, Shashi, 2014-11-17
revised	D'Eustachio, Peter, 2014-02-10
species	Homo sapiens
stableIdentifier	R-HSA-5250968.2
structureModified	
summation	Botulinum toxin type A (botA, also known as BoNT/A), a disul...

For disease pathways with a normal WT counterpart

These disease pathways need to be linked to the WT ELV to allow proper display of these events. This is accomplished by structuring the disease pathway appropriately and by sharing the WT diagram with the disease pathway. Both of these are explained in more detail below.

To start, make a record in the curator tool for your top level pathway as usual.

- Give your pathway a name. The top level pathway name is often very general.
- Add a disease term from DOID in the appropriate pathway record slot. See below for guidelines on selecting an appropriate disease term. Adding the disease term triggers the automated coloring of the event or entity in red in the curator tool and in the pathway browser.
- Add the relevant wild-type pathway to the "normal pathway" field on the pathway record. <u>This must be the "ELV-level" pathway</u> (i.e it is listed as a RepresentedPathway for a PathwayDiagram) <u>that contains the reaction-like events for which you intend to annotate corresponding disease events</u>. This is needed to allow proper overlay of your disease events on the greyed-out background of the WT pathway.

- (two other attributes, ‘entityFunctionalStatus’ and ‘normalReaction’ are also required for proper display of disease events, but these are applied at the RLE level not at the pathway level and will be described below.)
- To verify that you have properly set the NormalPathway, normalReaction, and representedPathway for your disease events, navigate to your pathway in the disease hierarchy, make sure the disease RLE overlay is correct. *Also verify that when you select the normalReaction in the details section, the pathway browser brings you to the correct normal pathway hierarchy and diagram.

The sub-structure of the disease pathway (children, grandchildren) can be quite varied, depending on what the biology needs to represent.

SIMPLEST PATHWAY STRUCTURE:

In the simplest case, there is one way a pathway can be altered during the course of a disease. This may be because a single variant in a single gene has been identified that affects the pathway in one way, or because there is a set of variants in a gene that have been identified that all have the same functional effect. In this case, all events can be shown directly under the parent disease pathway (grouped into subpathways if this grouping provides clarity to the user). All events will show up at the same time in a single view of the disease pathway.

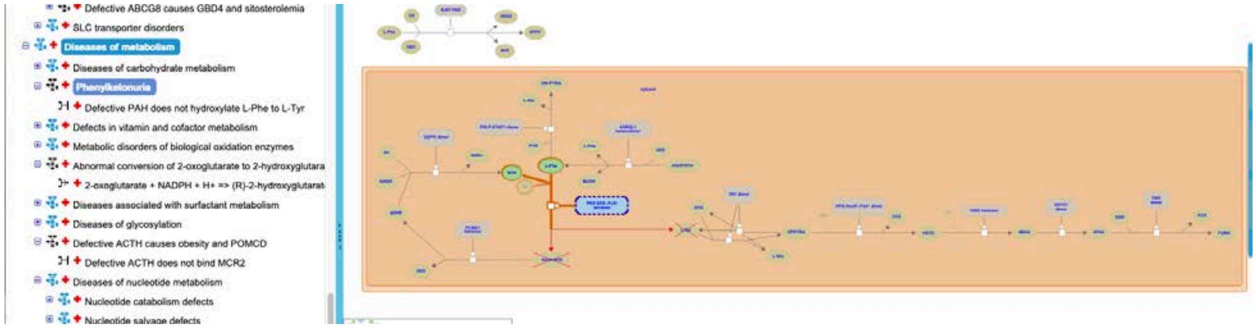
As an example of this simple disease structure, look at the Phenylketonuria (R-HSA-2160456) pathway (under Diseases of metabolism).

[Pathway:2160456] Phenylketonuria

Pathway Properties

Property Name	Value
deletedInstanceDB_ID	
hasEvent	Defective PAH does not hydroxylate L-Phe to L-Tyr
DB_ID	2160456
_displayName	Phenylketonuria
_doRelease	true
authored	D'Eustachio, P, 2012-03-04
compartment	
created	D'Eustachio, P, 2012-03-03
crossReference	
definition	
deletedStableIdentifier	
disease	phenylketonuria
name	Phenylketonuria
negativePrecedingEvent	
normalPathway	Metabolism of amino acids and derivatives
orthologousEvent	
precedingEvent	
previousReviewStatus	
relatedSpecies	
releaseDate	2012-06-12

Here, the parent disease pathway Phenylketonuria contains a single failed reaction showing the loss of function of the PAH variant PAH S40L. The pathway is tagged with the disease term “phenylketonuria” and the corresponding normal pathway (Metabolism of amino acids and derivatives) is added in the normalPathway attribute of the disease pathway record. The failed reaction (described more fully below) is shown in red automatically overlaid on the normal greyed-out pathway background (normal entities in the remainder of the WT pathway are non-interactive).



A simple disease pathway could also show the consequences of a gain-of-function mutation including the initiating event (for instance, ligand-independent dimerization and trans-autophosphorylation) and the downstream signaling events (binding to signaling partners, downstream phosphorylation events etc). All these RLE would be captured under the parent disease pathway and could be shown in the context of a single diagram, appropriately tagged with a disease term and the normal pathway.

More complex pathway structures

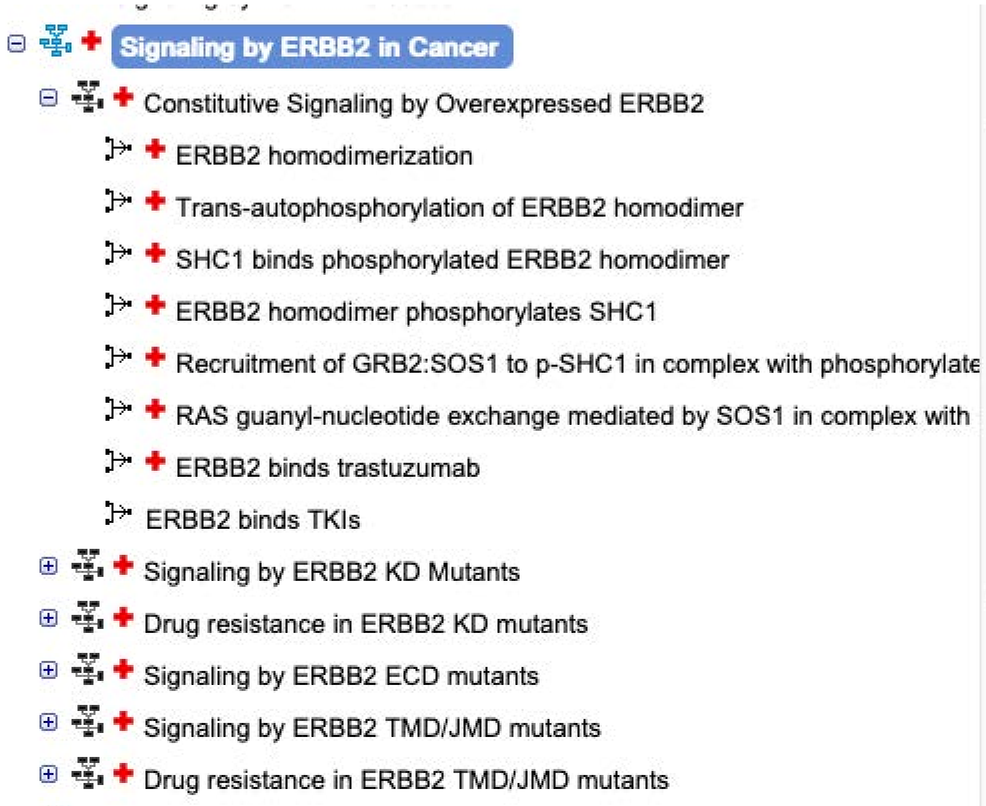
More often than not, however, a pathway can be altered in a variety of different ways: by mutation of a number of different proteins in the pathway, or by different types of mutations in the same protein that have different functional consequences, for instance. In these more complicated disease pathways, subpathways and sets of mutants become critical.

Reactome annotates the different functional consequences of different variants/sets of variants separately in their own view of the normal greyed-out WT pathway. For instance, a loss-of-function mutation in protein X would not be displayed in the same diagram view as a gain-of-function mutation in protein X that causes pathway activation- even though the affected pathway is (often, mostly) the same normal pathway.

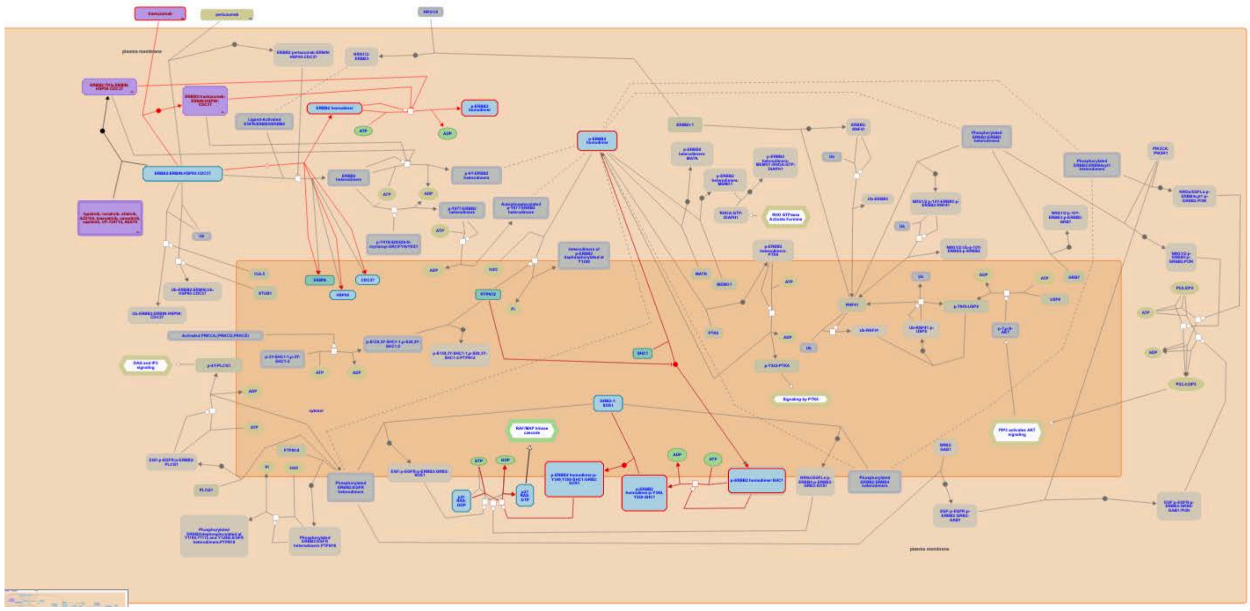
Variants (or sets of variants) that have the same functional consequence are grouped together and put in a sub-pathway that is distinct from sub-pathways showing the different functional consequences of other variants.

As an example, look at Signaling by ERBB2 in Cancer (R-HSA-1227990).

Here, the parent pathway is an EHLD. The pathway contains 6 child pathways, each describing different ways in which the same normal biology can be altered during cancer (Constitutive Signaling by Overexpressed ERBB2; Signaling by ERBB2 kinase domain (KD) mutants; Drug resistance in ERBB2 (KD) mutants (a special case, see below for structuring drug resistance annotations); Signaling by ERBB2 extracellular domain (ECD) mutants; Signaling by ERBB2 transmembrane domain/juxtamembrane domain (TMD/JMD) mutants; Drug resistance in ERBB2 TMD/JMD mutants).

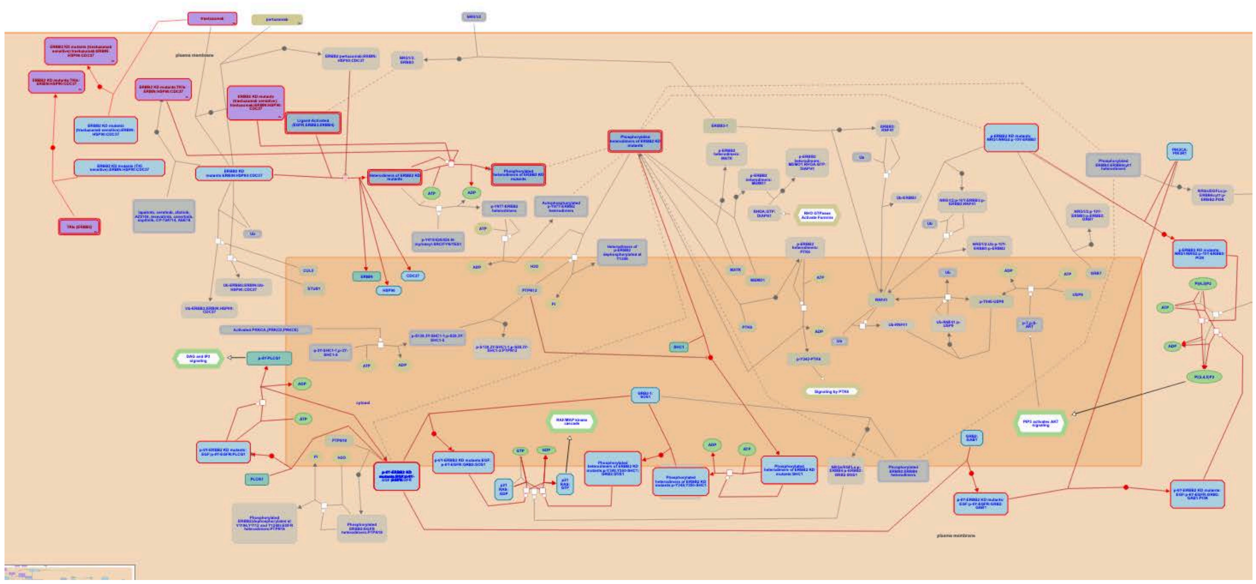


The disease reaction-like events of each of these child pathways (leaving aside the drug resistance pathways for the moment) are displayed in separate pathway views but overlaid on the same WT pathway diagram, as shown below for Constitutive Signaling by Overexpressed ERBB2:

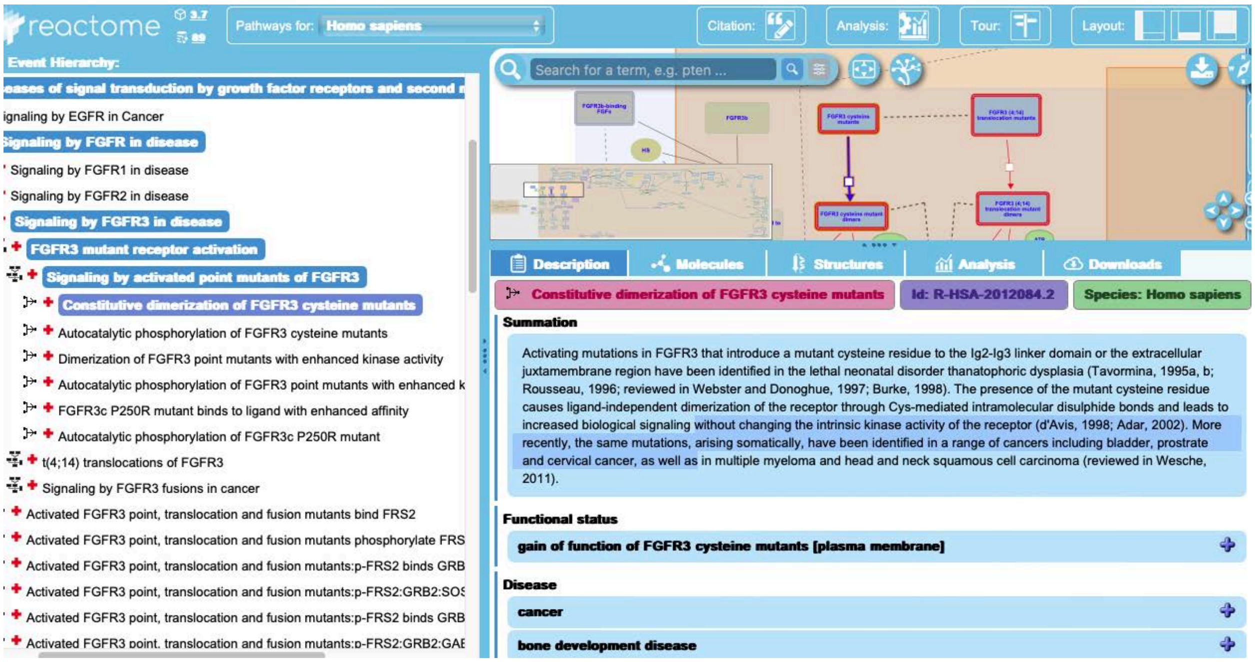


≠≠≠=

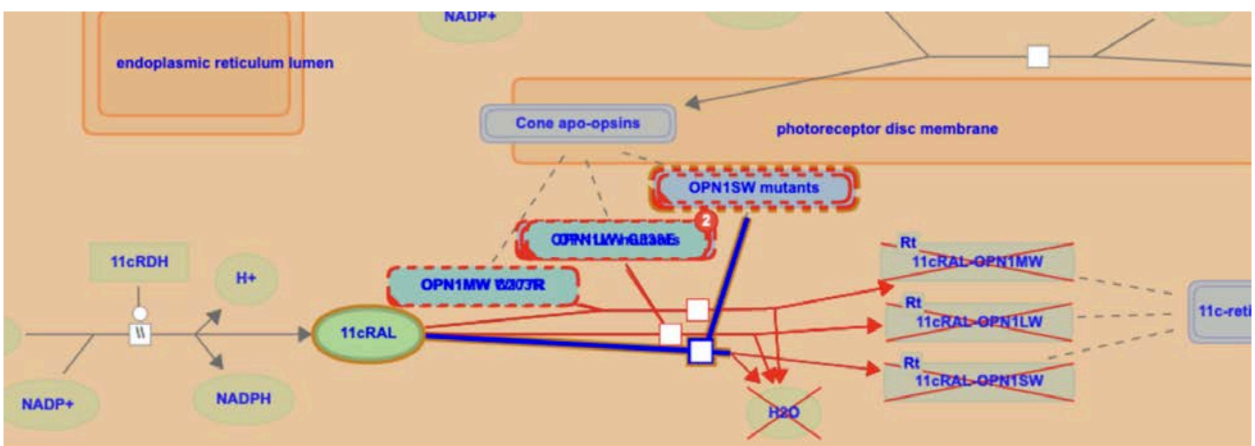
And for Signaling by ERBB2 KD mutants:



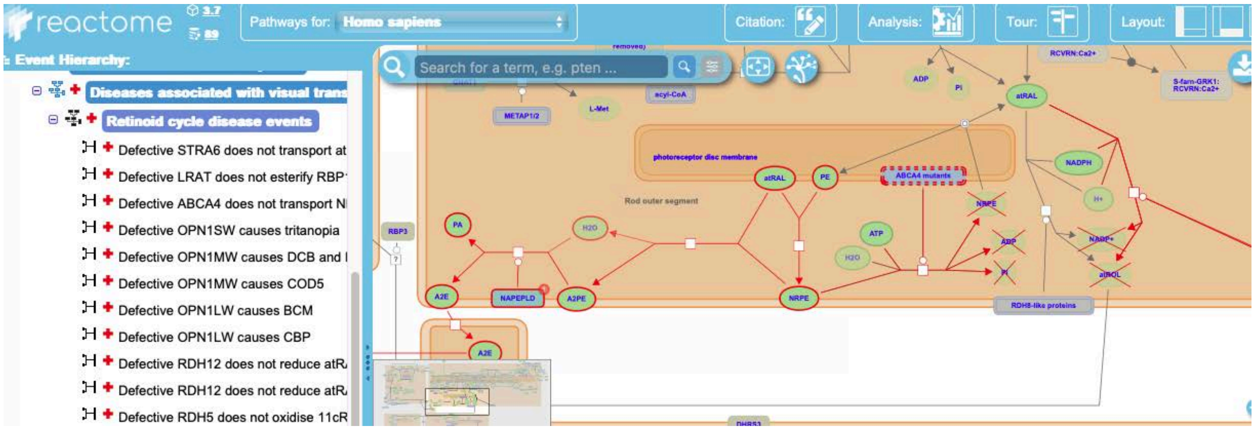
Separating functionally related variants into their own sets with their own diagram representations is clearer from a biological perspective, and should be done even if the variants in a given set arise in the context of multiple different diseases. For instance, mutations arise in FGFR3 in both bone development disorders and in cancer that allow the receptor to be constitutively active, dimerizing in the absence of ligands. These mutants, whether associated with bone disorders or cancer, are grouped together in a single set that is labeled with an appropriate disease term that encapsulates both diseases, or if that is so general a term as to be largely meaningless, with each of the relevant disease terms, as shown below:



Individual sets for groups of functionally related disease variants is also necessary to allow proper display of the disease events on the website. This is particularly true when the sets of variants affect the same WT RLE -in this case, separate sub-pathways are essential to allow proper display. If this is not done, the tool tries (successfully) to overlay multiple disease events onto a single reaction, making all of them illegible:



By consensus (as of August 2024), Reactome annotates loss-of-function failed reactions individually (ie each parent pathway containing a failed reaction should contain one and only one failed reaction). This is true even in cases where the failed reactions affect different reactions in the WT diagram and so could (unlike the case above) all be represented at the same time without ‘overlay clashing’. An old example with multiple failed reactions in a single parent pathway is shown in the pathway diagram below:



This convention was adopted to ensure consistent representation of failed reactions when taking into account the possibility of ‘overlay clashing’ when disease reactions need to be overlaid on the same WT reaction - and in spite of some user testing results that suggest that certain users would appreciate a way to combine different defective views into a single display to see all the weak points of a pathway.

In contrast to loss-of-function reactions, a gain-of-function pathway can contain multiple reactions, as the effects of a gain-of-function mutation (or set of gain-of-function mutations) may propagate over many steps of the wild-type pathway. For instance, after ligand-independent dimerization and trans-autophosphorylation of an RTK due to mutation, the activated mutant receptor may recruit intracellular signaling proteins to propagate the signal. By convention, the disease pathway is continued until the mutant entity is no longer directly involved in the events that are being depicted.

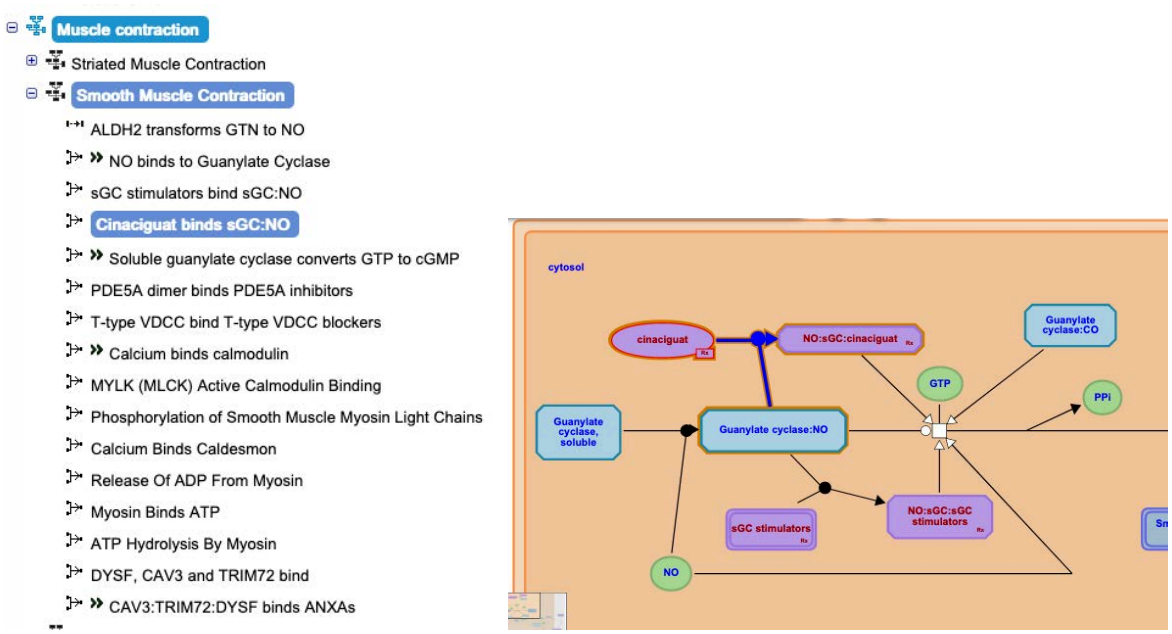
Structure of Pathways when disease reactions affect more than one normal pathway

If a disease variant affects a normal reaction that is a participant in multiple (say 2) normal pathways, the curator should create disease pathway names that reflect the process that is affected. For example, defects in SLC6A19 are known to cause Hartnup disorder. However, the SLC6A19 transporter functions in both the amino acid transport pathway and the neurotransmitter transport pathway. Therefore, two disease pathways need to be created: “Defective SLC6A19 disrupts the transport of amino acids causing Hartnup disorder” and “Defective SLC6A19 disrupts the transport of neurotransmitters causing Hartnup disorder”. In the former disease pathway, “SLC-mediated transport of amino acids” would be specified as the normalPathway and in the latter SLC-mediated transport of neurotransmitters would be specified as the normalPathway.

Structure of pathways involving drug-binding

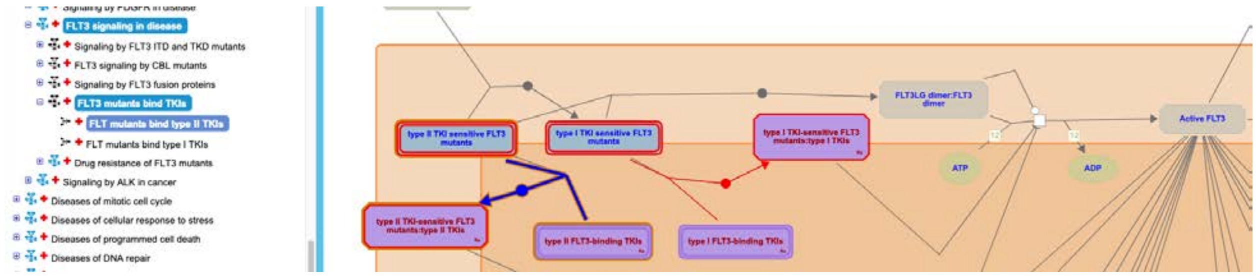
In Reactome, we annotate the action of drugs/therapeutics that affect pathway and protein activity. In general, drugs are shown binding to their target protein and then inhibiting or activating some other function of the target protein. See below for a full description of how to annotate drug:target interactions.

For drugs that affect the WT version of a protein, these binding and regulation events are shown in the context of the normal, WT pathway hierarchy and diagram. The drug physical entity and the binding event are labeled with an appropriate disease term, as described below. Drugs are also cross-referenced out to Guide to Pharmacology (GtoP) where possible, or to PubChem, as described below. Drug entities and related events are automatically colored in purple by virtue of these tags, and entities are decorated with the purple R_x icon at the bottom right, as shown below for the drug cinaciguat in the Smooth Muscle Contraction (R-HSA-9621179) pathway

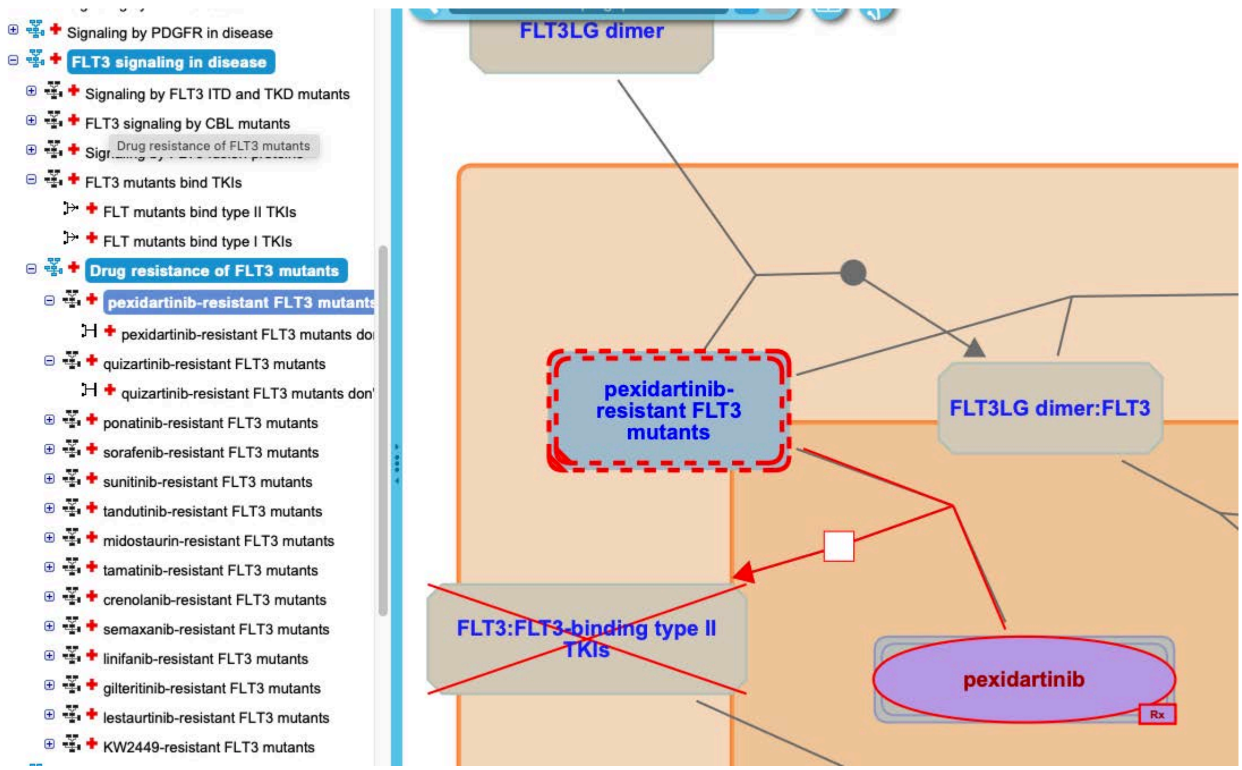


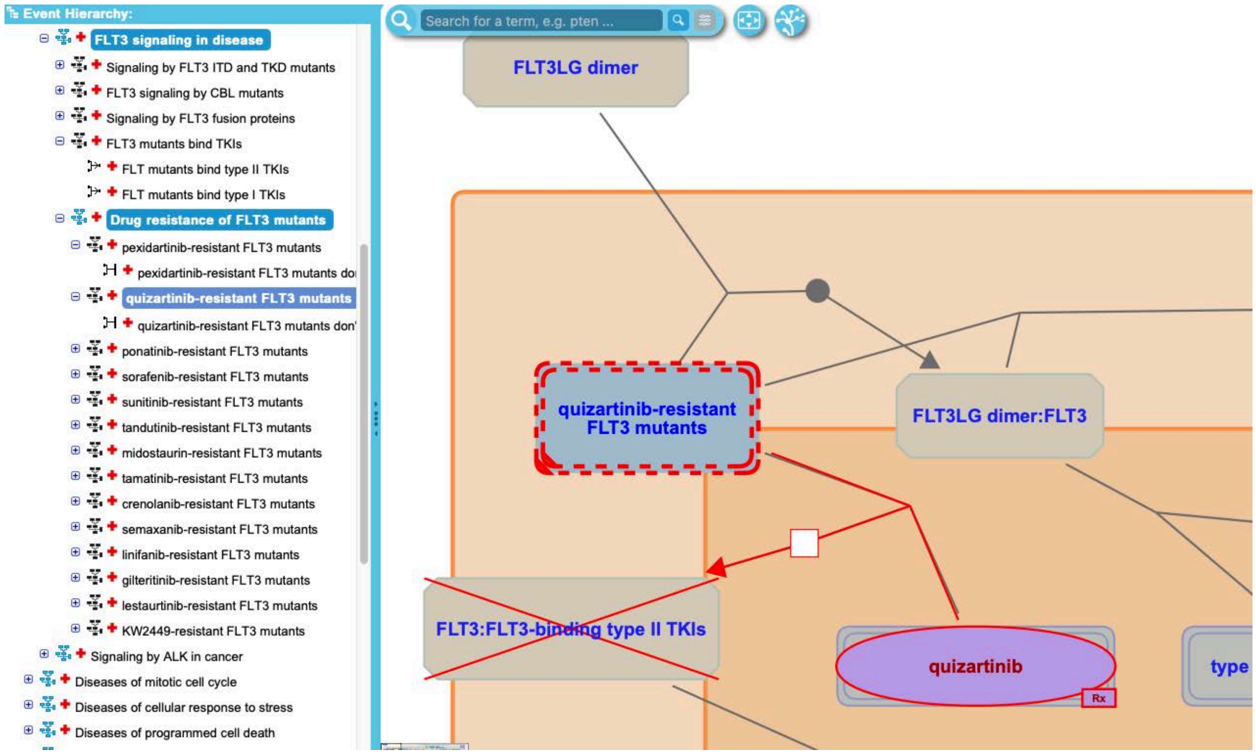
In cases where a drug has been shown to affect (or not affect) variant versions of proteins, Reactome disease pathways are able to capture this as well.

For drug-sensitive variants, the binding and regulation reactions are made in the same manner as for a drug that targets a wild-type protein, but the reactions are contained in the relevant disease pathway, as shown for FLT3 variants (R-HSA-9702510) in the screenshot below:

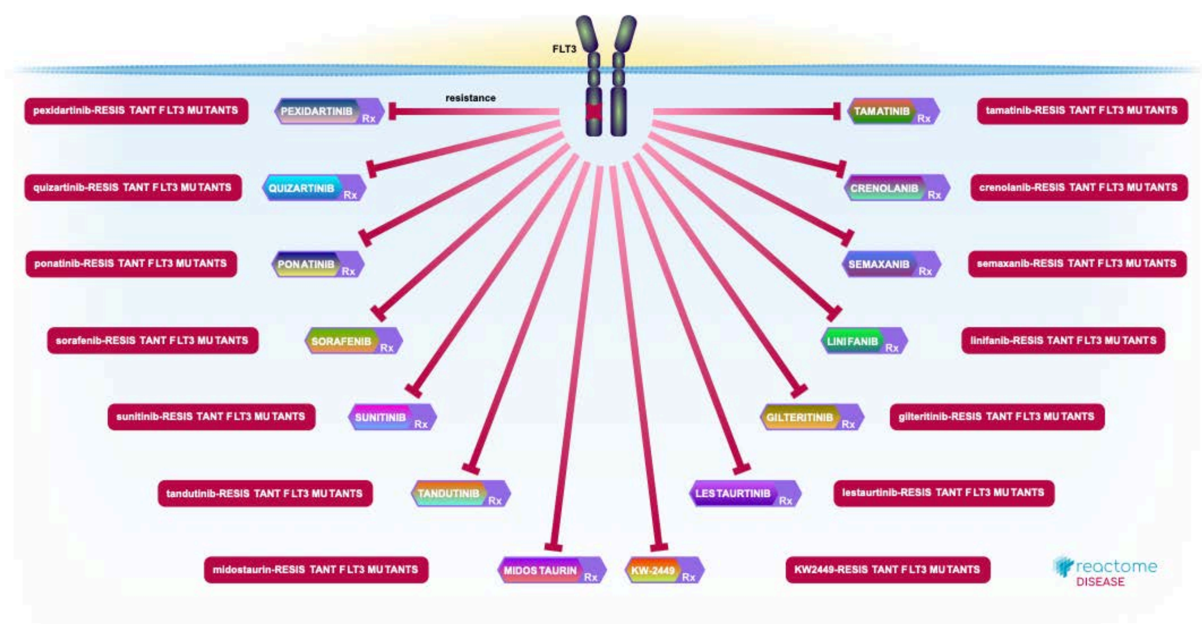


Drug resistant variants are shown as failed reactions, not binding to the relevant drug. Generally, sets of variants resistant to distinct drugs are represented individually in their own diagram views, as shown below for FLT3 variants resistant to pexidartinib and quizartinib, respectively:





In this case, the child pathway “Drug resistance of FLT3 mutants” is an EHLD level pathway.



The alternative to this representation would be to make a ‘superset’ of all the sub-sets of resistant mutants (“resistant to pexidartinib”, “resistant to quizartinib”, “resistant to ponitinib”) and make a single failed reaction where the superset was the input.

Sharing the diagram with disease pathways.

For disease pathways with no normal pathway

For infectious disease pathways, there is no sharing of WT diagrams. Each infectious disease process has its own diagram(s) with no ‘normal’ pathway slot filled in, and no need to share the WT pathway diagram.

For disease pathways with a normal WT counterpart

For disease pathways with a normal WT counterpart, you will need to share the WT pathway diagram with your disease pathway.

In principle, this is very simple. Go to the PathwayDiagram instance associated with your wild-type pathway and add your disease pathway name(s) to the multi-valued “representedPathway” slot.

In practice, which pathway to associate with the PathwayDiagram can be a bit more complicated. One way to think of the rule is: any disease pathways that directly contain disease RLE’s that need to be overlaid on the wild-type ELV need to be associated with the wild-type diagram.

Another way to think of this is: any RLEs that needs to be shown overlaid on the same WT reactions have to be in their own subpathway, and each of these subpathways need to be added as a ‘representedPathway’ in the PathwayDiagram record for the wild-type pathway.

In practice, the disease pathways that are added to the pathwayDiagram record are generally children or grandchildren pathways of your parent pathway.

Consider the hypothetical pathway above:

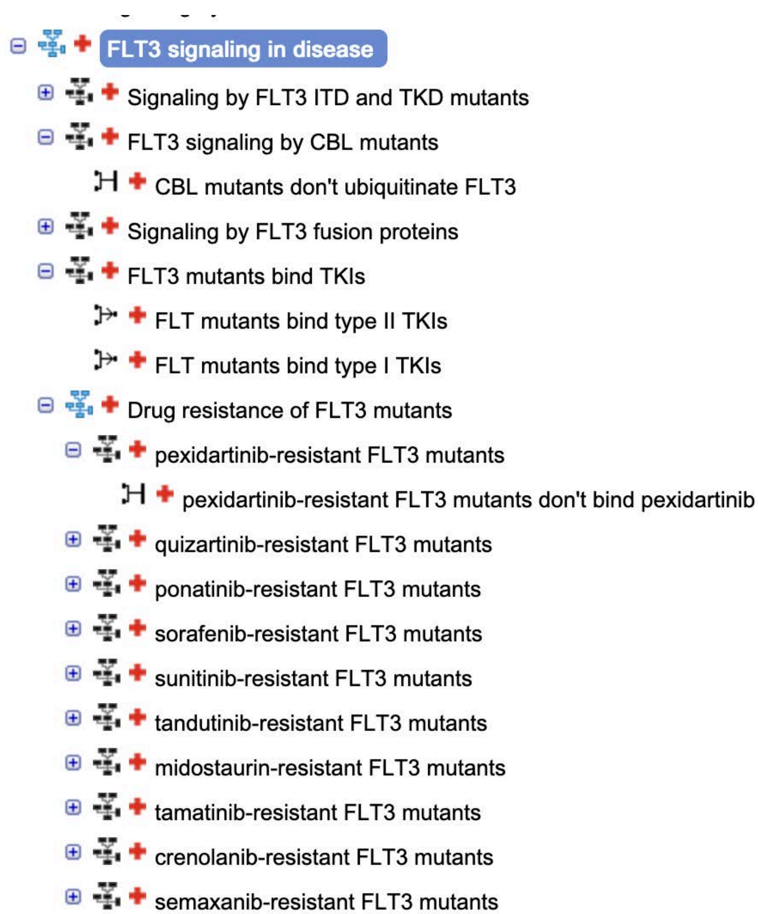
SIGNALLING BY RTK IN CANCER (parent pathway): this will often be an EHLD depicting the clickable subpathways. The parent pathway is **not** added to the WT pathwayDiagram record as a represented pathway

- constitutive signaling by overexpressed RTK (gain-of-function subpathway; pathway added to the WT pathwayDiagram record)
    - Individual RLEs, for instance
        - Constitutive dimerization of overexpressed RTK
        - Trans-autophosphorylation of overexpressed RTK etc

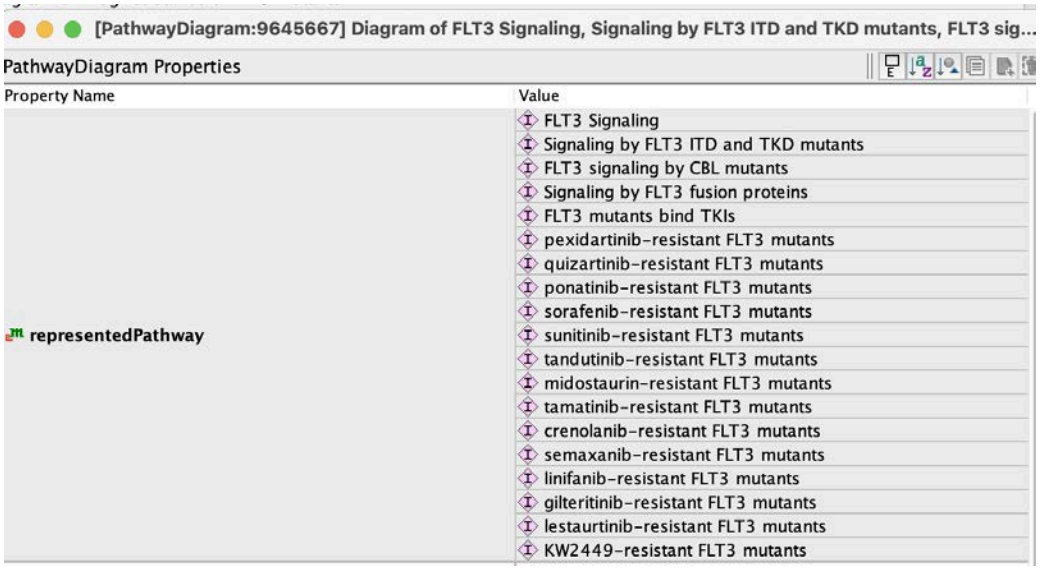
- constitutive signaling by RTK kinase domain mutants (gain-of-function subpathway with set of RTK mutants that have ligand-independent dimerization and activation; pathway added to the WT pathwayDiagram record)
    - Individual RLEs, for instance
        - Constitutive dimerization of KD mutants of RTK
        - Trans-autophosphorylation of KD mutants of RTK etc
- defective signaling by RTK kinase domain mutants (loss-of-function subpathway with set of RTK mutants that have lost kinase activity; pathway added to the WT pathwayDiagram record)
    - Individual RLEs (failedReactions), for instance
        - KD mutants of RTK don't trans-autophosphorylate
- signaling by RTK fusions (gain-of-function subpathway with set of fusion proteins that have ligand-independent dimerization and activation; pathway added to the WT pathwayDiagram record)
    - Individual RLEs for instance
        - Fusion mutants of RTK dimerize
        - etc
- RTK mutants bind tyrosine kinase inhibitors (essentially normal pathway showing binding of sensitive RTK mutants to known inhibitors; pathway added to the WTpathwayDiagram record)
    - RLEs for instance
        - TKI-sensitive RTK mutants bind TKIs
- Drug resistance of RTK mutants (loss-of-function subpathways showing inability of RTK mutants to bind and be inhibited by specified TKIs; this will have a new EHLD with subpathway icons, pathway not added to the WT pathwayDiagram record)
    - Subpathway for each drug such as
        - Resistance of RTK mutants to TKI 1 (pathway added to the WT pathwayDiagram record)
            - Failed reaction, "TKI 1- resistant RTK mutants don't bind TKI 1"
        - Resistance of RTK mutants to TKI 2 (pathway added to the WT pathwayDiagram record)

- Failed reaction “TKI 2-resistant RTK mutants don’t bind TKI 2”

For a real example:

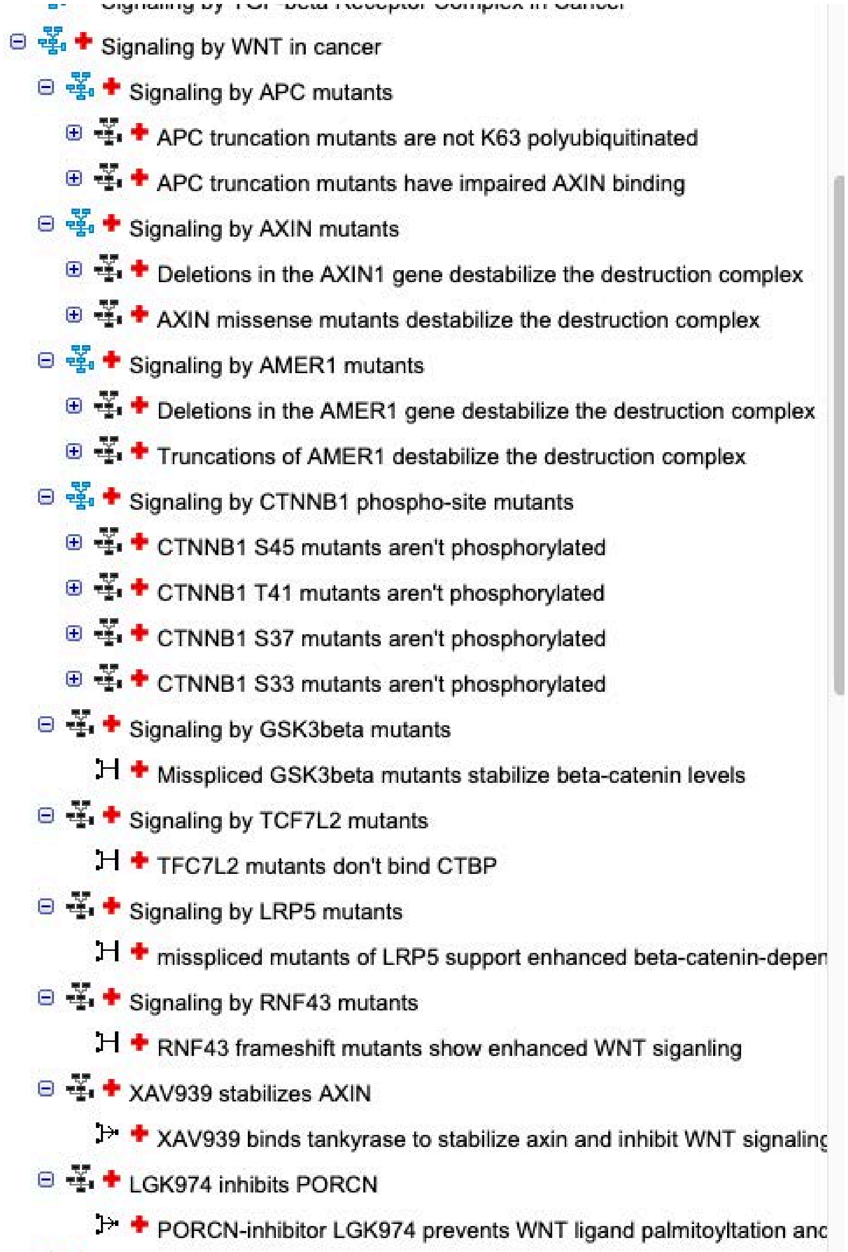


The corresponding PathwayDiagram record shown below lists first the WT pathway and then each of the disease RLE containing subpathways.



Note that the child pathway “Drug resistance of FLT3 mutants” has its own EHLD. It is a “grouping” pathway for all of the subpathways showing different drug resistant variants. This “grouping pathway itself does not have a normalpathway counterpart so is NOT associated with the WT pathway diagram. This is because each of the failed reactions (showing that the set of mutants is resistant to drug X, Y or Z etc) needs to be overlaid on the same WT reaction and therefore needs its own subpathway, as shown above.

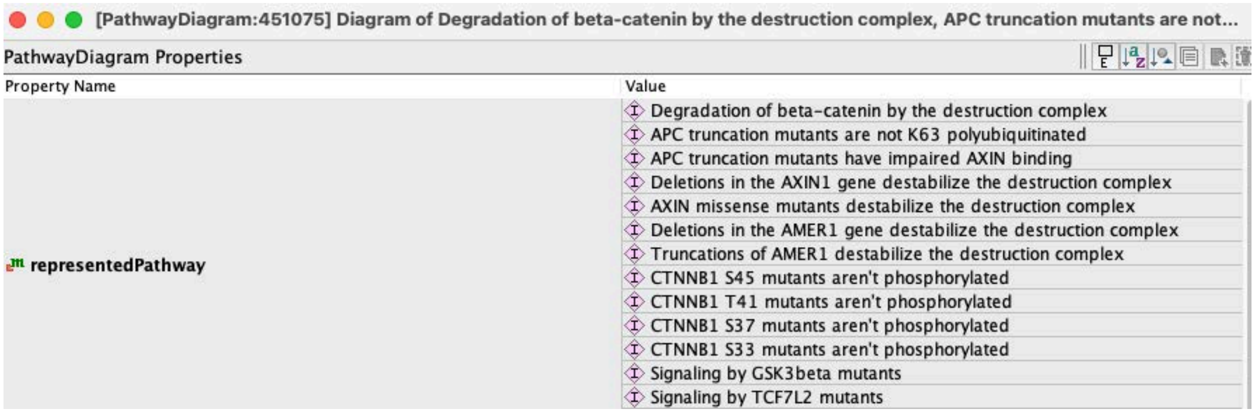
It is also possible to have different diagrams associated with different disease subpathways, as is shown in the “Signaling by WNT in cancer” pathway.



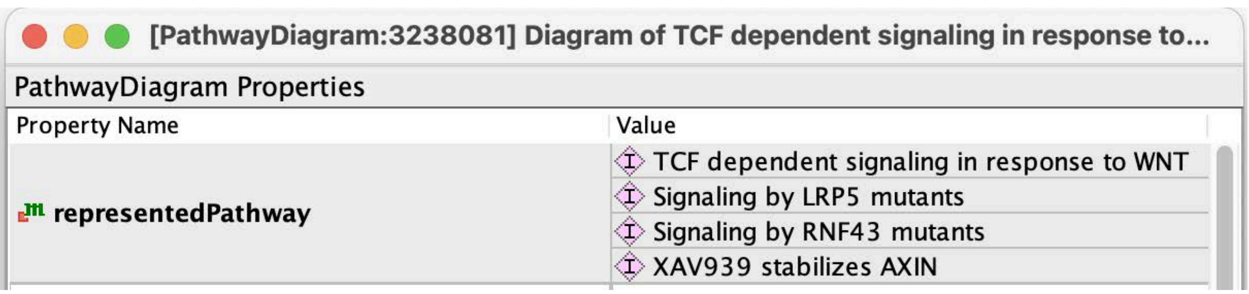
In this case, the wild-type WNT pathway has three associated normal pathway diagrams: Degradation of beta-catenin by the destruction complex, TCF-dependent signaling in response to WNT and WNT ligand biogenesis.

The parent disease WNT pathway, Signaling by WNT in cancer, has 10 subpathways containing a variety of gain-of-function and loss-of-function events.

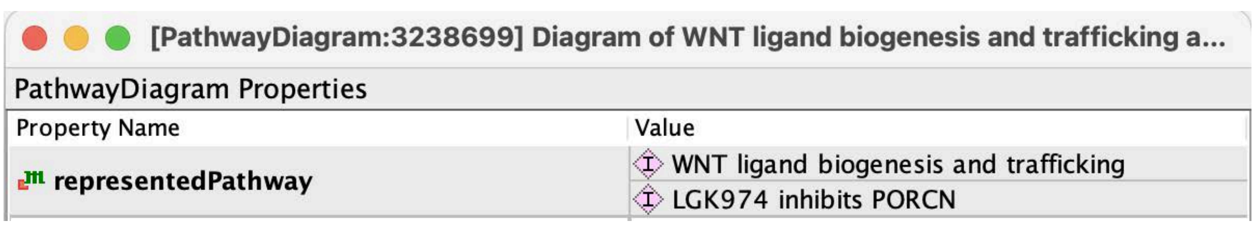
As shown below, the first 6 disease subpathways are all associated with the WT diagram “Degradation of beta-catenin by the destruction complex”:



The next 3 disease subpathways are associated with the WT diagram “TCF-dependent signaling in response to WNT”:



and the final disease subpathway is associated with the WT diagram “WNT ligand biogenesis and trafficking”:



Note: Any time you need to make revision to a "normal pathway" that has a disease counterpart, please be sure that you start off by "Building" (as opposed to"Checking out") the full disease pathway into a local project. Then open the Disease diagram and make any necessary edits there.

Assigning Disease term attributes

We want to use “disease” in a broad sense: if a process has a specific bad outcome we can associate it with that outcome even if it hasn’t yet progressed to the point of causing major disease. The association of Amyloidosis with the Amyloids pathway is an example. It is possible, for instance, that amyloid deposition, that is going on in all of us, is pathological, even though most of us will die of something else before the amyloidosis progresses to a point where it’s symptomatic. Targeting these early pathological processes (propathology?) will likely be very important: good for us as patients because early intervention slows or stops the process and keeps us healthy longer; good for the drug companies because that intervention will likely be a continuing one – an additional drug to be taken for life rather than a one-time course.

Reactome disease terms are taken from the [Disease Ontology](https://www.ebi.ac.uk/ols4/ontologies/doid) (https://www.ebi.ac.uk/ols4/ontologies/doid). If gk-central doesn’t have the disease term you need, you can create a new one by opting to create a new instance. Browse the hierarchy [here](https://www.ebi.ac.uk/ols4/ontologies/doid) (https://www.ebi.ac.uk/ols4/ontologies/doid), then create a new record in Reactome by entering the corresponding DOID identifier (number only) in the identifier slot. You will be asked if you want to import the entry. Say yes. If the Disease Ontology itself doesn’t have an appropriate term, we can request that one be made.

To pathways:

Please be sure to label the top-level disease pathway and its sub-level disease pathway with a disease attribute.

Note that disease terms applied to pathways and events should describe *all entities and events that are captured in that event. For this reason, disease terms applied at the pathway level are often very general (“cancer”, not “lung adenocarcinoma”, for instance). Disease terms tend to get more specific at the entity level, as described below.

To reactions:

A disease attribute should be added to all reactions involving disease-related physical entities, such as proteins of bacterial/viral/fungal pathogens, disease variant human proteins, and drugs used in disease management. Use the same procedure as the one described for the addition of disease attributes to pathways. If there are several related disease tags that are applicable to reactions and pathways, using the most general tag is preferable, and even if specific disease attributes are added, always include the most general attribute for a given disease type. For example, for cancer-related reactions and pathways, always include the cancer tag.

To entities:

Disease attributes should be added to mutant proteins associated with disease and may be very specific, referring to the specific disease type(s) in which a particular mutation was found. For example, EGFR L861Q mutant (DB_ID 1177542), in which L-leucine at position 861 is replaced with L-glutamine has been detected in non-small cell lung carcinoma and adult glioblastoma multiforme. Besides these specific disease tags, it may be advisable to also add more general disease tags to an EWAS, in this case, lung cancer, and cancer, to enable search of mutant EWASs using these general terms. When it comes to entity sets, such as EGFR KD mutants (DB_ID 1182966) which includes all kinase domain mutants of EGFR in cancer, a very general disease tag, such as cancer, is appropriate. This is because each member of the set has its own range of cancer types (while EGFR L861Q is found in lung cancer and glioblastoma, EGFR L858R is found in lung cancer, thymoma, thyroid cancer, breast cancer, and ovarian cancer), but their biological behavior is identical/similar. For the same reasons, only the general cancer disease tag should be used for events in which cancer disease entities participate.

To drugs:

When annotating drugs used to treat a particular disease, curators should add an appropriate disease tag to a drug entity. For a cancer drug, the general cancer disease attribute should definitely be added, but when it comes to more specific disease tags, it may be advisable to add only those cancer types for which the treatment by a given drug is approved.

Annotating Disease Reaction-Like-Events (RLEs)

Disease reaction-like events have 3 disease-specific fields that may need to be filled in:

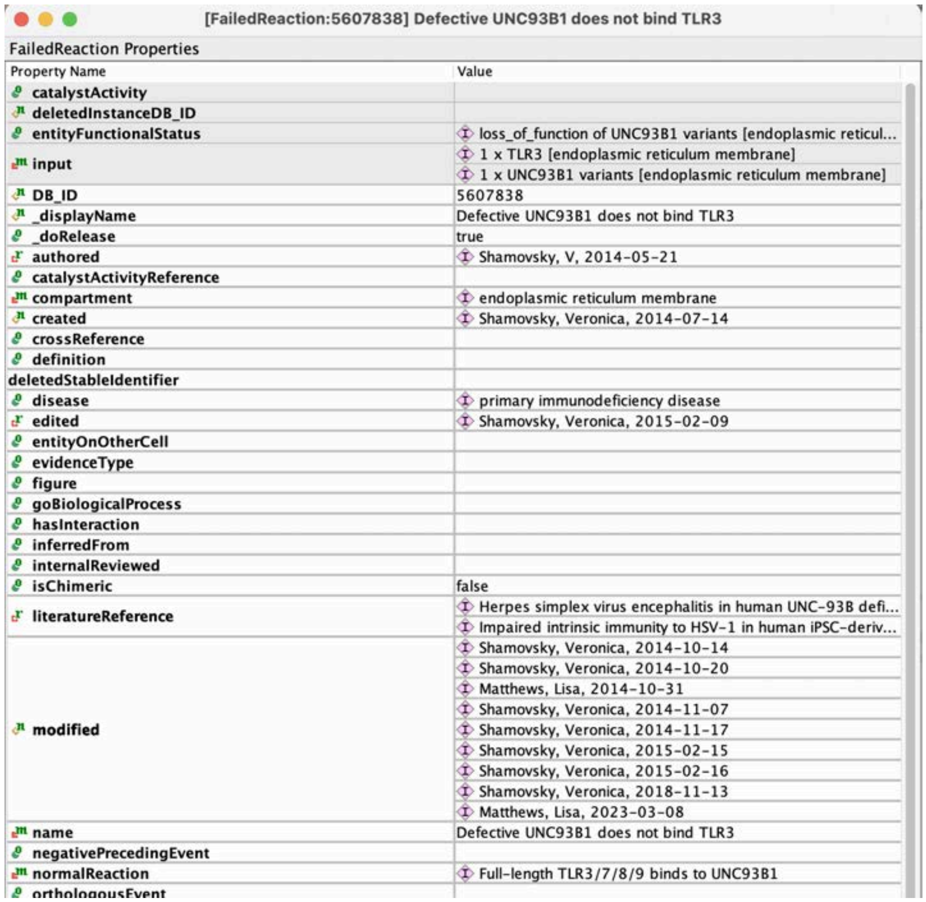
- disease (required for all disease RLE's; as described above)
- normalReaction (required for disease reactions that will be overlaid on a normal WT, described below)
- entityFunctionalStatus (EFS) (absolutely required for disease reactions that will be overlaid on a normal WT reaction, described below, but from a curation policy standpoint, required for *all disease reactions, regardless of overlay, as an additional source of information for users and to facilitate alignment with ACMG variant criteria).

Disease RLE's are either loss- or gain-of-function events. All loss-of-function are overlaid automatically on top of the corresponding WT reaction by virtue of their disease tag, normalReaction field and entityFunctionalStatus. Gain-of-function reactions may or may not be overlaid on normal WT reactions, as described below.

Annotating loss-of-function reactions

Loss-of-function 'reactions' are annotated as "FailedReaction". Failed reactions are labeled with a disease tag as described above but have only inputs (the disease variant entity or complex or set, plus any WT entities that are inputs in the 'normal' reaction), and do not have any outputs. See the note below about avoiding reaction participant mismatches.

The failedReaction record also lists the corresponding WT reaction in the "normalReaction" slot.

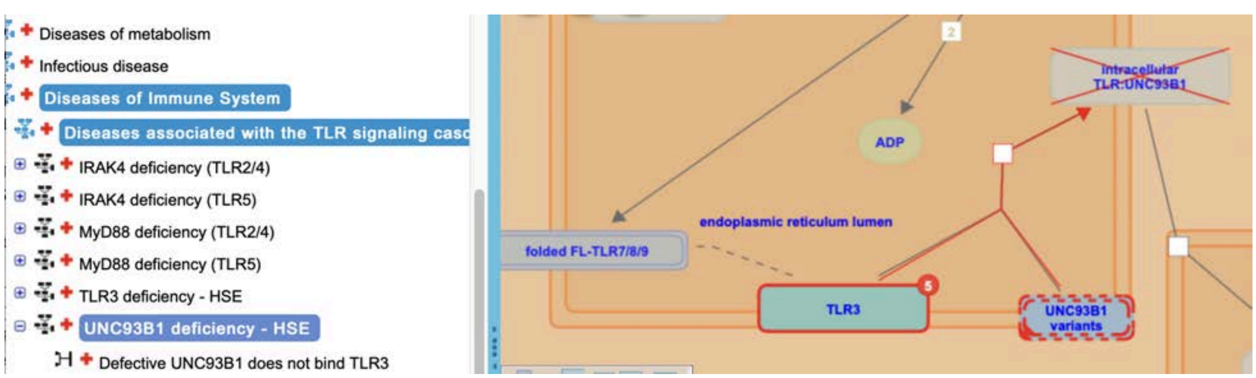


[FailedReaction:5607838] Defective UNC93B1 does not bind TLR3

FailedReaction Properties

Property Name	Value
catalystActivity	
deletedInstanceDB_ID	
entityFunctionalStatus	loss_of_function of UNC93B1 variants [endoplasmic reticul...
input	1 x TLR3 [endoplasmic reticulum membrane]
	1 x UNC93B1 variants [endoplasmic reticulum membrane]
DB_ID	5607838
_displayName	Defective UNC93B1 does not bind TLR3
_doRelease	true
authored	Shamovsky, V, 2014-05-21
catalystActivityReference	
compartment	endoplasmic reticulum membrane
created	Shamovsky, Veronica, 2014-07-14
crossReference	
definition	
deletedStableIdentifier	
disease	primary immunodeficiency disease
edited	Shamovsky, Veronica, 2015-02-09
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	
internalReviewed	
isChimeric	false
literatureReference	Herpes simplex virus encephalitis in human UNC-93B defi...
	Impaired intrinsic immunity to HSV-1 in human iPSC-deriv...
modified	Shamovsky, Veronica, 2014-10-14
	Shamovsky, Veronica, 2014-10-20
	Matthews, Lisa, 2014-10-31
	Shamovsky, Veronica, 2014-11-07
	Shamovsky, Veronica, 2014-11-17
	Shamovsky, Veronica, 2015-02-15
	Shamovsky, Veronica, 2015-02-16
	Shamovsky, Veronica, 2018-11-13
	Matthews, Lisa, 2023-03-08
name	Defective UNC93B1 does not bind TLR3
negativePrecedingEvent	
normalReaction	Full-length TLR3/7/8/9 binds to UNC93B1
orthologousEvent	

FailedReactions are placed in the appropriate disease pathway as usual, but are not dragged into the ELV by the curator. By virtue of the disease tag, the 'normalReaction' field, the EFS, and the appropriate sharing of pathway diagrams, FailedReactions are automatically overlaid on the background of the WT pathway and appear on top of the normal reaction with the disease input and the reaction node highlighted in red and the outputs of the WT reaction X'd out, as shown:



In general, FailedReactions should be named according to the structure “Defective [core protein name] doesn’t [some description of WT function]”, as in “Defective ALG9 does not add mannose to the N-glycan precursor”. The key is using the words ‘defective’ and ‘variant’ to describe the disease related entities.

To avoid reaction participant mismatches, all the inputs of the WT normal reaction must be present in the FailedReaction record (differences in stoichiometry are ignored). A good approach to making a FailedReaction can be to clone the normal WT reaction, switch type to FailedReaction, replace the WT entity with the relevant disease variant or set, and add disease tag, EFS and normal reaction to the relevant slots on the record. See the section below on avoiding participant mismatches for more considerations.

The molecular biology of the failed reaction should match the molecular biology of the normal reaction closely/exactly, both for biological coherence and to avoid mismatches. For instance, imagine a protein that binds to a substrate and then catalyzes a reaction. If there are subsets of variants identified that a) don’t bind or b) bind but are catalytically dead, these should be represented in two different failed reactions, the first showing a failed binding reaction and the second showing a failed catalysis. In some cases, this may require creating a new normal reaction for the WT pathway (if, for instance, the binding and catalysis are shown happening in a single step in the WT pathway). See here (Appendix C10, Multiple disease reactions and ENTITYFUNCTIONALSTATUS when using DISEASE VARIANT SUBSETS) for the original discussion of this).

Annotating gain-of-function reactions

There are two types of gain-of-function events in Reactome:

1. totally novel events that are not seen in the WT pathway. This may arise through the introduction of an infectious agent or by expression of a variant of a human protein that has an activity not displayed by the corresponding WT protein - for instance binding to a novel protein partner, or forming ligand-independent dimers.
2. gain-of-function events that happen at an elevated rate or efficiency compared to a corresponding event in a wild-type pathway.

All gain-of-function RLE’s are labeled with a disease tag as described above. This is needed for the automatic red line coloring in the ELV.

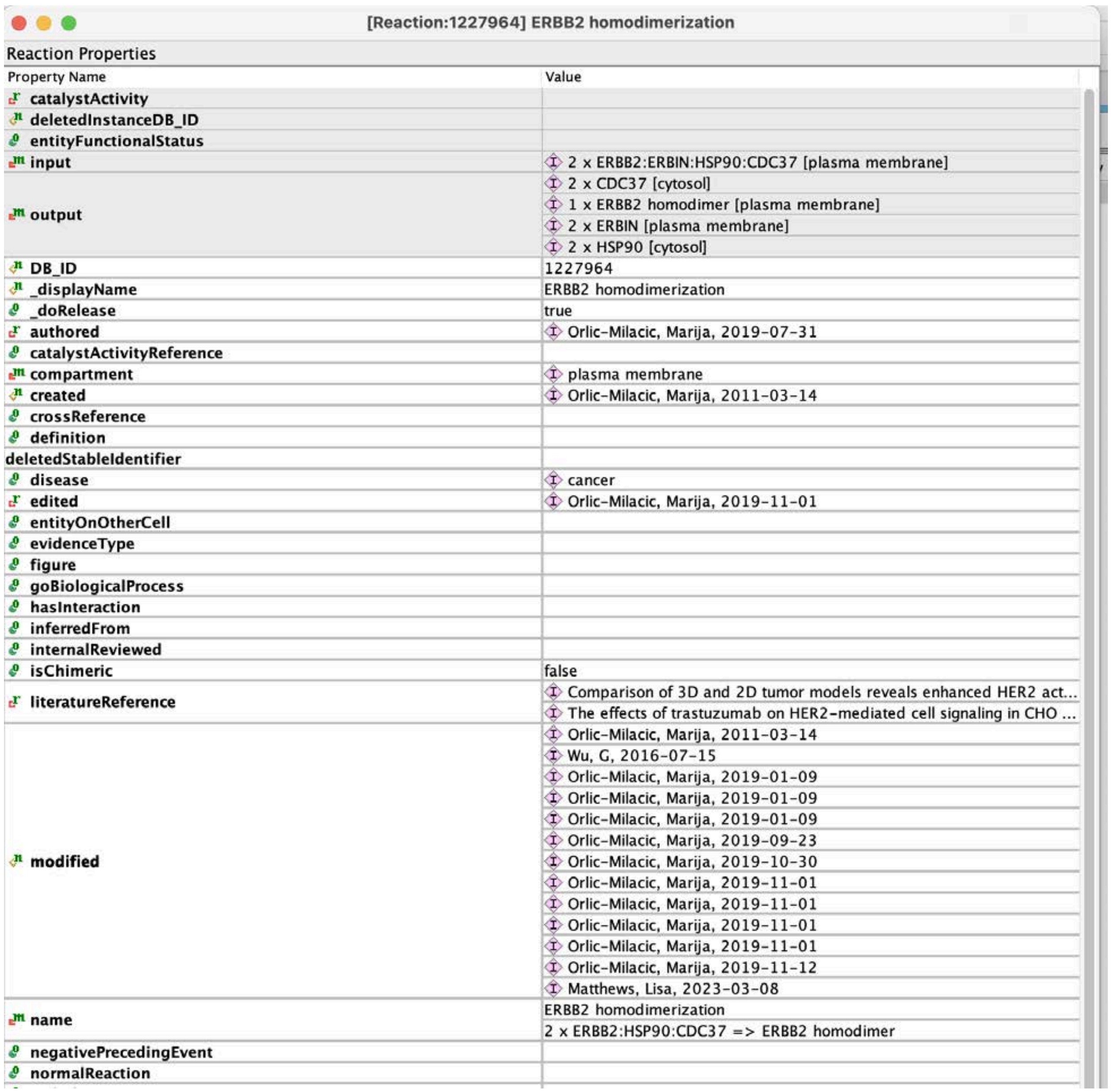
All gain-of-function RLEs should have their entity functional status slot filled out (described below).

Gain-of-function events not needing a ‘normalReaction’

Some gain-of-function events represent totally novel activities that are not seen in the WT pathway. Novel gain-of-function RLEs of this kind do not have their ‘normalReaction’ slot filled out and are not overlaid on top of a normal WT reaction.

These reactions are dragged into the pathway diagram manually by the curator as for WT reactions and occupy their own space in the pathway diagram.

Although the EntityFunctionalStatus slot is not required for display purposes for these RLEs, it should be filled out in any case. This is a new policy as of Feb 2023, so older records may not reflect this. This was introduced for consistency and to allow for better mapping of Reactome variants to ACMG variant criteria (see below).



[Reaction:1227964] ERBB2 homodimerization

Reaction Properties

Property Name	Value
catalystActivity	
deletedInstanceDB_ID	
entityFunctionalStatus	
input	2 x ERBB2:ERBIN:HSP90:CDC37 [plasma membrane]
output	2 x CDC37 [cytosol]
	1 x ERBB2 homodimer [plasma membrane]
	2 x ERBIN [plasma membrane]
	2 x HSP90 [cytosol]
DB_ID	1227964
_displayName	ERBB2 homodimerization
_doRelease	true
authored	Orlic-Milacic, Marija, 2019-07-31
catalystActivityReference	
compartment	plasma membrane
created	Orlic-Milacic, Marija, 2011-03-14
crossReference	
definition	
deletedStableIdentifier	
disease	cancer
edited	Orlic-Milacic, Marija, 2019-11-01
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	
internalReviewed	
isChimeric	false
literatureReference	Comparison of 3D and 2D tumor models reveals enhanced HER2 act...
	The effects of trastuzumab on HER2-mediated cell signaling in CHO ...
modified	Orlic-Milacic, Marija, 2011-03-14
	Wu, G, 2016-07-15
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-09-23
	Orlic-Milacic, Marija, 2019-10-30
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-12
	Matthews, Lisa, 2023-03-08
name	ERBB2 homodimerization
	2 x ERBB2:HSP90:CDC37 => ERBB2 homodimer
negativePrecedingEvent	
normalReaction	

Gain-of-function events with a ‘normalReaction’

If a disease variant (or set of variants) performs the same reaction as the WT counterpart at an elevated rate or efficiency, the disease RLE can be overlaid on the normal reaction, as for loss-of-function FailedReactions above. In this case, the ‘normalReaction’ field on the disease RLE must be filled in to allow for proper display.

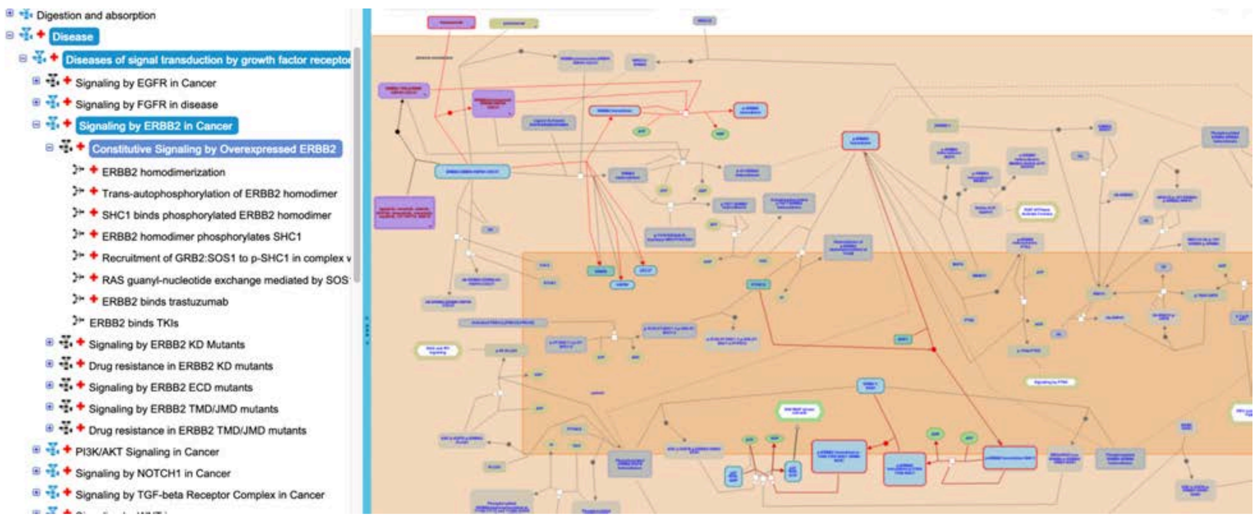
[Reaction:1248694] Trans-autophosphorylation of ERBB2 homodimer

Reaction Properties

Property Name	Value
catalystActivity	protein tyrosine kinase activity of ERBB2 homodimer [plasma...
deletedInstanceDB_ID	
entityFunctionalStatus	
input	12 x ATP [cytosol]
	1 x ERBB2 homodimer [plasma membrane]
output	12 x ADP [cytosol]
	1 x p-ERBB2 homodimer [plasma membrane]
DB_ID	1248694
_displayName	Trans-autophosphorylation of ERBB2 homodimer
_doRelease	true
authored	Orlic-Milacic, Marija, 2019-07-31
catalystActivityReference	
compartment	plasma membrane
	cytosol
	extracellular region
created	Orlic-Milacic, Marija, 2011-04-12
crossReference	
definition	
deletedStableIdentifier	
disease	cancer
edited	Orlic-Milacic, Marija, 2019-11-01
entityOnOtherCell	
evidenceType	
figure	
goBiologicalProcess	
hasInteraction	
inferredFrom	
internalReviewed	
isChimeric	false
literatureReference	Analysis of protein-protein interactions involved in the activa...
	Identification of autophosphorylation sites of HER2/neu
	The effects of trastuzumab on HER2-mediated cell signaling...
	Comparison of 3D and 2D tumor models reveals enhanced ...
modified	Wu, G, 2016-07-15
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-09
	Orlic-Milacic, Marija, 2019-01-10
	Orlic-Milacic, Marija, 2019-09-23
	Orlic-Milacic, Marija, 2019-10-30
	Orlic-Milacic, Marija, 2019-10-30
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-01
	Orlic-Milacic, Marija, 2019-11-12
	Matthews, Lisa, 2023-03-08
name	Trans-autophosphorylation of ERBB2 homodimer
negativePrecedingEvent	
normalReaction	Trans-autophosphorylation of ERBB2 heterodimers

Often, a signaling pathway for a particular variant or set of variants will start with a novel gain-of-function event (ligand-independent binding, for instance) and then be followed by a series of reactions that mirror the steps performed by the WT protein downstream of the initiating event. These ‘conserved’ downstream events are also tagged with the corresponding ‘normalReaction’ and an EFS, and are automatically overlaid on the normal pathway events.

This can be seen in the “Constitutive signaling by Overexpressed ERBB2” (R-HSA-9634285) pathway. In the first step, a novel homodimerization of ERBB2 is depicted, resulting from overexpression of the protein in cancer. This reaction has no ‘normalReaction’. Downstream of dimerization, the receptor trans-autophosphorylates, binds and phosphorylates SHC1, recruits GRB2:SOS1 and activates RAS, among other actions. Each of these downstream events has a normal counterpart in the WT pathway that is captured in the disease RLE record and each of these is automatically overlaid on the WT pathway diagram. All the RLEs related to overexpressed ERBB2 homodimers are shown at the same time in the same diagram representation:



FunctionalStatus

Overriding the automatic overlay

In some cases, a curator may wish to manually lay out a disease gain-of-function RLE even though it could theoretically be overlaid on a WT reaction. This may be to avoid reaction participant mismatches, described below, or for another reason. This is fine, and a curator would simply leave the ‘normalReaction’ slot empty and drag the reaction into the pathway diagram as usual. Disease tag and EntityFunctionalStatus fields should be filled out as usual.

EntityFunctionalStatus

EntityFunctionalStatus is used to mark a disease event as either a gain- or a loss-of-function, and to describe the underlying reason for the change in behavior. This tag, along with the “NormalReaction” attribute, is especially important for loss-of-function events, as it is needed for deploying the red lines and X’ing out of WT outputs in the website display. Note that EFS’s are applied at the reaction-like-event level and not to entities themselves. This is because a given variant/set of variants may play different functional roles in different disease or reaction contexts.

The screenshot below shows the EFS of AKT1 E17K in the disease RLE “AKT1 E17K mutant binds PIP2”.

[Reaction:2219536] AKT1 E17K mutant binds PIP2

Reaction Properties

Property Name	Value
catalystActivity	
deletedInstanceDB_ID	
entityFunctionalStatus	gain_of_function of AKT1 E17K [cytosol]
input	1 x AKT1 E17K [cytosol] 1 x PI(4,5)P2 [plasma membrane]
output	1 x AKT1 E17K mutant:PIP2 [plasma membrane]
DB_ID	2219536
_displayName	AKT1 E17K mutant binds PIP2

The EntityFunctionalStatus has three fields, diseaseEntity, functionalStatus and normalEntity:

[EntityFunctionalStatus:2362350] gain_of_function of AKT1 E17K [cytosol]

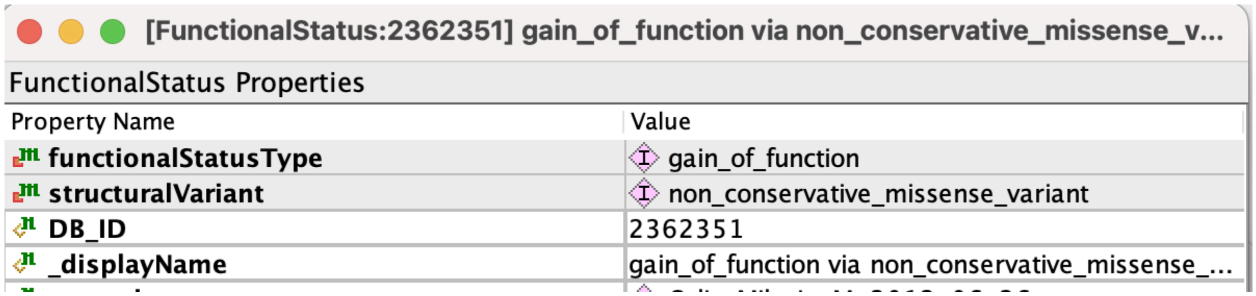
EntityFunctionalStatus Properties

Property Name	Value
diseaseEntity	AKT1 E17K [cytosol]
functionalStatus	gain_of_function via non_conservative_missen...
normalEntity	
DB_ID	2362350
_displayName	gain_of_function of AKT1 E17K [cytosol]

- The disease entity is the disease variant (or complex or set containing the relevant disease variant(s)) that is responsible for the phenotype.

- The normalEntity is the corresponding normal protein that takes part in the RLE captured in the ‘normalReaction’ slot of the disease RLE. This field must be filled out if a ‘normalReaction’ is specified in the disease RLE. This field helps the overlay find the appropriate normal entity to overlay the disease variant on and helps to avoid reaction participant mismatches, described below.

- The FunctionalStatus itself consists of 2 fields, FunctionalStatusType, and structural variant:


[FunctionalStatus:2362351] gain_of_function via non_conservative_missense_v...

FunctionalStatus Properties

Property Name	Value
functionalStatusType	gain_of_function
structuralVariant	non_conservative_missense_variant
DB_ID	2362351
_displayName	gain_of_function via non_conservative_missense_...

The structuralVariant field takes terms from [SequenceOntology](https://www.ebi.ac.uk/ols4/ontologies/so) (https://www.ebi.ac.uk/ols4/ontologies/so) and describes the nature of the underlying genetic change that gives rise to the variant.

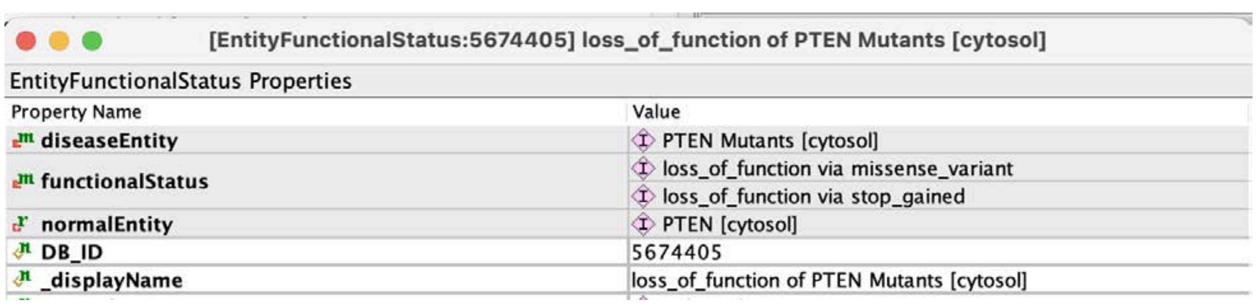
The functionalStatusType describes the functional consequence of this genetic change (gain-of-function, loss-of-function etc)

Many of the commonly used FunctionalStatus already exist in gk_central, but new combinations may be generated as required.

A given EFS may be used on as many disease RLEs as appropriate..

EFSs for variant sets:

The EFS shown above for AKT1 E17K is a simple case, where the disease variant is a single EWAS. Disease annotations often describe sets of mutants that function similarly. These sets can also be represented by a single EFS record that has multiple FunctionalStatus attributes to cover all the variants contained in the set, as shown in the screenshot below:

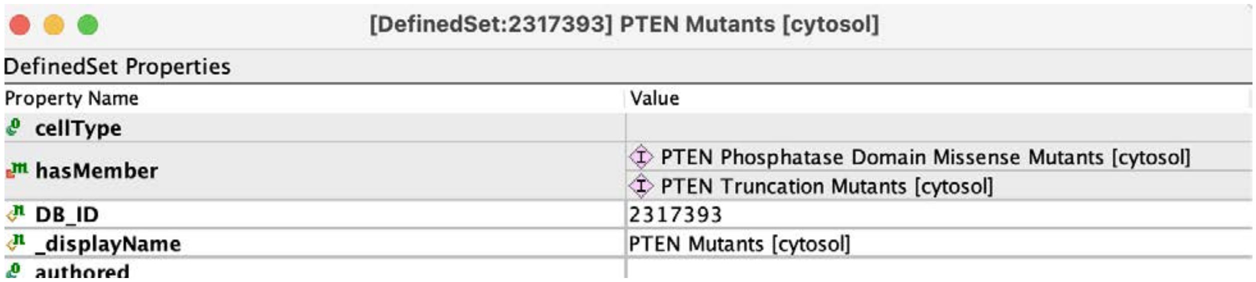

[EntityFunctionalStatus:5674405] loss_of_function of PTEN Mutants [cytosol]

EntityFunctionalStatus Properties

Property Name	Value
diseaseEntity	PTEN Mutants [cytosol]
functionalStatus	loss_of_function via missense_variant loss_of_function via stop_gained
normalEntity	PTEN [cytosol]
DB_ID	5674405
_displayName	loss_of_function of PTEN Mutants [cytosol]

Here, the disease entity is a set of PTEN mutants, some of which arise as a consequence of missense mutations and some as a result of novel stop mutations.

Best practice says that the set of PTEN mutants should itself contain subsets corresponding to each of the functional status types indicated in the EFS record, as shown below:



(Original discussion of this is here; Appendix C10)

AVOIDING REACTION PARTICIPANT MISMATCHES IN DISEASE CURATION

In order for the overlay function to work properly, there must be careful attention paid to aligning the inputs and outputs of disease and WT reaction participants. If this is not done properly, a Reaction Participant mismatch error will be generated during QA.

Tips to avoid reaction participant mismatches:

1. A disease reaction must have exactly one or zero 'normalReaction' counterparts.
2. The curator can always draw disease reactions directly onto pathway diagrams to bypass auto-generating reaction diagrams. However, to make sure auto-generation work, the curator must specify values in the entityFunctionalStatus slot. The disease PE wrapped in EFS will be mapped to normal reaction participant.
3. Mapping between disease reaction participants and normal reaction participants are always based on participants' referenceEntity values. If two normal reaction participants refer to the same ReferenceEntity instance, but having different roles (e.g. one input, and another catalyst), the mapping will not be reliable. Manual drawing is highly recommended for such a case.
4. If it is to be overlaid, a disease reaction, including FailedReaction and GoF reaction, must have exactly the same number of input entities as its normal reaction counterpart without considering stoichiometries (e.g. a disease reaction has input A, B, and its normal counterpart has A, A, B. This is fine, or vice versa). Creating failedReaction instances by cloning the normal event, switching the type of the clone to failedReaction, and then editing further as needed might be a good way to minimize this kind of mistake.
5. If a protein normally functions as part of a complex, there are two ways to annotate the disease reaction that occurs when a mutant form of the protein is expressed.

   a) If the mutant form of the protein still associates with the other members of the complex, but the complex does not function, then create a complex instance identical to the normal one except that the one affected normal protein is replaced with its disease counterpart. Then use that variant complex to create an entityFunctionalStatus instance.

b) If the mutant form of the protein is incapable of forming a complex at all, then create a preceding normal reaction to annotate formation of the normal complex (if that doesn't already exist), and create a failedReaction version of the normal complex-formation one, not of the later one in which the normal complex carries out its normal function. That yields the needed exact matching between the disease reaction and its normal counterpart.

Annotating Disease Entities

General approaches to disease entity annotation and naming

Naming

Reactome variant protein names are formulated in accordance with the HGVS nomenclature (https://hgvs-nomenclature.org/stable/), with the exception of using single letter abbreviations for amino acid residues.

When a descriptor is added (usually to set names), 'variant' is to be used rather than 'mutant' as in "PI3K variants". Specific rules for the naming of different variants are described more fully in the relevant sections below.

Aligning with ACMG variant guidelines

Where possible, we try to align our annotations with The American College of Medical Genetics and Genomics (ACMG) Standards and Guidelines for the interpretation of sequence variants (Richards et al. 2015). This framework classifies sequence variants into benign or pathogenic based on 1) population data, 2) computational and predictive data, 3) functional data, 4) segregation data, 5) de novo data, 6) allelic data, 7) other database, and 8) other data. Detailed criteria are listed for assessing the strength of evidence in each of the eight categories as supporting (BP1 to BP7) or strong (BS1 to BS4) for benign variants, and as supporting (PP1 to PP5), moderate (PM1 to PM6), strong (PS1 to PS4) or very strong (PVS1) for pathogenic variants.

Because Reactome's disease variant annotation focuses on annotating events at a molecular or biochemical level, (ie variants that have been functionally characterized) and on clinically relevant, pathogenic variants, benign variants are outside Reactome's scope.

In consequence, Reactome-annotated disease variants correspond to only 6 of the 27 ACMG criteria (PM1, PM4, PM5, PS1, PS3, and PVS1), as indicated in Table 1 and described more fully below. Of the remaining 21 ACMG criteria, BS1-4 and BP1-7, used for categorization of variants as benign, can be used for exclusion of potentially pathogenic variants from consideration when the relevant data is available. The criteria PP1-PP5, PM2, PM3, PM6, PS2, and PS4 are used for categorization of clinically relevant pathogenic variants but are not used in Reactome because they do not provide sufficient information to functionally annotate disease variants. These criteria can only be used to corroborate pathogenicity of a disease variant that conforms to criteria PM1, PM4, PM5, PS1, PS3 or PVS1.

Criterion	Evidence strength	Pathogenicity	Evidence type	Evidence definition	Suitability as direct Reactome evidence
BS1	strong	benign	population data	variant represents minor allele whose frequency is too high for disorder	-
BS2	strong	benign	population data	variant incidence in controls is inconsistent with disease penetrance	-
BS3	strong	benign	functional data	well-established functional studies show no deleterious effect for variant	-
BS4	strong	benign	segregation data	variant does not segregate with disease	-
BP1	supporting	benign	computational and predictive data	missense variant is found in gene in which only truncating variants cause disease	-
BP2	supporting	benign	allelic data	variant is observed in trans with dominant variant or in cis with pathogenic variant	-
BP3	supporting	benign	computational and predictive evidence data	variant is an in-frame indel in repeat without known function	-
BP4	supporting	benign	computational and predictive evidence data	multiple lines of computational evidence suggest variant has no impact on gene or gene product	-
BP5	supporting	benign	other data	variant is found in disease case with an alternate cause	-
BP6	supporting	benign	other database	a reputable source without shared data defines variant as benign	-
BP7	supporting	benign	computational and predictive	variant is silent variant with non predicted splice impact	-

			evidence data		
PP1	supporting	pathogenic	segregation data	variant co-segregates with disease in multiple affected family members	-
PP2	supporting	pathogenic	functional data	variant represents missense variant in gene in which there is low rate of benign missense variants and in which pathogenic missense variants are common	-
PP3	supporting	pathogenic	computational and predictive data	multiple lines of computational evidence support a deleterious effect of variant on gene/gene product	-
PP4	supporting	pathogenic	other data	variant is detected in a patient whose phenotype of family history are highly specific for disease gene	-
PP5	supporting	pathogenic	other database	a reputable source identifies variant as pathogenic	-
PM1	moderate	pathogenic	functional data	variant represents mutational hotspot or is within functional domain without benign variation	✓
PM2	moderate	pathogenic	population data	variant is absent in population databases	-
PM3	moderate	pathogenic	allelic data	for recessive disorders variant is detected in trans with known pathogenic variant	-f
PM4	moderate	pathogenic	computational and predictive data	variant represents protein length changing variant	✓
PM5	moderate	pathogenic	computational and predictive data	variant represents novel missense change at amino acid residue previously reported to be affected by pathogenic missense change	✓✓
PM6	moderate	pathogenic	de novo data	de novo variant is detected (without paternity or maternity confirmed)	-
PS1	strong	pathogenic	computational and predictive data	variant leads to same amino acid change as an established pathogenic variant	✓✓✓
PS2	strong	pathogeni	de novo	de novo variant is detected (with paternity	-

		c	data	and maternity confirmed)	
PS3	strong	pathogenic	functional data	well-established functional studies show variant's deleterious effect	✓✓✓
PS4	strong	pathogenic	population data	prevalence of variant in affected patients is statistically significantly increased over controls	-
PVS1	very strong	pathogenic	computational and predictive data	variant is predicted to be null variant in gene where LOF is known disease mechanism	✓✓

PS3, PVS1 and PS1

Of the 6 ACMG criteria suitable as direct evidence for annotation of disease variants in Reactome, PS3 is the gold standard - variants whose deleterious effect(s) are supported by well-established functional studies. An additional requirement for Reactome is that these functional studies are at the molecular mechanism resolution level. Variants with this classification may be of any type - missense, fusion, in-frame insertion-deletion (in-frame indel), nonsense, frameshift, etc. - so long as they have been functionally characterized. PS3 variants are annotated as direct disease reaction participants or as members of disease variant sets.

PM5

ACMG classifies as PM5 any variant that represents a novel missense change at an amino acid previously reported to be affected by a pathogenic missense change. Reactome extends this classification to include variants of any type (missense, fusion, in-frame indel, etc.) that have not been directly functionally characterized but contain mutations analogous or equivalent to those found in annotated PS3 variants. PM5 variants are annotated as "candidates" within Reactome's disease variant sets.

PM1 and PM4

Nonsense and frameshift variants not directly functionally studied that conform to the criteria PM1 (moderate pathogenic based on functional data where variant represents mutational hotspot or is within functional domain without benign variation) and PM4 (moderate pathogenic based on computational and predictive data where variant represents protein length changing variant) so that these variants are protein length changing variants that result in truncation or complete ablation of functional domains key to a particular protein function, are annotated as candidates of disease variants sets.

References

1. Richards S, Aziz N, Bale S, Bick D, Das S, Gastier-Foster J, Grody WW, Hegde M, Lyon E, Spector E, Voelkerding K, Rehm HL; ACMG Laboratory Quality Assurance Committee. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. Genet Med. 2015

May;17(5):405-24. doi: 10.1038/gim.2015.30. Epub 2015 Mar 5. PMID: 25741868; PMCID: PMC4544753.

Disease Entity Overview

When making a record for a disease variant, it is often most efficient to start by cloning the WT record and then changing the relevant fields and adding the disease-specific classes.

Like disease events, disease entities must be labeled with a disease tag (described above).

Where possible, disease entities will have a cross reference to an external database like COSMIC, OMIM, ClinGen or other, as shown. An entity can have as many cross-references as needed. COSMIC cross-referencing should make use of the Genomic Mutation ID where available, not the Legacy Identifier. ClinGen Canonical Allele (CA) identifiers are used wherever possible, but ClinVar IDs may be used if CA IDs are not available.

Disease entities should also have their literature field filled in citing the work where the variant was identified or characterized unless they are annotated as disease variant candidates.

[EntityWithAccessionedSequence:4791272] CTNNB1 S45P [cytosol]

EntityWithAccessionedSequence Properties

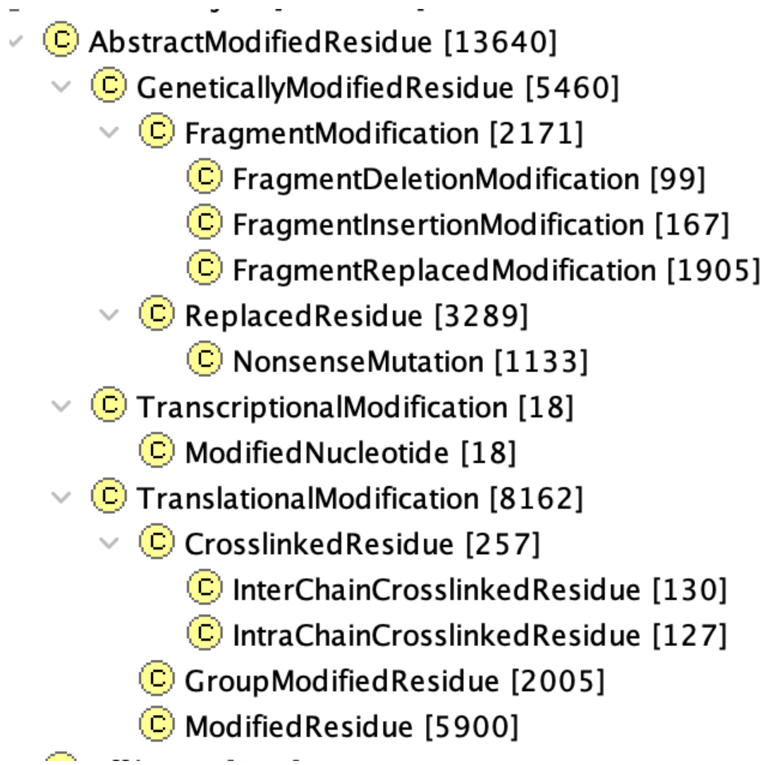
Property Name	Value
cellType	
compartment	cytosol
endCoordinate	781
hasModifiedResidue	L-serine 45 replaced with L-proline
referenceEntity	UniProt:P35222 CTNNB1
startCoordinate	1
DB_ID	4791272
_displayName	CTNNB1 S45P [cytosol]
authored	
created	Rothfels, K, 2013-10-28
crossReference	COSMIC:COSM12639 COSMIC:COSV62687880
definition	
disease	cancer hepatocellular carcinoma kidney cancer colorectal cancer adrenal gland cancer connective tissue cancer
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	AXIN1 mutations in hepatocellular carcinomas, and growt... Mutational spectrum of beta-catenin, AXIN1, and AXIN2 i... Nuclear accumulation of beta-catenin protein in Wilms' tu...
modified	Rothfels, K, 2013-11-01 Rothfels, K, 2014-02-04
name	CTNNB1 S45P Catenin beta-1 CTNB1_HUMAN
reviewed	
revised	
species	Homo sapiens
stableIdentifier	R-HSA-4791272.1
summation	
systematicName	

Data model classes for ‘hasModifiedResidue’

The changes in sequence that lead to the formation of a variant protein are captured by the field “hasModifiedResidue” using the modified residue class “GeneticallyModifiedResidue” to distinguish genetically encoded variants from post-translational modifications (TranslationalModifications class) and RNA editing (TranscriptionalModification class).

Note that in both its variant naming convention and in the annotation of ‘hasModifiedResidue’, Reactome focuses on amino acid (protein level) changes and does not capture changes at the nucleic acid level.

The data model includes the following classes of modified residues to capture the amino acid changes leading to a protein variant (under “GeneticallyModifiedResidue” to distinguish them from post-translational modifications).



The following sections will go through annotation and naming strategies for common classes of amino acid variations leading to variant protein forms.

Missense mutations

These are perhaps the most straightforward variants to annotate. These variants are the consequence of the replacement of a single amino acid codon with a non-synonymous amino acid codon without change in the reading frame.

Naming

Format: “prefix” + “wild-type amino_acid” + “position” + “new_amino_acid”

Where “prefix” is the HGNC gene name and the amino acids are represented by single letter codes

example, MTR P1173L: substitution of proline 1173 with leucine

How to annotate

Missense mutations are captured with the “ReplacedResidue” class of GeneticallyModifiedResidue.

Use PSI-MOD IDs to construct the replaced residue, identifying both the normal residue with an [amino acid] removal instance and its mutant replacement with an [amino acid] residue instance. Here, L-proline is replaced by L-leucine at coordinate 1173.

[ReplacedResidue:3321905] L-proline 1173 replaced with L-leucine

ReplacedResidue Properties

Property Name	Value
coordinate	1173
psiMod	L-proline removal [MOD:01645]
	L-leucine residue [MOD:00020]
referenceSequence	UniProt:Q99707 MTR
DB_ID	3321905
_displayName	L-proline 1173 replaced with L-leucine

[EntityWithAccessionedSequence:3321944] MTR P1173L [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	1265
hasModifiedResidue	L-proline 1173 replaced with L-leucine
referenceEntity	UniProt:Q99707 MTR
startCoordinate	1
DB_ID	3321944
_displayName	MTR P1173L [cytosol]
authored	
created	Jassal, B, 2013-05-02
crossReference	
definition	
disease	methylmalonic aciduria and homocystinuria type cblG
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	
modified	Jassal, B, 2013-06-07
name	MTR P1173L
	Methionine synthase
reviewed	
revised	
species	Homo sapiens
stableIdentifier	R-HSA-3321944.1

Nonsense mutations

Nonsense variants are the consequence of mutations that replace a single amino acid codon with a nonsense codon.

Naming

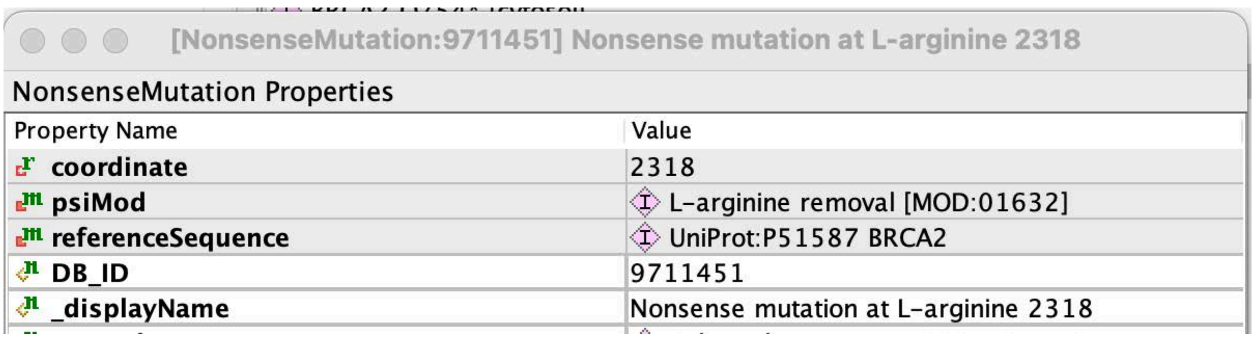
Format: “prefix” + “amino_acid” + “position” + “*” (stop codon asterisk shown in red just so it doesn’t get lost in the background)

Where “prefix” is the HGNC gene name, the amino acid is represented by single letter code and the stop codon is represented by an asterisk (*)

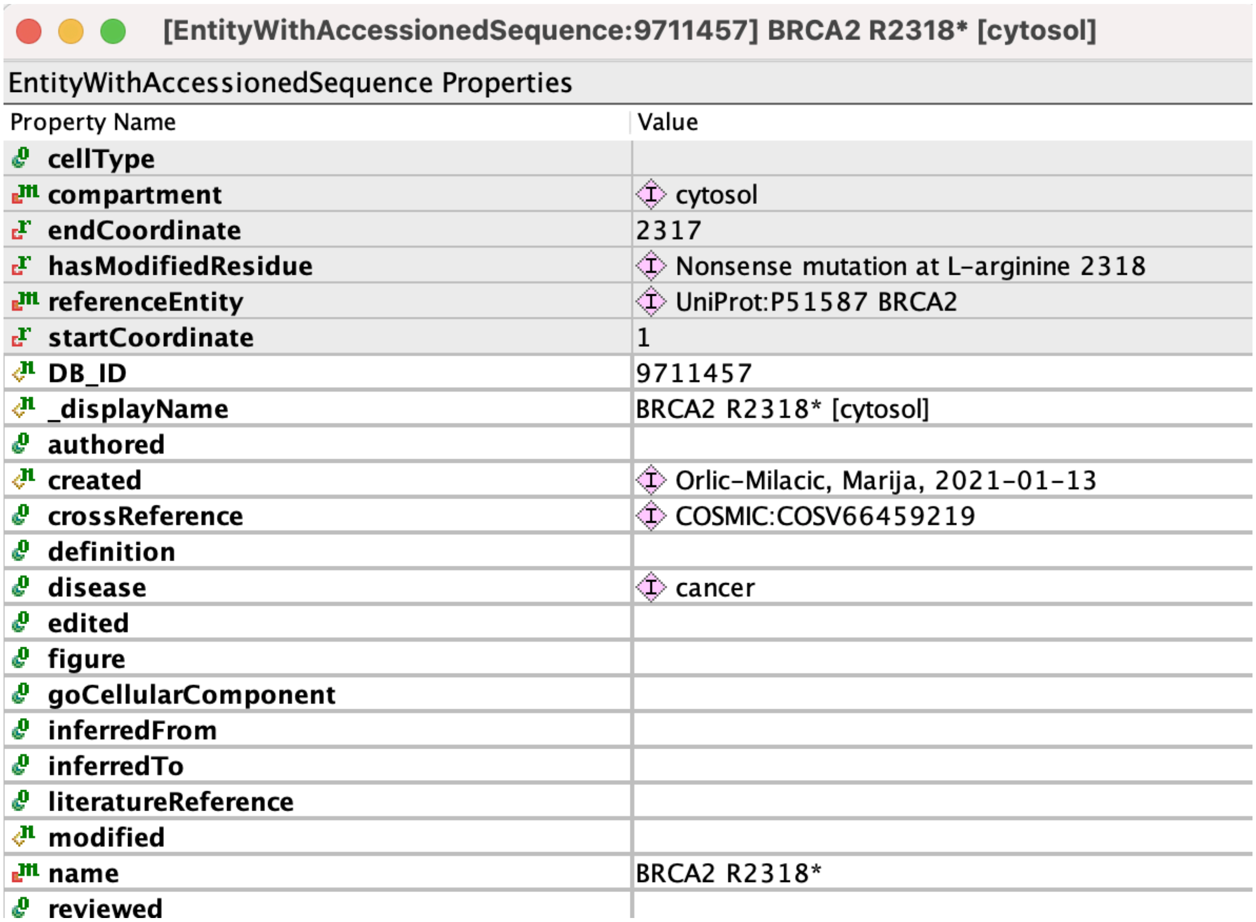
Example : BRCA2 R2318* - arginine 2318 replaced with stop codon

How to annotate

Nonsense mutations are captured using the “NonsenseMutation” subclass of “ReplacedResidue”. A single PSI MOD term is entered in the record indicating the removal of the WT amino acid. On the EWAS record, the endCoordinate is changed to be the last amino acid encoded by the variant (one less than the position of the stop codon).



On the EWAS record, the endCoordinate is changed to be the last amino acid encoded by the variant (one less than the position of the stop codon).



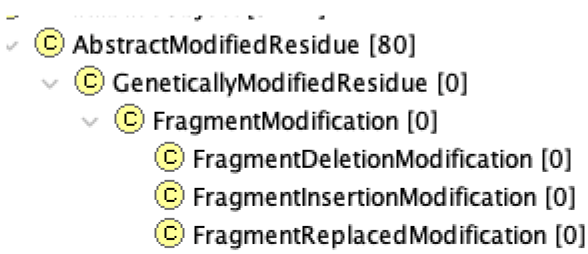
[EntityWithAccessionedSequence:9711457] BRCA2 R2318* [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	2317
hasModifiedResidue	Nonsense mutation at L-arginine 2318
referenceEntity	UniProt:P51587 BRCA2
startCoordinate	1
DB_ID	9711457
_displayName	BRCA2 R2318* [cytosol]
authored	
created	Orlic-Milacic, Marija, 2021-01-13
crossReference	COSMIC:COSV66459219
definition	
disease	cancer
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	
modified	
name	BRCA2 R2318*
reviewed	

FragmentModifications

The next section describes slightly more complicated variants involving changes to more than a single amino acid in the reference protein. These variants are captured under the AbstractModifiedResidue class “FragmentModification” that itself has three subclasses, as shown below:



In-frame deletions:

These variants are the consequence of the removal of a stretch of the WT amino acids between the initiator methionine and the stop codon without changing the reading frame of the protein.

Naming

Format: “prefix” + “first deleted amino_acid” +”first deleted amino acid position” + “_” +”last deleted amino acid” +”last amino acid position” + “del”

Where “prefix” is the HGNC gene name, the deleted amino acids are represented by single letter codes separated by underscore.

Example: HK1 H577_C672del: a deletion of residues H577 to C672, inclusive

How to annotate

Deletions are captured using the FragmentDeletionModification class specifying the start and stop coordinate of the deleted stretch of amino acids as shown below:

[FragmentDeletionModification:5621917] Deletion of residues 577 to 672

FragmentDeletionModification Properties

Property Name	Value
endPositionInReferenceSequence	672
referenceSequence	UniProt:P19367 HK1
startPositionInReferenceSequence	577
DB_ID	5621917
_displayName	Deletion of residues 577 to 672

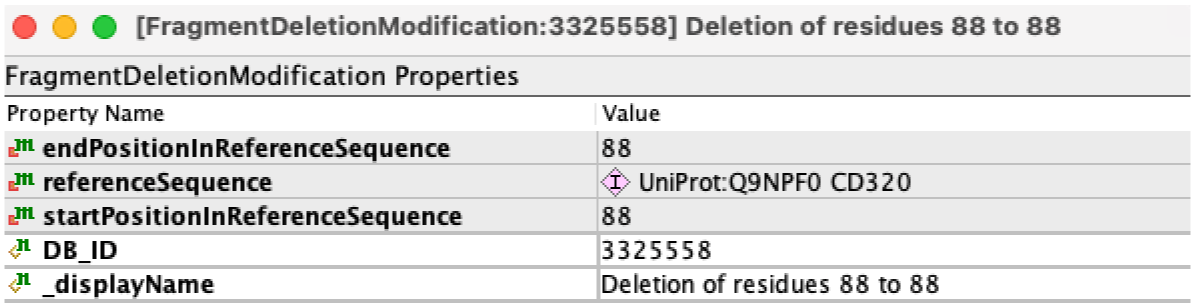
The corresponding EWAS record. Note that in this case, the end coordinate of the protein is not adjusted relative to the WT protein to account for the deleted stretch of amino acids.

[EntityWithAccessionedSequence:5621873] HK1 H577_C672del [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	917
hasModifiedResidue	Deletion of residues 577 to 672
referenceEntity	UniProt:P19367 HK1
startCoordinate	1
DB_ID	5621873
_displayName	HK1 H577_C672del [cytosol]
authored	
created	Jassal, Bijay, 2014-09-05
crossReference	
definition	
disease	congenital nonspherocytic hemolytic anemia

For deletion of a single amino acid, the start and end position of the FragmentDeletionModification are the same:

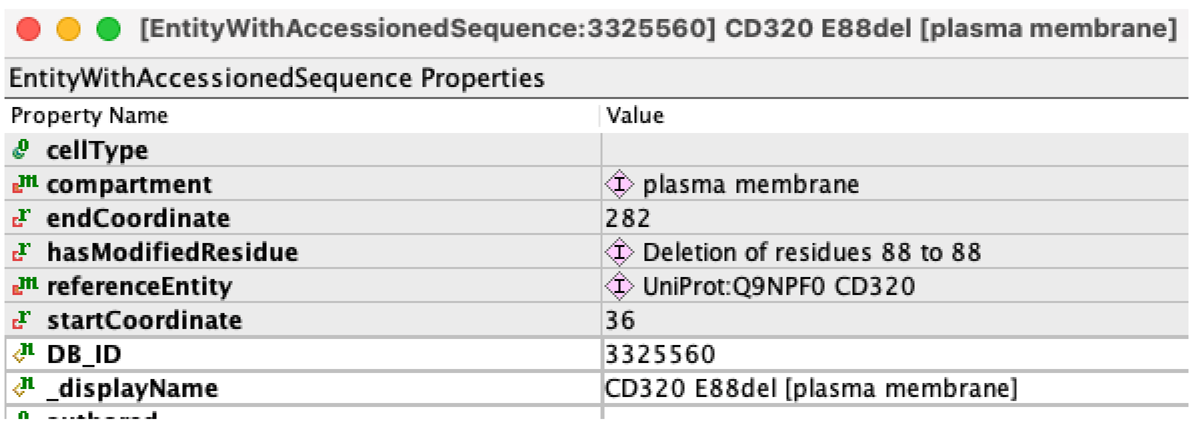


[FragmentDeletionModification:3325558] Deletion of residues 88 to 88

FragmentDeletionModification Properties

Property Name	Value
endPositionInReferenceSequence	88
referenceSequence	UniProt:Q9NPF0 CD320
startPositionInReferenceSequence	88
DB_ID	3325558
_displayName	Deletion of residues 88 to 88

And the EWAS is named without a range of amino acids (CD320 E88del) as shown below:



[EntityWithAccessionedSequence:3325560] CD320 E88del [plasma membrane]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	plasma membrane
endCoordinate	282
hasModifiedResidue	Deletion of residues 88 to 88
referenceEntity	UniProt:Q9NPF0 CD320
startCoordinate	36
DB_ID	3325560
_displayName	CD320 E88del [plasma membrane]

In-frame insertions

These variants are the consequence of adding a stretch of the amino acids between the initiator methionine and the stop codon without changing the reading frame of the protein. The inserted amino acids are NOT a copy of the sequence immediately upstream of the insertion site. If this is the case, these are annotated as duplications (below).

For example, for a sequence “YMVA”,

An insertion is YYMVAMVA - the YMVA in red is NOT a direct repeat of the upstream because the upstream sequence is interrupted by the insertion.

YMVA with a duplication is YMVAYMVA - the YMVA in red IS a direct repeat of the upstream sequence.

Naming

Format:

"prefix" + "N-terminal flanking amino acid" +"position of N-terminal flanking amino acid" + "_" +"C-terminal flanking amino acid" + "position of C-terminal flanking amino acid" +"ins" + "inserted sequence"

Where "prefix" is the HGNC gene name, the N- and C-terminal flanking amino acids are represented by single letter codes separated by underscore, and the inserted sequence is listed as single letter amino acid codes.

Example: - EGFR D770_N771insNPG - in this variant the three amino acids NPG are inserted between D770 and N771.

The WT protein sequence in this region is 770-DNPHV-774 and the variant sequence is 770-DNPGNPHV-(original sequence in blue, insertion in red). NPG is inserted at position 771.

How to annotate

Insertions are captured using FragmentReplacedModification class specifying the start and stop coordinates which overlap with positions of the N-terminal and C-terminal flanking amino acids, respectively, and specifying the alteredAminoAcidFragment, so that the first letter in the string is the one letter code for the N-terminal flanking amino acid, the last letter in the string is the one letter code for the D-terminal amino acid and the string in between correspond to the one letter codes of the inserted amino acids. The insertion site should be (StartPositioninReferenceSequence+1).

[FragmentReplacedModification:9831473] Replacement of residues 770 to 771 by DNPGN

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	DNPGN
endPositionInReferenceSequence	771
referenceSequence	UniProt:P00533 EGFR
startPositionInReferenceSequence	770
DB_ID	9831473
_displayName	Replacement of residues 770 to 771 by DNPGN

In the EWAS record, the end coordinate of the WT protein is not changed to accommodate the insertion.

[EntityWithAccessionedSequence:1996322] EGFR D770_N771insNPG [plasma membrane]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	plasma membrane
endCoordinate	1210
hasModifiedResidue	Replacement of residues 770 to 771 by DNPGN
referenceEntity	UniProt:P00533 EGFR
startCoordinate	25
DB_ID	1996322
_displayName	EGFR D770_N771insNPG [plasma membrane]
authored	
created	Orlic-Milacic, Marija, 2011-11-19
crossReference	
definition	
disease	cancer
	lung cancer
	non-small cell lung carcinoma
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	Structural, biochemical, and clinical characterization of...
modified	Orlic-Milacic, M, 2013-06-06
	Orlic-Milacic, Marija, 2023-03-20
name	EGFR D770_N771insNPG
	Epidermal Growth Factor Receptor D770_N771insNPG
	EGFR Asp770_Asn771insAsnProGly

In-frame duplications

These variants are the consequence of in-frame insertion of a stretch of the amino acid codons between the initiator methionine and the stop codon without changing the reading frame of the protein. The inserted amino acids ARE a copy of the sequence immediately upstream of the insertion site.

Naming

Format: “prefix” + “first amino acid in the duplicated string” + “position of the first amino acid in the duplicated string” + “_” +”last amino acid in the duplicated string” + “position of the last amino acid in the duplicated string” + “dup”

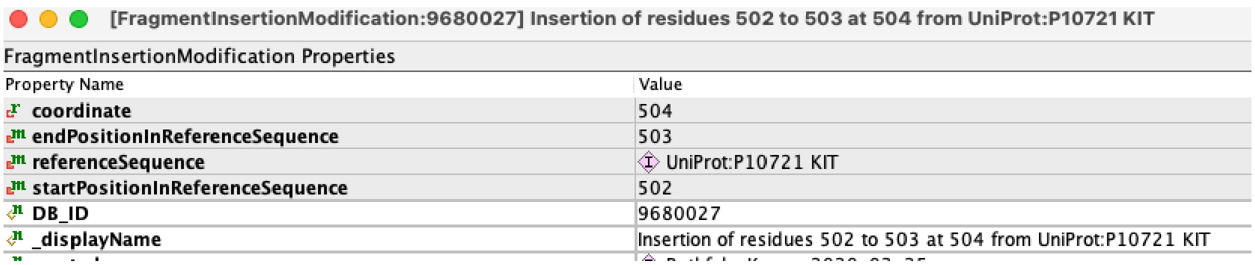
Where “prefix” is the HGNC gene name, the amino acids flanking the duplication are represented by single letter codes.

Example: KIT A502_Y503dup: a duplication of residues A502 and Y503. Because duplications are defined to happen at the C-terminal end of the sequence, the duplicated amino acids are located at coordinate 504, as shown below. For duplications, the site of insertion in the reference EWAS should always be (insertion end coordinate +1).

So in this example, WT KIT sequence is A502,Y503,F504,N505 etc and the variant sequence in A502,Y503,A502,Y503,F504,N505 etc, with duplicated residues shown in red and located at position 504 of WT sequence

How to annotate

Duplications are captured using the FragmentInsertionModification class specifying the start and end coordinates of the duplicated stretch of amino acids in the reference sequence, and with the coordinate of the insertion at (endPositioninReferenceSequence+1) as shown below:



Notice that the referenceSequence of the FragmentInsertionModification is the same as the referenceEntity of the disease variant EWAS itself. This is in contrast to the annotation of fusion proteins, which are also annotated with FragmentInsertionModfications, as described below - but in the case of fusion proteins, the referenceEntity of the FragmentInsertionModification is different from that of the disease variant EWAS referenceEntity.

As is the case with in-frame deletions and in-frame insertions, the end coordinate of the duplication variant EWAS is NOT changed relative to the WT protein.

[EntityWithAccessionedSequence:9680030] KIT A502_Y503dup [plasma membrane]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	plasma membrane
endCoordinate	976
hasModifiedResidue	Insertion of residues 502 to 503 at 504 from UniProt:P10721 KIT
referenceEntity	UniProt:P10721 KIT
startCoordinate	26
DB_ID	9680030
_displayName	KIT A502_Y503dup [plasma membrane]
authored	
created	Rothfels, Karen, 2020-03-25
crossReference	COSMIC:COSM12444
definition	
disease	cancer gastrointestinal stromal tumor
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	Pediatric mastocytosis is a clonal disease associated with D816V ...
modified	
name	KIT A502_Y503dup
reviewed	

Fusion proteins

Fusion proteins are the consequence of an in-frame fusion of coding sequences from two different genes, usually through chromosomal translocation. By convention, Reactome annotates these fusions as the insertion of a stretch of the C-terminal partner as a FragmentInsertionModification into the hasModifiedResidue slot of the N-terminal partner.

Naming

Format: “prefix (N-terminal protein)” + “(included amino acids from N-terminal protein)” + ”-” “prefix (C-terminal protein)” + “(included amino acids from C-terminal protein)” +”fusion”.

Where “prefix” is the HGNC gene name and (included amino acids) represents the start and end coordinates- but not the amino acid codes- of each fusion partner relative to their WT reference sequences, in accordance with Reactome’s controlled vocabulary for naming of protein fragments.

Example: ZMYM2(1-913)-FGFR1(429-822) - this is the insertion of FGFR1 residues 429-822 (relative to full-length WT FGFR1) after residue 913 of ZMYM2.

How to annotate

Fusions are captured as “FragmentInsertionModifications” class, specifying the start and stop codon of the C-terminal partner with coordinates from the WT C-terminal partner as shown below:

[FragmentInsertionModification:1604543] Insertion of residues 429 to 822 at 914 from UniProt:P11362 FGFR1

FragmentInsertionModification Properties

Property Name	Value
coordinate	914
endPositionInReferenceSequence	822
referenceSequence	UniProt:P11362 FGFR1
startPositionInReferenceSequence	429
DB_ID	1604543
_displayName	Insertion of residues 429 to 822 at 914 from UniProt:P11362 FGFR1

In the EWAS record, the end coordinate of the N-terminal partner is changed to the last amino acid of the N-terminal partner sequence that is contained in the fusion.

[EntityWithAccessionedSequence:1604544] ZMYM2(1-913)-FGFR1(429-822) fusion [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	913
hasModifiedResidue	Insertion of residues 429 to 822 at 914 from UniProt:P11362 FGFR1
referenceEntity	UniProt:Q9UBW7 ZMYM2
startCoordinate	1
DB_ID	1604544
_displayName	ZMYM2(1-913)-FGFR1(429-822) fusion [cytosol]
authored	
created	Rothfels, K, 2011-09-16
crossReference	
definition	
disease	subacute leukemia myelodysplastic myeloproliferative cancer precursor lymphoblastic lymphoma/leukemia
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	FGFR1 is fused with a novel zinc-finger gene, ZNF198, in the t(8;13) leukae... Consistent fusion of ZNF198 to the fibroblast growth factor receptor-1 in the... Fibroblast growth factor receptor 1 is fused to FIM in stem-cell myeloprolifer...
modified	Rothfels, K, 2011-09-21 Rothfels, K, 2011-10-05 Rothfels, K, 2011-11-22 Rothfels, Karen, 2014-11-20 Rothfels, Karen, 2021-09-13
name	ZMYM2(1-913)-FGFR1(429-822) fusion ZMYM2-FGFR1 fusion t(8;13) ZMYM2-FGFR1 fusion ZNF198-FGFR1 fusion ZNF198(1-913)-FGFR1(429-822) fusion t(8;13) ZNF198-FGFR1 fusion FIM-FGFR1 fusion RAMP-FGFR1 fusion

Post-translational modifications to either partner in the fusion protein are numbered according to each respective WT reference protein sequence and do not reflect the amino acid positions in the fusion. For instance, if in the context of the fusion protein, the FGFR1 partner is phosphorylated at (WT FGFR1 position) Y766, the fusion EWAS would be ZMYM2-pY766-FGFR1, despite the fact that in linear sequence the phosphorylation occurs at residue 1250 of the fusion (913+(822-429).

Fusion proteins with in-frame non-coding insertion sequences

If a fusion protein includes an in-frame insertion of amino acids encoded by an intronic or UTR sequence that are not present in the reference protein sequences of either of the two fusion partners, the insertion is defined in the name of the fusion protein in the format:

Naming

“prefix (N-terminal protein)”+ “(included amino acids from N-terminal protein)” + “ins” + “inserted sequence” + “-” + “prefix (C-terminal protein)” + “(included amino acids from C terminal protein)”+ fusion,

where the inserted sequence is listed as single letter amino acid codes.

Example: FIP1L1(1-391)insAQCP-PDGFRA(577-1089) fusion - this is an insertion of the amino acid string AQCP between the amino acid 391 of FIP1L1 and amino acid 577 of PDGFRA.

How to annotate:

An in-frame insertion of an amino acid string encoded by an intronic or UTR sequence that is not present in the reference protein sequences of either of the two fusion partners is annotated using the replacedFragmentModification, specifying that the last amino acid of the N-terminal fusion partner (proline at position 391 of FIP1L1 in this example) is replaced by a string of amino acids, where the first amino acid in the string is identical to the last amino acid in the N-terminal fragment, followed by the actual inserted string - PAQCP in this example, with the actual inserted string shown in bold.

[FragmentReplacedModification:9672470] Replacement of residues 391 to 391 b...

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	PAQCP
endPositionInReferenceSequence	391
referenceSequence	UniProt:Q6UN15 FIP1L1
startPositionInReferenceSequence	391
DB_ID	9672470
_displayName	Replacement of residues 391 to 391 by PAQCP

In the EWAS record, the end coordinate of the N-terminal partner is changed to the last amino acid of the N-terminal partner sequence that is contained in the fusion.

[EntityWithAccessionedSequence:9672502] FIP1L1(1-391)insAQCP-PDGFRA(577-1089) fusion [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	391
hasModifiedResidue	Replacement of residues 391 to 391 by PAQCP Insertion of residues 577 to 1089 at 392 from UniProt:P16234 P...
referenceEntity	UniProt:Q6UN15 FIP1L1
startCoordinate	1
DB_ID	9672502
_displayName	FIP1L1(1-391)insAQCP-PDGFRA(577-1089) fusion [cytosol]

Indels

In-frame indels are variants that are the consequence of simultaneous insertion and removal of a stretch of amino acids from the WT sequence without change in the reading frame..

Naming

Format: “prefix” + “first deleted amino acid” + “position of first deleted amino acid” + “_” + “last deleted amino acid” + “position of last deleted amino acid” + “delins” + “inserted sequence”

Where “prefix” is the HGNC gene name, amino acids+positions deleted specifies position and single letter code of the removed amino acids and inserted sequence specifies the new amino acids with their single letter codes.

Example: F8 V2144_Y2351delinsHGVLENGKNWEPSYRW - this is removal of residues V2144 to Y2351 from F8 and their replacement with the specified stretch of amino acids

How to annotate

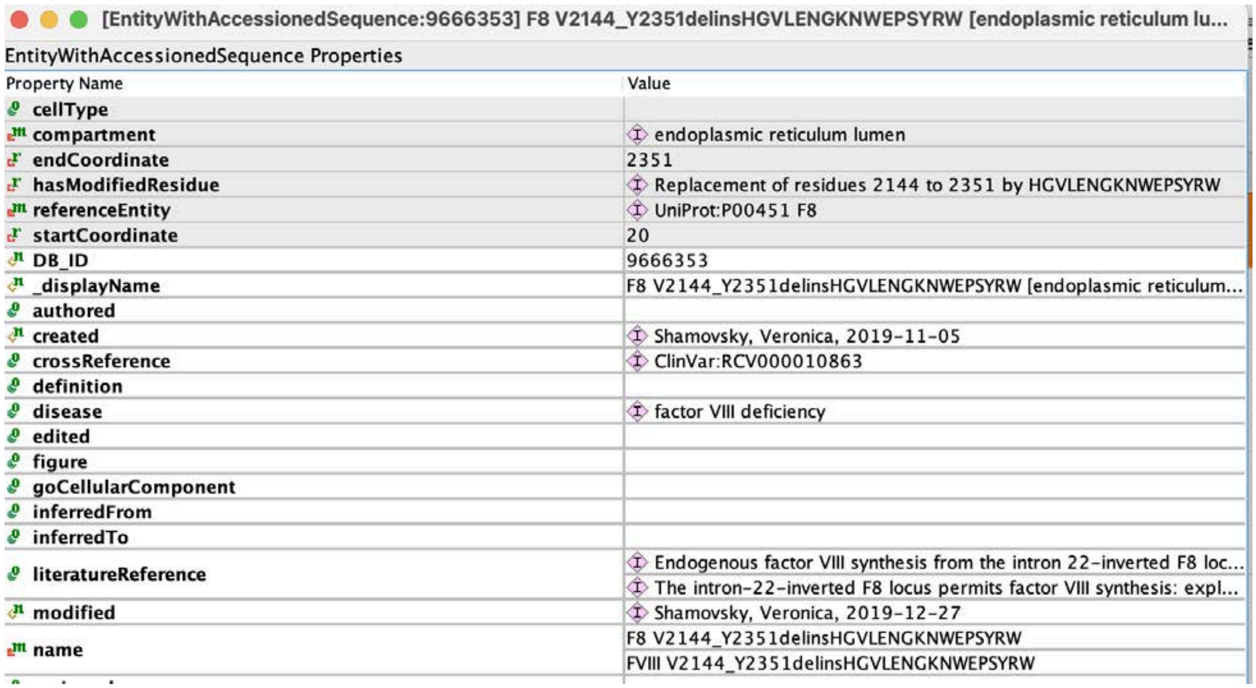
In-frame indels are captured with the FragmentReplacedModification class, specifying the start and end coordinates of the deleted amino acid stretch and specifying the inserted amino acid stretch that replaces them as an alteredAminoAcidfragment, using single letter codes entered as the start and stop positions, as shown below.

[FragmentReplacedModification:9666370] Replacement of residues 2144 to 2351 by HGVLENGKNWEPSYRW

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	HGVLENGKNWEPSYRW
endPositionInReferenceSequence	2351
referenceSequence	UniProt:P00451 F8
startPositionInReferenceSequence	2144
DB_ID	9666370
_displayName	Replacement of residues 2144 to 2351 by HGVLENGKNWEPSYRW

The end coordinate of the variant protein on the EWAS record is not changed relative to the WT protein to accommodate the indel.


[EntityWithAccessionedSequence:9666353] F8 V2144_Y2351delinsHGVLENGKNWEPSYRW [endoplasmic reticulum lu...

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	endoplasmic reticulum lumen
endCoordinate	2351
hasModifiedResidue	Replacement of residues 2144 to 2351 by HGVLENGKNWEPSYRW
referenceEntity	UniProt:P00451 F8
startCoordinate	20
DB_ID	9666353
_displayName	F8 V2144_Y2351delinsHGVLENGKNWEPSYRW [endoplasmic reticulum...
authored	
created	Shamovsky, Veronica, 2019-11-05
crossReference	ClinVar:RCV000010863
definition	
disease	factor VIII deficiency
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	Endogenous factor VIII synthesis from the intron 22-inverted F8 loc... The intron-22-inverted F8 locus permits factor VIII synthesis: expl...
modified	Shamovsky, Veronica, 2019-12-27
name	F8 V2144_Y2351delinsHGVLENGKNWEPSYRW FVIII V2144_Y2351delinsHGVLENGKNWEPSYRW

Frameshifts

Perhaps the most challenging to annotate! These are often incompletely described in papers, and in order to capture the amino acid changes that result, the curator may have to do quite a bit of leg work.

Frameshift variants are the consequence of mutations that lead to the insertion or deletion of nucleotides where the total number of inserted or deleted nucleotides is not a multiple of three, leading to a change in the reading frame.The published literature generally provides an exact description of the mutation at the *DNA* level, from which the curator may need to infer the altered C-terminal sequence of the mutated protein to create a “FragmentReplacedModifcation” in Reactome.

Before embarking on this exercise, however, consider the biology. Most (perhaps all) nonsense mutations that occur anywhere in a gene except its final exon trigger the process of nonsense-mediated decay. This process is part of mRNA maturation and its effect is to cause the mutant RNA to be degraded before it is ever translated. Thus, at the protein level annotated by Reactome, the result of the mutation is almost certainly disappearance of the protein, not appearance of an altered one. If the goal is to annotate the existence of a non-functional protein a better choice, other things being equal, would be to choose a missense mutation that has been shown to encode a protein whose normal activity (binding, catalysis, transport, etc.) is lost.

Naming

Format: "prefix" + "first amino acid altered due to frameshift" + "position of the first altered amino acid" + "fs" + "*" + "position of the stop codon relative to the first altered amino acid"

Where "prefix" is the HGNC gene name, and is followed by a single letter code description of the first affected amino acid, "fs" indicates frameshift, and "*#" indicates the length of amino acids before the first stop codon is reached.

Example: EXT1 L490Rfs*9 - in this protein, a frameshift mutation causes conversion of L490 to R, and is followed by a novel stretch of 7 amino acids (8 altogether, counting from the first substituted amino acid) before a stop codon is reached at the 9th position.

How to annotate

Frameshifts are captured using the FragmentReplacedModification class, specifying the start and end coordinates of the altered sequence, where the startPositionInReferenceSequence matches the position of the first amino acid substituted due to frameshift (490 in this example) and the endPositionInReferenceSequence is calculated by the formula:

endPositionInReferenceSequence = startPositionInReferenceSequence + frameshift length - 2

The frameshift length is specified in the name of the frameshift variant (the number after the asterisk and equals the length of the substituted amino acid string + 1 (as it takes into account the stop codon).

In the EXT1 L490Rfs*9 example,

endPositionInReferenceSequence = 490 + 9 - 2 = 497

The altered amino acid sequence itself is specified using the alteredAminoAcidFragment attribute. The end coordinate of the EWAS is changed to reflect the new C-terminal amino acid.

[FragmentReplacedModification:5357998] Replacement of residues 490 to 497...

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	RSLSPSQC
endPositionInReferenceSequence	497
referenceSequence	UniProt:Q16394 EXT1
startPositionInReferenceSequence	490
DB_ID	5357998
_displayName	Replacement of residues 490 to 497 by RSLSPSQC

If the paper does not completely describe all the details needed to annotate a frameshift variant, and if such details cannot be obtained from a disease variant database, they have to be deciphered.

The basic steps are

1. Alter the reference coding sequence for the wild-type protein to match the coding sequence of the variant. The reference coding sequence can be obtained from teh RefSeq mRNA cross-referenced by the UniProt record that is used as the reference for the wild-type proteins.
2. Translate the altered coding sequence in silico using EMBOSS TransSeq.
3. Align the resultant mutant peptide sequence with wild type protein sequence using EMBOSS Needle open software to determine the alteredAminoAcidFragment and the anime of the frameshift protein variant if not provided in the research article.

UniProt records provide the normal peptide sequence in FASTA format. They also provide a link to RefSeq (under the "Sequence databases" section) which links to the NCBI reference mRNA sequence. This is the starting point from where the nucleotide sequence is copied to Transeq below.

Several EMBOSS programs are available to translate and compare sequences.

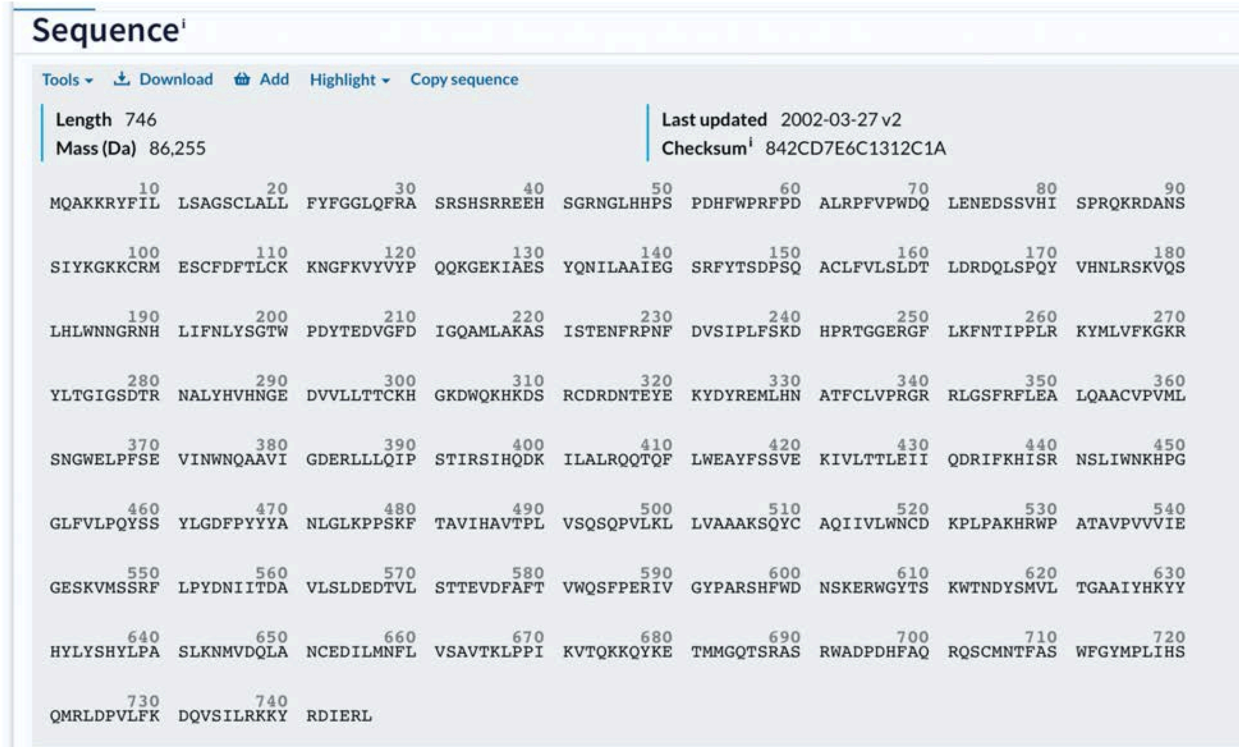
*Transeq (https://www.ebi.ac.uk/jdispatcher/st/emboss_transeq) translates nucleic acid sequences to their corresponding peptide sequences.

*Needle (https://www.ebi.ac.uk/jdispatcher/psa/emboss_needle) can align two peptide sequences to provide a global sequence alignment.

An example below shows how these steps are utilized to decipher a frameshift mutation.

The disorder "Hereditary multiple exostoses 1" (EXT1) is caused by loss of function mutations in Exostosin 1 (EXT1). One mutation is a 1-bp deletion at nucleotide 1469 in the EXT1 gene, resulting in a frameshift mutation with a premature stop codon (Philippe et al. 1997 , Ahn et al. 1995).

Go to the UniProt record for human EXT1 (https://www.uniprot.org/uniprotkb/Q16394/entry), and scroll to the FASTA sequence for the WT peptide sequence.



Sequence

Tools ▾ Download Add Highlight ▾ Copy sequence

Length 746
Mass (Da) 86,255

Last updated 2002-03-27 v2
Checksum 842CD7E6C1312C1A

10	20	30	40	50	60	70	80	90
MQAKKRYFIL	LSAGSCLALL	FYFGGLQFRA	SRSHSRREEH	SGRNGLHHPS	PDHFWPRFPD	ALRPFVPWDQ	LENEDSSVHI	SPRQKRDANS
100	110	120	130	140	150	160	170	180
SIYKGKKCRM	ESCFDFTLCK	KNGFKVYVYP	QQKGEKIAES	YQNILAAIEG	SRFYTSDPSQ	ACLFVLSLDT	LDRDQLSPQY	VHNLRSKVQS
190	200	210	220	230	240	250	260	270
LHLWNNGRNH	LIFNLYSGTW	PDYTEDVGFD	IGQAMLAKAS	ISTENFRPNF	DVSIPLFSKD	HPRTGGERGF	LKFNTIPPLR	KYMLVFKGKR
280	290	300	310	320	330	340	350	360
YLTGIGSDTR	NALYHVHNGE	DVVLLTTCKH	GKDWQKHKDS	RCDRDNTEYE	KYDYREMLHN	ATFCLVPRGR	RLGSFRFLEA	LQAACVPVML
370	380	390	400	410	420	430	440	450
SNGWELPFSE	VINWNQAAVI	GDERLLLQIP	STIRSIHQDK	ILALRQQTQF	LWEAYFSSVE	KIVLTTLEII	QDRIFKHISR	NSLIWNKHPG
460	470	480	490	500	510	520	530	540
GLFVLPQYSS	YLGDFPYYYA	NLGLKPPSKF	TAVIHAVTPL	VSQSQPVLKL	LVAAAKSQYC	AQIIVLWNCD	KPLPAKHRWP	ATAVPVVVIE
550	560	570	580	590	600	610	620	630
GESKVMSSRF	LPYDNIITDA	VLSLDEDTVL	STTEVDFAFT	VWQSFPERIV	GYPARSHFWD	NSKERWGYTS	KWTNDYSMVL	TGAAIYHKYY
640	650	660	670	680	690	700	710	720
HYLYSHYLPA	SLKNMVDQLA	NCEDILMNFL	VSAVTKLPPI	KVTQKKQYKE	TMMGQTSRAS	RWADPDHFAQ	RQSCMNTFAS	WFGYMPLIHS
730	740							
QMRLDPVLFK	DQVSILRKKY	RDIERL						

Click the “Download” link to obtain a text version of the sequence.

```
>sp|Q16394|EXT1_HUMAN Exostosin-1 OS=Homo sapiens OX=9606 GN=EXT1 PE=1 SV=2
MQAKKRYFILLSAGSCLALLFYFGGLQFRASRSHSRREEHSGRNGLHHPSPDHFWPRFPD
ALRPFVPWDQLENEDSSVHISPRQKRDANSSIYKGKKCRMESCFDFTLCKKNGFKVYVYP
QQKGEKIAESYQNILAAIEGSRFYTSDPSQACLFVLSLDTLDRDQLSPQYVHNLRSKVQS
LHLWNNGRNHLIFNLYSGTWPDYTEDVGFDIGQAMLAKASISTENFRPNFDVSIPLFSKD
HPRTGGERGFLKFNTIPPLRKYMLVFKGKRYLTGIGSDTRNALYHVHNGEDVVLLTTCKH
GKDWQKHKDSRCDRDNTEYEKYDYREMLHNATFCLVPRGRRLGSFRFLEALQAACVPVML
SNGWELPFSEVINWNQAAVIGDERLLLQIPSTIRSIHQDKILALRQQTQFLWEAYFSSVE
KIVLTTLEIIQDRIFKHISRNSLIWNKHPGGLFVLPQYSSYLGDFPYYYANLGLKPPSKF
TAVIHAVTPLVSQSQPVLKLLVAAAKSQYCAQIIVLWNCDKPLPAKHRWPATAVPVVVIE
GESKVMSSRFLPYDNIITDAVLSLDEDTVLSTTEVDFAFTVWQSFPERIVGYPARSHFWD
NSKERWGYTSKWTNDYSMVLTGAAIYHKYYHYLYSHYLPASLKNMVDQLANCEDILMNFL
VSAVTKLPPIKVTQKKQYKETMMGQTSRASRWADPDHFAQRQSCMNTFASWFGYMPLIHS
QMRLDPVLFKDQVSILRKKYRDIERL
```

Copy the peptide sequence to EMBOSS Needle and paste it into the first window.

A mutated nucleotide sequence needs to be created from which the translated mutant peptide sequence will be obtained.

In the UniProt record for EXT1, scroll to Sequence databases and select the nucleotide sequence from RefSeq (format NM_xxxxxx.x)

Sequence databases

CCDS | CCDS6324.1   RefSeq | NP_000118.2 NM_000127.2

SEQUENCE	PROTEIN	MOLECULE TYPE	STATUS
S79639 (EMBL | GenBank | DDBJ)	AAB62283.1 (EMBL | GenBank | DDBJ)	mRNA	
AK313129 (EMBL | GenBank | DDBJ)	BAG35949.1 (EMBL | GenBank | DDBJ)	mRNA	
CH471060 (EMBL | GenBank | DDBJ)	EAW91972.1 (EMBL | GenBank | DDBJ)	Genomic DNA	
BC001174 (EMBL | GenBank | DDBJ)	AAH01174.1 (EMBL | GenBank | DDBJ)	mRNA	
U70539 (EMBL | GenBank | DDBJ)	AAC51154.1 (EMBL | GenBank | DDBJ)	Genomic DNA	

This opens the NCBI nucleotide record for EXT1 mRNA. (https://www.ncbi.nlm.nih.gov/nuccore/NM_000127.2). Scroll down to the "ORIGIN" section where you will find the nucleotide sequence conveniently numbered. Copy the sequence and paste it into EMBOSS Transeq.

```
ORIGIN
        1 ggcgaccgaa cgcggcggtc ggcagcgttc gcgcgggggc ctgcgaagcg ctgctcgggg
       61 ccggcactgc ccgcggggag gacgcgccgc cgccgccacc cagcgccgcc gccgccgccg
      121 cctccagccg ggccgccgcg cgtcccgggg gccggccccg cgagcgcagg agtaaacacc
      181 gccggagtct tggagccgct gcagaaggga ataaagagag atgcagggat ttgtgaggtt
      241 acggcgcccc agctgcaaga tgcactagcc ggctgaaccc gggatcggct gacttgttgg
      301 aaccggagtg ctctgcacgg agagtggtgg atgagttgaa gttgccttcc cggggctcat
      361 tttccacgct gccgagagga atccgagagg caaggcaatc acttcgtctt gccattgatt
      421 gggtatcggg agcttttttt ttctcccctc tctctttctt ttcctccgtc ttgttgcatg
      481 caagaaaatt acagtccgct gctcgcccgc cctgggtgcg agatattcag ccccgctctc
      541 tcccgtgcat tgtgcaaccc aaagatgaaa gaccgaaggg gagaaagtta aagaaatcgc
      601 ccacatgcgc tggatcagtc cacggcttgg ggaaaggcat ccagagaagg tgggagcgga
      661 gagtttgaag tctttacagg cgggaagatg gcggactgga gctgaaagtg ttgattggga
      721 aacttgggtg attcttgtgt ttatttacaa tcctcttgac ccaggcagga cacatgcagg
      781 ccaaaaaacg ctatttcatc ctgctctcag ctggctcttg tctcgccctt ttgttttatt
      841 tcggaggctt gcagtttagg gcatcgagga gccacagccg gagagaagaa cacagcggta
      901 ggaatggctt gcaccacccc agtccggatc atttctggcc ccgcttcccg gacgctctgc
      961 gccccttcgt tccttgggat caattggaaa acgaggattc cagcgtgcac atttcccccc
     1021 ggcagaagcg agatgccaac tccagcatct acaaaggcaa gaagtgccgc atggagtcct
     1081 gcttcgattt caccccttgc aagaaaaacg gcttcaaagt ctacgtatac ccacagcaaa
     1141 aaggggagaa aatcgccgaa agttaccaaa acattctagc ggccatcgag ggctccaggt
     1201 tctacacctc ggaccccagc caggcgtgcc tctttgtcct gagtctggat actttagaca
     1261 gagaccagtt gtcacctcag tatgtgcaca atttgagatc caaagtgcag agtctccact
     1321 tgtggaacaa tggtaggaat catttaattt ttaatttata ttccggcact tggcctgact
     1381 acaccgagga cgtggggttt gacatcggcc aggcgatgct ggccaaagcc agcatcagta
     1441 ctgaaaactt ccgacccaac tttgatgttt ctattcccct cttttctaag gatcatccca
     1501 ggacaggagg ggagaggggg tttttgaagt tcaacaccat ccctcctctc aggaagtaca
     1561 tgctggtatt caaggggaag aggtacctga cagggatagg atcagacacc aggaatgcct
     1621 tatatcacgt ccataacggg gaggacgttg tgctcctcac cacctgcaag catggcaaag
     1681 actggcaaaa gcacaaggat tctcgctgtg acagagacaa caccgagtat gagaagtatg
     1741 attatcggga aatgctgcac aatgccactt tctgtctggt tcctcgtggt cgcaggcttg
     1801 ggtccttcag attcctggag gctttgcagg ctgcctgcgt ccctgtgatg ctcagcaatg
     1861 gatgggagtt gccattctct gaagtgatta attggaacca agctgccgtc ataggcgatg
     1921 agagattgtt attacagatt ccttctacaa tcaggtctat tcatcaggat aaaatcctag
     1981 cacttagaca gcagacacaa ttcttgtggg aggcttattt ttcttcagtt gagaagattg
     2041 tattaactac actagagatt attcaggaca gaatattcaa gcacatatca cgtaacagtt
     2101 taatatggaa caaacatcct ggaggattgt tcgtactacc acagtattca tcttatctgg
     2161 gagattttcc ttactactat gctaatttag gtttaaagcc cccctccaaa ttcactgcag
     2221 tcatccatgc ggtgaccccc ctggtctctc agtcccagcc agtgttgaag cttctcgtgg
     2281 ctgcagccaa gtcccagtac tgtgcccaga tcatagttct atggaattgt gacaagcccc
     2341 taccagccaa acaccgctgg cctgccactg ctgtgcctgt cgtcgtcatt gaaggagaga
     2401 gcaaggttat gagcagccgt tttctgccct acgacaacat catcacagac gccgtgctca
     2461 gccttgacga ggacacggtg ctttcaacaa cagaggtgga tttcgccttc acagtgtggc
     2521 agagcttccc tgagaggatt gtggggtacc ccgcgcgcag ccacttctgg gataactcta
     2581 aggagcggtg gggatacaca tcaaagtgga cgaacgacta ctccatggtg ttgacaggag
     2641 ctgctattta ccacaaatat tatcactacc tatactccca ttacctgcca gccagcctga
     2701 agaacatggt ggaccaattg gccaattgtg aggacattct catgaacttc ctggtgtctg
     2761 ctgtgacaaa attgcctcca atcaaagtga cccagaagaa gcagtataag gagacaatga
     2821 tgggacagac ttctcgggct tcccgttggg ctgaccctga ccactttgcc cagcgacaga
     2881 gctgcatgaa tacgtttgcc agctggtttg gctacatgcc gctgatccac tctcagatga
     2941 ggctcgaccc cgtcctcttt aaagaccagg tctctatttt gaggaagaaa taccgagaca
     3001 ttgagcgact ttgaggaatc cggctgagtg gggagggga agcaagaagg gatgggggtc
     3061 aagctgctct ctcttcccag tgcagatcca ctcatcagca gagccagatt gtgccaacta
     3121 tccaaaaact tagatgagca gaatgacaaa aaaaaaaagg ccaatgagaa ctcaactcct
     3181 ggctcctggg actgcaccag actgctccaa actcacctca ctggcttctg tgtcccaaga
     3241 ctaggttgtg tacagtttaa ttatggaaca ttaaataatt atttttgaaa tgattgctat
     3301 gcaggtttaa actttttaa tgatcaaaac tattaaaaac cagagttctt tgtttaatca
     3361 aaaaaaaaaa aaaaaa
//
```

In the Transeq window where the sequence is copied to, edit the nucleotide sequence according to evidence from literature. The coding sequence in the RefSeq record starts at position 774. The nucleotide change, from literature, is a 1-bp thymidine deletion at 1469 (Philippe et al. 1997] (table 2), Ahn et al. 1995] (Fig.6)).

Table 2

Characteristics of EXT1 and EXT2 Gene Defects in Patients with Multiple Exostoses

Family	Location of Mutation	Mutation	Consequence	Name of Mutation[a]
Familial cases:				
1	EXT1/Intron 5	TAGgta→TAGata	Altered splicing	1417 (+1G→A)
2	EXT2/Exon 4	TAC→TAG(Tyr→Stop)	Nonsense	Y222X
6	EXT2/Exon 4	GAT→AAT (Asp→Asn)	Missense	D227N
9	EXT1/Exon 3	TGG→TGA (Trp→Stop)	Nonsense	W374X
19	EXT2/Exon 2	TATATC→TATTATC	Frameshift	77- or 78insT
21	EXT1/Exon2	GGT→GAT (Gly→Asp)	Missense	G339D
25	EXT2/Exon4	GAT→AAT (Asp→Asn)	Missense	D227N
36	EXT1/Exon 6	CCCCCCTGG→CCCCCCGG	Frameshift	1469delT[b]
Isolated cases (*de novo* mutations):				
5	EXT1/Exon 6	CCCCCCTGG→CCCCCCGG	Frameshift	1469delT[b]
7	EXT2/ Exon 8	CAG→TAG (Glu→Stop)	Nonsense	Q401X
13	EXT1/Exon 1	GCTTCCCGG→GCTTCCGG	Frameshift	174-, 175- or 176delC
35	EXT1/Exon 2	CGC→TGC (Arg→Cys)	Missense	R340C

[a] According to the nomenclature defined by Beaudet and Tsui (1993). For both EXT1 and EXT2, base no. 1 corresponds to the first base of the initiation codon in the cDNA sequences (Ahn et al. 1995; Stickens et al. 1996; Wuyts et al. 1996).

[b] Already reported by Ahn et al. (1995) and by Wells et al. (1997).

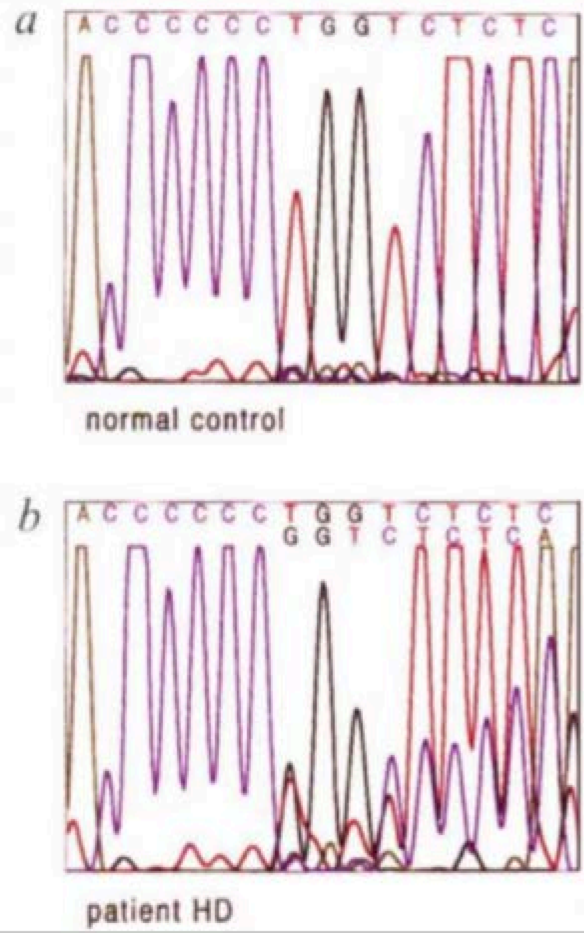
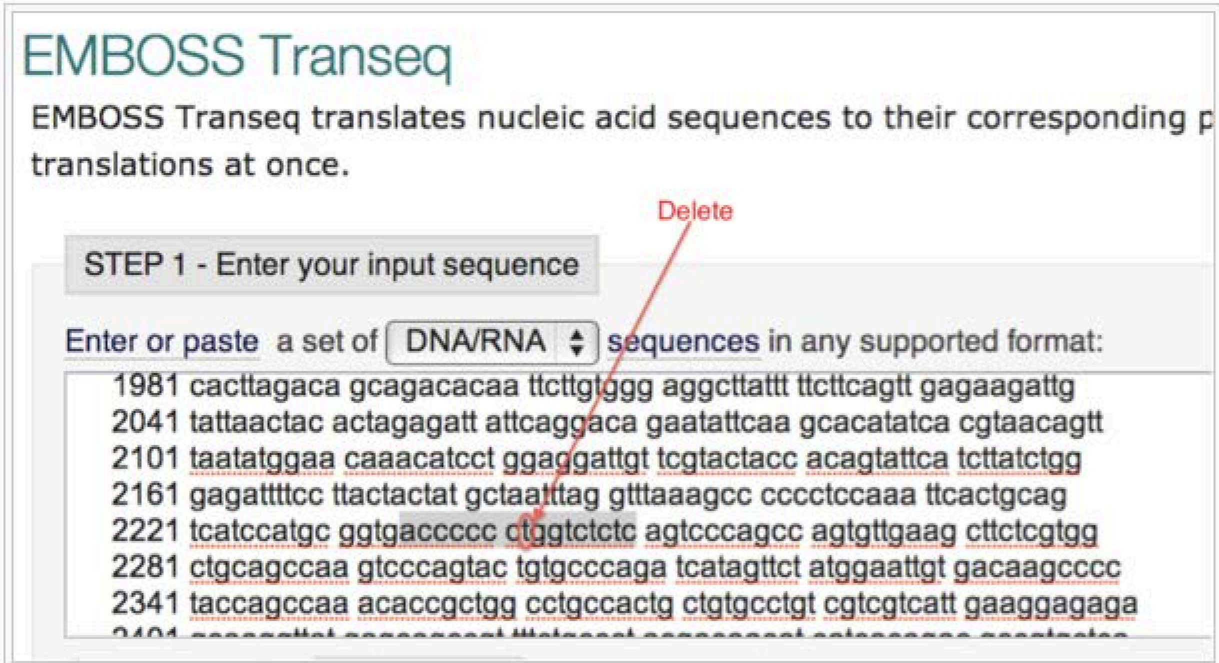


Fig. 6 Analysis of the *EXT1* mutation in family D. Direct sequencing of the PCR products amplified from genomic DNA with the 'forward' primer (see Methods). *a*, Sequence of a healthy individual corresponding to nucleotides 2113-2128 of the cDNA. *b*, Patient HD (individual I2) is heterozygous for the 1-bp deletion (T) at position 2120 of the cDNA. *c*, Segregation analysis in family D. The PCR products were digested with *Msp*I and separated on an agarose gel. DNAs of healthy family members remain uncut whereas approximately half of the PCR products from affected individuals are cut into 159- and 139-bp fragments. The point mutation cosegregates with the disease in family D and has been verified by direct sequencing in all family members.

Many times in literature, researchers only use the coding sequence in their experiments and count from that point onwards to the mutation site. As the coding sequence starts at 774, add that to 1469 (the deletion position) which equals 2243. Check this region in the sequence pasted into Transeq. The region matches that found in literature. Delete the t after the 6 c's and submit the job.

(Note, TranSeq can translate your sequence in any of the 6 reading frames. If you don't know which frame your initiator methionine is, you can translate the WT seq in all 3 forward reading frames and search the results for the amino acid sequence of the WT protein to orient yourself before submitting the variant allele DNA as described in the figure below.)

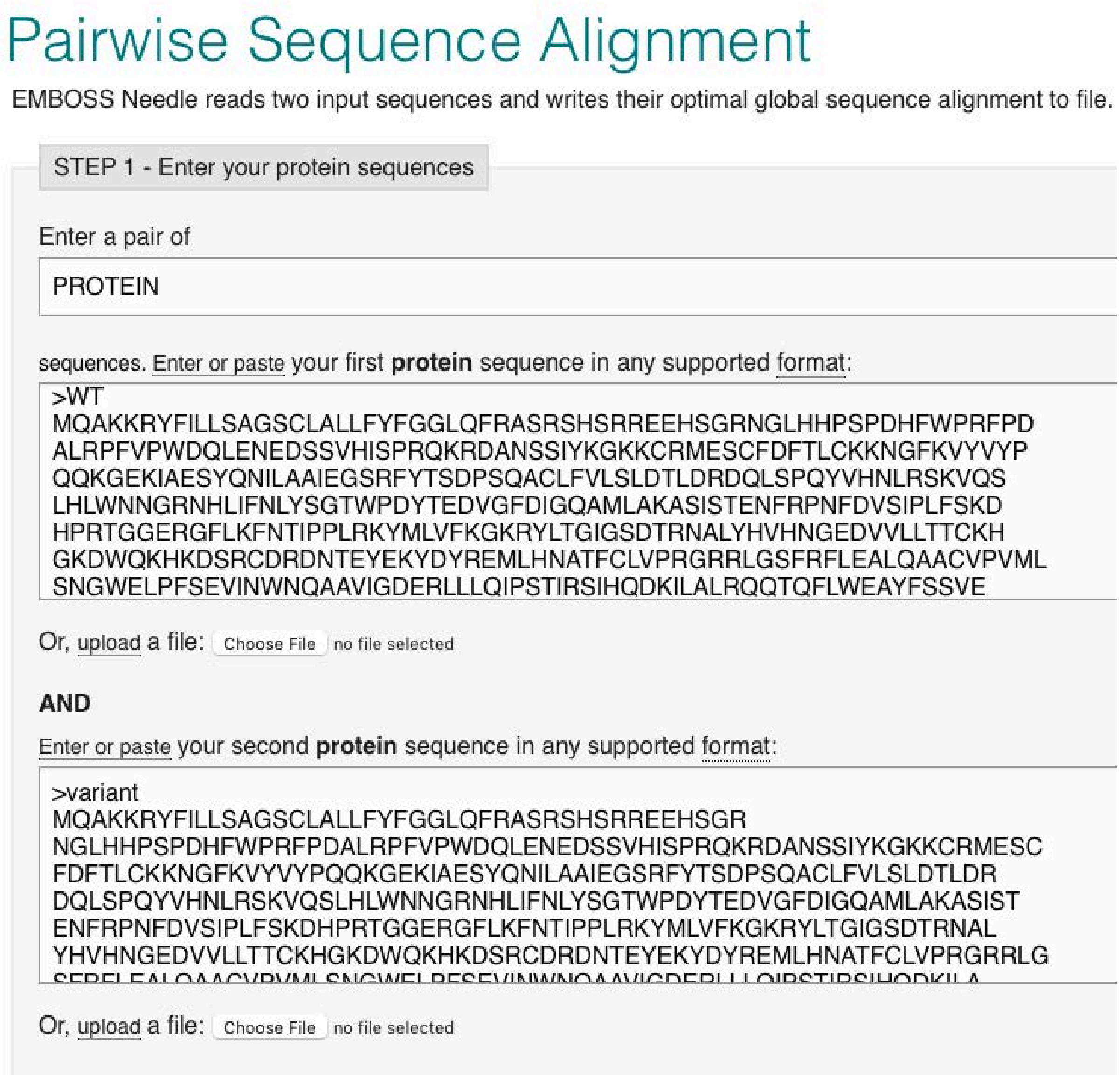


Deleting sequence in TranSeq

The result is below. Copy the mutant peptide sequence from the start to the first stop ("*") in the sequence

```
>EMBOSS_001_3
RPNAAVGSVRAGACEALLGAGTARGEDAPPPPPSAAAAAASSRAAARPGGRPRERRSKHR
RSLGAAAEGNKERCRDL*GYGAPAARCTSRLNPGSADLLEPECSARRVVDELKLPSRGSF
STLPRGIREARQSLRLAIDWVSGAFFFSPLSFFSSVLLHARKLQSAARPPWVRDIQPRSL
PCIVQPKDERPKGRKLKKSPTCAGSVHGLGKGIQRRWERRV*SLYRREDGGLELKVLIGK
LG*FLCLFTILLTQAGHMQAKKRYFILLSAGSCLALLFYFGGLQFRASRSHSRREEHSGR
NGLHHPSPDHFWPRFPDALRPFVPWDQLENEDSSVHISPRQKRDANSSIYKGKKCRMESC
FDFTLCKKNGFKVYVYPQQKGEKIAESYQNILAAIEGSRFYTSDPSQACLFVLSLDTLDR
DQLSPQYVHNLRSKVQSLHLWNNGRNHLIFNLYSGTWPDYTEDVGFDIGQAMLAKASIST
ENFRPNFDVSIPLFSKDHPRTGGERGFLKFNTIPPLRKYMLVFKGKRYLTGIGSDTRNAL
YHVHNGEDVVLLTTCKHGKDWQKHKDSRCDRDNTEYEKYDYREMLHNATFCLVPRGRRLG
SFRFLEALQAACVPVMLSNGWELPFSEVINWNQAAVIGDERLLLQIPSTIRSIHQDKILA
LRQQTQFLWEAYFSSVEKIVLTTLEIIQDRIFKHISRNSLIWNKHPGGLFVLPQYSSYLG
DFPYYYANLGLKPPSKFTAVIHAVTPRSLSPSQC*SFSWLQPSPSTVPRS*FYGIVTSPY
QPNTAGLPLLCLSSSLKERARL*AAVFCPTTTSSQTPCSALTRTRCFQQQRWISPSQCGR
ASLRGLWGTPRAATSGITLRSGGDTHQSGRTTTPWC*QELLFTTNIITTYTPITCQPA*R
TWWTNWPIVRTFS*TSWCLL*QNCLQSK*PRRSSIRRQ*WDRLLGLPVGLTLTTLPSDRA
A*IRLPAGLATCR*STLR*GSTPSSLKTRSLF*GRNTETLSDFEESG*VGEGKQEGMGVK
LLSLPSADPLISRARLCQLSKNLDEQNDKKKKANENSTPGSWDCTRLLQTHLTGFCVPRL
GCVQFNYGTLNNYF*NDCYAGLNFFNDQNY*KPEFFV*SKKKKKX
```

Paste into the second window of the Needle program. You had already pasted the w/t sequence (FASTA format) at the beginning of this exercise.



Click submit. An alignment result is returned that compares w/t to mutant sequences that you had input into Needle. Follow the sequence (vertical bars between the two sequences indicate similarity) till you reach the first dot (a difference between the two sequences).

```
WT              401 ILALRQQTQFLWEAYFSSVEKIVLTTLEIIQDRIFKHISRNSLIWNKHPG    450
                    ||||||||||||||||||||||||||||||||||||||||||||||||||
variant         401 ILALRQQTQFLWEAYFSSVEKIVLTTLEIIQDRIFKHISRNSLIWNKHPG    450

WT              451 GLFVLPQYSSYLGDFPYYYANLGLKPPSKFTAVIHAVTP-LVSQSQPVLK    499
                    ||||||||||||||||||||||||||||||||||||||| .:|.||.
variant         451 GLFVLPQYSSYLGDFPYYYANLGLKPPSKFTAVIHAVTPRSLSPSQC---    497
```

The first amino acid to change due to the frameshift is leucine to arginine at position 490. The altered mutant sequence after the frameshift is “RSLSPSQC” then the mutant sequence terminates. We can now fill in our FragmentReplacedModification instance.

[FragmentReplacedModification:5357998] Replacement of residues 490 to 497 by RSLSPSQC

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	RSLSPSQC
endPositionInReferenceSequence	497
referenceSequence	UniProt:Q16394 EXT1
startPositionInReferenceSequence	490
DB_ID	5357998
_displayName	Replacement of residues 490 to 497 by RSLSPSQC

After 8 altered amino acids, the 9th position is a stop (“*”). The name of the EWAS will reflect the mutation and fill in the hasModifiedResidue slot with the FragmentReplacedModification. Note that here, the end coordinate of the EWAS is changed to reflect the new C-terminal amino acid.

[EntityWithAccessionedSequence:3730975] EXT1 L490Rfs*9 [Golgi membrane]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	Golgi membrane
endCoordinate	497
hasModifiedResidue	Replacement of residues 490 to 497 by RSLSPSQC
referenceEntity	UniProt:Q16394 EXT1
startCoordinate	1
DB_ID	3730975
_displayName	EXT1 L490Rfs*9 [Golgi membrane]
authored	
created	Jassal, B, 2013-06-17
crossReference	
definition	
disease	hereditary multiple exostoses
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	
modified	Jassal, B, 2013-06-17
	Jassal, B, 2013-06-20
	Jassal, B, 2014-03-27
	Jassal, B, 2014-03-28
name	EXT1 L490Rfs*9
	Exostosin-1
	EXT1_HUMAN

Frameshifts that produce a stop in the first affected codon affected:

It is possible for a frameshift mutation to introduce an immediate stop codon. Although these should properly be annotated as frameshifts, Reactome is not able to capture a stop codon as a replaced residue, as would be required for FragmentReplacedModification as described above for lengthier frameshifts.

For this reason, frameshifts that produce a stop codon as the first affected codon are annotated as nonsense mutations as described above.

Frameshifts that produce a stop codon in the second affected codon:

Annotated as usual for a frameshift, as described above.

Mutations affecting the initiator methionine

Mutations creating a new downstream initiation site

A mutation that eliminates the initiator methionine and results in the aberrant initiation of translation from a downstream methionine is annotated as an in-frame deletion, as per HGVS recommendations, with the variant protein name specifying a deletion that stretches from the first amino acid in the reference sequence to the alternative initiator methionine.

Naming

Named in accordance with the deletion naming rules.

Format: “prefix” + “amino acid at position 2” + “2” + “_” + “M” +”position of methionine residue now used as initiator” + “del”

Where “prefix” is the HGNC gene name, the deleted amino acids are represented by single letter codes separated by underscore.

Example: SERPING1 A2_M53del - in this variant, a missense mutation substitutes the WT M1 with a leucine codon, leading to the usage of methionine at position 53 as an alternative start codon.

How to annotate

[FragmentDeletionModification:9651392] Deletion of residues 1 to 52

FragmentDeletionModification Properties

Property Name	Value
endPositionInReferenceSequence	52
referenceSequence	UniProt:P05155 SERPING1
startPositionInReferenceSequence	1
DB_ID	9651392
_displayName	Deletion of residues 1 to 52

Note that although by HGVS naming convention the variant EWAS is named SERPING1 A2_M53del, the FragmentDeletionModification record captures the biologically relevant deletion of residues M1-K52.

The protein variant EWAS record specifies the start coordinate as 53, without changing the end coordinate relative to the WT reference sequence.

[EntityWithAccessionedSequence:9651435] SERPING1 A2_M53del [endoplasmic reticu...

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	endoplasmic reticulum lumen
endCoordinate	500
hasModifiedResidue	Deletion of residues 1 to 52
referenceEntity	UniProt:P05155 SERPING1
startCoordinate	53
DB_ID	9651435
_displayName	SERPING1 A2_M53del [endoplasmic reticulum lumen]
authored	
created	Shamovsky, Veronica, 2019-06-25
crossReference	ClinVar:VCV000626353
definition	
disease	C1 inhibitor deficiency
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	Targeted next-generation sequencing for the mol...
	F12-46C/T polymorphism as modifier of the clini...
modified	Shamovsky, Veronica, 2020-01-10
	Orlic-Milacic, Marija, 2023-03-16
	Orlic-Milacic, Marija, 2023-03-17
	Orlic-Milacic, Marija, 2023-03-22
name	SERPING1 A2_M53del
	C1Inh A2_M53del
	plasma protease C1 inhibitor A2_M53del
	C1-In A2_M53del
	SERPING1 M1L
	SERPING1 Met1Leu

If a gene translocation fuses the promoter of one gene with the coding sequence of another gene, with accompanying truncation of the N' terminal coding sequence of the C-terminal fusion partner, the resulting disease variant is annotated as described above for mutations affecting the initiator methionine. The only such example in Reactome is the NOTCH1 variant NOTCH1 P2_M1580del (commonly known as TAN-1), which results from a translocation t(7;9) that fuses the promoter of T cell receptor beta to intron 24 of NOTCH1, resulting in the usage of an alternative start codon at position 1580 of the reference NOTCH1 sequence.

[FragmentDeletionModification:9830728] Deletion of residues 1 to 1579

FragmentDeletionModification Properties

Property Name	Value
endPositionInReferenceSequence	1579
referenceSequence	UniProt:P46531 NOTCH1
startPositionInReferenceSequence	1
DB_ID	9830728
_displayName	Deletion of residues 1 to 1579

[EntityWithAccessionedSequence:2220929] NOTCH1 P2_M1580del extracellula...

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	extracellular region
endCoordinate	1664
hasModifiedResidue	Deletion of residues 1 to 1579
referenceEntity	UniProt:P46531 NOTCH1
startCoordinate	1580
DB_ID	2220929
_displayName	NOTCH1 P2_M1580del extracellular fragment [e...

Mutations activating an upstream methionine initiation site.

A mutation that eliminates the initiator methionine and results in the aberrant initiation of translation from an upstream methionine is annotated as an insertion.

Naming

Named in accordance with insertion rules, above.

Format: “prefix” “amino_acids+positions_flanking” “ins” “inserted_sequence”

Where “prefix” is the HGNC gene name, the amino acids flanking the insertion are represented by single letter codes separated by underscore, and the inserted sequence is listed as single letter amino acid codes.

How to annotate

Annotate as for an insertion, with a FragmentInsertionModification, but preserve methionine 1 from the WT sequence and delete the novel initiator methionine to accommodate HGVS guidelines as described below:

As an example (made up, not annotated in Reactome)

WT coding sequence *MRST*MLRSQ… (WT initiating M in blue, upstream sequence in small italics)

Mutation in the initiator M changes it to V, and activates the methionine four amino acids upstream:

MRSTVLRSQ - this is what really happens, blue M converted to blue V, and inclusion in the coding sequence of upstream MRST.

By HGVS convention, this is actually annotated as:

MRSTVLRSQ - insertion of RSTV between the 'original' M and the second position amino acid (L).

Extension variants

The only currently annotated extension variant is IKBKG *420Wext*447, which represents a 3' extension, with replacement of the stop codon at position 420 with tryptophan codon and extension of the protein sequence by a total of 27 amino acids, with a new stop codon at position 447, counting from the start codon of the reference wild-type protein sequence.

How to annotate

Both 3' and 5' extensions can be annotated using FragmentReplacedModification class. In the case of 3' extensions, the wild-type stop codon coordinate is specified as the start and end coordinate of the replacement, and the added stretch of amino acids, in single letter abbreviations, is defined in the alteredAminoAcidFragment attribute. In the case of 5' extensions, the -1 would be specified as the start and end coordinate of the replacement.

[FragmentReplacedModification:5262861] Replacement of residues 420 to 420 by WGRPVQGHCLPRTCPGPCSLRPFLPPA

FragmentReplacedModification Properties

Property Name	Value
alteredAminoAcidFragment	WGRPVQGHCLPRTCPGPCSLRPFLPPA
endPositionInReferenceSequence	420
referenceSequence	UniProt:Q9Y6K9 IKBKG
startPositionInReferenceSequence	420
DB_ID	5262861
_displayName	Replacement of residues 420 to 420 by WGRPVQGHCLPRTCPGPCSLRPFLPPA
created	Shamovsky, V, 2014-02-09

On the EWAS record, the stop codon is changed relative to the WT reference sequence to reflect the extension.

[EntityWithAccessionedSequence:5262978] IKBKG *420Wext*447 [cytosol]

EntityWithAccessionedSequence Properties

Property Name	Value
cellType	
compartment	cytosol
endCoordinate	446
hasModifiedResidue	Replacement of residues 420 to 420 by WGRPVQGHCLPRTCPGPCSLRPFLPPA
referenceEntity	UniProt:Q9Y6K9 IKBKG
startCoordinate	1
DB_ID	5262978
_displayName	IKBKG *420Wext*447 [cytosol]
authored	
created	Shamovsky, V, 2014-02-09
crossReference	
definition	
disease	primary immunodeficiency disease
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	X-linked anhidrotic ectodermal dysplasia with immunodeficiency is caused b...
modified	Shamovsky, Veronica, 2014-07-14
	Shamovsky, Veronica, 2014-12-08
	Orlic-Milacic, Marija, 2023-02-06
name	IKBKG *420Wext*447
	NEMO *420Wext*447
	IKKG X420WextX447
	IKK-gamma X420W
	Inhibitor of nuclear factor kappa-B kinase gamma subunit
	IKBKG *420Trpext*447
	IKBKG Ter420TrpextTer447

Splicing variants

Splicing variants, if the exact effect of the mutation has been determined experimentally at the protein level, are annotated in one of the previously described categories, depending on the consequence of the splicing mutation on the amino acid sequence, and named accordingly, as HGVS currently does not provide guidelines for nomenclature of splicing variants at the protein level. This also applies to inversion variants. Of the 6 splicing variants currently annotated in Reactome, four are annotated as frameshift variants (EXT2 A361Pfs*44, NF1 Q616Gfs*4, NF1 R69Nfs*7, and UNC93B1 L230Afs*188), one is annotated as an in-frame insertion (POMT2 I444_N445insLLWQ), and one as an in-frame deletion (SLC11A2 L104_P142del). Two inversion variants currently annotated in Reactome, F8 A20_M2143delinsMRIQDPGK and F8 V2144_Y2351delinsHGVLENGKNWEPSYRW, are annotated as in-frame indels, as their names imply.

Variants with multiple in-cis mutations

These usually contain a primary mutation and a secondary mutation that was selected for during the course of treatment with target therapy as a drug-resistance mechanism. Each mutation is annotated independently and they are separated with a semicolon in the disease variant name, e.g. EGFR E746_A750del;T790M, where T790M is a mutation associated with resistance to small tyrosine kinase inhibitors.

Guideline for automated annotation of disease variants

Automated annotation of disease variants in Reactome proceeds through the following steps:

1) Starting from manually annotated disease variants grouped into functional sets, define external database-derived and/or literature-derived variants with matching disease association that conform to Reactome-expanded ACMG criterion PM5 or to combined ACMG criteria PM1 and PM4.

2) Categorize variants of interest by mutation type (missense, nonsense, etc.).

3) Run the script that creates individual disease variant records for automatic upload to the Reactome Curator Tool.

4) Import automatically annotated disease variants into existing disease variant functional sets.

5) Expanded disease variant functional sets are readily displayed in interactive pathway diagrams.

References

1. Philippe C, Porter DE, Emerton ME, Wells DE, Simpson AH, Monaco AP. Mutation screening of the EXT1 and EXT2 genes in patients with hereditary multiple exostoses. Am J Hum Genet. 1997 Sep;61(3):520-8. doi: 10.1086/515505. PMID: 9326317; PMCID: PMC1715939.
2. Ahn J, Lüdecke HJ, Lindow S, Horton WA, Lee B, Wagner MJ, Horsthemke B, Wells DE. Cloning of the putative tumour suppressor gene for hereditary multiple exostoses (EXT1). Nat Genet. 1995 Oct;11(2):137-43. doi: 10.1038/ng1095-137. PMID: 7550340.

Adding cross-references to disease entities

We cross reference disease entities to COSMIC (cancer-associated mutations; https://cancer.sanger.ac.uk/cosmic) and/or ClinGen (https://reg.clinicalgenome.org/redmine/projects/registry/genboree_registry/landing) where appropriate. An entity can have as many cross-references as needed. COSMIC cross-referencing should make use of the Genomic Mutation ID where available, not the Legacy Identifier. ClinGen Canonical Allele (CA) identifiers are used wherever possible, but ClinVar IDs may be used if CA IDs are not available.

Final steps

Once your curation is done, you will need to add your pathway into the Disease hierarchy in the relevant subpathway. Check out the disease chapter, add your pathway to the proper spot in the subpathways (or create a new child of Disease pathway, if appropriate) and commit the Disease chapter back into gk_central.

You will also have to check if any parent pathway EHLDs or the Disease parent EHLD requires updating to accommodate your new disease pathway. Coordinate with Cris to incorporate your updates to any EHLDs. Update any affected pathway summations to incorporate your new pathway.

Infectious Disease and Innate Immune System Pathway Curation Guide

The Reactome's concept of host-pathogen interactions refers to the relationship between a host organism and a pathogen (which can be any microorganism) at the molecular level. These interactions involve the recognition of the pathogen by the host's immune system, the activation of immune responses, and the evasion strategies employed by pathogens to avoid detection and destruction by the host's immune system.

We have set the disease / normal boundary by asking what the outcome of the host-pathogen interaction is. If the host wins and the pathogen is really neutralized, that counts as normal. If the pathogen wins to cause overt infection or to establish a stable latent state that could result in disease later, that counts as disease.

Entry and growth of pathogenic microorganisms within the host cell (e.g., HIV life cycle) also counts as disease and is annotated as a part of the "Infectious Diseases" module (see above).

In most cases, interactions involving host and pathogen entities are captured as a binding between the host PE (e.g., EWAS or a complex) and a pathogen-associated molecular pattern (PAMP), which could be a wide range of molecules found in pathogens. Examples of PAMPs include bacterial lipopolysaccharides, viral nucleic acids, and fungal cell wall components.

Annotating physical entities

1. Genomic, transcript, and protein sequences, including those of microbial origin, are annotated as EWASs which have their referenceEntity slot filled in with referenceSequence instance containing NCBI nucleotide, ENSEMBL, EMBL or UniProt identifier.
    - Note: Microbial nucleic acids can be annotated as direct children of GenomeEncodedEntitiy (GEE) class with microbial species assigned.
2. A simpleEntity should be used to annotate molecules such as bacterial lipopolysaccharides,peptidoglycans or fungal mannans. A referenceEntity is filled with a referenceMolecule instance containing a ChEBI identifier.

Assigning compartment for entities of microbial origin

We assign location on Reactome instances using GO compartment terms. The GO terms allow to distinguish Pathogen associated molecular patterns (PAMP) molecule locations in the pathogen compartments (e.g., "cell wall" or "virion membrane") and locations in infected host cells (e.g.,"host cell endoplasmic reticulum membrane”, "host cell cytosol"). The GO terms distinguish these locations of PAMPs from locations of endogenous host proteins in a normal cell (“endoplasmic reticulum membrane”, "cytosol" etc.).

In contrast to GO, we use normal cell terms (for example, “endoplasmic reticulum membrane” and **not** “host cell endoplasmic reticulum membrane”) to localize virus- or bacteria- or other pathogen-derived entities in an infected cell.

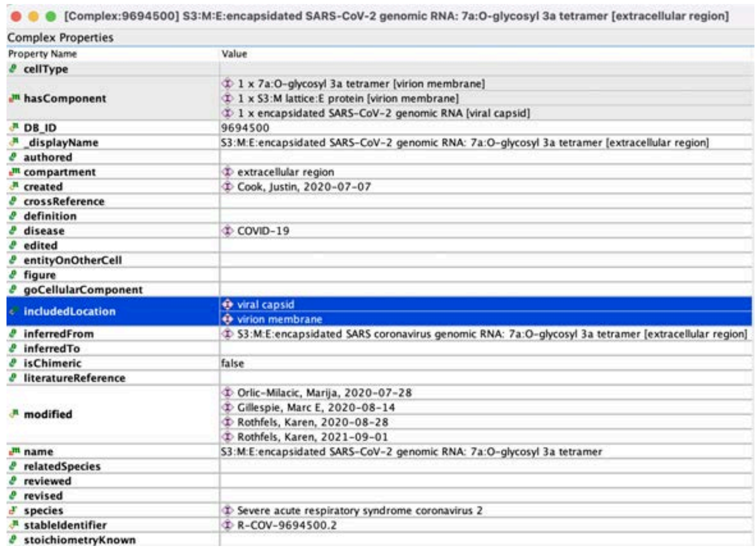
The GO terms that are used to localize proteins in a normal cell are listed in the Compartment class. Thus,

1. The compartment slot must be filled with a GO term from the Compartment list which includes those involving pathogen/symbiont patterns.

2. When the species slot value for a PE is a pathogen species and the compartment slot value is a normal cell term, then the **goCellularComponent** slot of that PE should also be filled with the GO "host cell [compartment]" term corresponding to the one chosen for the entity.

3. The Compartment list also includes GO terms that describe parts of bacteria or viruses ("cell wall", "viral capsid"). These terms can be also used to assign compartments for pathogen-derived entities.
    - Note: Individual pathogen proteins/entities localized in pathogen compartments (“viral capsid”, “virion membrane”) may form a complex, which functions in the host compartment. In this case, the record of the complex with microbial components should have a normal cell compartment term (e.g., “extracellular region”).

- The goCellularComponent slots in PE records of these pathogen-derived entities can be filled in with a GO “host cell



[Complex:9694500] S3:M:E:encapsidated SARS-CoV-2 genomic RNA: 7a:O-glycosyl 3a tetramer [extracellular region]

Complex Properties

Property Name	Value
cellType	
hasComponent	1 x 7a:O-glycosyl 3a tetramer [virion membrane] 1 x S3:M lattice:E protein [virion membrane] 1 x encapsidated SARS-CoV-2 genomic RNA [viral capsid]
DB_ID	9694500
_displayName	S3:M:E:encapsidated SARS-CoV-2 genomic RNA: 7a:O-glycosyl 3a tetramer [extracellular region]
authored	
compartment	extracellular region
created	Cook, Justin, 2020-07-07
crossReference	
definition	
disease	COVID-19
edited	
entityOnOtherCell	
figure	
goCellularComponent	
includedLocation	viral capsid virion membrane
inferredFrom	S3:M:E:encapsidated SARS coronavirus genomic RNA: 7a:O-glycosyl 3a tetramer [extracellular region]
inferredTo	
isChimeric	false
literatureReference	
modified	Orlic-Milacic, Marija, 2020-07-28 Gillespie, Marc E, 2020-08-14 Rothfels, Karen, 2020-08-28 Rothfels, Karen, 2021-09-01
name	S3:M:E:encapsidated SARS-CoV-2 genomic RNA: 7a:O-glycosyl 3a tetramer
relatedSpecies	
reviewed	
revised	
species	Severe acute respiratory syndrome coronavirus 2
stableIdentifier	R-COV-9694500.2
stoichiometryKnown	

compartment” term to depict their roles as invaders of a host cell.

- Note: A complex involving a pathogen protein/entity and host PE(s) should have a host compartment assigned, while microbial compartments assigned to individual components will be automatically added in the includedLocation slot.

Assigning species in PEs, pathways and reactions that involve host-pathogen interactions

1. The “species” slot on the event identifies the environment where the reaction is happening. This should almost always be human for human Reactome curation. The exceptions are when we have annotated reactions (RLEs) in other species for inference (e.g Mus) or when we are specifically annotating another species pathways as part of another project (e.g Gallus and Mycobacterium tuberculosis).

In disease or immune system processes that involve host-pathogen interactions the host would be listed as the “species” in the records of RLE and complexes, which are formed by host and pathogen entities.

- Note: When creating a chimeric reaction to be used for inference (i.e one that involves entities from multiple species and (usually) observed in vitro) ALL species should be listed in the species slot.
- Note: A complex involving only simpleEntities should have NO species listed. A complex involving a pathogen protein/entity and simpleEntity or an OtherEntity (i.e not having a species) should have the pathogen species listed in the species attribute and no relatedSpecies.

2. The “related species” is used to tag the source (or sources) of genome-encoded entities from “elsewhere” that act as “bystanders” in a process. In disease or immune system processes, the pathogen(s) would be listed as “relatedSpecies" in RLEs and complex instances.

- Note: relatedSpecies value(s) specified in RLEs should be assigned to a first parent pathway.

3. The use of "species" and "relatedSpecies" in sets follows the same rule as for complexes and events.

- Note: When the set members contain participants from different non-host pathogen species, ALL microbial species should be listed in the species slot.
- Note: If your annotations involve a species that is new to Reactome, please contact Peter D'Eustachio as it will need to have an abbreviation assigned by him.

Assigning a disease term for entities and reactions that involve host-pathogen interactions

1. For all kinds of pathogen entities in human (or other host) pathways the disease slots are filled in with a value corresponding to the disease they would cause if not neutralized. The effect of this rule is to automatically color all pathogenic entities red, both when they are involved in a disease and also when they are in the process of being detected and neutralized by the innate immune system.

2. Assigning a disease value for complexes and reactions depends on the outcome of the host-pathogen interaction.

2a. In the disease-preventing innate immune reactions (e.g., E. coli flagellin binds to human TLR5 (R-HSA-188025) as part of the human TLR cascade) the disease slot is not filled thus leaving a reaction with its default value (black, normal). Complexes formed between pathogen entities and the host ones that detect and neutralize them have empty disease slot, so in a normal reaction a red (disease) pathogen gets incorporated into a default (black, normal) pathogen : human receptor complex.

2b. In a disease-causing event (e.g., uptake of diphtheria toxin, Corynephage beta, which enters and destroys the human cell), a pathogen-derived entity (a disease slot is filled in) interacts with a normal host cell protein (default) to form an abnormal complex (a disease slot is filled in). A reaction-like event for an infectious disease process has a non-null disease slot value thus coloring its edges and nodes red. The entities will be automatically colored red if they are from a pathogen or complexes of human and pathogen proteins. Normal human entities that participate in these reactions are default (black).

2c. Disease-causing events are placed under disease, either infectious disease or diseases of a corresponding normal pathway. Disease host-pathogen interaction pathways have their own diagrams.

Examples of host-pathogen interactions:

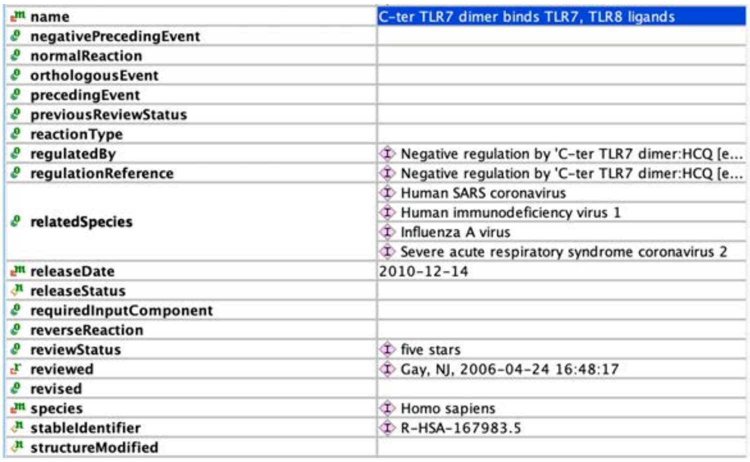
1. C-ter TLR7 dimer binds TLR7, TLR8 ligands (reaction) (R-HSA-167983)

Here a defined set of viral single stranded RNAs binds a human TLR7, setting off events that neutralize viruses and leave the human healthy – in our human-centric view, a normal process.

The reaction record lists all viruses in the ‘relatedSpecies’ slot, while human is indicated as the host in the species slot. The disease value is NULL.

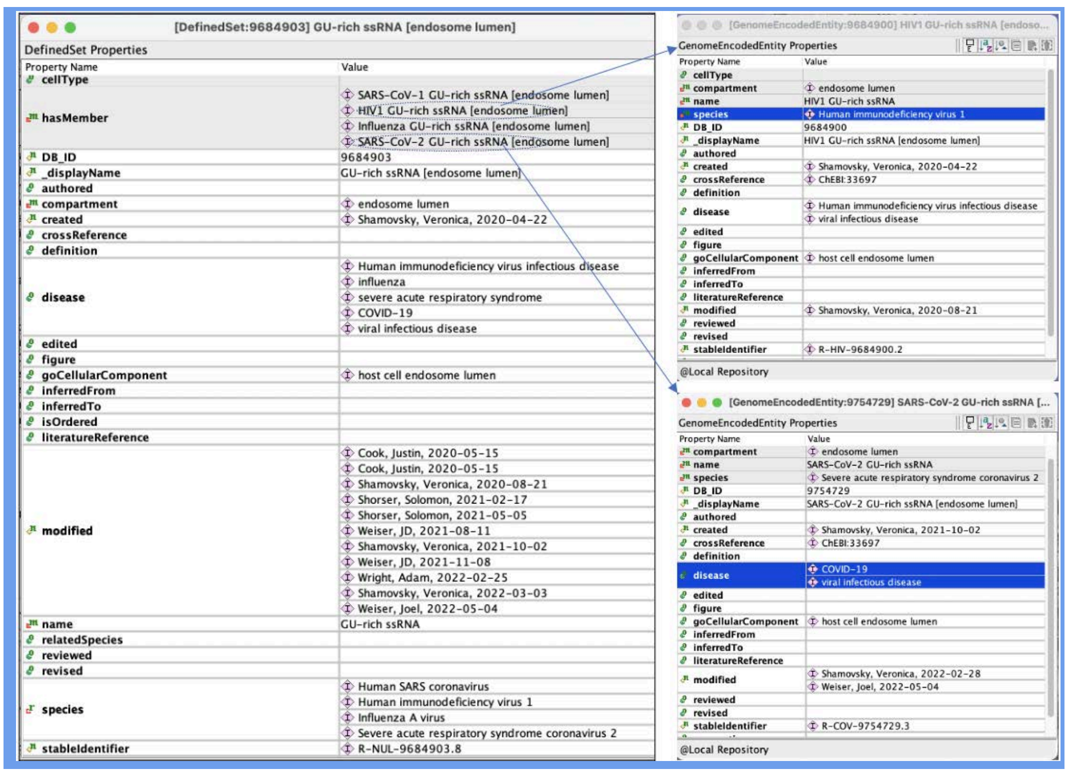
Reaction Properties

Property Name	Value
catalystActivity	
deletedInstanceDB_ID	
entityFunctionalStatus	
input	1 x C-ter TLR7 dimer [endosome membrane] 1 x TLR7, TLR8 ligands [endosome lumen]
output	1 x C-ter TLR7 dimer:TLR7, TLR8 ligands [endos...
DB_ID	167983
_displayName	C-ter TLR7 dimer binds TLR7, TLR8 ligands
_doRelease	true
authored	Luo, F, 2005-11-10 11:23:18
catalystActivityReference	
compartment	endosome membrane endosome lumen
created	Gillespie, ME, 2005-11-10 16:24:09
crossReference	
definition	
deletedStableIdentifier	
disease	
edited	Shamovsky, V, 2011-08-12
entityOnOtherCell	
evidenceType	



name	C-ter TLR7 dimer binds TLR7, TLR8 ligands
negativePrecedingEvent	
normalReaction	
orthologousEvent	
precedingEvent	
previousReviewStatus	
reactionType	
regulatedBy	Negative regulation by 'C-ter TLR7 dimer:HCQ [e...
regulationReference	Negative regulation by 'C-ter TLR7 dimer:HCQ [e...
relatedSpecies	Human SARS coronavirus Human immunodeficiency virus 1 Influenza A virus Severe acute respiratory syndrome coronavirus 2
releaseDate	2010-12-14
releaseStatus	
requiredInputComponent	
reverseReaction	
reviewStatus	five stars
reviewed	Gay, NJ, 2006-04-24 16:48:17
revised	
species	Homo sapiens
stableIdentifier	R-HSA-167983.5
structureModified	

In this reaction, a viral ssRNA is annotated as GenomeEncodedEntity (GEE) for each virus. Viral EWASs are combined in the defined set, listing all microbial species in the "spesies" slot, and all disease values in the disease slot. The compartment slot is filled in with "endosome lumen", while goCellularComponent indicates that this PE is located in "host cell endosome lumen".



[DefinedSet:9684903] GU-rich ssRNA [endosome lumen]

DefinedSet Properties

Property Name	Value
cellType	
hasMember	SARS-CoV-1 GU-rich ssRNA [endosome lumen] HIV1 GU-rich ssRNA [endosome lumen] Influenza GU-rich ssRNA [endosome lumen] SARS-CoV-2 GU-rich ssRNA [endosome lumen]
DB_ID	9684903
_displayName	GU-rich ssRNA [endosome lumen]
authored	
compartment	endosome lumen
created	Shamovsky, Veronica, 2020-04-22
crossReference	
definition	
disease	Human immunodeficiency virus infectious disease influenza severe acute respiratory syndrome COVID-19 viral infectious disease
edited	
figure	
goCellularComponent	host cell endosome lumen
inferredFrom	
inferredTo	
isOrdered	
literatureReference	
modified	Cook, Justin, 2020-05-15 Cook, Justin, 2020-05-15 Shamovsky, Veronica, 2020-08-21 Shorser, Solomon, 2021-02-17 Shorser, Solomon, 2021-05-05 Weiser, JD, 2021-08-11 Shamovsky, Veronica, 2021-10-02 Weiser, JD, 2021-11-08 Wright, Adam, 2022-02-25 Shamovsky, Veronica, 2022-03-03 Weiser, Joel, 2022-05-04
name	GU-rich ssRNA
relatedSpecies	
reviewed	
revised	
species	Human SARS coronavirus Human immunodeficiency virus 1 Influenza A virus Severe acute respiratory syndrome coronavirus 2
stableIdentifier	R-NUL-9684903.8

[GenomeEncodedEntity:9684900] HIV1 GU-rich ssRNA [endoso...

GenomeEncodedEntity Properties

Property Name	Value
cellType	
compartment	endosome lumen
name	HIV1 GU-rich ssRNA
species	Human immunodeficiency virus 1
DB_ID	9684900
_displayName	HIV1 GU-rich ssRNA [endosome lumen]
authored	
created	Shamovsky, Veronica, 2020-04-22
crossReference	ChEBI:33697
definition	
disease	Human immunodeficiency virus infectious disease viral infectious disease
edited	
figure	
goCellularComponent	host cell endosome lumen
inferredFrom	
inferredTo	
literatureReference	
modified	Shamovsky, Veronica, 2020-08-21
reviewed	
revised	
stableIdentifier	R-HIV-9684900.2

@Local Repository

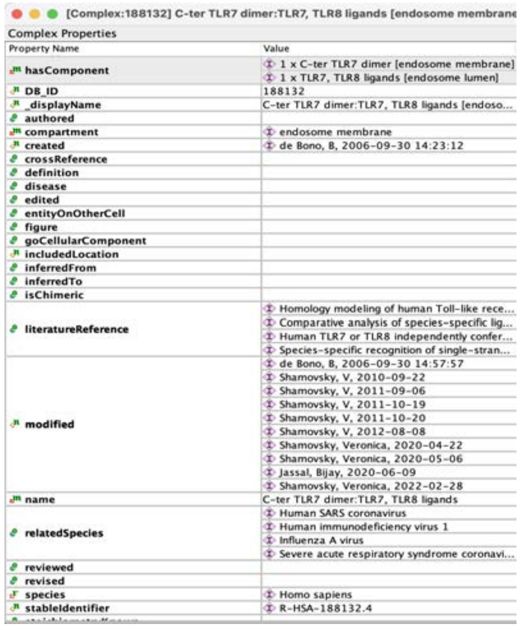
[GenomeEncodedEntity:9754729] SARS-CoV-2 GU-rich ssRNA [...

GenomeEncodedEntity Properties

Property Name	Value
compartment	endosome lumen
name	SARS-CoV-2 GU-rich ssRNA
species	Severe acute respiratory syndrome coronavirus 2
DB_ID	9754729
_displayName	SARS-CoV-2 GU-rich ssRNA [endosome lumen]
authored	
created	Shamovsky, Veronica, 2021-10-02
crossReference	ChEBI:33697
definition	
disease	COVID-19 viral infectious disease
edited	
figure	
goCellularComponent	host cell endosome lumen
inferredFrom	
inferredTo	
literatureReference	
modified	Shamovsky, Veronica, 2022-02-28 Weiser, Joel, 2022-05-04
reviewed	
revised	
stableIdentifier	R-COV-9754729.3

@Local Repository

The resulting complex of the human TLR7 homodimer and viral ssRNA has values similar to the reaction record:


[Complex:188132] C-ter TLR7 dimer:TLR7, TLR8 ligands [endosome membrane

Complex Properties

Property Name	Value
hasComponent	1 x C-ter TLR7 dimer [endosome membrane] 1 x TLR7, TLR8 ligands [endosome lumen]
DB_ID	188132
_displayName	C-ter TLR7 dimer:TLR7, TLR8 ligands [endoso...
authored	
compartment	endosome membrane
created	de Bono, B, 2006-09-30 14:23:12
crossReference	
definition	
disease	
edited	
entityOnOtherCell	
figure	
goCellularComponent	
includedLocation	
inferredFrom	
inferredTo	
isChimeric	
literatureReference	Homology modeling of human Toll-like rece... Comparative analysis of species-specific lig... Human TLR7 or TLR8 independently confer... Species-specific recognition of single-stran...
modified	de Bono, B, 2006-09-30 14:57:57 Shamovsky, V, 2010-09-22 Shamovsky, V, 2011-09-06 Shamovsky, V, 2011-10-19 Shamovsky, V, 2011-10-20 Shamovsky, V, 2012-08-08 Shamovsky, Veronica, 2020-04-22 Shamovsky, Veronica, 2020-05-06 Jassal, Bijay, 2020-06-09 Shamovsky, Veronica, 2022-02-28
name	C-ter TLR7 dimer:TLR7, TLR8 ligands
relatedSpecies	Human SARS coronavirus Human immunodeficiency virus 1 Influenza A virus Severe acute respiratory syndrome coronavi...
reviewed	
revised	
species	Homo sapiens
stableIdentifier	R-HSA-188132.4

2. Translocation of Vpr to the mitochondria (Reaction: R-HSA-180855)

Here an HIV protein is translocated to the mitochondria and while no human proteins are involved, the **species** listed for this reaction should still be human because it is a RLE occurring in the context of a human process.

The **relatedSpecies** should be Human immunodeficiency virus 1.

The **disease slot** is filled in with 'human immunodeficiency virus infectious disease':

3. Uptake and function of diphtheria toxin (Pathway: R-HSA-5336415)

As in case 1, the process is occurring in a human, so the pathway "species" is human involving a diphtheria protein (Corynephage beta). In this case, the result is that the bacterial protein enters and destroys the human cell – in our human-centric view, a disease process. The pathway records are filled in as follows:

species: Homo sapiens **relatedSpecies:** Corynephage beta **disease:** diphtheria

4. Host physical entity is transferred into a pathogen cell.

Consistent with the logic in case 1, 2 and 3, the following reactions are part of the human process, so it is reasonable to label them as human reactions, even in cases in which reactions are occurring inside the bacterium.

   a. Cathelicidin LL-37 localizes to bacterial cell wall (Reaction: R-HSA-1222685)

species: Homo sapiens **relatedSpecies**: Mycobacterium tuberculosis **disease:** NULL

b. Nitric oxide enters the bacterium (Reaction: R-HSA-1222662)

species: Homo sapiens **relatedSpecies**: Mycobacterium tuberculosis **disease:** NULL

- Note: When a reaction involves only SimpleEntities (e.g., case 4b), RLE instance goes on skip list because QA check will detect a conflict because there are no human entities specified as reaction components.

5. Cry proteins stabilize unphosphorylated Bmal1:Clock,Npas2 (Reaction: R-MMU-549451)

   Here, the reaction describes a process occurring in a non-human species (Mus) and *only* mouse proteins are involved.

   species: Mus musculus **relatedSpecies**: NULL **disease:** NULL

   The environment is mouse, the proteins are mouse, and success in the event keeps the mouse healthy. So this is a normal reaction whose species is mouse and whose related species is NULL.

[Additional information about these attributes and examples of how the species and relatedSpecies attributes are used in disease and normal curation can be found here.]

Final steps

Once your curation is done, you will need to add your pathway into the Disease hierarchy in the relevant subpathway. Check out the disease chapter, add your pathway to the proper spot in the subpathways (or create a new child of Disease pathway, if appropriate) and commit the Disease chapter back into gk_central.

You will also have to check if any parent pathway EHLDs or the Disease parent EHLD requires updating to accommodate your new disease pathway. Coordinate with Cris to incorporate your updates to any EHLDs. Update any affected pathway summations to incorporate your new pathway.

Drug Curation Guide

General points

When annotating drugs used to treat a particular disease, curators should add an appropriate disease tag to a drug entity. For a cancer drug, the general cancer disease attribute should definitely be added, but when it comes to more specific disease tags, it may be advisable to add only those cancer types for which the treatment by a given drug is approved.

In ***most*** cases, a drug will affect a target protein after it binds to it. Therefore, it will usually be the case that two steps are needed for any drug curated; a drug binding event to the target then whatever effect (in the form of a regulation) the drug has on the target (usually at an existing event). Sometimes (rarely), drugs will behave in other ways, for example, as group donors. The appropriate reaction must then be created.

The drug interaction file

The drug interaction file used in curation is a set of autogenerated drug reactions from GtP in an .rtpj. Bijay and Guanming devised a solution to create auto-generated drug reactions from GtP data. In a nutshell, the criteria used to create the file was to collect any approved drugs that bind the same human target, which is the primary target of those drugs, and create a Reactome-type reaction that has input of drug(s) and target and output of the drug(s): target complex. Also, collect all references from each drug page that describe them binding to this target into the Reactome reaction litRef slot.

See this file below.

Latest drug:target auto-generated GtP reaction file **IUPHAR_reactions_051821.rtpj** (in Reactome Team Drive > Curation > Drug curation).

Note most of these reactions are NOT checked into Reactome. Instead, identify the reaction you wish to curate and check that in ONLY from this file. Once in gk_central, check out into your local project and work on revising it.

This file provides a good starting point to revise any chosen reaction from this list you want to curate into Reactome. You'd still need to investigate whether these drugs are still in use or are experimental, for instance. As long as strong evidence exists to confirm pharmacological action, that drug can be curated.

Where you've found a drug in the literature that doesn't have a GtP ID, you can request new drug entries from GtP as long as you can provide evidence for the pharmacological effect of that drug. By consensus, Reactome primarily annotates drugs that have been approved for human use. If an exception is made in cases where a drug is biologically interesting / distinctive but not universally approved, it is crucial to explain in the summation why we have nevertheless chosen to annotate it.

Example - drug binds target, resultant complex regulates that target's event(s)

The drug **rivaroxaban** (Xarelto) functions as an anticoagulant that binds to and inhibits Factor Xa in the blood clotting cascade. This has the effect of slowing down the rate of blood coagulation, helping to prevent the formation of blood clots.

Find the pathway of the target protein where this drug acts and download the ELV-level pathway into your tool. The relevant pathway here is Formation of Fibrin Clot (Clotting Cascade R-HSA-140877)

You need to determine what type of drug this is, so you can create the relevant drug instance. Reading some literature and/or checking its identifiers (Chebi, GtP) will determine this.

There are two ways to create the drug reaction; either use the auto-generated reaction (recommended, Steps 1a, 2a) or create from scratch (Steps 1b, 2b).

Step 1a

Find the reaction from the auto-generated drug reaction file (above) which should have the entity and target information already filled. Check it into gk_central then check out into your local project.

Step 1b

Below is the process to create new drug instances (steps 1b and 2b) if there is no auto-generated reaction.

This example is a chemical drug so create an instance of chemicalDrug in the curator tool. The instance has mandatory slots

name - enter the common name and a brand name (optional)
rivaroxaban, Xarelto

compartment - same as the target protein in most cases (exception for example target is a membrane protein, drug will be extracellular)
extracellular region

disease - A DOID identifier that most closely matches the disease this drug is involved in
pulmonary embolism (DOID:9477)

referenceEntity - a referenceTherapeutic instance containing an GtP identifier. Any drug you create **must** have an GtP identifier so check their website first for one before curating
GtP:6388

(On the rare occasion that an GtP ID is not available for your drug, you can use other resources like PubChem Compound, DrugBank or Chebi as alternative identifiers. Remember to fill in the 'approved' slot in the referenceEntity instance when using these alternatives as this slot is mandatory. If you can't find a suitable identifier, see instructions below describing how to request a Chebi identifier.).

crossReference - currently optional but desirable - Add a Chebi identifier. If one is not available, easy to request one (see Best Practices- Using Chemical Compound Databases, Appendix C8).
Chebi:68579

literatureReference - currently optional but desirable - Indicate the specific references for the drug instance. This is useful for data dumps as well as users wanting specific references for a drug, rather than trying to match a specific reference from the summation for a set of drugs

reactionType - currently optional but desirable. Choose an item from this list (taken from the IUPHAR / Guide to Pharmacology glossary, https://www.guidetopharmacology.org/helpPage.jsp#glossary) to classify the effect of the drug on the reaction:

Activator - "A ligand that activates an ion channel by binding directly to it, including ligands that cause a change in the voltage dependence of channel gating that favours activation."

Agonist - "A ligand that binds to a receptor and alters the receptor state resulting in a biological response. Conventional agonists stabilise an active conformation of a receptor and increase receptor activity. When the receptor stimulus induced by an agonist reaches the maximal response capability of the system (tissue), then it will produce the system maximal response and be a full agonist in that system."
Allosteric modulator (is_a modulator) - "A ligand that increases (positive) or decreases (negative) the action of an agonist or antagonist by combining with a distinct site on the receptor macromolecule. Neutral allosteric ligands bind to an allosteric site without affecting the binding or function of orthosteric ligands but can still block the action of other allosteric modulators that act via the same allosteric site."
Antagonist - "A drug that reduces the action of another drug, generally an agonist"

Antibody binding - A drug that is an antibody (immunoglobulin) and acts by binding the epitope for which it is specific.*
Channel blocker - “A ligand that blocks ion movement through an ion channel.”
Channel opener - A ligand that facilitates ion movement through an ion channel.*
Inhibitor - “A ligand that inhibits ion channel gating by directly binding to the channels, including ligands that inhibit ion channel activation and those that inhibit ion channel inactivation.”
Modulator - A ligand that changes the activity of its target protein on binding
Positive modulator (is_a modulator) - A ligand that increases the activity of its target protein on binding*.
Allosteric antagonist - (is_a antagonist) A ligand that decreases the action of an agonist or antagonist by combining with a distinct site on the receptor macromolecule*.
Antisense inhibitor - (is_a inhibitor) An RNA oligonucleotide ligand that base-pairs with an endogenous mRNA, to form a rapidly degraded complex.*
Gating inhibitor - (is_a inhibitor) ???
Inverse agonist - (is_a agonist) A ligand that by binding to unoccupied receptors reduces the fraction of them in an active conformation. For nuclear receptors, an inverse agonist refers to ligands that can promote co-repressor recruitment. A drug that binds to the same receptor as an agonist but induces a pharmacological response opposite to that of the agonist.
Negative allosteric modulator - (is_a allosteric modulator) A ligand that decreases the action of an agonist or antagonist by combining with a distinct site on the receptor macromolecule.*
Partial agonist - (is_a agonist) Agonists that in a given tissue, under specific conditions, cannot elicit as large an effect (even when applied at high concentration, so that all the receptors should be occupied) as can another agonist acting through the same receptors in the same tissue.
Positive allosteric modulator - (is_a allosteric modulator) A ligand that increases the action of an agonist or antagonist by combining with a distinct site on the receptor macromolecule.*

rivaroxaban chemicalDrug entry

ChemicalDrug Properties

Property Name	Value
cellType	
compartment	extracellular region
disease	pulmonary embolism
referenceEntity	rivaroxaban [IUPHAR:6388]
DB_ID	9015055
_displayName	rivaroxaban [extracellular region]
authored	
created	Jassal, Bijay, 2017-08-08
crossReference	ChEBI:68579
definition	
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
literatureReference	Discovery of the novel antithrombotic ...
	Rivaroxaban: a novel, oral, direct fact...
	The discovery and development of riv...
	Pharmacology of anticoagulants used i...
modified	Jassal, Bijay, 2017-08-09
	Jassal, Bijay, 2017-08-15
	Weiser, JD
	Jassal, Bijay, 2018-03-14
	Weiser, JD
	Shorser, Solomon, 2018-08-13
	Wu, G, 2018-09-19
	Jassal, Bijay, 2018-10-18
	Jassal, Bijay, 2018-10-18
	Wu, G, 2018-11-05
	Wu, G, 2018-12-12
name	rivaroxaban
	Xarelto
reviewed	
revised	
stableIdentifier	R-ALL-9015055.1
summation	
systematicName	

Rivaroxaban is one of a number of ‘xabans’ that are direct oral anticoagulants (DOACs) and all act on Factor Xa in an inhibitory way. Since they all act in the same way, a set of these drugs can be constructed.

[DefinedSet:9038743] Factor Xa inhibitors [extracellular region]

DefinedSet Properties

Property Name	Value
cellType	
hasMember	apixaban [extracellular region]
	betrixaban [extracellular region]
	edoxaban [extracellular region]
	rivaroxaban [extracellular region]
	eribaxaban [extracellular region]
DB_ID	9038743
_displayName	Factor Xa inhibitors [extracellular region]
authored	
compartment	extracellular region
created	Jassal, Bijay, 2018-03-14
crossReference	
definition	
disease	
edited	
figure	
goCellularComponent	
inferredFrom	
inferredTo	
isOrdered	
literatureReference	
modified	Jassal, Bijay, 2018-03-14
	Jassal, Bijay, 2018-03-23
name	Factor Xa inhibitors
relatedSpecies	
reviewed	
revised	
species	
stableIdentifier	R-ALL-9038743.1
summation	
systematicName	

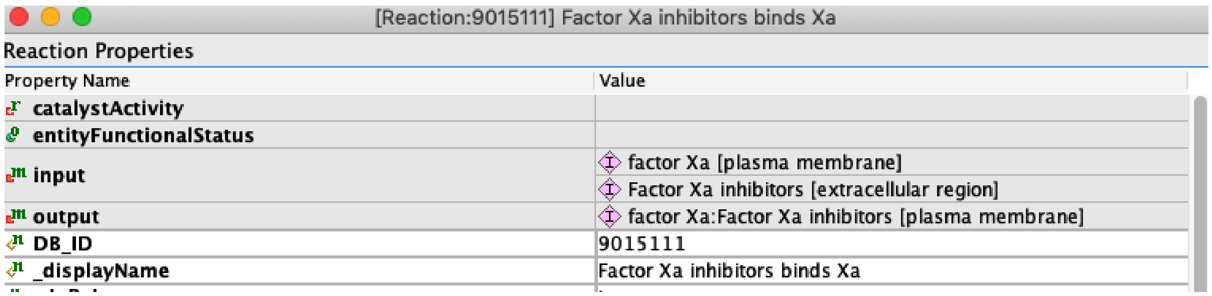
STEP 2A

In the autogenerated reaction, check the 'functional form' of the target. In pharmacological databases, the target is described as a single protein. This is what will be populated in the autogenerated reaction. However, in Reactome, the target protein may be in a complex, a cleaved form or PTM'd, etc etc. You will need to replace the target protein in the autogenerated reaction IF IT IS DIFFERENT to what is curated in Reactome. Check the compartments of all entities at this stage.

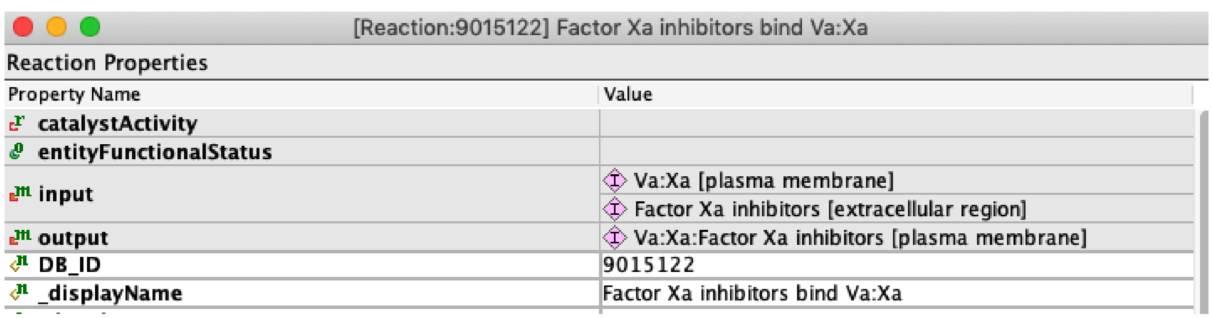
Step 2b

If creating a new reaction from scratch, find the ‘functional form’ of the target in the relevant Reactome reaction to be used in the drug reaction.

Factor Xa inhibitors (“xabans”) binds Factor Xa. The resulting complex will act on Factor Xa event. Fill in the disease slot (same as for the drug).



Factor Xa inhibitors also bind complexed Factor Xa (Va:Xa). The resulting complex will act on the Va:Xa event. Fill in the disease slot (same as for the drug).



Step 3

(will most likely be existing event/s or if not, you’ll need to create one to be the regulated event).

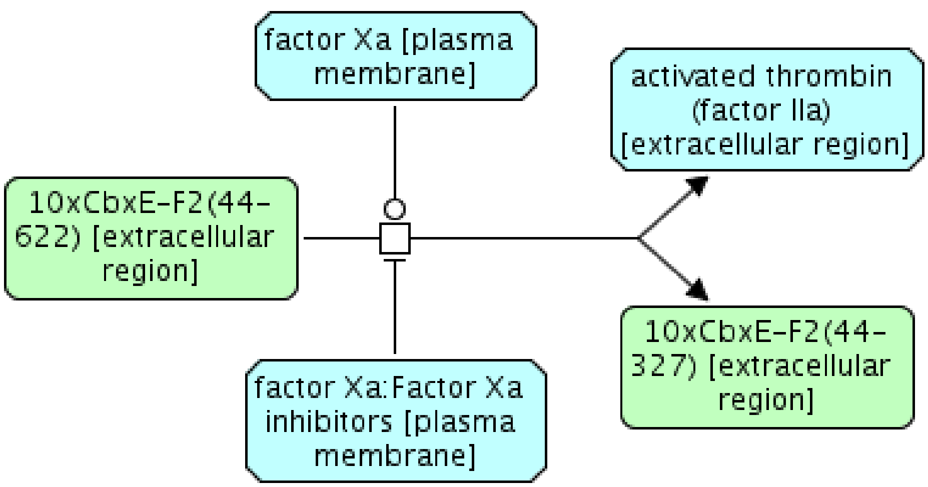
Find the event that Factor Xa is involved in and create a negative regulation of that event with the complex “Factor Xa:Factor Xa inhibitors”.

Create the negative regulation instance as below

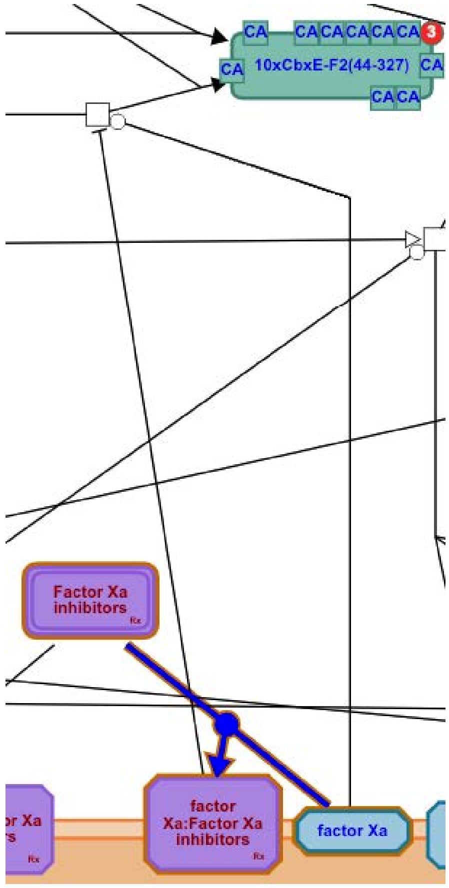
NegativeRegulation Properties

Property Name	Value
regulator	factor Xa:Factor Xa inhibitors [plasma membrane]
DB_ID	9015052
_displayName	Negative regulation by 'factor Xa:Factor Xa inhibitors [plasma membrane]'
activeUnit	
activity	
created	Jassal, Bijay, 2017-08-08
goBiologicalProcess	
literatureReference	
modified	Jassal, Bijay, 2017-08-08
	Jassal, Bijay, 2017-08-08
	Jassal, Bijay, 2017-08-08
	Jassal, Bijay, 2017-10-25
	Jassal, Bijay, 2018-03-02
	Jassal, Bijay, 2018-03-14
	Wu, G, 2018-05-08
stableIdentifier	R-HSA-9015052.3
summation	Factor Xa is a target for direct oral anticoagulant (DOAC) d...

...as seen in the curator tool central panel



...and as seen in the web display. Any entity containing a drug will be coloured purple.



Note - the actual regulation 'line' will be color-sensitive in the future* (ie purple for drugs).

*[This feature is currently on hold]

Add appropriate supporting references to both events.

Curator Tool Protocol

To support the new gk_central schema for drug annotation, a new feature has been added to the tool: auto-fetch the required information for the new ReferenceTherapeutic class based on the drug identifier in [GtP](https://www.guidetopharmacology.org/) (https://www.guidetopharmacology.org/). Here is an example of how to use it:

1. On the GtP web page, http://www.guidetopharmacology.org, search for imatinib in their search box.
2. On the result page, imatinib is the first entry. Click it.
3. On the imatinib page, find Ligand ID for it: 5687.
4. In the curator tool, create a new ReferenceTherapeutic instance, and enter 5687 as the identifier.
    - The curator tool will ask you if you want to fetch information from GtP.

- Choose “Yes”. Information for 5687 will be fetched automatically.

New molecule request in ChEBI

To request new identifiers for chemicals (and drugs) from ChEBI, follow this submission process. To submit to Guide to Pharmacology, you need to contact one of their curators to ask if the drug you want curated in GtP has an appropriate pharmacological action they are willing to add. Literature references always help your cause. The person to contact is Elena Faccenda, a senior curator at GtP.

Checklist

Here is a checklist for the whole drug curation process:

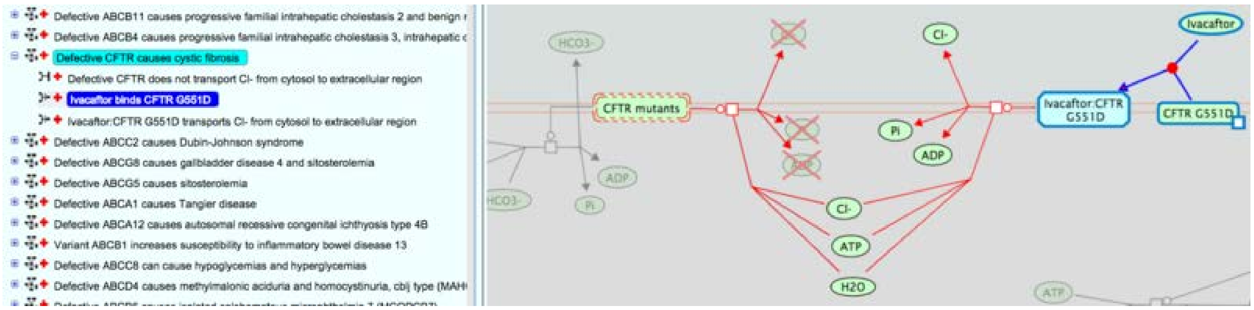
- Drugs in Guide to Pharmacology (GtP, formerly GtP) must be target-specific and approved (can be special cases eg. cancer therapeutics). See the GtP file here (Appendix C11).
- Check if the drug already exists in gk_central (NOT Reactome’s public website)
- Build projects (multiple at once if necessary) for all diagrams to be changed
- Create Drug entities, if necessary for different compartments
- Add ChEBI cross-references
- Drug entities have their disease attribute filled in
- Literature goes in the (binding) reaction
- Set the reaction’s doRelease=false
- In the reaction summary, mention all participating drugs
- In the ELV, the drug reactions need to be type changed to “Association”
- Only after review can the Regulation/Reference be added to a reviewed physiological reaction
- Fill in the tracking sheet, one item for every new drug reaction

Complexities of Disease and Drug curation: Special cases

Example 1 - drug-target interaction in disease:

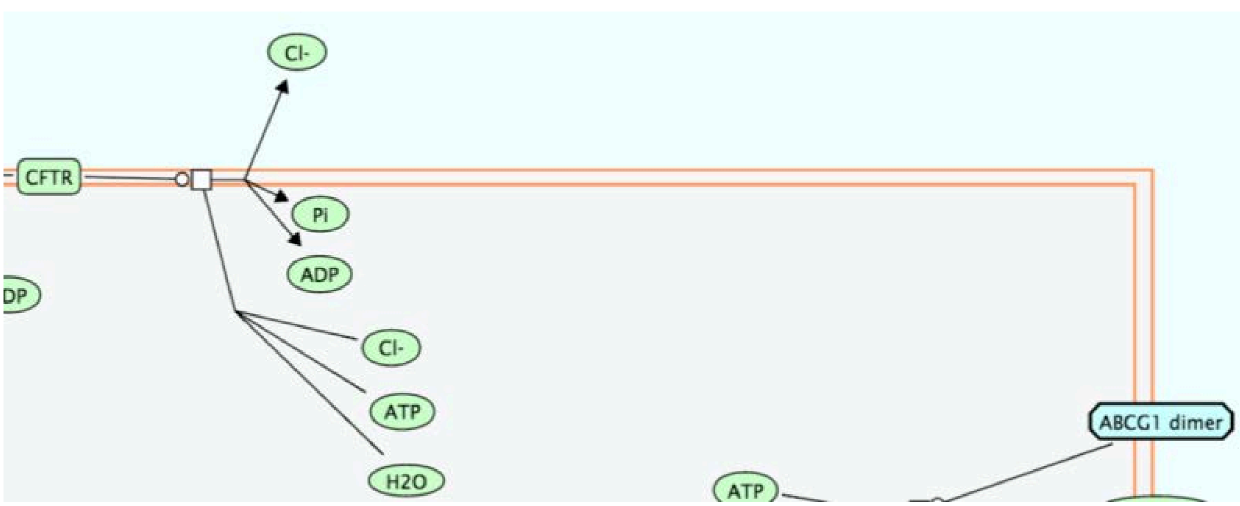
The cystic fibrosis transmembrane conductance regulator (CFTR) is a low conductance chloride-selective channel that mediates the transport of chloride ions in human airway epithelial cells. Chloride ions plays a key role in maintaining homeostasis of epithelial secretions in the lungs. Defects in CFTR can cause cystic fibrosis (CF), resulting in an ionic imbalance that impairs clearance of secretions, not only in the lung, but also in the pancreas, gastrointestinal tract and liver. More than 1500 mutations in the CFTR gene have been identified. CF patients with a particular mutation, G551D, have shown lung function improvements when given the drug Ivacaftor.

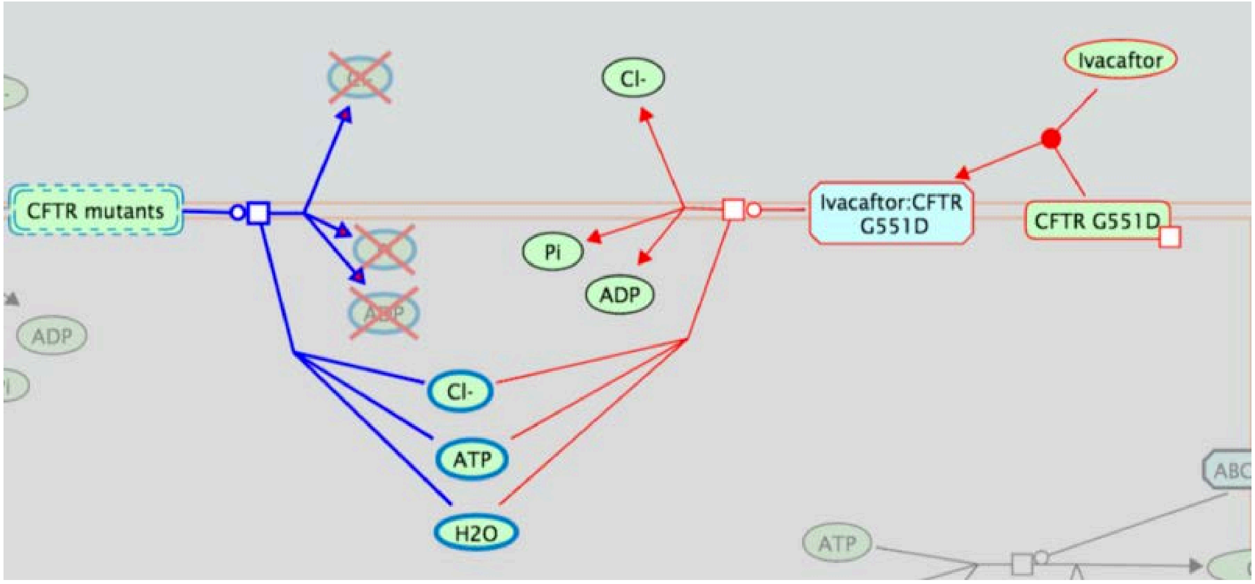
To curate the mutation reversal effect of the drug, create two events; first the drug binding to mutant protein and second, this complex reviving the function equivalent to the w/t protein.



Reactions showing ivacaftor binding the mutant protein (highlighted blue) and the following reaction (centre) showing channel functionality restored. The left-side reaction shows defective CFTR which cannot transport Cl- ions.

The construct in the hierarchy is complicated by having the loss-of-function (LoF) reaction (the left side) as part of the display. LoF reactions are generated automatically on the website. To add the reactions involving drugs, they must be drawn manually on the diagram shared by the disease pathway and the normal pathway, in this case, the PathwayDiagram instance lists the normal pathway (ABC-family proteins mediated transport) and the disease pathway (Defective CFTR causes cystic fibrosis), together with other disease pathways relevant to this diagram as representedPathway. The two disease drug reactions are drawn in this diagram and their edges coloured red. The cystic fibrosis disease attribute is associated with both.



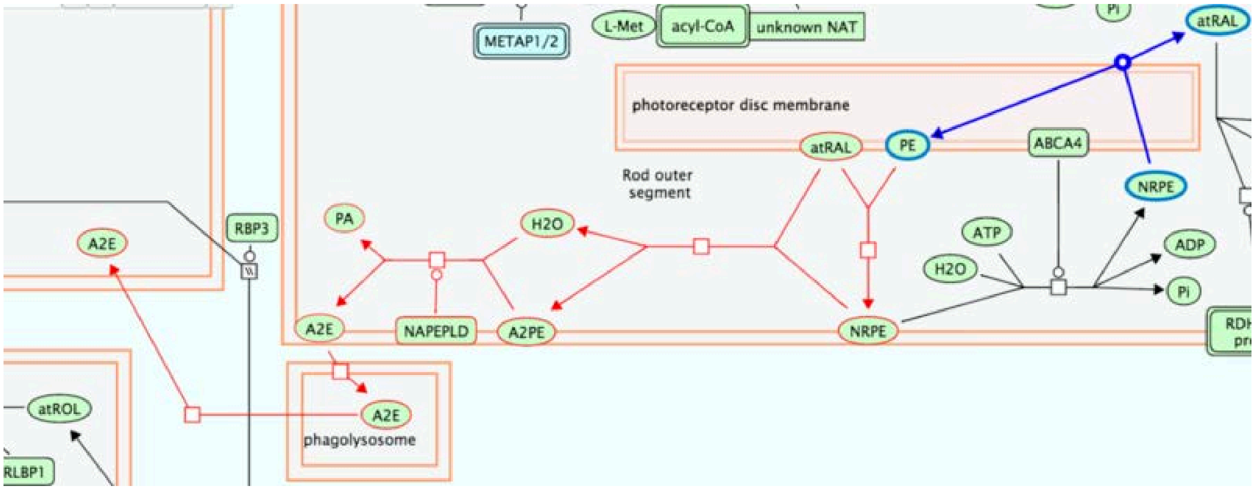


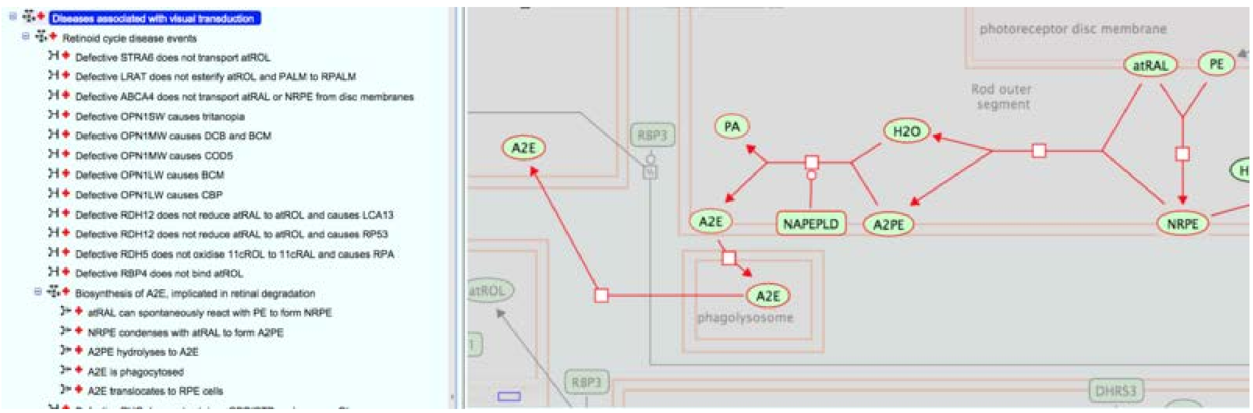
Now the LoF reaction can be viewed being switched between it and its normal counterpart and will work independently of these two drug-mutant reactions which only appear when looking at the disease counterpart.

Example 2 - diseases caused by accumulation of substrate over time:

Special case in Reactome where the reaction describing a particular substrate's metabolism occurs in normal physiology but over time, accumulation of the substrate leads to toxic consequences which can lead to disease. Examples are neurodegenerative diseases such as Alzheimer's and Parkinson's diseases, chronic obstructive pulmonary disease and retinal macular degeneration.

The bisretinoid A2E (di-retinoid-pyridinium-ethanolamine) is a major component of lipofuschin, a yellow-brown pigment grain composed mainly of lipids but also sugars and certain metals whose accumulation is associated with degenerative diseases. In the eye, A2E is the end-product of the condensation of 2 molecules of all-trans-retinal and phosphatidylethanolamine in photoreceptor outer disc membranes. Once formed, A2E is phagocytosed, together with outer segments, to retinal pigment epithelial (RPE) cells where it accumulates. There is no evidence as yet to indicate that A2E can be catabolised. The relationship between lipofuscin accumulation and retinal degeneration is illustrated by Stargardt disease type 1. Because the reactions can be considered "normal" (they occur as part of the normal metabolism of retinoids) as well as disease-causing, they are drawn in the normal pathway diagram and coloured red. To display in the disease view of the diagram, these reactions are added as components of the disease container pathway "Diseases associated with visual transduction". These reactions will now display in both normal and disease diagrams.



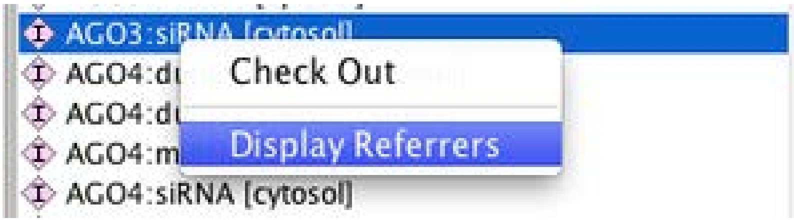


Example 3 - Disease-Specific Gene Expression

See discussion here (Appendix 12) of how to handle cases where gene expression is stimulated downstream of a disease-specific event, but the gene expression doesn't inherently include a disease entity.

Modifying, Deleting and Merging instances

If you locally modify or delete an instance you checked out from gk_central, it will also be **modified/deleted in gk_central** when you synchronize. You must **ALWAYS CHECK FIRST** that the instance you intend to modify/delete is not in use elsewhere, outside your local project. The best way to do this is to search for it in gk_central using the Database Browser Schema View, and select "display referrers". If it has any you didn't know about, do not modify or delete it!



Always use the curatortool database browser to check gk_central for referrers. Right clicking on a database object will allow you to look for referrers.
There may be circumstances when you think something should be modified or removed but if it has been used by another curator, check with them first, or contact an experienced curator for advice.

Procedure for deleting instances (Entities, Events or Regulation) that have been released:

A _deleted instances **MUST** be created any time a **released** Event or Entity is retired/deleted from the database. For example, when event restructuring occurs, any previously released events should NOT simply be flipped to do_release false. When this happens. this instance "disappears without a trace" to users. Before removing the instance, it should be replaced (by a replacement instance if available) in all referrers (where relevant) AND THEN deleted with the creation of a "_deleted" instance that gives a reason for deletion, a replacement instance (where relevant) and an explanation of the deletion in the curatorComment.

*Please note if you are deleting a pathway because it has been restructured and that pathway has a DOI, it is *essential* to specify the replacement instances in the _deleted instance. The historic author and reviewer instances should also be transferred to the replacement instance.

The procedure for choosing the appropriate "reason" and "replacement" instances is described below. This rule applies not only if an EWAS is retired or deleted, but also if the referenceGeneSequence associated with the EWAS is changed, e.g., because a UniProt entry is made obsolete and a different UniProt entry is to be used in its place.

Deleted Instance Options - When you opt to delete an instance you will be presented with a "Deleted Instance Dialog". If the instance has been released, a _deleted instance MUST be created. You can determine if the entity/event has been released by checking if the "released" attribute of its stableID is set to "True". **Note:** _deleted instances MUST have "reason" attribute filled. There are several options:

- **Killed and not replaced** - Used for events or physical entities that are simply wrong and must be removed. The event/entity is deleted and killed and not replaced in all of its referrers. It is CRITICAL that you also identify all diagrams in which the deleted instance has been used and make sure that the entity/event is deleted there as well. If you have the diagram open at the time that you delete the deprecated instance the diagram may updated automatically, but you should verify this. When selecting Killed and not replaced as "reason", the curator must also fill in a "curatorComments" that provides an explanation.

- **Duplicate** - When two separate but duplicate events or physical entities (A and B) are discovered within the data set. These events or physical entities must be exactly the same. Chose the duplicate instance with the greatest number of referrers (A)as the one to retain. B will be deleted. BEFORE deleting B, you will need to replace it with "A" in all of the referrers of "B". It is CRITICAL that you also identify all diagrams in which the deleted instance has been used and make sure that the entity/event is replaced there as well. If you have the diagram open at the time that you make the replacement of the deprecated instance the diagram may be updated automatically, but you should verify this.
When ready to delete B, use "Duplicate" as the reason in the _deleted instance. In this case, you MUST specify the replacement instance DB_ID in the "replacement" attribute. ****Note for discussion: it is also possible to resolve duplicate instances via a merge in the curator tool as long as ALL of the referrers of both A and B are in the project during the merge. However, in the case of the merge, it is not possible to specify which instance is preserved as far as I know. Also, with a merge the history is lost (there is no record of the deleted instance). Question: should merge be used for "unreleased" duplicates?
- **Split** - When a single entity is found to represent two separate and distinct entities. If the entity is to be split as a result of an external database change (UniProt) this is handled by the managing editor (L.Mattthews), and clean up instructions will be provided. If splitting an event such as a high level pathway, the deletion must be discussed with the editors. If the instance to be split is an event, the common practice is to duplicate and then partition the components accordingly. Generally we feel that split should not be used for reactions and low-level pathway events at this time.
- **Merged** - Very similar to duplicate. When a number of similar (but not exactly the same) events/entities A and B are merged to a single instance A and B is deleted. These could be steps in a pathway that are collapsed to a single step. When Uniprot merges entries (say 2 proteins thought to be different turn out to be the same), this is reported in the Uniprot update and is managed by the release manager. Again, BEFORE deleting B, you will need to replace it with "A" in all of the referrers of "B"! It is CRITICAL that you also identify all diagrams in which the deleted instance has been used and make sure that the entity/event is replaced there as well. If you have the diagram open at the time that you make the replacement of the deprecated instance the diagram may be updated automatically, but you should verify this.
When ready to delete B, use "Merged" as the reason in the _deleted instance. In this case, you MUST specify the replacement instance (A) DB_ID in the "replacement" attribute.
- **Obsolete** - Instance A is "killed" but has a replacement B. BEFORE deleting A, you will need to replace it with "B" in all of the referrers of "B"! It is CRITICAL that you also identify all diagrams in which the deleted instance has been used and

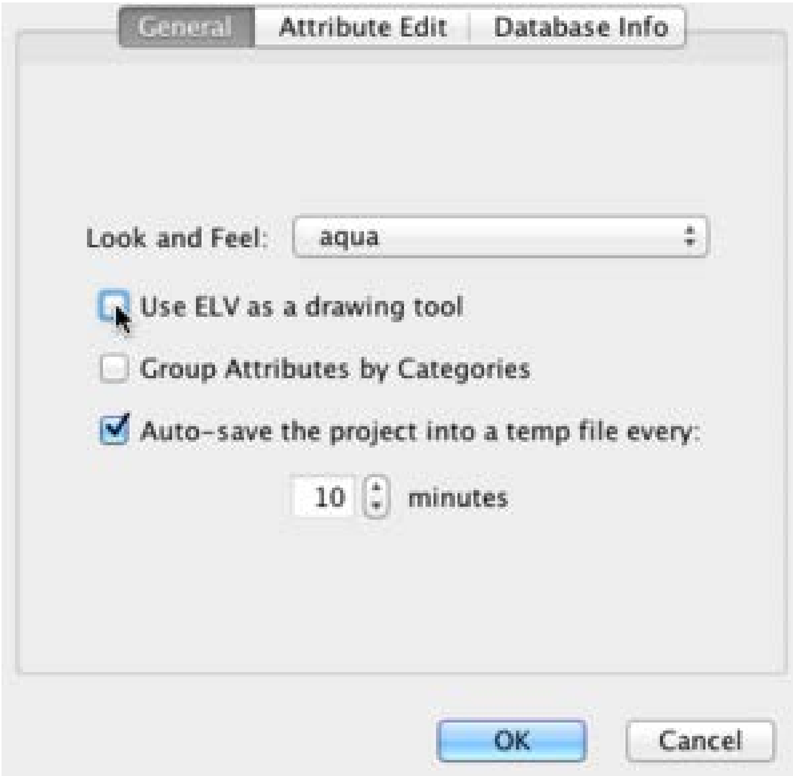
make sure that the entity/event is replaced there as well. If you have the diagram open at the time that you make the replacement of the deprecated instance the diagram may be updated automatically, but you should verify this. When ready to delete A, use "Obsolete" as the reason. In this case, you MUST specify the replacement instance (B) in the "replacement" attribute. Provide an explanation for replacement in the "curatorComments" attribute.

Updating modified reactions in the ELV (critical!)

When modifying existing reactions that have already been drawn into diagrams and deployed, be aware that not all changes made to participants in reactions will be automatically reflected in a diagram. For example, if you plan to replace the input of a reaction that is an EWAS with a set, the set will be drawn into the diagram, but the original EWAS will not disappear. It will have to be manually removed as described in the example below.

Example:
The original reaction had a single EWAS (FDX1L) as an input. I then made a defined set of 2 FDX proteins (FDX1, FDX1L) and substituted the defined set for the single EWAS as an input for the reaction. The ELV correctly showed the change (use of the set as an input) but also continued to show FDX1L (as a member of the defined set). I don't really want the EWAS FDX1L on the ELV anymore so I tried to delete it but received the error message "In the ELV drawing mode an entity cannot be deleted if it is linked to a reaction...").

To remove the EWAS (From the diagram only!!!) deselect the "Use ELV as a drawing tool" within the curatortool preferences panel, select to delete the EWAS from the ELV but retain the EWAS in the project. **After you delete these unwanted objects, please turn on this checkbox again to avoid any side-effects in the future.**



Toggling the "Use ELV as a drawing tool" option to off allows the drawing tool to directly modify the data in your project. Be very careful and be sure to reselect this option once you are done.

Please note that if an unused entity icon is still connected by lines to other entities (e.g., an orphan EWAS is connected to a set that includes it), you need to delete these connecting lines before the curator tool will allow you to delete the unused entity icon.

The bottom line is that you should always review and update the pathway diagram after modifying the reaction, and if the reaction is used in multiple diagrams ALL diagrams will need to be updated, so check for referrers of any reaction that is being modified.

Merging duplicate entities

To perform merging (for a complex for example), do the following:

1). In the database schema view, get these two complexes together in the instance list (the central panel) by doing REGEXP search: list DB_IDs together as this: DB_ID|DB_ID (Note | between two DB_IDs)

2). Check out these two instances.

3). While still in the view from step 1, search for Referrers for one complex, and check out all referrers in the Referrer dialog. Do this for the other instance.

4). In the local project schema view, merge these two complexes.

5). Find all instances that have been changed based on flag “>”.

6). Find pathway diagrams containing changed reactions. You may have to do this manually for the time being. In the db schema view, get pathways containing these changed reactions and see if they have diagrams (if the pathway listed does NOT have a diagram, look upwards at parent pathways until you find one that DOES have a diagram. Check these diagrams out by using “Build Project”. Re-open diagrams should fix references automatically. But you may need to re-position auto-inserted complex in the diagrams. So please check it manually to make sure.

7). Commit all changes into gk_central.
8). Delete the deprecated instance AFTER all other changes. Do NOT commit the deletion first. By committing the deletion first, the curator tool automatically updated referrers of the deleted instance by removing the deleted instance from their attributes and attached an InstanceEdit to these referrers. Consequently, you will see conflicts if the deletion is done first.

Deleting diagrams

If you are deleting a pathway diagram, you must delete the corresponding PathwayDiagram instance, you don't need to delete SearchableVertex instances that are associated with that diagram. In the future, this will be handled automatically.

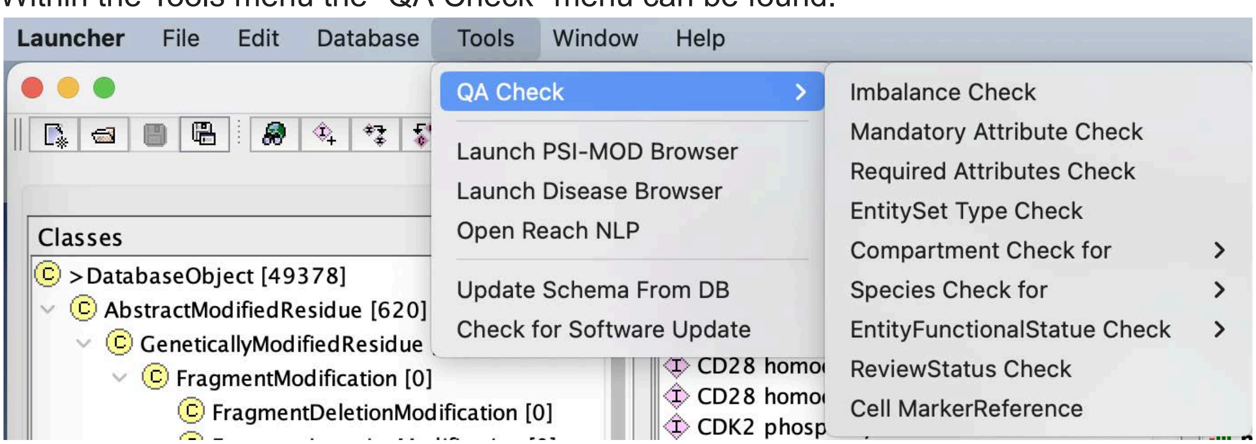
Diagram checks after deleting entities or reactionlikeEvents

If and when you need to delete an entity from gk_central, you must run the deleted object in diagram check over gk_central to find any diagrams that have used those instances.

QA

Project QA using the Curator Tool QA checks

Within the Tools menu the "QA Check" menu can be found.



QA Check Menu

This menu has nine separate QA script items within it.

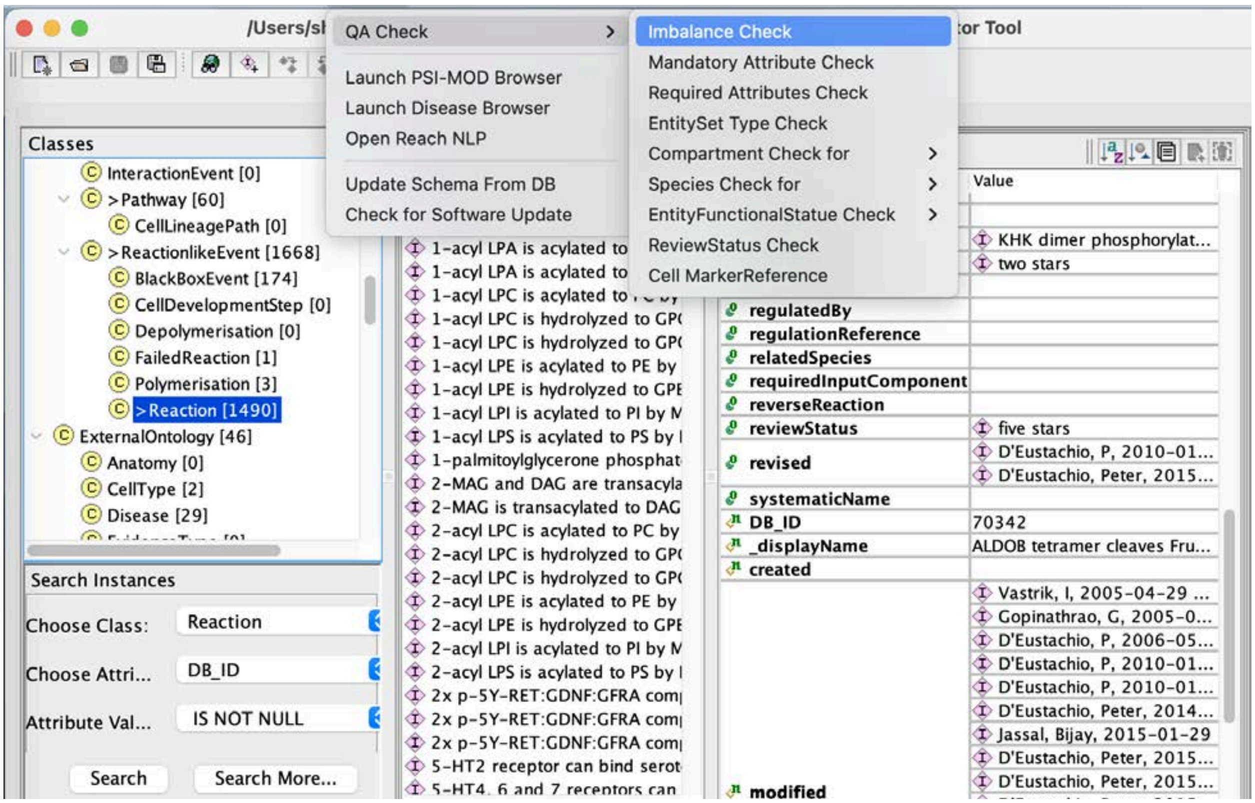
- Imbalance Check: checks that the EWAS referenceEntities present as input are also present as output and in the correct stoichiometry.
- Mandatory Attributes Check: Checks that the mandatory attributes for a class have been entered.
- Required Attributes Check: Checks that the required attributes for a class have been entered.
- EntitySet Type Check: Checks that set members belong to the same data model class.
- Compartment Check:
  - EntitySet: Looks for conflicts between compartment of a set and the compartment of members of that set.
  - Complex: Looks for conflicts between compartment of a complex and the compartment of individual components.
  - Reaction: Looks for conflicts between compartment of a reaction and the compartment(s) of individual participants.
- Species Check for
  - Entity Set: Looks for conflicts between the species of set members and the species of the set containing them..
  - Complex: Looks for conflicts between species of a complex and the species of individual components of that complex.
  - Reaction: Looks for conflicts between species of reactionLikeEvent and the species of all reaction participants.
  - Pathway: Looks for conflicts between the species of a pathway and the species of its contained events.
- EntityFunctionalStatus check
  - DiseaseEntity: Verifies that the diseaseEntity specified in the EntityFunctional status of an disease RLE is a participant of the disease RLE
  - NormalEntity:Verifies that the normalEntity specified in the EntityFunctional status of the disease RLE is a participant of the normalReaction specified in the disease RLE.
  -
- ReviewStatus Check: Flags Events have not been reviewed after structureModified instanceEdit date stamp. This should only be run on the sliceDB (editorial release manager does this at release time).

- Cell MarkerReference: Checks for proper assignment of Marker references in Cell instances.

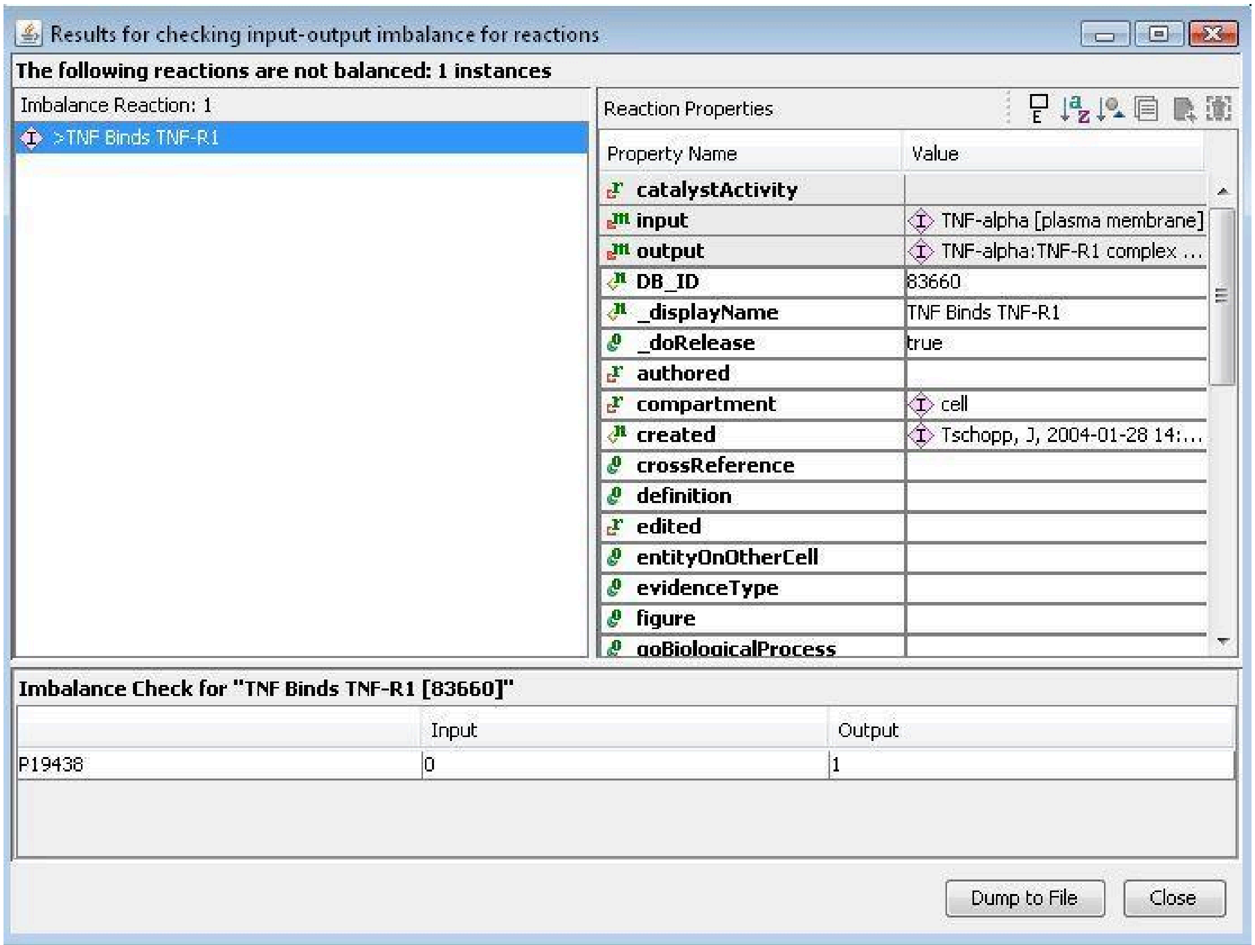
Note: In order for the QA checks to effectively pick up errors, the project that you are working on must be fully extracted from the database. Instructions on how to do a full extraction can be found [here](#) (Appendix C13).

Imbalance Check

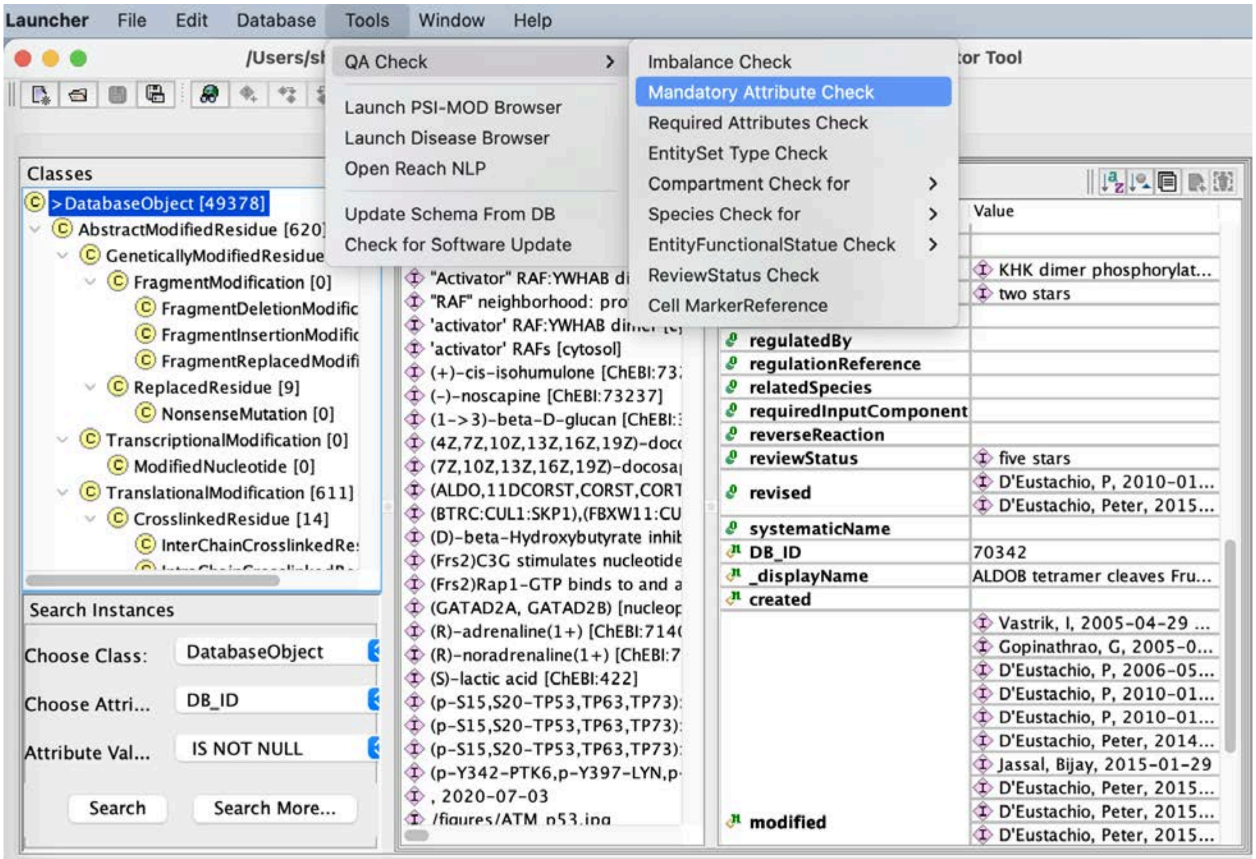
You must select Reactions in the hierarchy to perform this check.



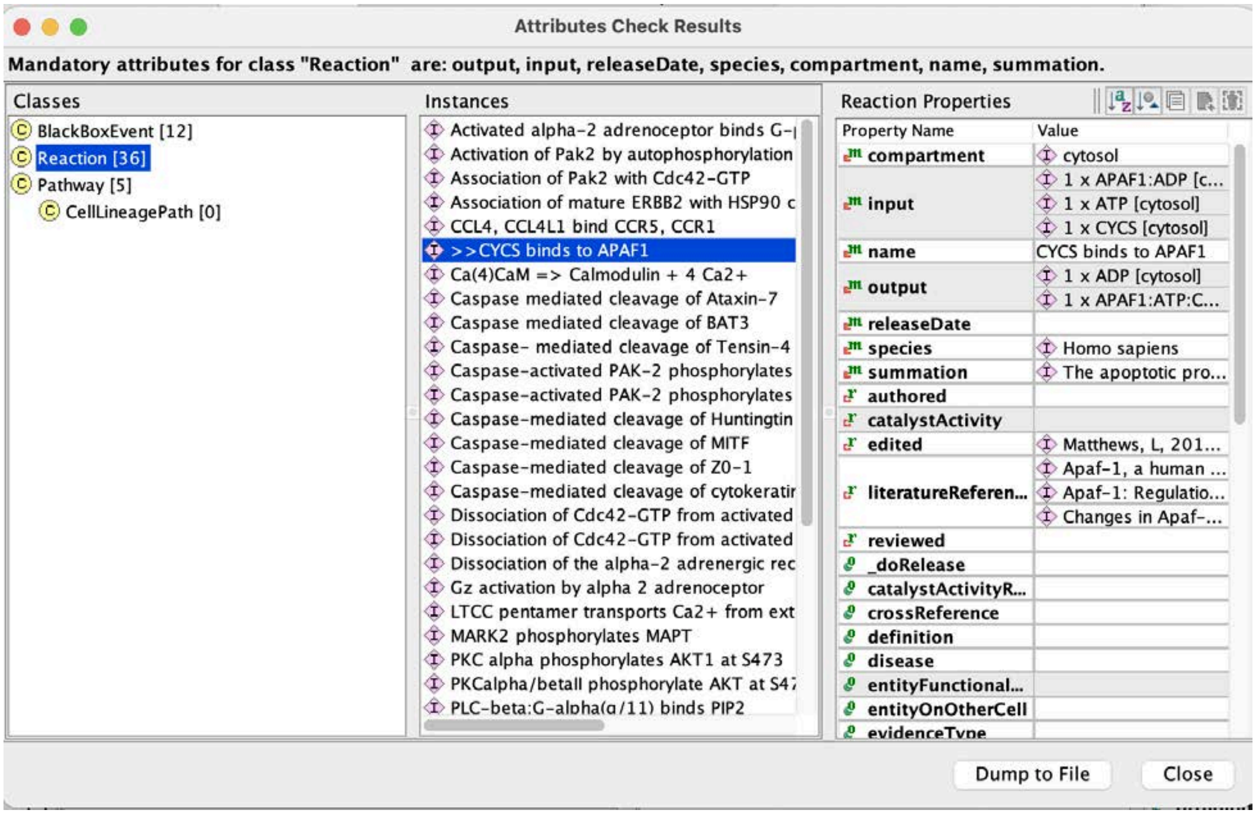
Reactions are flagged as cleavage reactions if the output differs from the input only that the output contains "fragments" of the input molecule. A true imbalance is shown below:



MANDATORY ATTRIBUTE CHECK



A list of instances missing mandatory attributes (by class) is shown. To make the missing attributes of the instance easier to see, you can use the "order attribute" button (downward arrow with circle and triangle) in the upper right side of the tool. This orders the instances by type (mandatory, required, optional...etc)



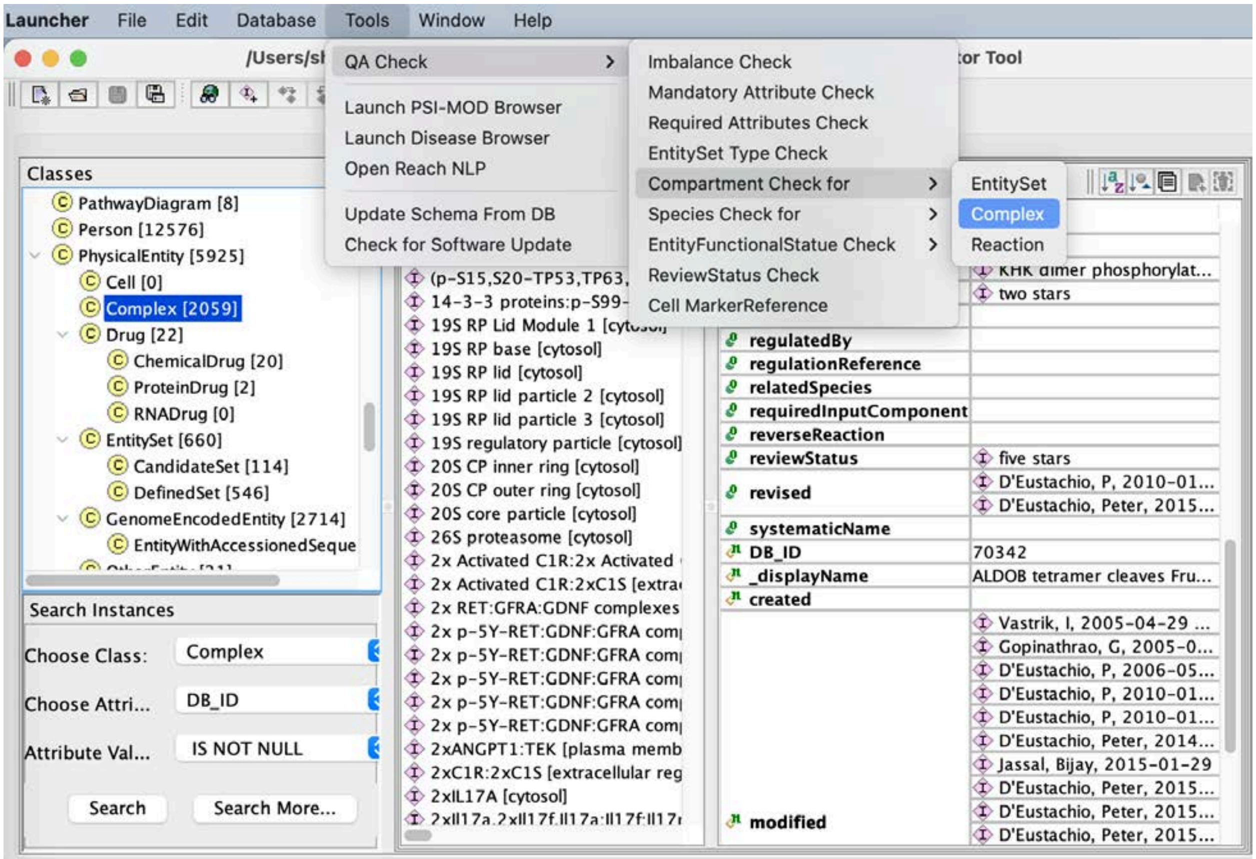
Required attribute check

This checks work in the same way that the mandatory attribute check works.

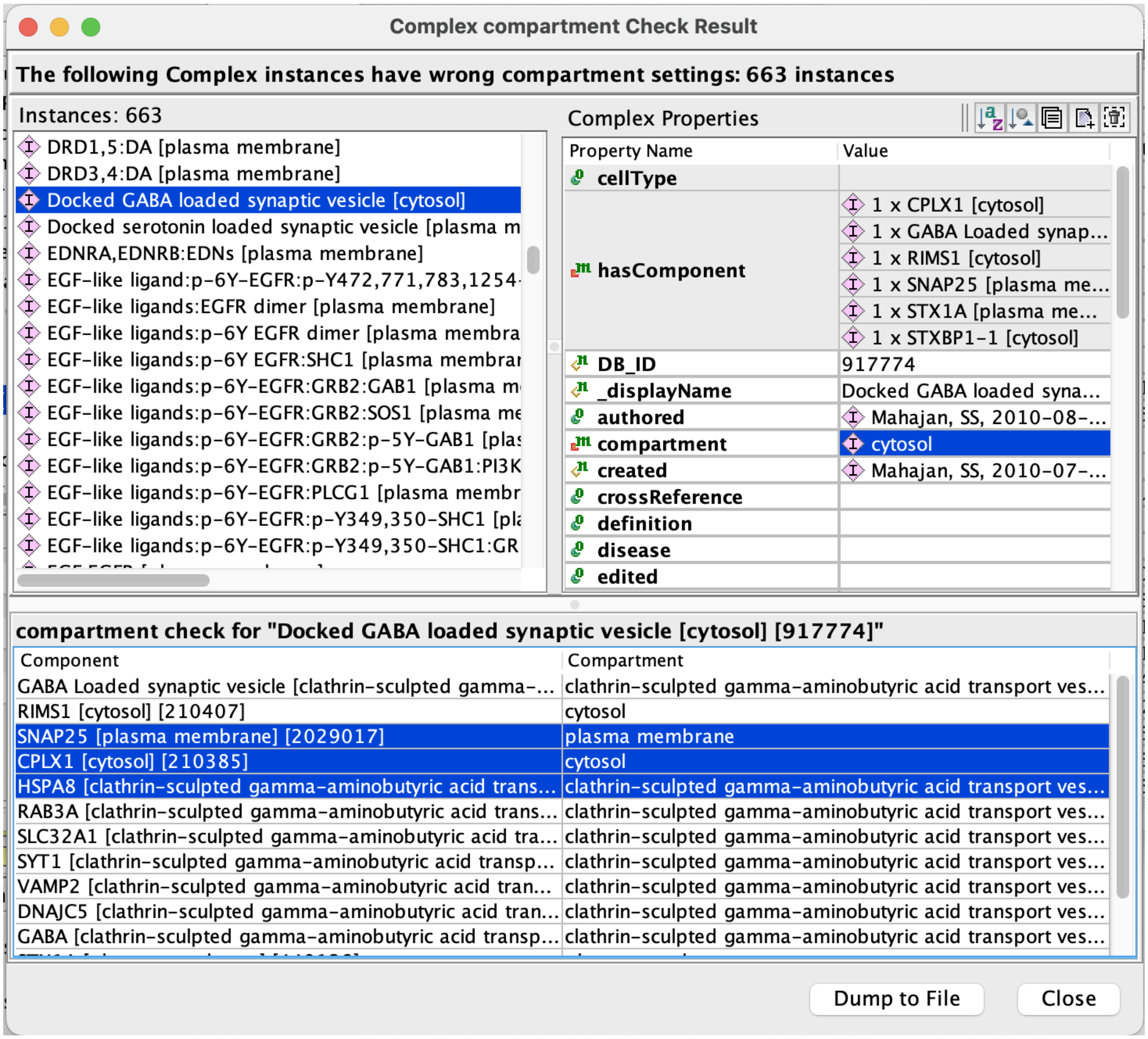
Compartment check

Select the class you want to check in the hierarchy panel

Curator tool QA check menu bar.



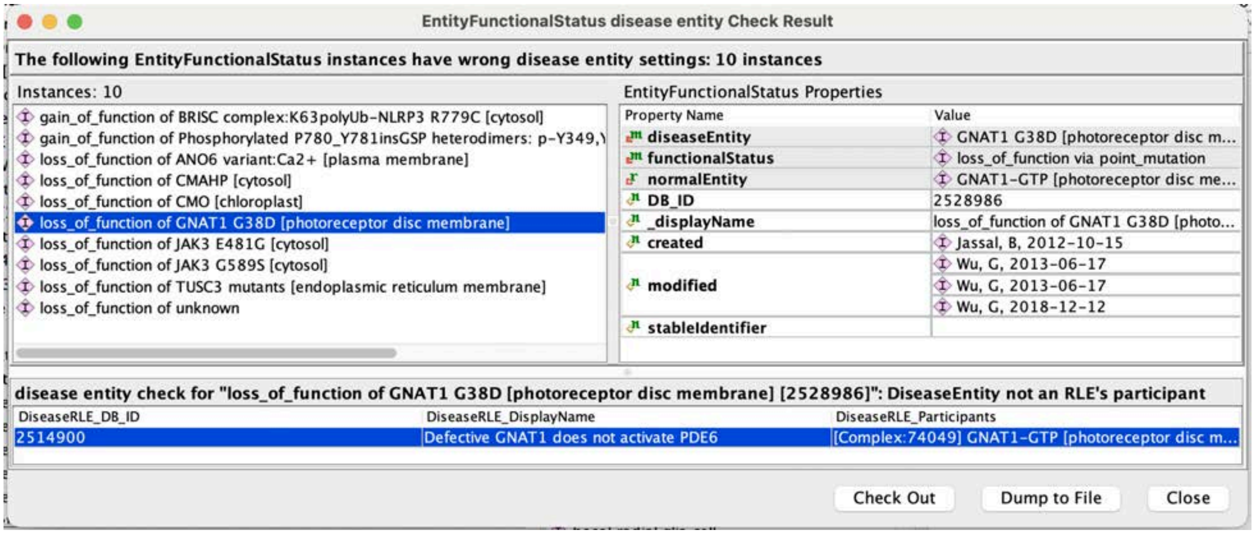
Compartment conflicts are indicated in the bottom of the dialog box.

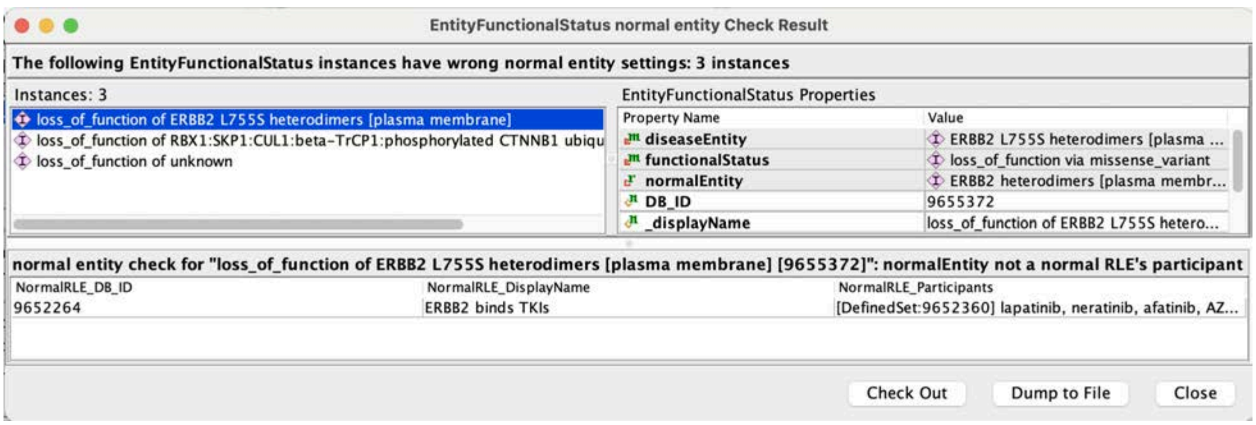


Species checks

These checks work in the same way that compartment checks work with the exception that Pathways are also checked for species conflicts.

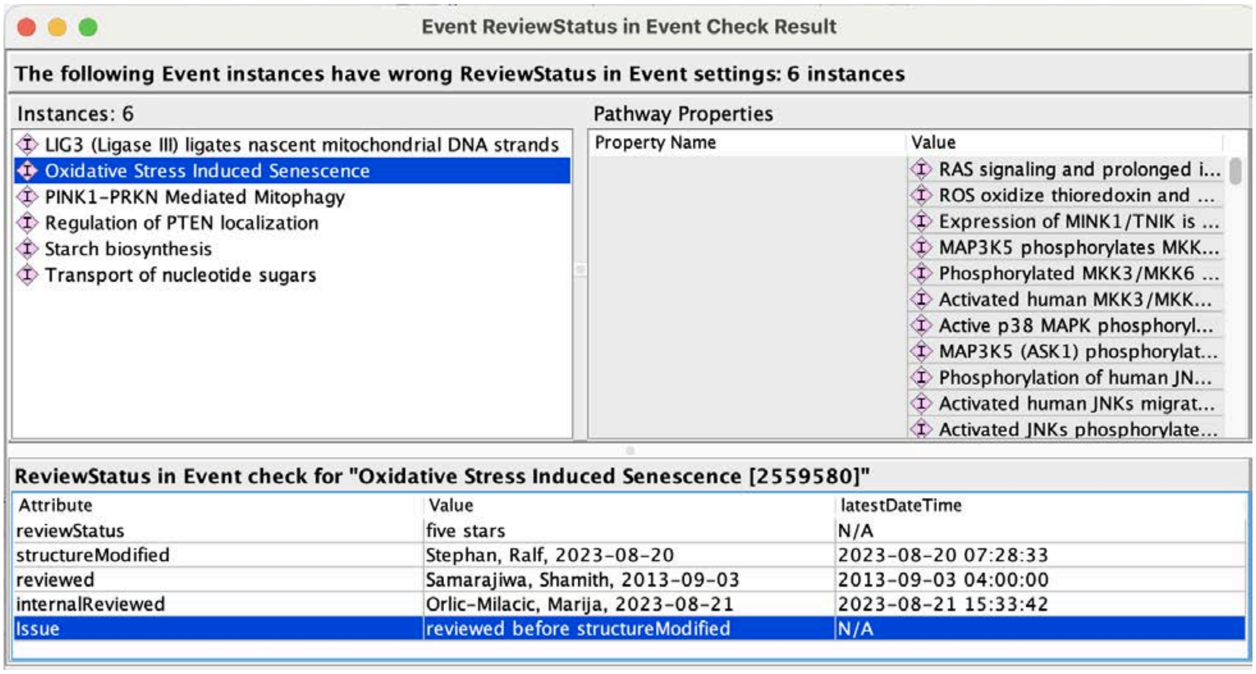
EntityFunctionalStatus checks





ReviewStatus checks

This Event check flags instances that have a structureModified date after the last reviewed date as displayed in the dialog box below:



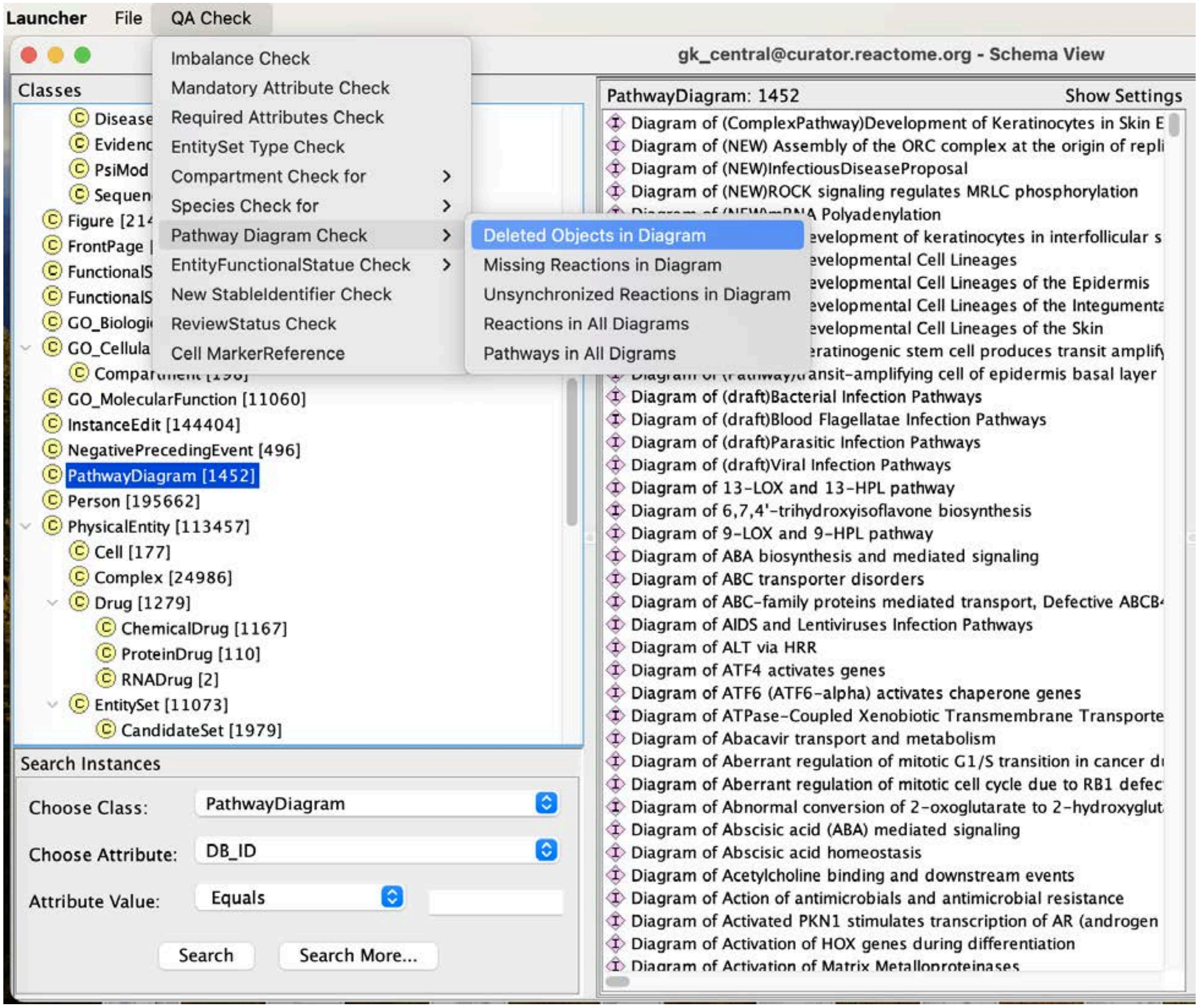
Cell MarkerReference

Diagram checks

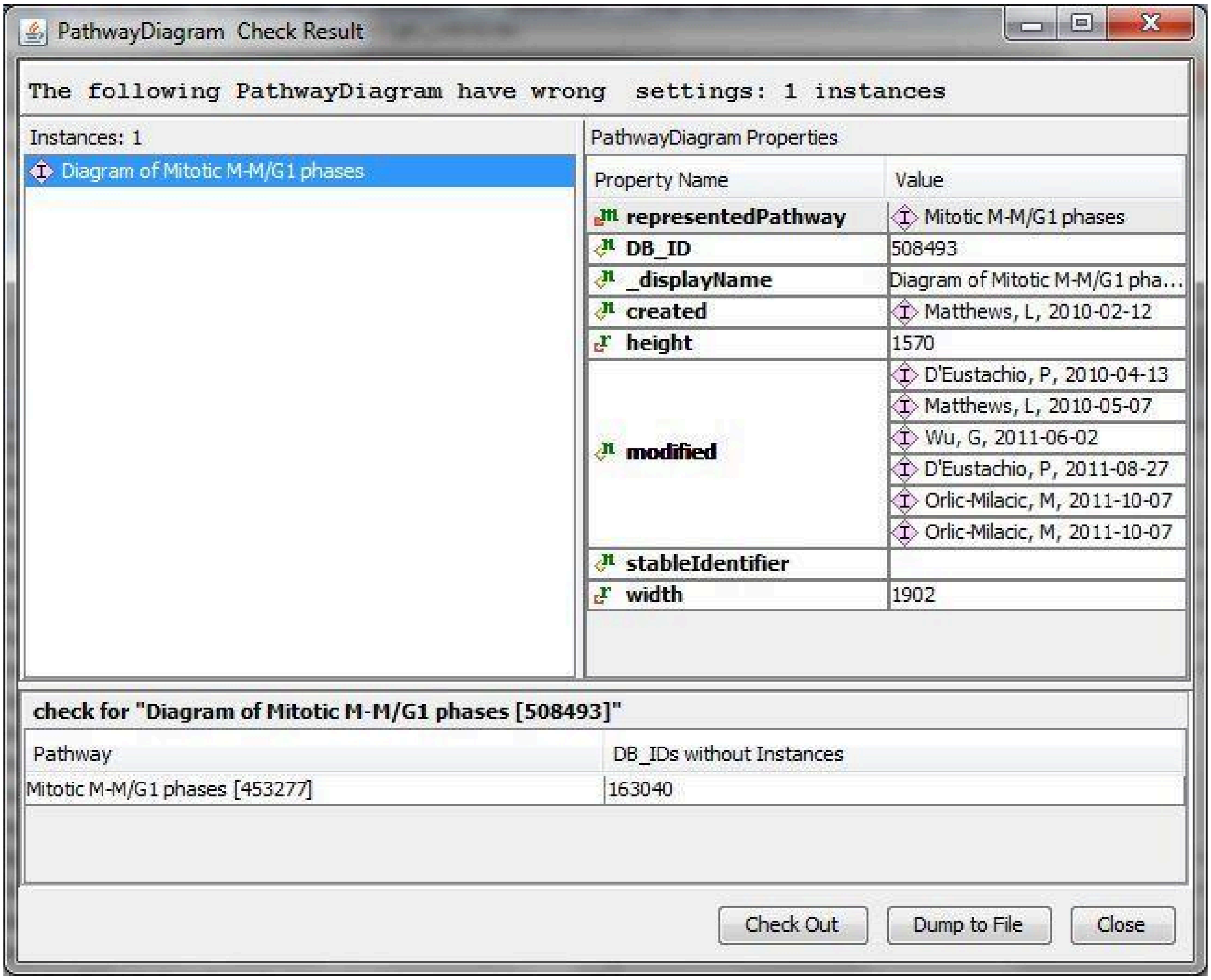
- Deleted objects in diagrams

When participants in a reaction are modified (or a reaction is deleted) in the instanceview of the curator tool, curators must verify that these changes have propagated to the diagram(s) in which the reaction has been drawn. This does not happen automatically. It is best to have the diagram open while modifying reaction instances in the instance browser. Make sure, before committing changes to reactions, that you have opened and scanned the diagram(s) housing any modified reactions to check for hanging entities.

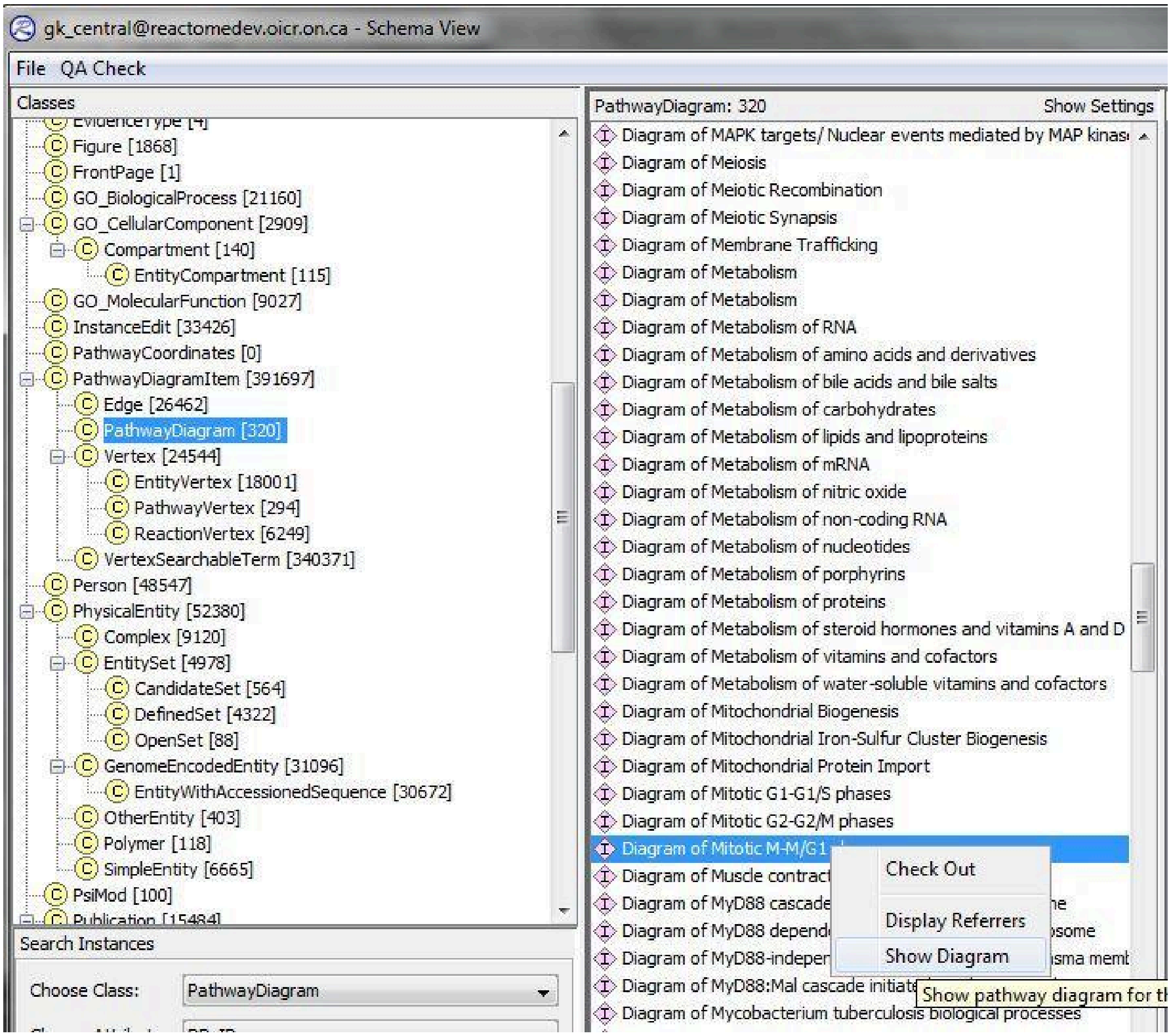
The “Deleted objects in diagram check” will help verify that you haven’t missed anything after committing. Running this check over all of gk_central will identify <u>any</u> diagram that has “unconnected entities” due to changes in the reaction or ReactionLikeEvent participants You may also select and check individual pathway diagrams of interest this way, but running across the whole database ensures you haven’t missed/forgotten other diagrams in which your modified reactions may be drawn.



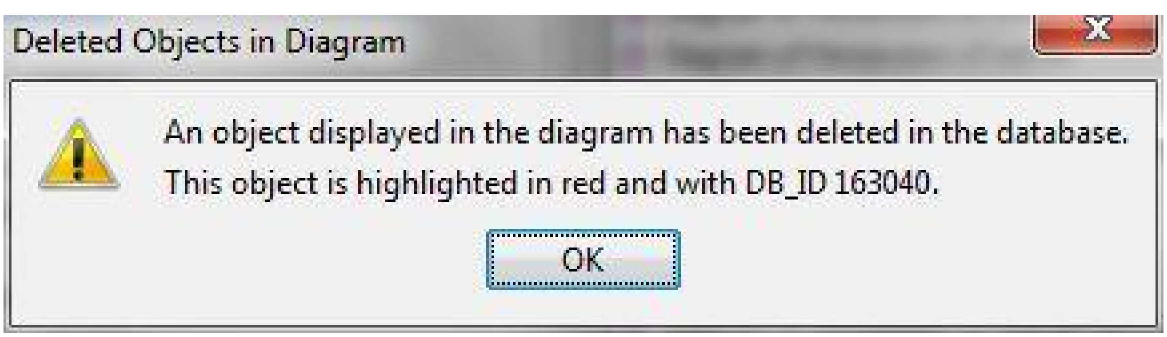
Any affected diagrams will be flagged. The DB_ID of the deleted instance will be displayed, but to see where the affected "objects" are in the diagram, you will need to view the diagram in gk_central.



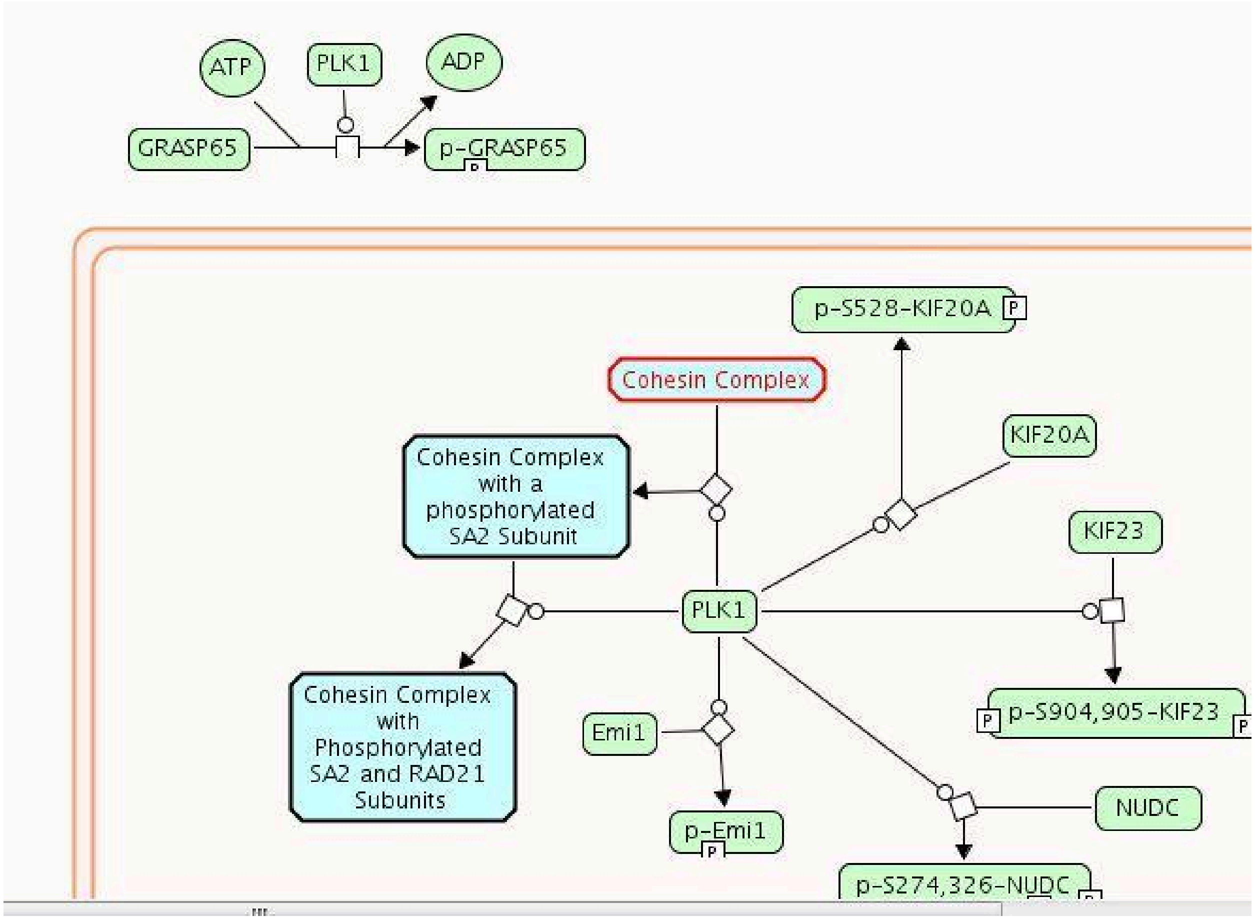
Select that diagram in gk_central, right click and opt to "Show diagram".



Affected objects will be flagged



and the objects in the diagrams highlighted in red.



QA of projects before release

Curators can reduce the number of QA fixes required at release time by running a number of QA checks in the curator tool before committing the final version of their projects.

- Top Six List Of Problems Identified During the Slice

    - Missing mandatory attributes
    - Species conflicts
    - Unbalanced reactions

Open the curator tool QA check under to "tool menu bar" at the top of the tool as shown above

- Missing modifiedResidue nodes in diagram

Curator tool QA checks:

1. All of the instances must be updated from gk_central in order for these checks to be meaningful. This checks out any shell attributes for the instances you want to check.

Everyday QA includes:

- Complete check-outs (No shell instances)
- Match instance in DB
- QA Tools

Curation Tools/Automation

LitBall tool for Exhaustive Literature Reference Searching

The LitBall tool is described here. A link to the slides describing its use can be found here.

Remote Attribute Search Tool

See here (Appendix C14) for description of the remote attribute search tool.

To search Reactome curator, use:
http://curator.reactome.org/cgi-bin/remoteattsearch2?DB=gk_central

To search reactomerelease, use:
https://release.reactome.org/cgi-bin/remoteattsearch2?DB=slice_test

Using IDG to find interacting pathways for gene lists:

Still have to ask Guanming to do this with a list you provide. To see results of the analyase of an example set of IBD related genes, see below:
Please see the results uploaded in this folder:

https://drive.google.com/drive/folders/1YzmrDrtnHMhX0VJg2A7tdAHi5GMxJiJd

Please read READM.txt first to see what they are. I used the scatter pathway plot (see the two plots in that HTML file) we developed at OHSU to show the hits of pathways. Please use that CSV file to see what genes each pathway is associated with.

I tried to overlay to ReactomeFoam, but it is just too complicated. You may check with Chuqiao if she has an easier way to overlay the counts using the CSV file. But I think these scatter plots should serve the purpose.

Using Reach NLP feature in the Curator Tool to process a personalized list of PMCIDs.

The Reach tool embedded in the curator tool uses automated NLP technology to extract pathways information from journal articles and suggest potentially annotateable interactions. The below protocol allows the curator to specify the list of PMCIDs through which REACH will search.

1. Formulate a PubMed search term, e.g. “(CDH2 OR "cadherin-2" OR "N-cadherin") AND (expression or transcription or promoter or binding or complex or interaction or regulation) and adhesion”

2. Note the number of results: e.g. 2,312 results for the above search term (on June 16, 2022). If a gene is poorly studied, it’s best to just use gene name + synonyms as a search term. If it’s a well-studied, popular gene, there will be too many results to process, most of which will be of no use for Reactome. In the latter case, it’s best to add additional words that imply some mechanistic evidence (second bracket in the above search term) and the annotation context (in the case of CDH2 it’s cell adhesion). You can build the search term interactively, depending on whether the number of results seems manageable and if the content of results seems appropriate.

3. Filter the results with the Free Full Text filter, the left-hand tab in PubMed: 1,230 results.

4. Change Display options FORMAT in the upper right corner from Summary (default) to PMID.

5. Save the list of PMIDs for your reference.

6. Go to https://www.ncbi.nlm.nih.gov/pmc/tools/idconv/ . This tool allows conversion of PMIDs to PMCIDs that need to be fed to NLP Reach. Paste the list of PMIDs and select csv as the output format (underneath the entry window). Unfortunately, the tool can process no more than 200 PMIDs at a time, unless an API is used – perhaps curators could get some training on how to do this. Otherwise, a long list of PMIDs will have to be processed in more than one run.

7. Look at the output file – if no PMCID is available for a PMID, then the PMCID column will stay blank. The PMIDs with no matching PMCID have to be manually evaluated for relevance by a curator.

8. Copy-paste the list of PMCIDs into a text editor such as Notepad and submit for processing (at the moment, forward to Guanming; the processing time is substantial, so it's best to do it well in advance).

9. After obtaining the results file in the .rrtf format (the file can be quite big, so a Dropbox needs to be used for sharing), follow Guanming's instructions for opening them in the Curator Tool:

    a. In the Curator Tool, use menu Tools/Open Reach NLP to open the window for Reach
    b. Find a button called "Import" at the bottom. Please use Import to load the results from the rrtf file

10. After importing the NLP results, you can visualize them using different filters:
    a. The number of rows will appear overwhelming. However, most rows do not contain proteins/genes/RNAs as their Entities (please note Entity A and Entity B). Filter the results to contain your gene of interest or one of its synonyms as either Entity A or Entity B. Currently, it's not possible to filter for several things at once using the OR statement, so you have to filter the results for keywords one-by-one. Please note that the filter is case-sensitive. For example, if you filter for "cdh2", you may get 0 results, while you will get quite a few if you filter with "CDH2".
    b. You can filter by the number of statements that support a relationship. Please note that more than 1 statement can originate from a single publication. Manual inspection is required to verify that a relationship is supported by at least 2 publications.

Examples of how to use the remoteattsearch tool can be found here (Appendix C14).

Identifying list members that are unique to one of two lists using Microsoft Excel

Here (Appendix C14) is a procedure that describes how to take two lists and compare entries to identify those that are present in only one of the two lists. This procedure can be useful, for example, to compare the list of proteins in the gk_central vs. live site to find those that are unreleased.

Preparing a pathway report for Internal and external review

When you prepare a word document for external review, include (or highlight within the document) ONLY the events that you are asking to be reviewed. Longer documents tend to be harder to get reviewed. If you need to include previously reviewed content for context, "gray" this out (using lighter gray text or highlighting) and make reference to this fact out at the top of the word document. Also, point this fact out to the managing editor sending out reviewer requests so that this fact can be pointed out in the letter as well.

Reporting Bugs

Please use JIRA to report bugs in the tool

Appendices

Appendix C1: Overview of Curation

Appendix C2: Quick Guide for New Curators

Appendix C3: Evidence Confidence

Appendix C4: UpdateTracker

Appendix C5: Data Model Glossary

Appendix C6: Small_molecule_renaming - small mols short names

Appendix C7: Review of the PrecedingEvent Suggestion Feature

Appendix C8: Best Practices: Using Chemical Compound Databases

Appendix C9: Cytomics documentation

Appendix C10: Multiple disease and EntityFunctionalStatus when using disease variant subsets

Appendix C11: GtP_all_interactions

Appendix C12. Disease-specific gene expression

Appendix C13: Deep Extraction of Events

Appendix C14: Remote Attribute Search and examples

Appendix C15: Remove duplicates MSExcel

Appendix C16: RHEA-GO alignment

V95_Dec 2025
- Added a description of naming guideline for disease pathways where the same defective RLE affects more than one pathway.

V94_Sept 2025
- Added a note to verify normalPathway has been assigned to a disease pathway.

V93_March 2025
- Added a link to the LitBall comprehensive literature search tool in the Literature reference description and in the Tool section.

V92_March 2025
- Added missing description of ActiveUnit of CatalystActivity and its intended use.
- Added precisions about handling deletion/replacement of pathways with DOIs
- Added Appendix C16: RHEA-GO alignment
- Updated outdated regulation instance descriptions.
- Added a note about how not to create a empty pathway diagram instance by accident under: Editing existing pathways and corresponding diagrams

V91_Dec 2024
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- Added: Under Curation Tools, added a description of how to identify interacting pathways for a list of genes using the IDG tool.

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- incorporation of Cell lineage curation guide
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